

A MODERN ALMAGEST:  
An Updated Version of Ptolemy's Model of the Solar  
System

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# Table of Contents

|                                                         |           |
|---------------------------------------------------------|-----------|
| <b>Preface</b>                                          | <b>v</b>  |
| <b>1 Introduction</b>                                   | <b>1</b>  |
| 1.1 Euclid's Elements and Ptolemy's Almagest . . . . .  | 1         |
| 1.2 Ptolemy's model of the Solar System . . . . .       | 2         |
| 1.3 Copernicus's model of the Solar System . . . . .    | 6         |
| 1.4 Kepler's model of the Solar System . . . . .        | 8         |
| 1.5 Purpose of book . . . . .                           | 9         |
| <b>2 Spherical astronomy</b>                            | <b>11</b> |
| 2.1 The celestial sphere . . . . .                      | 11        |
| 2.2 Celestial motions . . . . .                         | 11        |
| 2.3 Celestial coordinates . . . . .                     | 11        |
| 2.4 The ecliptic circle . . . . .                       | 13        |
| 2.5 Ecliptic coordinates . . . . .                      | 14        |
| 2.6 The signs of the zodiac . . . . .                   | 17        |
| 2.7 Ecliptic declinations and right ascensions. . . . . | 17        |
| 2.8 Local horizon and meridian . . . . .                | 18        |
| 2.9 Horizontal coordinates . . . . .                    | 21        |
| 2.10 Meridian transits . . . . .                        | 22        |
| 2.11 Principal terrestrial latitude circles . . . . .   | 23        |
| 2.12 Equinoxes and solstices . . . . .                  | 23        |
| 2.13 Terrestrial climes . . . . .                       | 25        |
| 2.14 Ecliptic ascensions . . . . .                      | 25        |
| 2.15 Azimuth of ecliptic ascension point . . . . .      | 28        |
| 2.16 Ecliptic altitude and orientation . . . . .        | 29        |
| 2.17 Tables . . . . .                                   | 33        |
| <b>3 Dates and times</b>                                | <b>61</b> |
| 3.1 Julian and Gregorian calendars . . . . .            | 61        |
| 3.2 Julian day number . . . . .                         | 61        |
| 3.3 Determination of Julian day numbers . . . . .       | 61        |
| 3.4 Tables . . . . .                                    | 62        |
| <b>4 Planetary orbit models</b>                         | <b>65</b> |
| 4.1 The model of Kepler . . . . .                       | 65        |
| 4.2 The model of Hipparchus . . . . .                   | 68        |
| 4.3 The model of Ptolemy . . . . .                      | 69        |
| 4.4 The model of Copernicus . . . . .                   | 70        |

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|          |                                                           |            |
|----------|-----------------------------------------------------------|------------|
| <b>5</b> | <b>The Sun</b>                                            | <b>73</b>  |
| 5.1      | Solar ecliptic longitude model . . . . .                  | 73         |
| 5.2      | Determination of solar ecliptic longitude . . . . .       | 75         |
| 5.3      | Example solar longitude calculations . . . . .            | 75         |
| 5.4      | Determination of equinox and solstice dates . . . . .     | 77         |
| 5.5      | The equation of time . . . . .                            | 79         |
| 5.6      | Solar distance model . . . . .                            | 80         |
| 5.7      | Tables . . . . .                                          | 81         |
| <br>     |                                                           |            |
| <b>6</b> | <b>The Moon</b>                                           | <b>87</b>  |
| 6.1      | Lunar ecliptic longitude model . . . . .                  | 87         |
| 6.2      | Determination of lunar ecliptic longitude . . . . .       | 88         |
| 6.3      | Example lunar longitude calculations . . . . .            | 89         |
| 6.4      | Lunar ecliptic latitude model . . . . .                   | 91         |
| 6.5      | Determination of lunar ecliptic latitude . . . . .        | 92         |
| 6.6      | Example lunar latitude calculations . . . . .             | 92         |
| 6.7      | The length of a month . . . . .                           | 93         |
| 6.8      | Lunar distance model . . . . .                            | 94         |
| 6.9      | Lunar parallax . . . . .                                  | 96         |
| 6.10     | Tables . . . . .                                          | 98         |
| <br>     |                                                           |            |
| <b>7</b> | <b>Syzygies and Eclipses</b>                              | <b>107</b> |
| 7.1      | Syzygies . . . . .                                        | 107        |
| 7.2      | Lunar-solar elongation model . . . . .                    | 108        |
| 7.3      | Determination of lunar-solar elongation . . . . .         | 108        |
| 7.4      | Example syzygy calculations . . . . .                     | 109        |
| 7.5      | Solar and lunar eclipses . . . . .                        | 111        |
| 7.6      | Example eclipse calculations . . . . .                    | 114        |
| 7.7      | Eclipse statistics . . . . .                              | 115        |
| 7.8      | Eclipse cycles . . . . .                                  | 116        |
| 7.9      | Tables . . . . .                                          | 118        |
| <br>     |                                                           |            |
| <b>8</b> | <b>The superior planets</b>                               | <b>123</b> |
| 8.1      | Planetary ecliptic longitude model . . . . .              | 123        |
| 8.2      | Determination of the ecliptic longitude of Mars . . . . . | 127        |
| 8.3      | Example martian ecliptic longitude calculations . . . . . | 128        |
| 8.4      | Conjunction, opposition, and station dates . . . . .      | 131        |
| 8.5      | Determination of ecliptic longitude of Jupiter . . . . .  | 133        |
| 8.6      | Determination of ecliptic longitude of Saturn . . . . .   | 135        |
| 8.7      | Tables . . . . .                                          | 137        |

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|                                                                      |                |
|----------------------------------------------------------------------|----------------|
| <b>9 The inferior planets</b>                                        | <b>151</b>     |
| 9.1 Determination of ecliptic longitude . . . . .                    | 151            |
| 9.2 Determination of ecliptic longitude of Venus . . . . .           | 153            |
| 9.3 Conjunction and greatest elongation dates . . . . .              | 156            |
| 9.4 Determination of ecliptic longitude of Mercury . . . . .         | 157            |
| 9.5 Tables . . . . .                                                 | 160            |
| <br><b>10 Planetary latitudes</b>                                    | <br><b>169</b> |
| 10.1 Determination of ecliptic latitude of superior planet . . . . . | 169            |
| 10.2 Determination of ecliptic latitude of Mars . . . . .            | 171            |
| 10.3 Determination of ecliptic latitude of Jupiter . . . . .         | 173            |
| 10.4 Determination of ecliptic latitude of Saturn . . . . .          | 174            |
| 10.5 Determination of ecliptic latitude of inferior planet . . . . . | 175            |
| 10.6 Determination of ecliptic latitude of Venus . . . . .           | 177            |
| 10.7 Determination of ecliptic latitude of Mercury . . . . .         | 179            |
| 10.8 Tables . . . . .                                                | 180            |
| <br><b>Technical terms</b>                                           | <br><b>191</b> |
| <br><b>Index of symbols</b>                                          | <br><b>197</b> |
| <br><b>Bibliography</b>                                              | <br><b>199</b> |



## Preface

This book is devoted to a re-examination of Claudius Ptolemy's *Almagest*, one of the foundational scientific works of antiquity. Although often contrasted unfavorably with Euclid's *Elements*, and frequently dismissed as overly complex, and misguided in its geocentric approach, the *Almagest* remains a remarkable achievement: namely, a mathematically sophisticated and observationally grounded attempt to describe the motions of the heavens. The aim of my work is to reassess the scientific merits of Ptolemy's model by reconstructing it in a modern framework. Using contemporary mathematical methods, as well as standard astronomical conventions and terminology, I seek to render the structure and logic of the Ptolemaic system more accessible to the modern reader. In the process, I demonstrate that many common criticisms of the *Almagest* are either overstated or based on misunderstandings, and that, when properly interpreted, Ptolemy's model can be viewed as a surprisingly accurate approximation to the later Keplerian description of planetary motion. I do not attempt to reproduce every aspect of the original *Almagest*. Rather than revisiting the construction of trigonometric tables or ancient observational techniques, I focus on the essential geometric and kinematic ideas underlying the model. Certain known deficiencies in Ptolemy's system are corrected, and, where appropriate, his methods are reformulated in terms of equivalent but more transparent schemes. The result is a streamlined yet faithful reconstruction of the Ptolemaic system—one that preserves its historical character while demonstrating its continued mathematical and scientific interest. It is my hope that this approach will allow readers to better appreciate both the ingenuity of Ptolemy's achievement and its place in the development of astronomical science.

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# 1. Introduction

## 1.1 *Euclid's Elements and Ptolemy's Almagest*

The modern world inherited two major scientific treatises from the civilization of ancient Greece. The first of these, the *Elements* (Στοιχεία) of Euclid (Εὐκλείδης), is a large compendium of mathematical theorems concerning geometry, proportion, and number theory. These theorems were not necessarily discovered by Euclid himself. In fact, the theorems were largely the work of earlier mathematicians, such as Pythagoras (Πυθαγόρας) and his school, Eudoxos of Cnidus (Εὐδόξος ὁ Κνίδιος), and Theaetetus of Athens (Θεαίτητος ὁ Ἀθηναῖος). However, Euclid is credited with arranging the theorems in a logical manner, such that a given theorem only depends on ones appearing before it in the treatise. Euclid also demonstrated that all of the theorems follow from a set of definitions combined with five simple, but unproved, axioms. The *Elements* is rightly regarded as the first, largely successful, attempt to construct an axiomatic system in mathematics, and is still held in high esteem within the scientific community.

The second treatise, the *Almagest*<sup>1</sup> of Claudius Ptolemy (Κλαύδιος Πτολεμαῖος), is an attempt to find a simple geometric explanation for the apparent motions of the Sun, the Moon, and the five visible (to the naked eye) planets in the Earth's sky. On the basis of his own naked-eye observations, combined with those of earlier astronomers such as Hipparchus of Nicaea (Ἰππάρχος ὁ Νικαεύς), Ptolemy proposed a model of the Solar System in which the Earth is stationary. According to this model, the Sun moves in a circular orbit, (nearly) centered on the Earth, that maintains a fixed inclination of about 23° to the terrestrial equator. Furthermore, each planet moves on the rim of a small circle called an *epicycle* (ἐπίκυκλος in Greek), whose center revolves around the Earth on a large eccentric circle called a *deferent* (ἀποφορά in Greek). (See Figure 8.2.) The planetary deferents and epicycles also maintain fixed inclinations,<sup>2</sup> which are all fairly close to 23°, to the terrestrial equator.

The scientific reputation of the *Almagest* has not fared as well as that of Euclid's *Elements*. Nowadays, it is a commonly held belief, even amongst scientists, that Ptolemy's mistaken adherence to the tenets of Aristotelian philosophy—in particular, the immovability of the Earth, and the necessity for heavenly bodies to move uniformly in circles—led him to construct an over-complicated, unwieldy, and faintly ridiculous model of planetary motion. As is well known, Ptolemy's model was superseded in 1543 AD by the heliocentric model of Nicolaus Copernicus, in which the planets revolve about the Sun in circular orbits.<sup>3</sup> The Copernican model was, in turn, superseded in the early 1600's AD by the, ultimately correct, model of Johannes Kepler, in which the planets revolve about the Sun in eccentric elliptical orbits.

The aim of this treatise is to re-examine the scientific merits of Ptolemy's *Almagest*.

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<sup>1</sup> The true title of this work is Μαθηματικὴ Σύνταξις, which means "Mathematical Treatise". The treatise was later called Ἡ Μεγάλη Σύνταξις, meaning "The Great Treatise", and the superlative form of the adjective, μεγίστη, with the Arabic article "al" prepended, lies behind the Arabic name from which the English name *Almagest* derives.

<sup>2</sup> Actually, Ptolemy erroneously allowed the inclinations of the deferents and epicycles to vary slightly.

<sup>3</sup> In fact, the planets revolve on small circular epicycles, whose centers revolve around the Sun on eccentric circular deferents.

## 1.2 *Ptolemy's model of the Solar System*

Claudius Ptolemy lived and worked in the city of Alexandria, capital of the Roman province of Egypt, during the reigns of the later Flavian and the Antonine emperors. Ptolemy was heir—via the writings of Euclid, and later mathematicians such as Apollonius of Perga (Ἀπολλώνιος ὁ Περγαῖος), and Archimedes of Syracuse (Ἀρχιμήδης ὁ Συρακούσιος)—to the considerable mathematical knowledge of geometry and arithmetic acquired by the civilization of ancient Greece. Ptolemy also inherited an extensive Babylonian and ancient Greek body of knowledge concerning observational and theoretical astronomy. The most important astronomer prior to Ptolemy was undoubtedly Hipparchus of Nicaea (second century BC), who produced the first circular chord table (which is key to Ptolemy's trigonometry), compiled the first star catalog, developed the theory of solar motion used by Ptolemy, discovered the precession of the equinoxes, and collected an extensive set of astronomical observations—some of which he made himself, and some of which he acquired from Babylonian records—which were available to Ptolemy (possibly via the famous Library of Alexandria). Other astronomers who made significant contributions prior to Ptolemy include Meton of Athens (Μέτων ὁ Ἀθηναῖος, 5th century BC), Eudoxos of Cnidus (5th/4th century BC), Callipus of Cyzicus (Καλλίππος ὁ Κυζικίος, 4th century BC), Aristarchus of Samos (Ἀρίσταρχος ὁ Σάμιος, 4th/3rd century BC), Eratosthenes of Cyrene (Ἐρατοσθένης ὁ Κυρηναῖος, 3rd/2nd century BC), and Menelaus of Alexandria (Μενέλαος ὁ Ἀλεξανδρεὺς, 1st century AD).

Ptolemy's aim in the *Almagest* is to construct a kinematic model of the Solar System, as seen from the Earth. In other words, the *Almagest* outlines a relatively simple geometric model that describes the apparent motions of the Sun, Moon, and planets, relative to the Earth, but does not attempt to explain why these motions occur. (The models of Copernicus and Kepler are similar to that of Ptolemy in this respect.) As such, the fact that the model described in the *Almagest* is geocentric in nature is a non-issue, because the Earth is stationary in its own frame of reference. This is not to say that the heliocentric hypothesis is without advantages. As we shall see, the assumption of heliocentricity allowed Copernicus to determine, for the first time, the ratios of the mean radii of the various planets in the Solar System.

We now know, from the work of Kepler, that planetary orbits are actually ellipses that are confocal with the Sun. Such orbits possess two main properties. First, they are eccentric; that is, the Sun is displaced from the geometric center of the orbit. Second, they are elliptical; that is, the orbit is elongated along a particular axis. Now, Keplerian orbits are characterized by a quantity,  $e$ , known as the *eccentricity*, that measures their deviation from circularity. It is easily demonstrated that the eccentricity of a Keplerian orbit scales as  $e$ , whereas the corresponding degree of elongation scales as  $e^2$ . Because the orbits of the visible planets in the Solar System all possess relatively small values of  $e$  (that is,  $e \leq 0.21$ ), it follows that, to an excellent approximation, these orbits can be represented as eccentric circles; that is, circles that are not quite concentric with the Sun. In other words, we can neglect the ellipticities of planetary orbits compared to their eccentricities. This is exactly what Ptolemy does in the *Almagest*. It follows that Ptolemy's assumption that heavenly bodies move in circles is actually one of the main strengths of his model, rather than being the main weakness, as is commonly supposed.

Kepler's second law of planetary motion states that the radius vector connecting a planet to the Sun sweeps out equal areas in equal time intervals. In the approximation in which planetary

orbits are represented as eccentric circles, this law implies that a typical planet revolves around the Sun at a non-uniform rate. However, it is easily demonstrated that the non-uniform rotation of the radius vector connecting the planet to the Sun implies a uniform rotation of the radius vector connecting the planet to the so-called *equant*; that is, the point diagrammatically opposite to the Sun with respect to the geometric center of the orbit. (See Figure 1.1.) Ptolemy discovered the equant scheme empirically, and used it to control the non-uniform rotation of the planets in his model. In fact, this discovery is one of Ptolemy's main claims to fame.

It follows, from the previous discussion, that the geocentric model of Ptolemy is equivalent to a heliocentric model in which the various planetary orbits are represented as eccentric circles, and in which the radius vector connecting a given planet to its corresponding equant revolves at a uniform rate. In fact, Ptolemy's model of planetary motion can be thought of as a version of Kepler's model that is accurate to first order in the planetary eccentricities. (See Chapter 4.) According to the Ptolemaic scheme, from the point of view of the Earth, the orbit of the Sun is described by a single circular motion, whereas that of a planet is described by a combination of two circular motions. In reality, the single circular motion of the Sun represents the (approximately) circular motion of the Earth around the Sun, whereas the two circular motions of a typical planet represent a combination of the planet's (approximately) circular motion around the Sun, and the Earth's motion around the Sun. Incidentally, the popular story that Ptolemy's scheme requires an absurdly large number of circles in order to fit the observational data to any degree of accuracy has no basis in fact. Actually, Ptolemy's model of the Sun and the planets, which fits the data very well, only contains 12 circles (that is, 6 deferents and 6 epicycles). (There are 6 epicycles because Mercury possesses an additional spurious epicycle.) Ptolemy's lunar model contains an additional 3 circles. The additional common tale that medieval astronomers added more and more epicycles to Ptolemy's model in a vain attempt to make it fit the observations better also has no basis in fact.

Ptolemy is often accused of slavish adherence to the tenants of Aristotelian philosophy, to the overall detriment of his model. However, despite Ptolemy's conventional geocentrism, his model of the Solar System deviates from orthodox Aristotelian philosophy in a number of crucially important respects. In his treatise "On the Heavens" (*Περὶ Οὐρανοῦ*) Aristotle (*Ἀριστοτέλης*) argues, from a purely philosophical standpoint, that heavenly bodies should move in single uniform circles. However, in the Ptolemaic system, the motion of the planets is a combination of two circular motions. Moreover, at least one of these motions is non-uniform. Aristotle also argues, again from purely philosophical grounds, that the Earth is located at the exact center of the universe, about which all heavenly bodies orbit in concentric circles. However, in the Ptolemaic system, the Earth is slightly displaced from the center of the universe. Indeed, there is no unique center of the universe, because the circular orbit of the Sun and the circular planetary deferents all have slightly different geometric centers, none of which coincide with the Earth. As described in the *Almagest*, the non-orthodox (from the point of view of Aristotelian philosophy) aspects of Ptolemy's model were ultimately dictated by observations. This suggests that, although Ptolemy's world-view was based on Aristotelian philosophy, he did not hesitate to deviate from this standpoint when required to by observational data.

From our heliocentric point of view, it is easily appreciated that the epicycles of the *superior planets* (that is, the planets farther from the Sun than the Earth) in Ptolemy's model actually

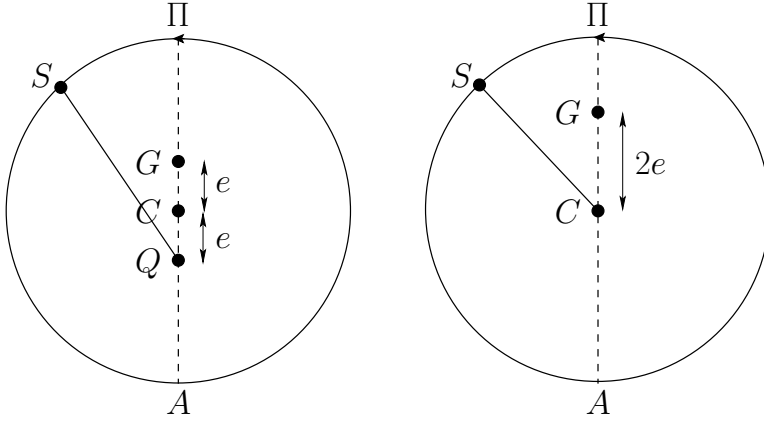


Figure 1.1: Hipparchus's (and Ptolemy's) model of the Sun's apparent orbit about the Earth (right) compared to the optimal model (left). The radius vectors in both models rotate uniformly. Here,  $S$  is the Sun,  $G$  the Earth,  $C$  the geometric center of the orbit,  $Q$  the equant,  $\Pi$  the perigee, and  $A$  the apogee. The radius of the orbit is normalized to unity.

represent the Earth's orbit around the Sun, whereas the deferents represent the planets' orbits around the Sun. (See Figure 8.1.) It follows that the epicycles of the superior planets should all be the same size (that is, the size of the Earth's orbit), and that the radius vectors connecting the centers of the epicycles to the planets should always all point in the same direction as the vector connecting the Earth to the Sun.

We can also appreciate that the deferents of the *inferior planets* (that is, the planets closer to the Sun than the Earth) in Ptolemy's model actually represent the Earth's orbit around the Sun, whereas the epicycles represent the planets' orbits around the Sun. (See Figure 9.1.) It follows that the deferents of the inferior planets should all be the same size (that is, the size of the Earth's orbit), and that the centers of the epicycles (relative to the Earth) should all correspond to the position of the Sun (relative to the Earth).

The geocentric model of the Solar System outlined previously represents a perfected version of Ptolemy's model, constructed with a knowledge of the true motions of the planets around the Sun. Not surprisingly, the model actually described in the *Almagest* deviates somewhat from this ideal form. In the following, we shall refer to these deviations as "errors", but this should not be understood in a pejorative sense.

Ptolemy's first error lies in his model of the Sun's apparent motion around the Earth, which he inherited from Hipparchus. Figure 1.1 compares what Ptolemy actually did, in this respect, compared to what he should have done in order to be completely consistent with the rest of his model. Let us normalize the mean radius of the Sun's apparent orbit to unity, for the sake of clarity. Ptolemy should have adopted the model shown on the left in Figure 1.1, in which the Earth is displaced from the center of the Sun's orbit a distance  $e = 0.0167$  (the eccentricity of the Earth's orbit around the Sun) towards the perigee (the point of the Sun's closest approach to

the Earth), and the equant is displaced the same distance in the opposite direction. The instantaneous angular position of the Sun is then obtained by allowing the radius vector connecting the equant to the Sun to rotate uniformly at the Sun's mean orbital angular velocity. Of course, this implies that the Sun rotates non-uniformly about the Earth. Ptolemy actually adopted the Hipparchian model shown on the right in Figure 1.1. In this model, the Earth is displaced a distance  $2e$  from the center of the Sun's orbit in the direction of the perigee, and the Sun rotates at a uniform rate (that is, the radius vector  $CS$  rotates uniformly). It turns out that, to first order in  $e$ , these two models are equivalent in terms of their ability to predict the angular position of the Sun relative to the Earth. (See Chapter 4.) Nevertheless, the Hipparchian model is incorrect, because it predicts too large (by a factor of two) a variation in the radial distance of the Sun from the Earth (and, hence, the angular size of the Sun) during the course of a year. (See Chapter 4.) Ptolemy probably adopted the Hipparchian model because his Aristotelian leanings prejudiced him in favor of uniform circular motion whenever this was consistent with observations. (It should be noted that Ptolemy was not especially interested in explaining the relatively small variations in the angular size of the Sun during the year; presumably, because this effect would have been almost impossible for him to accurately measure.)

Ptolemy's next error was to neglect the non-uniform rotation of the superior planets on their epicycles. This is equivalent to neglecting the orbital eccentricity of the Earth (recall that the epicycles of the superior planets actually represent the Earth's orbit) compared to those of the superior planets. It turns out that this is a fairly good approximation, because the superior planets all have significantly greater orbital eccentricities than the Earth. Nevertheless, neglecting the non-uniform rotation of the superior planets on their epicycles has the unfortunate effect of obscuring the tight coupling between the apparent motions of these planets, and that of the Sun. The radius vectors connecting the epicycle centers of the superior planets to the planets themselves should always all point exactly in the same direction as that of the Sun relative to the Earth. When the aforementioned non-uniform rotation is neglected, the radius vectors instead point in the direction of the mean Sun relative to the Earth. The *mean Sun* is a fictitious body that has the same apparent orbit around the Earth as the real Sun, but that circles the Earth at a uniform rate. The mean Sun only coincides with the real Sun twice a year.

Ptolemy's third error is associated with his treatment of the inferior planets. As we have seen, in going from the superior to the inferior planets, deferents and epicycles effectively swap roles. For instance, it is the deferents of the inferior planets, rather than the epicycles, that represent the Earth's orbit. Hence, for the sake of consistency with his treatment of the superior planets, Ptolemy should have neglected the non-uniform rotation of the epicycle centers around the deferents of the inferior planets, and retained the non-uniform rotation of the planets themselves around the epicycle centers. Instead, he did exactly the opposite. This is equivalent to neglecting the inferior planets' orbital eccentricities relative to that of the Earth. It follows that this approximation only works when an inferior planet has a significantly smaller orbital eccentricity than that of the Earth. It turns out that this is indeed the case for Venus, which has the smallest eccentricity of any planet in the Solar System. Thus, Ptolemy was able to successfully account for the apparent motion of Venus. Mercury, on the other hand, has a much larger orbital eccentricity than the Earth. Moreover, it is particularly difficult to obtain good naked-eye positional data for Mercury, because this planet always appears very close to the Sun in the sky.

Consequently, Ptolemy's Mercury data was highly inaccurate. Not surprisingly, Ptolemy was not able to account for the apparent motion of Mercury using his standard deferent-epicycle approach. Instead, in order to fit the data, he was forced to introduce an additional, and entirely spurious, epicycle into his model of Mercury's orbit.

Ptolemy's fourth, and possibly largest, error is associated with his treatment of the Moon. It should be noted that the Moon's motion around the Earth is extremely complicated in nature, because it is strongly perturbed by the Sun, and was not fully understood until the early 20th century AD. Ptolemy constructed an ingenious geometric model of the Moon's orbit that was capable of predicting the lunar ecliptic longitude to reasonable accuracy. Unfortunately, this model necessitates a monthly variation in the Earth-Moon distance by a factor of about two, which implies a similarly large variation in the Moon's angular diameter. However, the observed variation in the Moon's diameter is very much smaller than this. Hence, Ptolemy's model of lunar motion is not even approximately correct.

Ptolemy's fifth error is associated with his treatment of planetary ecliptic latitudes. Given that the deferents and epicycles of the superior planets represent the orbits of the planets themselves around the Sun, and the Sun's apparent orbit around the Earth, respectively, it follows that one should take the slight inclination of planetary orbits to the ecliptic plane (that is, the plane of the Sun's apparent orbit) into account by tilting the deferents of superior planets, while keeping their epicycles parallel to the ecliptic. Similarly, given that the epicycles and deferents of inferior planets represent the orbits of the planets themselves around the Sun, and the Sun's apparent orbit around the Earth, respectively, one should tilt the epicycles of inferior planets, while keeping their deferents parallel to the ecliptic. Finally, because the inclination of planetary orbits are all essentially constant in time, the inclinations of the epicycles and deferents should also be constant. Unfortunately, when Ptolemy constructed his theory of planetary latitudes, he tilted the both deferents and epicycles of all of the planets. Even worse, he allowed the inclinations of the epicycles to the ecliptic plane to vary in time. The net result is a theory that is far more complicated than is necessary.

The final failing in Ptolemy's model of the Solar System lies in its scale invariance. Using angular position data alone, Ptolemy was able to determine the ratio of the epicycle radius to that of the deferent for each planet, but was not able to determine the relative sizes of the deferents of different planets. In order to break this scale invariance it is necessary to make an additional assumption; namely, that the Earth orbits the Sun. This brings us to Copernicus.

### *1.3 Copernicus's model of the Solar System*

The Polish astronomer Nicolaus Copernicus (Mikołaj Kopernik, 1473–1543 AD) studied the *Almagest* assiduously, but eventually became dissatisfied with Ptolemy's approach. The main reason for this dissatisfaction was not the geocentric nature of Ptolemy's model, but rather the fact that it mandates that heavenly bodies execute non-uniform circular motion. Copernicus, like Aristotle, was convinced that the supposed perfection of the heavens requires such bodies to execute uniform circular motion only. Copernicus was thus spurred to construct his own model of the Solar System, which was described in his book "On the Revolutions of the Heavenly Spheres" (*De Revolutionibus Orbium Coelestium*), published in the year of his death.

The most well-known aspect of Copernicus's model is the fact that it is heliocentric. As has already been mentioned, when describing the motion of the Sun, Moon, and planets relative to the Earth, it makes little practical difference whether one adopts a geocentric or a heliocentric model of the Solar System. Having said this, the heliocentric approach does have one large advantage. If we accept that the Sun, and not the Earth, is stationary, then it immediately follows that the epicycles of the superior planets, and the deferents of the inferior planets, represent the Earth's orbit around the Sun. Hence, all of these circles must be the same size. This realization allows us to break the scale invariance which is one of the main failings of Ptolemy's model. Thus, the ratio of the deferent radius to that of the epicycle for a superior planet, which is easily inferred from observations, actually corresponds to the ratio of planet's orbital radius to that of the Earth. Likewise, the ratio of the epicycle radius to that of the deferent for an inferior planet, which is again easily determined observationally, also corresponds to the ratio of the planet's orbital radius to that of the Earth. Using this type of reasoning, Copernicus was able to construct the first accurate scale model of the Solar System, and to firmly establish the order in which the planets orbit the Sun. In some sense, this was his main achievement.

Copernicus's insistence that heavenly bodies should only move in uniform circles lead him to reject Ptolemy's equant scheme, and to replace it with the scheme illustrated in Figure 1.2. According to Copernicus, a heliocentric planetary orbit is a combination of two circular motions. The first is the motion of the planet around a small circular epicycle, and the second is the motion of the center of the epicycle around the Sun on a circular deferent. Both motions are uniform, and in the same direction. However, the former motion is twice as fast as the latter. In addition, the Sun is displaced from the center of the deferent in the direction of the perihelion, the displacement being proportional to the orbital eccentricity. Furthermore, the Sun's displacement is three times greater than the radius of the epicycle. Finally, the radius of the deferent is equal to the major radius of the planetary orbit. It turns out that Copernicus's scheme is a marginally less accurate approximation than Ptolemy's to a low eccentricity Keplerian orbit. (See Chapter 4.)

Copernicus modeled the orbit of the Earth around the Sun using an Hipparchian scheme (see Figure 1.1) in which the Earth moves uniformly around an eccentric circle. Unfortunately, such a scheme exaggerates the variation in the radial distance between the Earth and the Sun during the course of a year by a factor of two, and so introduces significant errors into the calculation of the parallax of the planets due to the motion of the Earth. On the other hand, Copernicus's model of the Moon's orbit around the Earth is a considerable improvement on Ptolemy's, because it does not grossly exaggerate the monthly variation in the Earth-Moon distance. Like Ptolemy, Copernicus introduced an additional spurious epicycle into his model of Mercury's orbit, and erroneously allowed the inclination of his planetary orbits to vary slightly in time.

In summary, Copernicus's model of the Solar System actually contains more circles than Ptolemy's, the only difference being that Copernicus's epicycles are much smaller than Ptolemy's. Indeed, the model of Copernicus is about as complicated, and not appreciably more accurate (except in the case of the Moon's radial distance from the Earth), than that described in the *Almagest*. In this respect, Copernicus cannot be said to have demonstrated the correctness of his heliocentric approach on the basis of observational data.

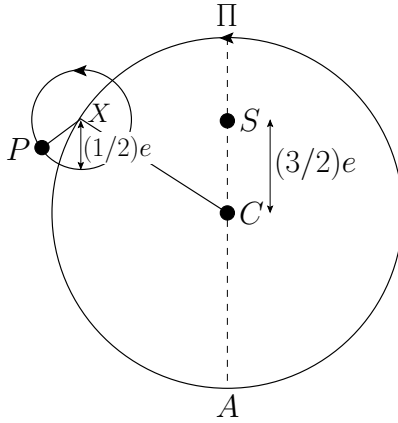


Figure 1.2: Copernicus's model of a heliocentric planetary orbit. Here,  $S$  is the Sun,  $P$  the planet,  $C$  the geometric center of the deferent,  $X$  the center of the epicycle,  $\Pi$  the perihelion, and  $A$  the aphelion. The radius vectors  $CX$  and  $XP$  both rotate uniformly in the same direction, but  $XP$  rotates twice as fast as  $CX$ . The major radius of the orbit is normalized to unity.

#### 1.4 Kepler's model of the Solar System

Johannes Kepler (1571–1630 AD) was very fortunate to inherit an extensive set of naked-eye solar, lunar, and planetary angular position data from the Danish astronomer Tycho Brahe (1546–1601 AD). This data extended over many decades, and was of unprecedented accuracy.

Although Kepler adopted the heliocentric approach of Copernicus, what he effectively first did was to perfect Ptolemy's model of the Solar System (or, rather, its heliocentric equivalent). Thus, Kepler replaced Ptolemy's erroneous equantless model of the Sun's apparent orbit around the Earth with a corrected version containing an equant; in the process, halving the eccentricity of the orbit. (See Figure 1.1.) Kepler also introduced equants into the epicycles of the superior and inferior planets. Once he had perfected Ptolemy's model, the heliocentric nature of the Solar System became manifestly apparent to Kepler. For instance, he found that the epicycles of the superior planets, the Sun's apparent orbit around the Earth, and the deferents of the inferior planets, all had exactly the same eccentricity. The obvious implication is that these circles all correspond to some common motion within the Solar System; in fact, the motion of the Earth around the Sun.

Once Kepler had corrected the Almagest model, he compared its predictions with his observational data. In particular, Kepler investigated the apparent motion of Mars in the night sky. Kepler found that his model performed extremely well, but that there remained small differences between its predictions and the observational data. The maximum discrepancy was about  $8'$ ; that is, about one quarter of the apparent size of the Sun. By the standards of naked-eye astronomy, this was a very small discrepancy. Nevertheless, given the incredible accuracy of Tycho Brahe's observations, the discrepancy was still significant. Thus, Kepler embarked on



an epic new series of calculations which eventually lead him to the conclusion that the planetary orbits are actually eccentric ellipses, rather than eccentric circles. Kepler published the results of his research in his treatise “New Astronomy” (*Astronomia Nova*) in 1609 AD. It is interesting to note that had Tycho’s data been a little less accurate, or had the orbit of Mars been a little less eccentric, Kepler might well have settled for a model which was kinematically equivalent to a perfected version of the model described in the *Almagest*. We can also appreciate that, given the far less accurate observational data available to Ptolemy, there was no way in which he could have discerned the very small difference between elliptical planetary orbits and the eccentric circular orbits employed in the *Almagest*.

### 1.5 *Purpose of book*

As we have seen, misconceptions abound regarding the details of Ptolemy’s model of the Solar System, as well as its scientific merit. Part of the reason for this is that the *Almagest* is an extremely difficult book for a modern reader to comprehend. For instance, virtually all of its theoretical results are justified via lengthy and opaque geometric proofs. Moreover, the plane and spherical trigonometry employed by Ptolemy is of a rather primitive nature, and, consequently, somewhat unwieldy. Dates are also a major stumbling block, because three different systems are used in the *Almagest*, all of which are archaic, and essentially meaningless to the modern reader. Another difficulty is the unfamiliar, and far from optimal, ancient Greek method of representing numbers and fractions. Finally, the terminology employed in the *Almagest* is, in many instances, significantly different to that used in modern astronomy textbooks.

The aim of this book is to reconstruct Ptolemy’s model of the Solar System employing modern mathematical methods, standard dates, and conventional astronomical terminology. It is hoped that the resulting model will enable the reader to comprehend the full extent of Ptolemy’s scientific achievement. In fact, the model described in this work is a somewhat improved version of Ptolemy’s, in that all of the previously mentioned deficiencies have been corrected. Furthermore, Ptolemy’s equant scheme has been replaced by a Keplerian scheme, expanded to second order in the planetary eccentricities. It should be noted, however, that these two schemes are essentially indistinguishable for small eccentricity orbits. Certain aspects of the *Almagest* have not been reproduced. For instance, it was not thought necessary to instruct the reader on how to construct trigonometric tables, or primitive astronomical instruments. Furthermore, no attempt has been made to derive any of the model parameters directly from observational data, because the orbital elements and physical properties of the Sun, Moon, and planets are, by now, extremely well established. Any detailed discussion of the fixed stars has also been omitted, because stellar positions are also very well established, and the apparent motion of the stars in the sky is comparatively straightforward compared to those of the Sun, the Moon, and the planets. What remains is a mathematical model of the Solar System that is surprisingly accurate (the maximum errors in the ecliptic longitudes of the Sun, Moon, Mercury, Venus, Mars, Jupiter, and Saturn during the years 1995–2006 AD are 0.7′, 14′, 28′, 10′, 14′, 4′, and 1′, respectively), yet sufficiently simple that all of the necessary calculations can be performed by hand, with the aid of tables. The form of the calculations, as well as the layout of the tables, is, for the most part, fairly similar to those found in the *Almagest*. Many examples of the use of the tables

are provided.

## 2. Spherical astronomy

### 2.1 *The celestial sphere*

It is often helpful to imagine that celestial objects are attached to a vast sphere centered on the Earth. This fictitious construction is known as the *celestial sphere*. The Earth's dimensions are assumed to be infinitesimally small compared to those of the sphere. (Because the distance of a typical celestial object from the Earth is very much larger than the Earth's radius.) It follows that only half of the sphere is visible from any particular observation site on the Earth's surface. Furthermore, the angular position of a given celestial object (relative to some fixed celestial reference) is the same at all such sites. In other words, there is negligible parallax associated with viewing the same celestial object from different observation sites on the surface of the Earth.<sup>1</sup>

### 2.2 *Celestial motions*

Celestial objects exhibit two distinct types of motion. The first motion is such that the whole celestial sphere, and all of the celestial objects attached to it, rotates uniformly from east to west once every 24 (sidereal) hours, about a fixed axis passing through the Earth's north and south poles. This type of motion is called *diurnal motion*, and is a consequence of the Earth's daily rotation. Diurnal motion preserves the relative angular positions of all celestial objects. However, certain celestial objects, such as the Sun, the Moon, and the planets, possess a second motion, superimposed on the first, that causes their angular positions to slowly change relative to one another, and to the fixed stars. This *intrinsic motion* of objects in the Solar System is due to a combination of the Earth's orbital motion about the Sun, and the orbital motions of the Moon and the planets about the Earth and the Sun, respectively.

Incidentally, the terms for the Earth, the Sun, and the Moon in Greek are ἡ γῆ, ὁ ἥλιος, and ἡ σελήνη, respectively. Likewise, the terms for the planets Mercury, Venus, Mars, Jupiter, and Saturn are ὁ τοῦ Ἑρμοῦ [the (star) of Hermes], ὁ τῆς Ἀφροδίτης [the (star) of Aphrodite], ὁ τοῦ Ἄρεως [the (star) of Ares], ὁ τοῦ Διός [the (star) of Zeus], and ὁ τοῦ Κρόνου [the (star) of Cronus]. Note that the word star was usually taken as read in Greek.

### 2.3 *Celestial coordinates*

Consider Figure 2.1. The celestial sphere rotates about the celestial axis,  $PP'$ , which is the imagined extension of the Earth's axis of rotation. This axis intersects the celestial sphere at the *north celestial pole*,  $P$ , and the *south celestial pole*,  $P'$ . It follows that the two celestial poles are unaffected by diurnal motion, and remain fixed in the sky.

The *celestial equator*,  $VUV'U'$ , is the intersection of the Earth's equatorial plane with the celestial sphere, and is therefore perpendicular to the celestial axis. The so-called *vernal equinox*,  $V$ , is a particular point on the celestial equator that is used as the origin of celestial longitude. Furthermore, the *autumnal equinox*,  $V'$ , is a point that lies directly opposite the vernal equinox

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<sup>1</sup>The one exception to this rule is the Moon, which is sufficiently close to the Earth that its diurnal parallax is significant. See Section 6.9.

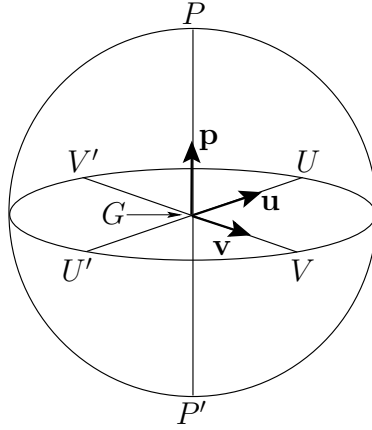


Figure 2.1: The celestial sphere. Here,  $G$ ,  $P$ ,  $P'$ ,  $V$ , and  $V'$  represent the Earth, the north celestial pole, the south celestial pole, the vernal equinox, and the autumnal equinox, respectively. Moreover,  $VUV'U'$  is the celestial equator, and  $PP'$  the celestial axis.

on the celestial equator. Let the line  $UU'$  lie in the plane of the celestial equator such that it is perpendicular to  $VV'$ , as shown in the figure.

It is helpful to define three, right-handed, mutually perpendicular, unit vectors;  $\mathbf{v}$ ,  $\mathbf{u}$ , and  $\mathbf{p}$ . Here,  $\mathbf{v}$  is directed from the Earth to the vernal equinox,  $\mathbf{u}$  from the Earth to point  $U$ , and  $\mathbf{p}$  from the Earth to the north celestial pole. See Figure 2.1.

Let  $G$  be the Earth. Consider a general celestial object,  $R$ . See Figure 2.2. The location of  $R$  on the celestial sphere is conveniently specified by two angular coordinates,  $\delta$  and  $\alpha$ . Let  $GR'$  be the projection of  $GR$  onto the equatorial plane. The coordinate  $\delta$ , which is known as *declination*, is the angle subtended between  $GR'$  and  $GR$ . Objects north of the celestial equator have positive declinations, and vice versa. It follows that objects on the celestial equator have declinations of  $0^\circ$ , whereas the north and south celestial poles have declinations of  $+90^\circ$  and  $-90^\circ$ , respectively. The coordinate  $\alpha$ , which is known as *right ascension*, is the angle subtended between  $GV$  and  $GR'$ . Right ascension increases from west to east (that is, in the opposite direction to the celestial sphere's diurnal rotation). Thus, the vernal and autumnal equinoxes have right ascensions of  $0^\circ$  and  $180^\circ$ , respectively. Note that  $\alpha$  lies in the range  $0^\circ$  to  $360^\circ$ . Right ascension is sometimes measured in hours, instead of degrees, with one hour corresponding to  $15^\circ$  (because it takes 24 (sidereal) hours for the celestial sphere to complete one diurnal rotation). In this scheme, the vernal and autumnal equinoxes have right ascensions of 0 hours and 12 hours, respectively. Moreover,  $\alpha$  lies in the range 0 to 24 hours. (Incidentally, in this book,  $\alpha$  is measured relative to the mean equinox at date, unless otherwise specified.) Finally, let  $\mathbf{r}$  be a unit vector that is directed from the Earth to  $R$ . See Figure 2.2. It is easily demonstrated that

$$\mathbf{r} = \cos \delta \cos \alpha \mathbf{v} + \cos \delta \sin \alpha \mathbf{u} + \sin \delta \mathbf{p}, \quad (2.1)$$

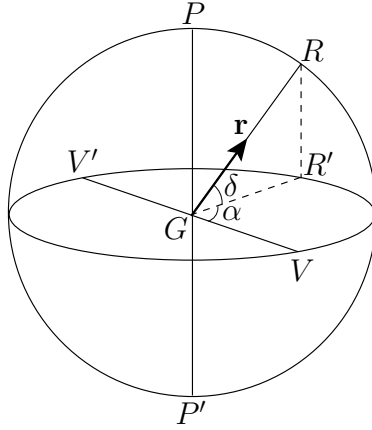


Figure 2.2: Celestial coordinates. Here,  $G$  is the Earth,  $R$  a celestial object, and  $R'$  the object's projection onto the plane of the celestial equator,  $VR'V'$ .

and

$$\sin \delta = \mathbf{r} \cdot \mathbf{p}, \quad (2.2)$$

$$\tan \alpha = \frac{\mathbf{r} \cdot \mathbf{u}}{\mathbf{r} \cdot \mathbf{v}}. \quad (2.3)$$

## 2.4 The ecliptic circle

During the course of a year, the Sun's intrinsic motion causes it to trace out a fixed circle that bisects the celestial sphere. This circle is known as the *ecliptic*. The Sun travels around the ecliptic from west to east (that is, in the opposite direction to the celestial sphere's diurnal rotation). Moreover, the ecliptic circle is inclined at a fixed angle of  $\epsilon = 23^\circ 26'$  to the celestial equator. This angle actually represents the fixed inclination of the Earth's axis of rotation to the normal to its orbital plane.<sup>2</sup>

The vernal equinox,  $V$ , is defined as the point at which the ecliptic crosses the celestial equator from south to north (in the direction of the Sun's ecliptic motion). See Figure 2.3. Likewise, the autumnal equinox,  $V'$ , is the point at which the ecliptic crosses the celestial equator from north to south. In addition, the *summer solstice*,  $S$ , is the point on the ecliptic that is furthest north of the celestial equator, whereas the *winter solstice*,  $S'$ , is the point that is furthest south. It follows that the lines  $VV'$  and  $SS'$  are perpendicular. Let  $QQ'$  be the normal to the plane of the ecliptic that passes through the Earth, as shown in Figure 2.3. Here,  $Q$  is termed the

<sup>2</sup>In fact,  $\epsilon$  is very slowly decreasing in time. The value of  $\epsilon$  used in the Almagest is  $23^\circ 51'$ . However, the true value of  $\epsilon$  in Ptolemy's day was  $23^\circ 41'$ .

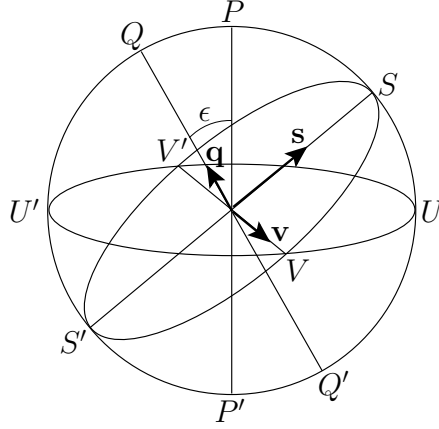


Figure 2.3: The ecliptic circle. Here,  $P$ ,  $P'$ ,  $Q$ ,  $Q'$ ,  $V$ ,  $V'$ ,  $S$ , and  $S'$  denote the north celestial pole, the south celestial pole, the north ecliptic pole, the south ecliptic pole, the vernal equinox, the autumnal equinox, the summer solstice, and the winter solstice, respectively. Moreover,  $VUV'U'$  is the celestial equator,  $VS'V'S'$  the ecliptic, and  $PP'$  the celestial axis.

*northern ecliptic pole*, and  $Q'$  the *southern ecliptic pole*. It is easily demonstrated that

$$\mathbf{s} = \cos \epsilon \mathbf{u} + \sin \epsilon \mathbf{p}, \quad (2.4)$$

$$\mathbf{q} = -\sin \epsilon \mathbf{u} + \cos \epsilon \mathbf{p}, \quad (2.5)$$

where  $\mathbf{s}$  is a unit vector that is directed from the Earth to the summer solstice, and  $\mathbf{q}$  a unit vector that is directed from the Earth to the north ecliptic pole. See Figure 2.3. We can also write

$$\mathbf{u} = \cos \epsilon \mathbf{s} - \sin \epsilon \mathbf{q}, \quad (2.6)$$

$$\mathbf{p} = \sin \epsilon \mathbf{s} + \cos \epsilon \mathbf{q}. \quad (2.7)$$

Thus,  $\mathbf{v}$ ,  $\mathbf{s}$ , and  $\mathbf{q}$  constitute another right-handed, mutually perpendicular, set of unit vectors.

## 2.5 Ecliptic coordinates

It is convenient to specify the positions of the Sun, Moon, and planets in the sky using a pair of angular coordinates,  $\beta$  and  $\lambda$ , that are measured with respect to the ecliptic, rather than the celestial equator. Let  $G$  and  $R$  denote the Earth and a celestial object, respectively, and let  $GR'$  denote the projection of the line  $GR$  onto the plane of the ecliptic,  $VR'V'$ . See Figure 2.4. The coordinate  $\beta$ , which is known as *ecliptic latitude*, is the angle subtended between  $GR'$  and  $GR$ . Objects north of the ecliptic plane have positive ecliptic latitudes, and vice versa. The coordinate  $\lambda$ , which is known as *ecliptic longitude*, is the angle subtended between  $GV$  and  $GR'$ . Ecliptic longitude increases from west to east (that is, in the same direction that the Sun

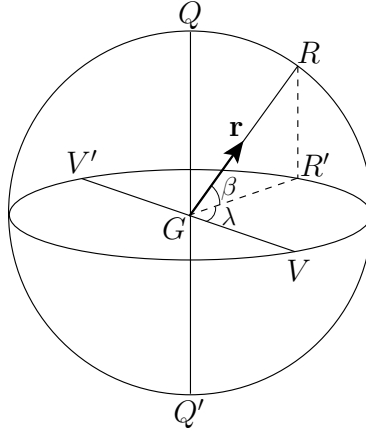


Figure 2.4: Ecliptic coordinates. Here,  $G$  is the Earth,  $R$  a celestial object, and  $R'$  the object's projection onto the ecliptic plane,  $VR'V'$ .

travels around the ecliptic). (Again, in this book,  $\lambda$  is measured relative to the mean equinox at date, unless specified otherwise.) Note that the basis vectors in the ecliptic coordinate system are  $\mathbf{v}$ ,  $\mathbf{s}$ , and  $\mathbf{q}$ , whereas the corresponding basis vectors in the celestial coordinate system are  $\mathbf{v}$ ,  $\mathbf{u}$ , and  $\mathbf{p}$ . See Figures 2.1 and 2.3. By analogy with Equations (2.1)–(2.3), we can write

$$\mathbf{r} = \cos \beta \cos \lambda \mathbf{v} + \cos \beta \sin \lambda \mathbf{s} + \sin \beta \mathbf{q}, \quad (2.8)$$

$$\sin \beta = \mathbf{r} \cdot \mathbf{q}, \quad (2.9)$$

$$\tan \lambda = \frac{\mathbf{r} \cdot \mathbf{s}}{\mathbf{r} \cdot \mathbf{v}}, \quad (2.10)$$

where  $\mathbf{r}$  is a unit vector that is directed from  $G$  to  $R$ . Hence, it follows from Equations (2.1), (2.4), and (2.5) that

$$\sin \beta = \cos \epsilon \sin \delta - \sin \epsilon \cos \delta \sin \alpha, \quad (2.11)$$

$$\tan \lambda = \frac{\cos \epsilon \cos \delta \sin \alpha + \sin \epsilon \sin \delta}{\cos \delta \cos \alpha}. \quad (2.12)$$

These expressions specify the transformation from celestial to ecliptic coordinates. The inverse transformation follows from Equations (2.2), (2.3), and (2.6)–(2.8):

$$\sin \delta = \cos \epsilon \sin \beta + \sin \epsilon \cos \beta \sin \lambda, \quad (2.13)$$

$$\tan \alpha = \frac{\cos \epsilon \cos \beta \sin \lambda - \sin \epsilon \sin \beta}{\cos \beta \cos \lambda}. \quad (2.14)$$

Figure 2.5 shows all stars of visible magnitude less than +6 lying within  $15^\circ$  of the ecliptic. Table 2.1 gives the ecliptic longitudes (relative to the start of the appropriate zodiacal sign; see

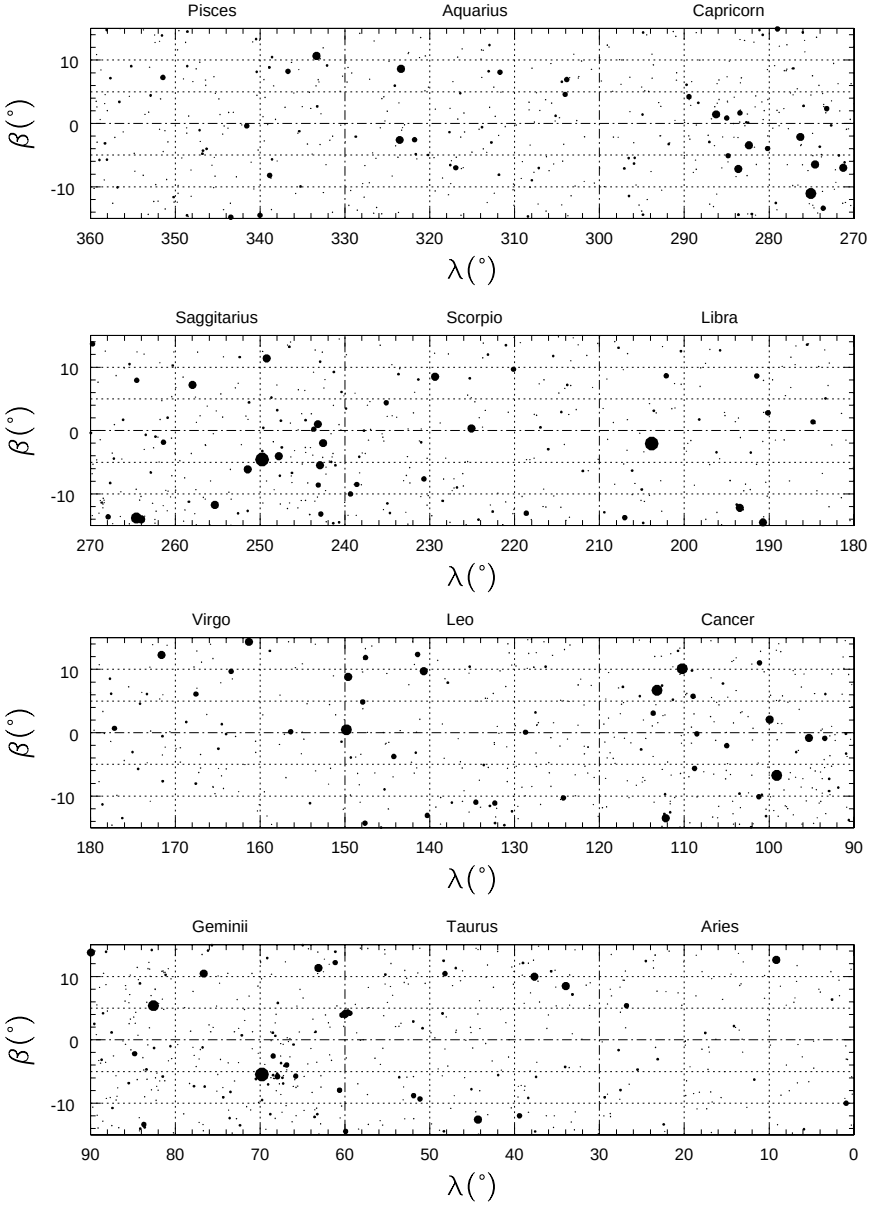


Figure 2.5: Map showing all stars of visual magnitude less than +6 lying within  $15^{\circ}$  of the ecliptic plane.



next section), ecliptic latitudes, and visible magnitudes of a selection of these stars that lie within  $10^\circ$  of the ecliptic. The figure and the table can be used to convert ecliptic longitude and latitude into approximate position in the sky against the backdrop of the fixed stars.

Figure 2.5 and Table 2.1 contain more or less the same information as that contained in Section 5 of Book VII of the *Almagest* (although we have restricted our attention to stars that lie relatively close to the ecliptic). Incidentally, the ancient Greeks did not individually name stars, describing them instead in figurative terms, such as “the bright, reddish star on the right shoulder of Orion” (Betelgeuse).

## 2.6 *The signs of the zodiac*

The *signs of the zodiac* are a well-known set of names given to  $30^\circ$  long segments of the ecliptic circle. Thus, the sign of Aries extends over the range of ecliptic longitudes  $0^\circ$ – $30^\circ$ , the sign of Taurus over the range  $30^\circ$ – $60^\circ$ , and so on. Note that, as a consequence of the precession of the equinoxes, the signs of the zodiac no longer coincide with the constellations of the same name. See Figure 2.5. (Incidentally, Ptolemy was well aware of this fact because, even in his day, the signs of the zodiac did not coincide with the constellations. The signs coincided with the constellations in the time of the ancient Mesopotamians; that is, about 1500 BC.) The 12 zodiacal signs are listed in the following table. It can be seen from the table that ecliptic longitude  $72^\circ$  corresponds to the twelfth degree of Gemini, and ecliptic longitude  $242^\circ$  to the second degree of Sagittarius, et cetera.

| Sign   | Abbr. | Longitude                 | Sign    | Abbr. | Longitude                 | Sign        | Abbr. | Longitude                 |
|--------|-------|---------------------------|---------|-------|---------------------------|-------------|-------|---------------------------|
| Aries  | AR    | $000^\circ$ – $030^\circ$ | Leo     | LE    | $120^\circ$ – $150^\circ$ | Sagittarius | SG    | $240^\circ$ – $270^\circ$ |
| Taurus | TA    | $030^\circ$ – $060^\circ$ | Virgo   | VI    | $150^\circ$ – $180^\circ$ | Capricorn   | CP    | $270^\circ$ – $300^\circ$ |
| Gemini | GE    | $060^\circ$ – $090^\circ$ | Libra   | LI    | $180^\circ$ – $210^\circ$ | Aquarius    | AQ    | $300^\circ$ – $330^\circ$ |
| Cancer | CN    | $090^\circ$ – $120^\circ$ | Scorpio | SC    | $210^\circ$ – $240^\circ$ | Pisces      | PI    | $330^\circ$ – $360^\circ$ |

Incidentally, the term zodiac comes from the Greek ζῳδιακός κύκλος, meaning “circle of little animals”. The Greek names for the zodiacal signs were Κριός (Ram), Ταῦρος (Bull), Δίδυμοι (Twins), Κάρκινος (Crab), Λέων (Lion), Παρθένος (Virgin), Ζυγός (Balance/scale), Σκορπιός (Scorpion), Τοξότης (Archer), Αἰγόκερως (Goat-horned), Ὑδροχόος (Water-pourer), and Ἰχθύες (Fishes). The familiar names listed in the previous table are merely latinized versions of the Greek names.

## 2.7 *Ecliptic declinations and right ascensions.*

According to Equations (2.13) and (2.14), the celestial coordinates of a point on the ecliptic circle (which corresponds to  $\beta = 0$ ) that has ecliptic longitude  $\lambda$  are specified by

$$\sin \delta = \sin \epsilon \sin \lambda, \quad (2.15)$$

$$\tan \alpha = \cos \epsilon \tan \lambda. \quad (2.16)$$

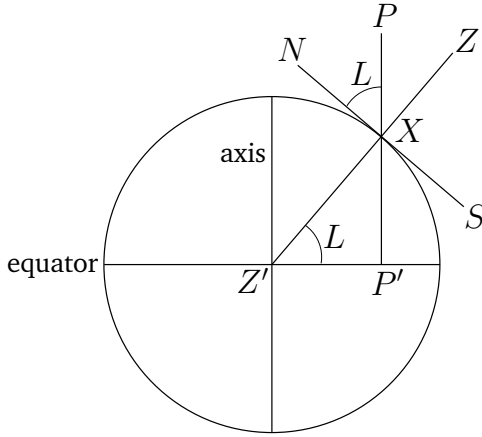


Figure 2.6: A general observation site  $X$ , of latitude  $L$ , on the surface of the Earth. Here,  $P$ ,  $P'$ ,  $Z$ , and  $Z'$  denote the directions to the north celestial pole, south celestial pole, zenith, and nadir, respectively. The line  $NS$  represents the local horizon.

The previous formulae have been used to construct Table 2.2, which lists the declinations and right ascensions of a set of equally spaced points on the ecliptic circle. Note that  $\lambda$  is measured relative to the start of the appropriate zodiacal sign.

Table 2.2 contains essentially the same information as is contained in the “Table of inclinations” (Κανόνιον λοξώσεως) that appears in Section 15 of Book I of the *Almagest*. The slight difference between the entries appearing in our tables and those appearing in Ptolemy’s table is due to the fact that Ptolemy adopted the value  $23^{\circ}51'20''$  for the obliquity of the ecliptic, whereas we are using the modern (and correct) value  $23^{\circ}26'$ .

## 2.8 Local horizon and meridian

Consider a general observation site  $X$  located on the surface of the Earth. (Note that, in the following, it is tacitly assumed that the site lies the Earth’s northern hemisphere. However, the analysis also applies to sites situated in the the southern hemisphere.) The local *zenith*,  $Z$ , is the point on the celestial sphere that is directly overhead at  $X$ , whereas the *nadir*,  $Z'$ , is the point that is directly underfoot. See Figure 2.6. The *horizon* is the tangent plane to the Earth at  $X$ , and divides the celestial sphere into two halves. The upper half, containing the zenith, is visible from site  $X$ , whereas the lower half is invisible.

Figure 2.7 shows the visible half of the celestial sphere at observation site  $X$ . Here,  $NE\!SW$  is the local horizon, and  $N$ ,  $E$ ,  $S$ , and  $W$  are the north, east, south, and west compass points, respectively. The plane  $NPZS$ , which passes through the north and south compass points, as well as the zenith, is known as the local *meridian*. The meridian is perpendicular to the horizon. The north celestial pole lies in the meridian plane, and is elevated an angular distance  $L$  above the north compass point. See Figures 2.6 and 2.7. Here,  $L$  is the terrestrial *latitude* of

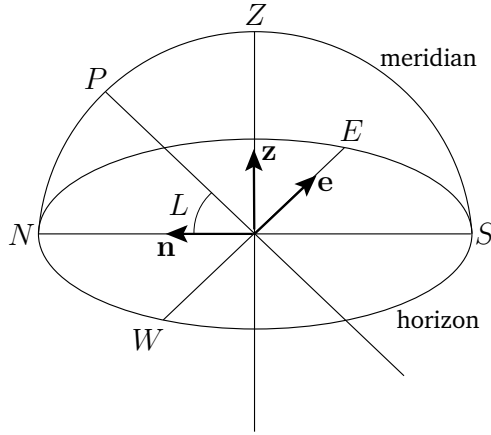


Figure 2.7: The local horizon and meridian. Here,  $N$ ,  $S$ ,  $E$ ,  $W$  denote the north, south, east, and west compass points,  $Z$  the zenith, and  $P$  the north celestial pole. Moreover,  $NE\!SW$  is the horizon, and  $NP\!ZS$  the meridian.

observation site  $X$ . It is helpful to define three, right-handed, mutually perpendicular, local unit vectors;  $\mathbf{e}$ ,  $\mathbf{n}$ , and  $\mathbf{z}$ . Here,  $\mathbf{e}$  is directed toward the east compass point,  $\mathbf{n}$  toward the north compass point, and  $\mathbf{z}$  toward the zenith. See Figures 2.7.

Figure 2.8 shows the meridian plane at  $X$ . Let the line  $MM'$  lie in this plane such that it is perpendicular to the celestial axis,  $PP'$ . Moreover, let  $M$  lie in the visible hemisphere. It is helpful to define the unit vector  $\mathbf{m}$  which is directed toward  $M$ , as shown in the diagram. It is easily seen that

$$\mathbf{n} = \cos L \mathbf{p} - \sin L \mathbf{m}, \quad (2.17)$$

$$\mathbf{z} = \sin L \mathbf{p} + \cos L \mathbf{m}. \quad (2.18)$$

Figure 2.9 shows the celestial equator viewed from observation site  $X$ . Here,  $\alpha_0$  is the right ascension of the celestial objects culminating (that is, attaining their highest altitude in the sky) on the meridian at the time of observation. Incidentally, it is easily demonstrated that all objects culminating on the meridian at any instant in time have the same right ascension. Note that the angle  $\alpha_0$  increases uniformly in time, at the rate of  $15^\circ$  a (sidereal) hour, due to the diurnal motion of the celestial sphere. It can be seen from the diagram that

$$\mathbf{m} = \sin \alpha_0 \mathbf{u} + \cos \alpha_0 \mathbf{v}, \quad (2.19)$$

$$\mathbf{e} = \cos \alpha_0 \mathbf{u} - \sin \alpha_0 \mathbf{v}. \quad (2.20)$$

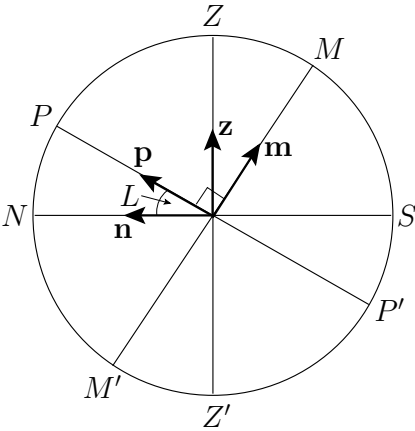


Figure 2.8: The local meridian.

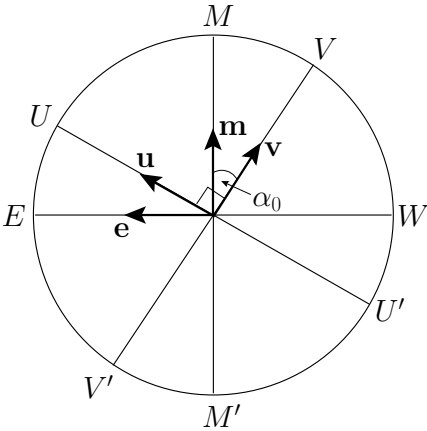


Figure 2.9: The local celestial equator.

Thus, from Equations (2.17) and (2.18),

$$\mathbf{e} = -\sin \alpha_0 \mathbf{v} + \cos \alpha_0 \mathbf{u}, \quad (2.21)$$

$$\mathbf{n} = -\sin L \cos \alpha_0 \mathbf{v} - \sin L \sin \alpha_0 \mathbf{u} + \cos L \mathbf{p}, \quad (2.22)$$

$$\mathbf{z} = \cos L \cos \alpha_0 \mathbf{v} + \cos L \sin \alpha_0 \mathbf{u} + \sin L \mathbf{p}. \quad (2.23)$$

Similarly, from Equations (2.6) and (2.7),

$$\mathbf{e} = -\sin \alpha_0 \mathbf{v} + \cos \epsilon \cos \alpha_0 \mathbf{s} - \sin \epsilon \cos \alpha_0 \mathbf{q}, \quad (2.24)$$

$$\begin{aligned} \mathbf{n} = & -\sin L \cos \alpha_0 \mathbf{v} + (\cos L \sin \epsilon - \sin L \cos \epsilon \sin \alpha_0) \mathbf{s} \\ & + (\cos L \cos \epsilon + \sin L \sin \epsilon \sin \alpha_0) \mathbf{q}, \end{aligned} \quad (2.25)$$

$$\begin{aligned} \mathbf{z} = & \cos L \cos \alpha_0 \mathbf{v} + (\sin L \sin \epsilon + \cos L \cos \epsilon \sin \alpha_0) \mathbf{s} \\ & + (\sin L \cos \epsilon - \cos L \sin \epsilon \sin \alpha_0) \mathbf{q}. \end{aligned} \quad (2.26)$$

## 2.9 Horizontal coordinates

It is convenient to specify the positions of celestial objects in the sky, when viewed from a particular observation site,  $X$ , on the Earth's surface, using a pair of angular coordinates,  $a$  and  $A$ , that are measured with respect to the local horizon. Let  $R$  denote a celestial object, and  $XR'$  the projection of the line  $XR$  onto the horizontal plane,  $NESW$ . See Figure 2.10. The coordinate  $a$ , which is known as *altitude*, is the angle subtended between  $XR'$  and  $XR$ . Objects above the horizon have positive altitudes, whereas objects below the horizon have negative altitudes. The zenith has altitude  $90^\circ$ , and the horizon altitude  $0^\circ$ . The coordinate  $A$ , which is known as *azimuth*, is the angle subtended between  $XN$  and  $XR'$ . Azimuth increases from the north towards the east. Thus, the north, east, south, and west compass points have azimuths of  $0^\circ$ ,  $90^\circ$ ,  $180^\circ$ , and  $270^\circ$ , respectively. Note that the basis vectors in the horizontal coordinate system are  $\mathbf{e}$ ,  $\mathbf{n}$ , and  $\mathbf{z}$ , whereas the corresponding basis vectors in the celestial coordinate system are  $\mathbf{v}$ ,  $\mathbf{u}$ , and  $\mathbf{p}$ . See Figures 2.1 and 2.7. By analogy with Equations (2.1)–(2.3), we can write

$$\mathbf{r} = \cos a \sin A \mathbf{e} + \cos a \cos A \mathbf{n} + \sin a \mathbf{z}, \quad (2.27)$$

$$\sin a = \mathbf{r} \cdot \mathbf{z}, \quad (2.28)$$

$$\tan A = \frac{\mathbf{r} \cdot \mathbf{e}}{\mathbf{r} \cdot \mathbf{n}}, \quad (2.29)$$

where  $\mathbf{r}$  is a unit vector directed from  $X$  to  $R$ . Hence, it follows from Equations (2.1), and (2.22)–(2.23), that

$$\sin a = \sin L \sin \delta + \cos L \cos \delta \cos(\alpha - \alpha_0), \quad (2.30)$$

$$\tan A = \frac{\cos \delta \sin(\alpha - \alpha_0)}{\cos L \sin \delta - \sin L \cos \delta \cos(\alpha - \alpha_0)}. \quad (2.31)$$

These expressions allow us to calculate the altitude and azimuth of a celestial object of declination  $\delta$  and right ascension  $\alpha$  that is viewed from an observation site on the Earth's surface of

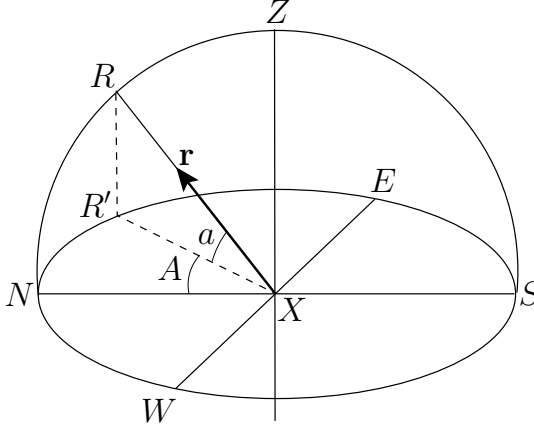


Figure 2.10: Horizontal coordinates. Here,  $R$  is a celestial object, and  $R'$  its projection onto the horizontal plane,  $NESW$ .

terrestrial latitude  $L$  at an instance in time when celestial objects of right ascension  $\alpha_0$  are culminating at the meridian.. According to Equations (2.8), and (2.25)–(2.26), the altitude and azimuth of a similarly viewed point on the ecliptic (which corresponds to  $\beta = 0$ ) of ecliptic longitude  $\lambda$  are given by

$$\begin{aligned} \sin a &= \cos L \cos \lambda \cos \alpha_0 + \sin L \sin \epsilon \sin \lambda \\ &+ \cos L \cos \epsilon \sin \lambda \sin \alpha_0, \end{aligned} \quad (2.32)$$

$$\tan A = \frac{\cos \epsilon \sin \lambda \cos \alpha_0 - \cos \lambda \sin \alpha_0}{\cos L \sin \epsilon \sin \lambda - \sin L \cos \lambda \cos \alpha_0 - \sin L \cos \epsilon \sin \lambda \sin \alpha_0}. \quad (2.33)$$

### 2.10 Meridian transits

Consider a celestial object, of declination  $\delta$  and right ascension  $\alpha$ , that is viewed from an observation site on the Earth's surface of terrestrial latitude  $L$ . According to Equation (2.30), the object culminates, or attains its highest altitude in the sky, when  $\alpha_0 = \alpha$ . This event is known as an *upper transit*. Furthermore, the object attains its lowest altitude in the sky when  $\alpha_0 = 180^\circ + \alpha$ . This event is known as a *lower transit*. Both upper and lower transits take place as the object in question passes through the meridian plane.

According to Equation (2.30), the altitude of a celestial object at its upper transit satisfies  $\sin a_+ = \cos(L - \delta)$ , implying that

$$a_+ = 90^\circ - |L - \delta|. \quad (2.34)$$

Likewise, the altitude at its lower transit satisfies  $\sin a_- = -\cos(L + \delta)$ , giving

$$a_- = |L + \delta| - 90^\circ. \quad (2.35)$$

The previous two expressions allow us to group celestial objects into three classes. Objects with declinations satisfying  $|L + \delta| > 90^\circ$  never set; that is, their lower transits lie above the horizon. Objects with declinations satisfying  $|L - \delta| > 90^\circ$  never rise; that is, their upper transits lie below the horizon. Finally, objects with declinations that satisfy neither of the two previous inequalities both rise and set during the course of a day. It follows that all celestial objects appear to rise and set when viewed from an observation site on the terrestrial equator (which corresponds to  $L = 0^\circ$ ). On the other hand, when viewed from an observation site at the north pole (which corresponds to  $L = 90^\circ$ ), objects north of the celestial equator never set, while objects south of the celestial equator never rise, and vice versa for objects viewed from the south pole. All three classes of celestial object are present when the sky is viewed from an observation site on the Earth's surface of intermediate latitude.

### 2.11 Principal terrestrial latitude circles

According to Equation (2.15), the Sun's declination varies between  $-\epsilon$  and  $+\epsilon$  during the course of a year. It follows from Equation (2.34) that it is only possible for the Sun to have an upper transit at the zenith in a region of the Earth whose latitude lies between  $-\epsilon$  and  $\epsilon$ . The circles of latitude bounding this region are known as the *tropics*. Thus, the *tropic of Capricorn*—so-called because the Sun is at the winter solstice, and, therefore, at the first point of Capricorn (that is, the zeroth degree of Capricorn), when it culminates at the zenith at this latitude—lies at  $L = -23^\circ 26'$ . Moreover, the *tropic of Cancer*—so-called because the Sun is at the summer solstice, and, therefore, at the first point of Cancer, when it culminates at the zenith at this latitude—lies at  $L = +23^\circ 26'$ . Incidentally, the word “tropic” derives from the Greek τροπή, meaning a turn or change in direction.

Equations (2.34) and (2.35) imply that the Sun does not rise for part of the year, and does not set for part of the year, in two regions of the Earth whose terrestrial latitudes satisfy  $|L| > 90^\circ - \epsilon$ . These two regions are bounded by the poles and two circles of latitude known as the *arctic circles*. The *south arctic circle* lies at  $L = -66^\circ 34'$ . Likewise, the *north arctic circle* lies at  $L = +66^\circ 34'$ . Incidentally, the word “arctic” comes from the Greek ἀρκτικός, meaning “of the bear”, and is so called because the most prominent northern constellations are the “great bear” and the “little bear”.

The equator, the two tropics, and the two arctic circles constitute the five *principal latitude circles* of the Earth, and are shown in Figure 2.11.

### 2.12 Equinoxes and solstices

The ecliptic longitude of the Sun when it reaches the vernal equinox is  $\lambda = 0^\circ$ . It follows, from Equation (2.32), that the altitude of the Sun on the day of the equinox is given by  $\sin a = \cos L \cos \alpha_0$ . Thus, the Sun rises when  $\alpha_0 = -90^\circ$ , culminates at an altitude of  $90^\circ - |L|$  when  $\alpha_0 = 0^\circ$ , and sets when  $\alpha_0 = 90^\circ$ . We conclude that the length of the equinoctial day is 180 time-degrees, which is equivalent to 12 hours (because  $15^\circ$  of right ascension cross the meridian in one hour). Thus, day and night are equally long on the day of the vernal equinox. (The word equinox is derived from the Latin word *aequinoctium*, which means “equal night”).

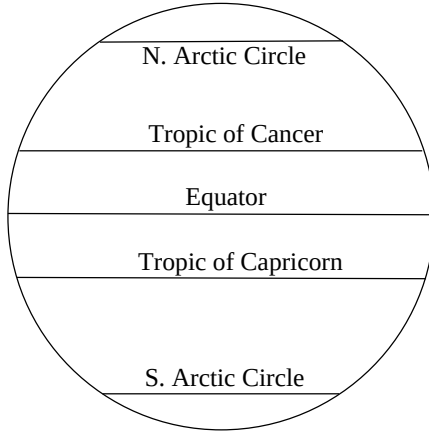


Figure 2.11: The principal latitude circles of the Earth.

It is easily demonstrated that the same is true on the day of the autumnal equinox.

The ecliptic longitude of the Sun when it reaches the summer solstice is  $\lambda = 90^\circ$ . It follows that the altitude of the Sun on the day of the solstice is given by  $\sin a = \sin L \sin \epsilon + \cos L \cos \epsilon \sin \alpha_0$ . Thus, the Sun rises when  $\alpha_0 = -\sin^{-1}(\tan L \tan \epsilon)$ , culminates at an altitude of  $90^\circ - |L - \epsilon|$  when  $\alpha_0 = 90^\circ$ , and sets when  $\alpha_0 = 180^\circ + \sin^{-1}(\tan L \tan \epsilon)$ . We conclude that the length of the longest day of the year in the Earth's northern hemisphere (which, of course, occurs when the Sun reaches the summer solstice) is  $180^\circ + 2 \sin^{-1}(\tan L \tan \epsilon)$  time-degrees. Likewise, the length of the shortest night (which also occurs at the summer solstice) is  $180 - 2 \sin^{-1}(\tan L \tan \epsilon)$  time-degrees. These formulae are only valid for northern latitudes below the arctic circle. At higher latitudes, the Sun never sets for part of the year, and the longest "day" is consequently longer than 24 hours. It is easily demonstrated that the shortest day in the Earth's northern hemisphere, which takes place when the Sun reaches the winter solstice, is equal to the shortest night, and the longest night (which also occurs at the winter solstice) to the longest day. Moreover, the Sun culminates at an altitude of  $90^\circ - |L + \epsilon|$  on day of the winter solstice. Again, at latitudes above the arctic circle, the Sun never rises for part of the year, and the longest "night" is consequently longer than 24 hours.

Consider an observation site on the Earth's surface of latitude  $L$  that lies above the northern arctic circle. The declination of the Sun on the first day after the spring equinox on which it fails to set is  $\delta = 90^\circ - L$ . According to Equation (2.15), the Sun's ecliptic longitude on this day is  $\sin^{-1}(\cos L / \sin \epsilon)$ . Likewise, the declination of the Sun on the day when it starts to set again is  $\delta = 90^\circ - L$ , and its ecliptic longitude is  $180^\circ - \sin^{-1}(\cos L / \sin \epsilon)$ . Assuming that the Sun travels around the ecliptic circle at a uniform rate (which is approximately true), the fraction of a year that the Sun stays above the horizon in summer is

$$0.5 - \sin^{-1}(\cos L / \sin \epsilon) / 180^\circ.$$

It is easily demonstrated that the fraction of a year that the Sun stays below the horizon in winter



is also

$$0.5 - \sin^{-1}(\cos L / \sin \epsilon) / 180^\circ.$$

### 2.13 *Terrestrial climes*

Table 2.3 specifies the length of the longest day, as well as the altitude of the Sun when it culminates at the meridian on the days of the equinoxes and solstices, calculated for a set of observation sites in the northern hemisphere with equally spaced terrestrial latitudes. This table was constructed using the formulae in the previous section. The table can be adapted to observation sites in the Earth's southern hemisphere via the following simple transformation:  $L \rightarrow -L$ , Summer  $\leftrightarrow$  Winter, N  $\leftrightarrow$  S. For instance, at a latitude of  $-10^\circ$ , the longest day, which corresponds to the winter solstice, is of length  $12^h 35^m$ . Moreover, on this day, the Sun's upper transit is south of the zenith, at an altitude of  $+76^\circ 34'$ .

Table 2.3 contains essentially the same information as is contained in the discussion that appears in Section 6 of Book II of the *Almagest*. Incidentally, the word “clime”, meaning a region defined by its weather, comes from the Greek *κλίμα*, meaning slope of inclination. Of course, the inclination in question is the inclination of the Sun's rays to the vertical at a given location on the Earth.

### 2.14 *Ecliptic ascensions*

Consider the rising, or *ascension*, of celestial objects at the eastern horizon, as viewed from a particular observation site on the Earth's surface. If the observation site lies on the terrestrial equator then all celestial objects appear to ascend at right angles to the horizon. This process is known as *right ascension*. On the other hand, if the observation site does not lie at the equator then celestial objects appear to ascend at an oblique angle to the horizon. This process is known as *oblique ascension*. For the case of right ascension, it is easily demonstrated that all celestial objects with the same celestial longitude ascend simultaneously. Indeed, celestial longitude is generally known as “right ascension” because, in the case of right ascension, the celestial longitude of an object (in hours) is simply the time elapsed between the ascension of the vernal equinox, and the ascension of the object in question.

Let us now consider the ascension of points on the ecliptic. Applying Equation (2.30) to a point on the celestial equator (which corresponds to  $\delta = 0$ ) of right ascension  $\alpha$ , we obtain

$$\sin a = \cos L \cos(\alpha - \alpha_0) = \cos L \sin(\alpha_0 - \alpha + 90^\circ). \quad (2.36)$$

It follows that we can write

$$\alpha_0 = \alpha - 90^\circ, \quad (2.37)$$

where  $\alpha$  is the right ascension of the point on the celestial equator that ascends at the eastern horizon (that is,  $a = 0$  and  $da/dt > 0$ ) at the same time that celestial objects of right ascension  $\alpha_0$  are culminating at the meridian. Substituting this result into Equation (2.32), we get

$$\sin a = \cos L \cos \lambda \sin \alpha + \sin L \sin \epsilon \sin \lambda - \cos L \cos \epsilon \sin \lambda \cos \alpha, \quad (2.38)$$

which implies that if  $a = 0$  and  $da/dt > 0$  then

$$\tan \lambda = \frac{\cos L \sin \alpha}{\cos L \cos \epsilon \cos \alpha - \sin \epsilon \sin L}. \quad (2.39)$$

This expression specifies the ecliptic longitude,  $\lambda$ , of the point on the ecliptic circle that ascends simultaneously with a point on the celestial equator of right ascension  $\alpha$ . Note, incidentally, that points on the celestial equator ascend at a uniform rate of  $15^\circ$  an hour at all viewing sites on the Earth's surface (except the poles, where the celestial equator does not ascend at all). The same is not true of points on the ecliptic. Expression (2.39) can be inverted to give

$$\alpha = \tan^{-1}(\tan \lambda \cos \epsilon) - \sin^{-1} \left[ \frac{\sin \lambda \sin \epsilon \tan L}{(1 - \sin^2 \lambda \sin^2 \epsilon)^{1/2}} \right]. \quad (2.40)$$

The solution of Equation (2.40) for observation sites lying above the arctic circle is complicated by the fact that, at such sites, a section of the ecliptic never sets, or descends, and a section never ascends. It is easily demonstrated that the section that never descends lies between ecliptic longitudes  $\lambda_c$  and  $180^\circ - \lambda_c$ , whereas the section that never ascends lies between longitudes  $180^\circ + \lambda_c$  and  $360^\circ - \lambda_c$ . Here,  $\lambda_c = \sin^{-1}(\cos L / \sin \epsilon)$ . Points on the ecliptic of longitude  $\lambda_c$ ,  $180^\circ - \lambda_c$ ,  $180^\circ + \lambda_c$ , and  $360^\circ - \lambda_c$  ascend simultaneously with points on the celestial equator of right ascension  $360^\circ - \alpha_c$ ,  $\alpha_c$ ,  $360^\circ - \alpha_c$ , and  $\alpha_c$ , respectively. Here,  $\alpha_c = \cos^{-1}(1 / \tan L \tan \epsilon)$ .

Tables 2.4–2.16 list the ascensions of a series of equally spaced points on the ecliptic circle, as viewed from a set of observation sites in the Earth's northern hemisphere with different terrestrial latitudes. The tables were calculated with the aid of formula (2.40). Let us now illustrate the use of these tables.

Consider a day on which the Sun is at ecliptic longitude 14LE00 (that is,  $14^\circ 00'$  into the sign of Leo). What is the length of the day (that is, the period between sunrise and sunset) at an observation site on the Earth's surface of latitude  $+30^\circ$ . Consulting Table 2.7, we find that the Sun ascends simultaneously with a point on the celestial equator of right ascension  $126^\circ 32'$ . Now, the ecliptic is a great circle on the celestial sphere. Hence, exactly half of the ecliptic is visible from any observation site on the Earth's surface. This implies that when a given point on the ecliptic circle is ascending, the point directly opposite it on the circle is descending, and vice versa. Let us term the directly opposite point the *complementary point*. By definition, the difference in ecliptic longitude between a given point on the ecliptic circle and its complementary point is  $180^\circ$ . Thus, the complementary point to 14LE00 is 14AQ00. It follows that 14AQ00 ascends at the same time that 14LE00 descends. In other words, the Sun sets when 14AQ00 ascends. Consulting Table 2.7, we find that the Sun sets at the same time that a point on the celestial equator of right ascension  $326^\circ 23'$  rises. Thus, in the time interval between the rising and setting of the Sun, a  $326^\circ 23' - 126^\circ 32' = 199^\circ 51'$  section of the celestial equator ascends at the eastern horizon. However, points on the celestial equator ascend at the uniform rate of  $15^\circ$  an hour. Thus, the length, in hours, of the period between the rising and setting of the Sun is  $199^\circ 51' / 15^\circ = 13^h 19^m$ . In other words, the length of the day in question is  $13^h 19^m$ .

The previous calculation is slightly inaccurate for a number of reasons. Firstly, it neglects the fact that the Sun is continuously moving on the ecliptic circle at the rate of about  $1^\circ$  a day.

Secondly, it neglects the fact that the celestial equator ascends at the rate of  $15^\circ$  per sidereal, rather than solar, hour. A sidereal hour is  $1/24$ th of a *sidereal day*, which is the time between successive upper transits of a fixed celestial object, such as a star. On the other hand, a solar hour is  $1/24$ th of a *solar day*, which is the mean time between successive upper transits of the Sun. A sidereal day is shorter than a solar day by 4 minutes. Fortunately, it turns out that these first two inaccuracies largely cancel one another out. Another source of inaccuracy is the fact that, due to refraction of light by the atmosphere, the Sun is actually  $1^\circ$  below the horizon when it appears to rise or set. The final source of inaccuracy is the fact that the Sun has a finite angular extent (of about half a degree), and that, strictly speaking, dawn and dusk commence when the Sun's upper limb rises and sets, respectively. Of course, our calculation only deals with the rising and setting of the center of the Sun. Taken together, the previously mentioned inaccuracies can cause the true length of a day differ from that calculated from the ascension tables by up to 15 minutes.

Tables 2.4–2.16 also effectively list the descents of a series of equally spaced points on the ecliptic circle, as viewed from a set of observation sites in the Earth's southern hemisphere with different terrestrial latitudes (which are minus those specified in the various tables). For instance, Table 2.5 gives the right ascensions of points on the celestial equator that set simultaneously with points on the ecliptic, as seen from an observation site at latitude  $-10^\circ$ .

Consider a day on which the Sun is at ecliptic longitude 08SC00. Let us calculate the length of the day at an observation site on the Earth's surface of latitude  $-50^\circ$ . Consulting Table 2.9, we find that the Sun sets simultaneously with a point on the celestial equator of right ascension  $233^\circ 09'$ . Now, the complementary point on the ecliptic to 08SC00 is 08TA00. Consulting Table 2.9 again, we find that this point sets simultaneously with a point on the celestial equator of right ascension  $18^\circ 07'$ . It follows that the Sun rises simultaneously with the latter point. Thus, the time interval between the rising and setting of the Sun is  $233^\circ 09' - 18^\circ 07' = 215^\circ 02'$  time-degrees, or  $14^h 20^m$ .

The *ascendent*, or *horoscope*, is defined as the point on the ecliptic that is ascending at the eastern horizon. Suppose that we wish to find the ascendent 2.6 hours after sunrise, as seen from an observation site of latitude  $+55^\circ$ , on a day on which the Sun has ecliptic longitude 16SC00. Of course, knowledge of the ascendent at the time of birth is key to drawing up a natal chart in astrology. Hence, this type of calculation was of great importance to the ancient Greeks. (It was of particular importance to Ptolemy, because he wrote the standard treatise on Greek natal astrology; the so-called *Tetrabiblos*.) Consulting Table 2.10, we find that, on the day in question, the Sun rises simultaneously with a point on the celestial equator of right ascension  $248^\circ 46'$ . Now, 2.6 hours corresponds to  $39^\circ 00'$ . Thus, the ascendent rises simultaneously with a point on the celestial equator of right ascension  $248^\circ 46' + 39^\circ 00' = 287^\circ 46'$ . Consulting Table 2.10 again, we find that, to the nearest degree, the ascendent at the time in question has ecliptic longitude 13SG00.

Suppose, next, that we wish to find the right ascension,  $\alpha$ , of the point on the celestial equator that culminates simultaneously with a given point on the ecliptic of ecliptic longitude  $\lambda$ . From Equation (2.33), we can see that if  $A = 180^\circ$  then  $\tan A = 0$ , and  $\tan \lambda \cos \epsilon = \tan \alpha$ , or

$$\alpha = \tan^{-1}(\tan \lambda \cos \epsilon). \quad (2.41)$$

However, this expression is identical to expression (2.40), when the latter is evaluated for the special case  $L = 0^\circ$ . It follows that our problem can be solved by consulting Table 2.4, which is the ascension table for the case of right ascension. For instance, on a day on which the ecliptic longitude of the Sun is 08TA00, we find from Table 2.4 that the right ascension of the point on the celestial equator that culminates simultaneously with the Sun (in other words, that culminates at local noon) is  $35^\circ 38'$ . Moreover, this is the case for observation sites at all terrestrial latitudes. Note that we have effectively calculated the right ascension of the Sun on the day in question.

Suppose, finally, that we wish to find the point on the ecliptic that culminates 7 hours after local noon on the aforementioned day. Because 7 hours corresponds to  $105^\circ$ , the right ascension of the point on the celestial equator that culminates simultaneously with the point in question is  $35^\circ 38' + 105^\circ 00' = 140^\circ 38'$ . Consulting Table 2.4 again, we find that, to the nearest degree, the ecliptic longitude of the point in question is 18LE00.

Note that Tables 2.4–2.16 contain essentially the same information as is contained in the “Table of ascensions by tenth parts” (Κανόνιον τῶν κατὰ δεκαμοιρίαν ἀναφορῶν) that appears in Section 8 of Book II of the *Almagest*.

### 2.15 *Azimuth of ecliptic ascension point*

Consider the azimuth of the point on the ecliptic circle that is ascending at the eastern horizon. According to Equation (2.27), the azimuth of any point on the horizon (which corresponds to  $a = 0^\circ$ ) satisfies  $\cos A = \mathbf{r} \cdot \mathbf{n}$ . It follows from Equations (2.8) and (2.25) that

$$\begin{aligned} \cos A &= -\cos \lambda \sin L \sin \alpha \\ &\quad + \sin \lambda \cos L \sin \epsilon + \sin \lambda \sin L \cos \epsilon \cos \alpha. \end{aligned} \quad (2.42)$$

Here, we have made use of the fact that the point in question also lies on the ecliptic (which corresponds to  $\beta = 0$ ), as well as the fact that  $\alpha_0 = \alpha - 90^\circ$ , where  $\alpha$  is the right ascension of the simultaneously rising point on the celestial equator. Moreover,  $\lambda$  is the ecliptic longitude of the point in question, and  $L$  the terrestrial latitude of the observation site. Now,  $\lambda$  and  $\alpha$  satisfy Equation (2.39), as well as the previous equation. Thus, eliminating  $\alpha$  between these two equations, we obtain

$$\cos A = \frac{\sin \lambda \sin \epsilon}{\cos L}. \quad (2.43)$$

This expression gives the azimuth,  $A$ , of the ascending point of the ecliptic as a function of its ecliptic longitude,  $\lambda$ , and the latitude,  $L$ , of the observation site.

For instance, suppose that we wish to find the azimuth of the point at which the Sun rises on the eastern horizon at an observation site of terrestrial latitude  $+60^\circ$ , on a day on which the Sun's ecliptic longitude is 08PI00. It follows from Equation (2.43) that

$$A = \cos^{-1}[\sin(338^\circ) \sin(23^\circ 26') / \cos(60^\circ)] = 107^\circ 20'.$$

We conclude that the Sun rises  $17^\circ 20'$  to the south of the east compass point on the day in question. It is easily demonstrated that the Sun sets  $17^\circ 20'$  south of the west compass point

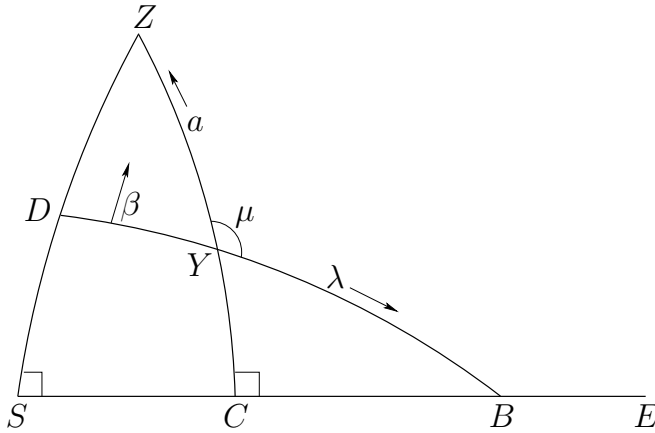


Figure 2.12: Parallaxic angle in the case where increasing altitude corresponds to increasing ecliptic latitude. Here,  $SCBE$  is the southern horizon, with  $S$  and  $E$  the south and east compass points, respectively. Moreover,  $DYB$  is the ecliptic,  $ZDS$  the meridian,  $Z$  the zenith, and  $ZYC$  an altitude circle.

on the same day (neglecting the slight change in the Sun’s ecliptic latitude during the course of the day.) Likewise, it can easily be shown that, at an observation site of terrestrial latitude  $-60^\circ$ , the Sun also rises  $17^\circ 20'$  to the south of the east compass point on the day in question, and sets  $17^\circ 20'$  to the south of the west compass point.

### 2.16 Ecliptic altitude and orientation

Consider a point on the ecliptic circle of ecliptic longitude  $\lambda$ . We wish to determine the altitude of this point, as well as the angle subtended there between the ecliptic and the vertical,  $t$  hours before or after it culminates at the meridian, as seen from an observation site on the Earth’s surface of latitude  $L$ .

The situation is as shown in Figure 2.12. Here,  $Y$  is the point in question, and  $ZYC$  an altitude circle (that is, a great circle passing through the zenith) drawn through it. We wish to determine the altitude  $a \equiv CY$  of point  $Y$ , as well as the angle  $\mu \equiv ZYB$ . Note that  $\mu$  is defined such that it lies between the ecliptic in the direction of increasing ecliptic longitude and the altitude circle in the direction of increasing altitude. Moreover,  $\mu$  is acute when increasing altitude,  $a$ , corresponds to increasing ecliptic latitude,  $\beta$ , and obtuse when increasing  $a$  corresponds to decreasing  $\beta$ . See Figures 2.12 and 2.13. Incidentally, this definition is adopted in order to simplify the calculation of lunar parallax. See Section 6.9. In the following, we shall refer to  $\mu$  as the *parallaxic angle*. However, it should be noted that, according to the modern definition, the parallaxic angle is  $90^\circ - \mu$ .

From Equations (2.15) and (2.16), the declination and right ascension of point  $Y$  are given

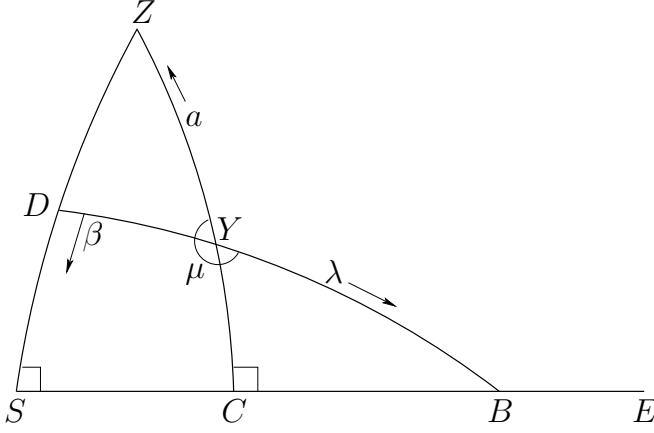


Figure 2.13: Parallactic angle in the case where increasing altitude corresponds to decreasing ecliptic latitude. Here,  $SCBE$  is the southern horizon, with  $S$  and  $E$  the south and east compass points, respectively. Moreover,  $DYB$  is the ecliptic,  $ZDS$  the meridian,  $Z$  the zenith, and  $ZYC$  an altitude circle.

by

$$\sin \delta = \sin \epsilon \sin \lambda, \quad (2.44)$$

$$\tan \alpha = \cos \epsilon \tan \lambda, \quad (2.45)$$

respectively. We can also write  $\alpha_0 = \alpha - t$ , where  $\alpha_0$  is the right ascension of the point on the ecliptic that is culminating (that is, point  $D$  in the diagram), and  $t$  is measured in time-degrees. Note that if  $t$  is positive then it measures time before culmination, whereas if it is negative then its magnitude measures time after culmination. It follows from Equations (2.30) and (2.31) that the altitude and azimuth of point  $Y$  satisfy

$$\sin a = \sin L \sin \delta + \cos L \cos \delta \cos t, \quad (2.46)$$

$$\tan A = \frac{\cos \delta \sin t}{\cos L \sin \delta - \sin L \cos \delta \cos t}, \quad (2.47)$$

respectively.

From Equation (2.8), the unit vector

$$\mathbf{r} = \cos \lambda \mathbf{v} + \sin \lambda \mathbf{s} \quad (2.48)$$

is directed from the observation site to point  $Y$ . Furthermore, the unit vector

$$\frac{\partial \mathbf{r}}{\partial \lambda} = -\sin \lambda \mathbf{v} + \cos \lambda \mathbf{s} \quad (2.49)$$

is tangent to the ecliptic circle, at point  $Y$ , in the direction of increasing ecliptic longitude. From Equation (2.27), the unit vector

$$\mathbf{r} = \cos a \sin A \mathbf{e} + \cos a \cos A \mathbf{n} + \sin a \mathbf{z} \quad (2.50)$$

is directed from the observation site to point  $Y$ . Here,  $a$  and  $A$  are the altitude and azimuth, respectively, of this point in the sky. Moreover, the unit vector

$$\begin{aligned} \frac{\partial \mathbf{r}}{\partial a} &= -\sin a \sin A \mathbf{e} - \sin a \cos A \mathbf{n} + \cos a \mathbf{z} \\ &\equiv (\cos A \mathbf{e} - \sin A \mathbf{n}) \times \mathbf{r} \end{aligned} \quad (2.51)$$

is a tangent to the altitude circle passing through point  $Y$  in the direction of increasing altitude. It follows from the definition of parallactic angle, and elementary vector algebra, that

$$\begin{aligned} \cos \mu &= \frac{\partial \mathbf{r}}{\partial \lambda} \cdot \frac{\partial \mathbf{r}}{\partial a} = -\sin \lambda \cos A \mathbf{v} \times \mathbf{e} \cdot \mathbf{r} + \sin \lambda \sin A \mathbf{v} \times \mathbf{n} \cdot \mathbf{r} \\ &\quad + \cos \lambda \cos A \mathbf{s} \times \mathbf{e} \cdot \mathbf{r} - \cos \lambda \sin A \mathbf{s} \times \mathbf{n} \cdot \mathbf{r}. \end{aligned} \quad (2.52)$$

However, according to Equations (2.24), (2.25), and (2.48),

$$\mathbf{v} \times \mathbf{e} \cdot \mathbf{r} = \sin \lambda \sin \epsilon \cos \alpha_0, \quad (2.53)$$

$$\mathbf{v} \times \mathbf{n} \cdot \mathbf{r} = -\sin \lambda (\cos L \cos \epsilon + \sin L \sin \epsilon \sin \alpha_0), \quad (2.54)$$

$$\mathbf{s} \times \mathbf{e} \cdot \mathbf{r} = -\cos \lambda \sin \epsilon \cos \alpha_0, \quad (2.55)$$

$$\mathbf{s} \times \mathbf{n} \cdot \mathbf{r} = \cos \lambda (\cos L \cos \epsilon + \sin L \sin \epsilon \sin \alpha_0). \quad (2.56)$$

The previous five equations can be combined to give

$$\begin{aligned} \cos \mu &= -\cos A \sin \epsilon \cos(\alpha - t) \\ &\quad - \sin A [\cos L \cos \epsilon + \sin L \sin \epsilon \sin(\alpha - t)]. \end{aligned} \quad (2.57)$$

Now, it follows from Equation (2.26) that

$$\mathbf{z} \cdot \mathbf{q} = \sin L \cos \epsilon - \cos L \sin \epsilon \sin(\alpha - t). \quad (2.58)$$

This quantity is significant because if  $\mathbf{z} \cdot \mathbf{q} > 0$  then increasing altitude corresponds to increasing ecliptic latitude, whereas if  $\mathbf{z} \cdot \mathbf{q} < 0$  then increasing altitude corresponds to decreasing ecliptic latitude. Thus, in the former case,  $\mu$  is the solution of Equation (2.57) that lies in the range  $0^\circ \leq \mu \leq 180^\circ$ , whereas in the latter case it is the solution that lies in the range  $180^\circ \leq \mu \leq 360^\circ$ .

According to Equation (2.46), the critical value of  $t$  at which point  $Y$  reaches the horizon is given by

$$\cos t_h = -\tan L \tan \delta. \quad (2.59)$$

Of course, the previous equation is only soluble if  $|\tan L \tan \delta| < 1$ . However, it is easily demonstrated that if  $\tan L \tan \delta < -1$  then point  $Y$  never sets, whereas if  $\tan L \tan \delta > 1$  then point  $Y$  never rises.

Note that the value of  $\mu$  at  $t = 0$  represents the inclination of the ecliptic to the vertical as point  $Y$  culminates. Furthermore, the values of  $\mu$  at  $t = t_h$  (corresponding to  $a = 0^\circ$ ) represent the inclination of the ecliptic to the vertical as point  $Y$  rises and sets.

Tables 2.17–2.25 show the altitudes of twelve equally spaced points on the ecliptic, as well as the parallactic angle at these points, as functions of time, calculated for a series of observation sites in the Earth's northern hemisphere with equally spaced terrestrial latitudes. The twelve points correspond to the start of the twelve zodiacal signs, and are named accordingly. Thus, "Aries" corresponds to ecliptic longitude  $0^\circ$ , "Taurus" to ecliptic longitude  $30^\circ$ , et cetera. For each point, four columns of data are provided. The first column corresponds to the time (in hours and minutes) either before or after the culmination of the point, the second column gives the altitude of the point (which is the same in both cases), the third column gives the parallactic angle,  $\mu$ , for the case in which the first column indicates time prior to the culmination of the point, and the fourth column gives the parallactic angle for the opposite case. Data is only provided for cases in which the various points on the ecliptic lie on or above the horizon.

Now, it can be seen, from the previous analysis, that if  $L \rightarrow -L$ ,  $t \rightarrow t$ ,  $\lambda \rightarrow \lambda + 180^\circ$  then  $\delta \rightarrow -\delta$ ,  $\alpha \rightarrow \alpha + 180^\circ$ ,  $A \rightarrow 180^\circ - A$ ,  $\cos \mu \rightarrow \cos \mu$ ,  $\mathbf{z} \cdot \mathbf{q} \rightarrow -\mathbf{z} \cdot \mathbf{q}$ , and so  $a \rightarrow a$ ,  $\mu \rightarrow 360^\circ - \mu$ . It follows that Tables 2.17–2.25 can also be used to calculate altitudes and parallactic angles of points on the ecliptic, as functions of time, for observation sites in the Earth's southern hemisphere. For example, suppose that we wish to determine the altitude and parallactic angle of the first point of Gemini (that is,  $\lambda = 60^\circ$ ), as seen from an observation site of terrestrial latitude  $-10^\circ$ , 3 hours before and after it culminates at the meridian. In order to do this, we must examine the Sagittarius (that is,  $\lambda = 240^\circ$ ) entry in the  $L = +10^\circ$  ecliptic altitude table; that is, Table 2.18 (because  $\lambda \rightarrow \lambda + 180^\circ$  when  $L \rightarrow -L$ ). The fourth row of this entry tells us that,  $t = 03:00$  hours before culmination, the altitude and parallactic angle of the first point of Gemini are  $a = 36^\circ 26'$  and  $\mu = 360^\circ - 162^\circ 11' = 197^\circ 49'$ , respectively (because  $a \rightarrow a$  and  $\mu \rightarrow 360^\circ - \mu$  as  $L \rightarrow -L$ ). This row also tells us that,  $t = 03:00$  hours after culmination, the altitude and parallactic angle of the first point of Gemini are  $a = 36^\circ 26'$  and  $\mu = 360^\circ - 042^\circ 17' = 317^\circ 43'$ , respectively.

Note that Tables 2.17–2.25 contain essentially the same information as is contained in the "Tabulation of angles and arcs according to parallels" (*\*Εκθεσις τῶν κατὰ παράλληλον γωνιῶν καὶ περιφερειῶν*) that appears in Section 13 of Book II of the *Almagest*.



## 2.17 Tables

| Aries     |         |      |              | Libra       |         |      |              |
|-----------|---------|------|--------------|-------------|---------|------|--------------|
| $\lambda$ | $\beta$ | Mag. | Name         | $\lambda$   | $\beta$ | Mag. | Name         |
| 09°09'    | +12°36' | +2.8 | $\gamma$ PEG | 04°50'      | +01°22' | +3.9 | $\eta$ VIR   |
| 26°49'    | +05°23' | +3.6 | $\eta$ PSC   | 10°08'      | +02°46' | +3.5 | $\gamma$ VIR |
|           |         |      |              | 11°28'      | +08°37' | +3.4 | $\delta$ VIR |
|           |         |      |              | 22°08'      | +08°38' | +3.4 | $\zeta$ VIR  |
|           |         |      |              | 23°51'      | −02°03' | +1.0 | $\alpha$ VIR |
| Taurus    |         |      |              | Scorpio     |         |      |              |
| $\lambda$ | $\beta$ | Mag. | Name         | $\lambda$   | $\beta$ | Mag. | Name         |
| 03°58'    | +08°29' | +2.6 | $\beta$ ARI  | 15°5'       | +00°20' | +2.8 | $\alpha$ LIB |
| 07°39'    | +09°58' | +2.0 | $\alpha$ ARI | 19°22'      | +08°30' | +2.6 | $\beta$ LIB  |
| Gemini    |         |      |              | Sagittarius |         |      |              |
| $\lambda$ | $\beta$ | Mag. | Name         | $\lambda$   | $\beta$ | Mag. | Name         |
| 00°00'    | +04°03' | +2.9 | $\eta$ TAU   | 02°34'      | −01°59' | +2.3 | $\delta$ SCO |
| 09°47'    | −05°28' | +0.9 | $\alpha$ TAU | 02°56'      | −05°29' | +2.9 | $\pi$ SCO    |
| 22°34'    | +05°23' | +1.7 | $\beta$ TAU  | 03°11'      | +01°00' | +2.6 | $\beta$ SCO  |
|           |         |      |              | 07°48'      | −04°02' | +2.9 | $\sigma$ SCO |
|           |         |      |              | 09°46'      | −04°34' | +1.0 | $\alpha$ SCO |
|           |         |      |              | 11°27'      | −06°08' | +2.8 | $\tau$ SCO   |
|           |         |      |              | 17°58'      | +07°12' | +2.4 | $\eta$ OPH   |

| Cancer    |         |      |                | Capricorn |         |      |               |
|-----------|---------|------|----------------|-----------|---------|------|---------------|
| $\lambda$ | $\beta$ | Mag. | Name           | $\lambda$ | $\beta$ | Mag. | Name          |
| 05°18′    | −00°49′ | +2.9 | $\mu$ GEM      | 01°16′    | −07°00′ | +3.0 | $\gamma$ SGR  |
| 09°06′    | −06°44′ | +1.9 | $\gamma$ GEM   | 04°34′    | −06°28′ | +2.7 | $\delta$ SGR  |
| 09°56′    | +02°04′ | +3.0 | $\epsilon$ GEM | 06°19′    | −02°08′ | +2.8 | $\lambda$ SGR |
| 20°14′    | +10°06′ | +2.0 | $\alpha$ GEM   | 12°23′    | −03°27′ | +2.0 | $\sigma$ SGR  |
| 23°13′    | +06°41′ | +1.1 | $\beta$ GEM    | 13°38′    | −07°11′ | +2.6 | $\zeta$ SGR   |
|           |         |      |                | 16°15′    | +01°26′ | +2.9 | $\pi$ SGR     |
| Leo       |         |      |                | Aquarius  |         |      |               |
| $\lambda$ | $\beta$ | Mag. | Name           | $\lambda$ | $\beta$ | Mag. | Name          |
| 20°47′    | +09°43′ | +3.0 | $\epsilon$ LEO | 23°24′    | +08°37′ | +2.9 | $\beta$ AQR   |
| 29°37′    | +08°49′ | +2.6 | $\gamma$ LEO   | 23°33′    | −02°36′ | +2.9 | $\delta$ CAP  |
| 29°50′    | +00°28′ | +1.4 | $\alpha$ LEO   |           |         |      |               |
| Virgo     |         |      |                | Pisces    |         |      |               |
| $\lambda$ | $\beta$ | Mag. | Name           | $\lambda$ | $\beta$ | Mag. | Name          |
| 06°23′    | +00°09′ | +3.9 | $\rho$ LEO     | 06°43′    | +08°14′ | +3.8 | $\gamma$ AQR  |
| 13°25′    | +09°40′ | +3.3 | $\theta$ LEO   | 08°52′    | −08°12′ | +3.3 | $\delta$ AQR  |
| 17°34′    | +06°06′ | +3.9 | $\iota$ LEO    | 11°34′    | −00°23′ | +3.7 | $\lambda$ AQR |
| 27°10′    | +00°42′ | +3.6 | $\beta$ VIR    | 21°27′    | +07°15′ | +3.7 | $\gamma$ PSC  |

Table 2.1: Ecliptic longitudes (relative to the mean equinox at the J2000 epoch), ecliptic latitudes, and visual magnitudes of selected bright stars lying within 10° of the ecliptic plane.

| Aries     |          |          | Taurus    |          |          | Gemini    |          |          |
|-----------|----------|----------|-----------|----------|----------|-----------|----------|----------|
| $\lambda$ | $\delta$ | $\alpha$ | $\lambda$ | $\delta$ | $\alpha$ | $\lambda$ | $\delta$ | $\alpha$ |
| 00°       | +00°00′  | 000°00′  | 00°       | +11°28′  | 027°55′  | 00°       | +20°09′  | 057°49′  |
| 02°       | +00°48′  | 001°50′  | 02°       | +12°10′  | 029°50′  | 02°       | +20°33′  | 059°54′  |
| 04°       | +01°35′  | 003°40′  | 04°       | +12°51′  | 031°45′  | 04°       | +20°57′  | 062°00′  |
| 06°       | +02°23′  | 005°30′  | 06°       | +13°31′  | 033°41′  | 06°       | +21°18′  | 064°07′  |
| 08°       | +03°10′  | 007°21′  | 08°       | +14°10′  | 035°38′  | 08°       | +21°38′  | 066°14′  |
| 10°       | +03°58′  | 009°11′  | 10°       | +14°49′  | 037°36′  | 10°       | +21°57′  | 068°22′  |
| 12°       | +04°45′  | 011°02′  | 12°       | +15°26′  | 039°34′  | 12°       | +22°13′  | 070°30′  |
| 14°       | +05°31′  | 012°53′  | 14°       | +16°02′  | 041°33′  | 14°       | +22°28′  | 072°39′  |
| 16°       | +06°18′  | 014°44′  | 16°       | +16°37′  | 043°32′  | 16°       | +22°42′  | 074°48′  |
| 18°       | +07°04′  | 016°36′  | 18°       | +17°11′  | 045°32′  | 18°       | +22°54′  | 076°57′  |
| 20°       | +07°49′  | 018°28′  | 20°       | +17°44′  | 047°33′  | 20°       | +23°03′  | 079°07′  |
| 22°       | +08°34′  | 020°20′  | 22°       | +18°16′  | 049°35′  | 22°       | +23°12′  | 081°17′  |
| 24°       | +09°19′  | 022°13′  | 24°       | +18°46′  | 051°38′  | 24°       | +23°18′  | 083°28′  |
| 26°       | +10°02′  | 024°07′  | 26°       | +19°15′  | 053°41′  | 26°       | +23°22′  | 085°39′  |
| 28°       | +10°46′  | 026°00′  | 28°       | +19°43′  | 055°45′  | 28°       | +23°25′  | 087°49′  |
| 30°       | +11°28′  | 027°55′  | 30°       | +20°09′  | 057°49′  | 30°       | +23°26′  | 090°00′  |

| Cancer    |          |          | Leo       |          |          | Virgo     |          |          |
|-----------|----------|----------|-----------|----------|----------|-----------|----------|----------|
| $\lambda$ | $\delta$ | $\alpha$ | $\lambda$ | $\delta$ | $\alpha$ | $\lambda$ | $\delta$ | $\alpha$ |
| 00°       | +23°26′  | 090°00′  | 00°       | +20°09′  | 122°11′  | 00°       | +11°28′  | 152°05′  |
| 02°       | +23°25′  | 092°11′  | 02°       | +19°43′  | 124°15′  | 02°       | +10°46′  | 154°00′  |
| 04°       | +23°22′  | 094°21′  | 04°       | +19°15′  | 126°19′  | 04°       | +10°02′  | 155°53′  |
| 06°       | +23°18′  | 096°32′  | 06°       | +18°46′  | 128°22′  | 06°       | +09°19′  | 157°47′  |
| 08°       | +23°12′  | 098°43′  | 08°       | +18°16′  | 130°25′  | 08°       | +08°34′  | 159°40′  |
| 10°       | +23°03′  | 100°53′  | 10°       | +17°44′  | 132°27′  | 10°       | +07°49′  | 161°32′  |
| 12°       | +22°54′  | 103°03′  | 12°       | +17°11′  | 134°28′  | 12°       | +07°04′  | 163°24′  |
| 14°       | +22°42′  | 105°12′  | 14°       | +16°37′  | 136°28′  | 14°       | +06°18′  | 165°16′  |
| 16°       | +22°28′  | 107°21′  | 16°       | +16°02′  | 138°27′  | 16°       | +05°31′  | 167°07′  |
| 18°       | +22°13′  | 109°30′  | 18°       | +15°26′  | 140°26′  | 18°       | +04°45′  | 168°58′  |
| 20°       | +21°57′  | 111°38′  | 20°       | +14°49′  | 142°24′  | 20°       | +03°58′  | 170°49′  |
| 22°       | +21°38′  | 113°46′  | 22°       | +14°10′  | 144°22′  | 22°       | +03°10′  | 172°39′  |
| 24°       | +21°18′  | 115°53′  | 24°       | +13°31′  | 146°19′  | 24°       | +02°23′  | 174°30′  |
| 26°       | +20°57′  | 118°00′  | 26°       | +12°51′  | 148°15′  | 26°       | +01°35′  | 176°20′  |
| 28°       | +20°33′  | 120°06′  | 28°       | +12°10′  | 150°10′  | 28°       | +00°48′  | 178°10′  |
| 30°       | +20°09′  | 122°11′  | 30°       | +11°28′  | 152°05′  | 30°       | +00°00′  | 180°00′  |

| Libra     |          |          | Scorpio   |          |          | Sagittarius |          |          |
|-----------|----------|----------|-----------|----------|----------|-------------|----------|----------|
| $\lambda$ | $\delta$ | $\alpha$ | $\lambda$ | $\delta$ | $\alpha$ | $\lambda$   | $\delta$ | $\alpha$ |
| 00°       | −00°00′  | 180°00′  | 00°       | −11°28′  | 207°55′  | 00°         | −20°09′  | 237°49′  |
| 02°       | −00°48′  | 181°50′  | 02°       | −12°10′  | 209°50′  | 02°         | −20°33′  | 239°54′  |
| 04°       | −01°35′  | 183°40′  | 04°       | −12°51′  | 211°45′  | 04°         | −20°57′  | 242°00′  |
| 06°       | −02°23′  | 185°30′  | 06°       | −13°31′  | 213°41′  | 06°         | −21°18′  | 244°07′  |
| 08°       | −03°10′  | 187°21′  | 08°       | −14°10′  | 215°38′  | 08°         | −21°38′  | 246°14′  |
| 10°       | −03°58′  | 189°11′  | 10°       | −14°49′  | 217°36′  | 10°         | −21°57′  | 248°22′  |
| 12°       | −04°45′  | 191°02′  | 12°       | −15°26′  | 219°34′  | 12°         | −22°13′  | 250°30′  |
| 14°       | −05°31′  | 192°53′  | 14°       | −16°02′  | 221°33′  | 14°         | −22°28′  | 252°39′  |
| 16°       | −06°18′  | 194°44′  | 16°       | −16°37′  | 223°32′  | 16°         | −22°42′  | 254°48′  |
| 18°       | −07°04′  | 196°36′  | 18°       | −17°11′  | 225°32′  | 18°         | −22°54′  | 256°57′  |
| 20°       | −07°49′  | 198°28′  | 20°       | −17°44′  | 227°33′  | 20°         | −23°03′  | 259°07′  |
| 22°       | −08°34′  | 200°20′  | 22°       | −18°16′  | 229°35′  | 22°         | −23°12′  | 261°17′  |
| 24°       | −09°19′  | 202°13′  | 24°       | −18°46′  | 231°38′  | 24°         | −23°18′  | 263°28′  |
| 26°       | −10°02′  | 204°07′  | 26°       | −19°15′  | 233°41′  | 26°         | −23°22′  | 265°39′  |
| 28°       | −10°46′  | 206°00′  | 28°       | −19°43′  | 235°45′  | 28°         | −23°25′  | 267°49′  |
| 30°       | −11°28′  | 207°55′  | 30°       | −20°09′  | 237°49′  | 30°         | −23°26′  | 270°00′  |

| Capricorn |          |          | Aquarius  |          |          | Pisces    |          |          |
|-----------|----------|----------|-----------|----------|----------|-----------|----------|----------|
| $\lambda$ | $\delta$ | $\alpha$ | $\lambda$ | $\delta$ | $\alpha$ | $\lambda$ | $\delta$ | $\alpha$ |
| 00°       | −23°26′  | 270°00′  | 00°       | −20°09′  | 302°11′  | 00°       | −11°28′  | 332°05′  |
| 02°       | −23°25′  | 272°11′  | 02°       | −19°43′  | 304°15′  | 02°       | −10°46′  | 333°60′  |
| 04°       | −23°22′  | 274°21′  | 04°       | −19°15′  | 306°19′  | 04°       | −10°02′  | 335°53′  |
| 06°       | −23°18′  | 276°32′  | 06°       | −18°46′  | 308°22′  | 06°       | −09°19′  | 337°47′  |
| 08°       | −23°12′  | 278°43′  | 08°       | −18°16′  | 310°25′  | 08°       | −08°34′  | 339°40′  |
| 10°       | −23°03′  | 280°53′  | 10°       | −17°44′  | 312°27′  | 10°       | −07°49′  | 341°32′  |
| 12°       | −22°54′  | 283°03′  | 12°       | −17°11′  | 314°28′  | 12°       | −07°04′  | 343°24′  |
| 14°       | −22°42′  | 285°12′  | 14°       | −16°37′  | 316°28′  | 14°       | −06°18′  | 345°16′  |
| 16°       | −22°28′  | 287°21′  | 16°       | −16°02′  | 318°27′  | 16°       | −05°31′  | 347°07′  |
| 18°       | −22°13′  | 289°30′  | 18°       | −15°26′  | 320°26′  | 18°       | −04°45′  | 348°58′  |
| 20°       | −21°57′  | 291°38′  | 20°       | −14°49′  | 322°24′  | 20°       | −03°58′  | 350°49′  |
| 22°       | −21°38′  | 293°46′  | 22°       | −14°10′  | 324°22′  | 22°       | −03°10′  | 352°39′  |
| 24°       | −21°18′  | 295°53′  | 24°       | −13°31′  | 326°19′  | 24°       | −02°23′  | 354°30′  |
| 26°       | −20°57′  | 297°60′  | 26°       | −12°51′  | 328°15′  | 26°       | −01°35′  | 356°20′  |
| 28°       | −20°33′  | 300°06′  | 28°       | −12°10′  | 330°10′  | 28°       | −00°48′  | 358°10′  |
| 30°       | −20°09′  | 302°11′  | 30°       | −11°28′  | 332°05′  | 30°       | −00°00′  | 360°00′  |

Table 2.2: Declinations and right ascensions of points on the ecliptic circle.

| $L$  | Longest Day                      | Summer Solstice Noon<br>Altitude of Sun | Equinoctial Noon<br>Altitude of Sun | Winter Solstice Noon<br>Altitude of Sun |
|------|----------------------------------|-----------------------------------------|-------------------------------------|-----------------------------------------|
| +00° | 12 <sup>h</sup> 00 <sup>m</sup>  | +66°34' N                               | +90°00' S                           | +66°34' S                               |
| +05° | 12 <sup>h</sup> 17 <sup>m</sup>  | +71°34' N                               | +85°00' S                           | +61°34' S                               |
| +10° | 12 <sup>h</sup> 35 <sup>m</sup>  | +76°34' N                               | +80°00' S                           | +56°34' S                               |
| +15° | 12 <sup>h</sup> 53 <sup>m</sup>  | +81°34' N                               | +75°00' S                           | +51°34' S                               |
| +20° | 13 <sup>h</sup> 13 <sup>m</sup>  | +86°34' N                               | +70°00' S                           | +46°34' S                               |
| +25° | 13 <sup>h</sup> 33 <sup>m</sup>  | +88°26' S                               | +65°00' S                           | +41°34' S                               |
| +30° | 13 <sup>h</sup> 56 <sup>m</sup>  | +83°26' S                               | +60°00' S                           | +36°34' S                               |
| +35° | 14 <sup>h</sup> 21 <sup>m</sup>  | +78°26' S                               | +55°00' S                           | +31°34' S                               |
| +40° | 14 <sup>h</sup> 51 <sup>m</sup>  | +73°26' S                               | +50°00' S                           | +26°34' S                               |
| +45° | 15 <sup>h</sup> 25 <sup>m</sup>  | +68°26' S                               | +45°00' S                           | +21°34' S                               |
| +50° | 16 <sup>h</sup> 09 <sup>m</sup>  | +63°26' S                               | +40°00' S                           | +16°34' S                               |
| +55° | 17 <sup>h</sup> 06 <sup>m</sup>  | +58°26' S                               | +35°00' S                           | +11°34' S                               |
| +60° | 18 <sup>h</sup> 29 <sup>m</sup>  | +53°26' S                               | +30°00' S                           | +06°34' S                               |
| +65° | 21 <sup>h</sup> 07 <sup>m</sup>  | +48°26' S                               | +25°00' S                           | +01°34' S                               |
| +70° | 62 <sup>d</sup> 06 <sup>h</sup>  | +43°26' S                               | +20°00' S                           | −03°26' S                               |
| +75° | 100 <sup>d</sup> 06 <sup>h</sup> | +38°26' S                               | +15°00' S                           | −08°26' S                               |
| +80° | 130 <sup>d</sup> 02 <sup>h</sup> | +33°26' S                               | +10°00' S                           | −13°26' S                               |
| +85° | 156 <sup>d</sup> 22 <sup>h</sup> | +28°26' S                               | +05°00' S                           | −18°26' S                               |
| +90° | 182 <sup>d</sup> 15 <sup>h</sup> | +23°26' S                               | +00°00' S                           | −23°26' S                               |

Table 2.3: Terrestrial climes in the Earth's northern hemisphere. The superscripts  $h$ ,  $m$ , and  $d$  stand for hours, minutes, and days, respectively. The symbols S and N indicated that the upper transit of the Sun occurs to the south and north of the zenith, respectively.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 000°00′  | 00°       | 027°55′  | 00°       | 057°49′  | 00°       | 090°00′  | 00°       | 122°11′  | 00°       | 152°05′  |
| 02°       | 001°50′  | 02°       | 029°50′  | 02°       | 059°54′  | 02°       | 092°11′  | 02°       | 124°15′  | 02°       | 154°00′  |
| 04°       | 003°40′  | 04°       | 031°45′  | 04°       | 062°00′  | 04°       | 094°21′  | 04°       | 126°19′  | 04°       | 155°53′  |
| 06°       | 005°30′  | 06°       | 033°41′  | 06°       | 064°07′  | 06°       | 096°32′  | 06°       | 128°22′  | 06°       | 157°47′  |
| 08°       | 007°21′  | 08°       | 035°38′  | 08°       | 066°14′  | 08°       | 098°43′  | 08°       | 130°25′  | 08°       | 159°40′  |
| 10°       | 009°11′  | 10°       | 037°36′  | 10°       | 068°22′  | 10°       | 100°53′  | 10°       | 132°27′  | 10°       | 161°32′  |
| 12°       | 011°02′  | 12°       | 039°34′  | 12°       | 070°30′  | 12°       | 103°03′  | 12°       | 134°28′  | 12°       | 163°24′  |
| 14°       | 012°53′  | 14°       | 041°33′  | 14°       | 072°39′  | 14°       | 105°12′  | 14°       | 136°28′  | 14°       | 165°16′  |
| 16°       | 014°44′  | 16°       | 043°32′  | 16°       | 074°48′  | 16°       | 107°21′  | 16°       | 138°27′  | 16°       | 167°07′  |
| 18°       | 016°36′  | 18°       | 045°32′  | 18°       | 076°57′  | 18°       | 109°30′  | 18°       | 140°26′  | 18°       | 168°58′  |
| 20°       | 018°28′  | 20°       | 047°33′  | 20°       | 079°07′  | 20°       | 111°38′  | 20°       | 142°24′  | 20°       | 170°49′  |
| 22°       | 020°20′  | 22°       | 049°35′  | 22°       | 081°17′  | 22°       | 113°46′  | 22°       | 144°22′  | 22°       | 172°39′  |
| 24°       | 022°13′  | 24°       | 051°38′  | 24°       | 083°28′  | 24°       | 115°53′  | 24°       | 146°19′  | 24°       | 174°30′  |
| 26°       | 024°07′  | 26°       | 053°41′  | 26°       | 085°39′  | 26°       | 118°00′  | 26°       | 148°15′  | 26°       | 176°20′  |
| 28°       | 026°00′  | 28°       | 055°45′  | 28°       | 087°49′  | 28°       | 120°06′  | 28°       | 150°10′  | 28°       | 178°10′  |
| 30°       | 027°55′  | 30°       | 057°49′  | 30°       | 090°00′  | 30°       | 122°11′  | 30°       | 152°05′  | 30°       | 180°00′  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00′  | 00°       | 207°55′  | 00°         | 237°49′  | 00°       | 270°00′  | 00°       | 302°11′  | 00°       | 332°05′  |
| 02°       | 181°50′  | 02°       | 209°50′  | 02°         | 239°54′  | 02°       | 272°11′  | 02°       | 304°15′  | 02°       | 333°60′  |
| 04°       | 183°40′  | 04°       | 211°45′  | 04°         | 242°00′  | 04°       | 274°21′  | 04°       | 306°19′  | 04°       | 335°53′  |
| 06°       | 185°30′  | 06°       | 213°41′  | 06°         | 244°07′  | 06°       | 276°32′  | 06°       | 308°22′  | 06°       | 337°47′  |
| 08°       | 187°21′  | 08°       | 215°38′  | 08°         | 246°14′  | 08°       | 278°43′  | 08°       | 310°25′  | 08°       | 339°40′  |
| 10°       | 189°11′  | 10°       | 217°36′  | 10°         | 248°22′  | 10°       | 280°53′  | 10°       | 312°27′  | 10°       | 341°32′  |
| 12°       | 191°02′  | 12°       | 219°34′  | 12°         | 250°30′  | 12°       | 283°03′  | 12°       | 314°28′  | 12°       | 343°24′  |
| 14°       | 192°53′  | 14°       | 221°33′  | 14°         | 252°39′  | 14°       | 285°12′  | 14°       | 316°28′  | 14°       | 345°16′  |
| 16°       | 194°44′  | 16°       | 223°32′  | 16°         | 254°48′  | 16°       | 287°21′  | 16°       | 318°27′  | 16°       | 347°07′  |
| 18°       | 196°36′  | 18°       | 225°32′  | 18°         | 256°57′  | 18°       | 289°30′  | 18°       | 320°26′  | 18°       | 348°58′  |
| 20°       | 198°28′  | 20°       | 227°33′  | 20°         | 259°07′  | 20°       | 291°38′  | 20°       | 322°24′  | 20°       | 350°49′  |
| 22°       | 200°20′  | 22°       | 229°35′  | 22°         | 261°17′  | 22°       | 293°46′  | 22°       | 324°22′  | 22°       | 352°39′  |
| 24°       | 202°13′  | 24°       | 231°38′  | 24°         | 263°28′  | 24°       | 295°53′  | 24°       | 326°19′  | 24°       | 354°30′  |
| 26°       | 204°07′  | 26°       | 233°41′  | 26°         | 265°39′  | 26°       | 298°00′  | 26°       | 328°15′  | 26°       | 356°20′  |
| 28°       | 206°00′  | 28°       | 235°45′  | 28°         | 267°49′  | 28°       | 300°06′  | 28°       | 330°10′  | 28°       | 358°10′  |
| 30°       | 207°55′  | 30°       | 237°49′  | 30°         | 270°00′  | 30°       | 302°11′  | 30°       | 332°05′  | 30°       | 360°00′  |

Table 2.4: Right ascensions of the ecliptic for latitude 0°.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 000°00′  | 00°       | 025°52′  | 00°       | 054°07′  | 00°       | 085°37′  | 00°       | 118°28′  | 00°       | 150°02′  |
| 02°       | 001°42′  | 02°       | 027°39′  | 02°       | 056°07′  | 02°       | 087°48′  | 02°       | 120°38′  | 02°       | 152°04′  |
| 04°       | 003°23′  | 04°       | 029°27′  | 04°       | 058°08′  | 04°       | 089°59′  | 04°       | 122°47′  | 04°       | 154°06′  |
| 06°       | 005°05′  | 06°       | 031°16′  | 06°       | 060°10′  | 06°       | 092°11′  | 06°       | 124°56′  | 06°       | 156°07′  |
| 08°       | 006°47′  | 08°       | 033°05′  | 08°       | 062°13′  | 08°       | 094°23′  | 08°       | 127°05′  | 08°       | 158°08′  |
| 10°       | 008°29′  | 10°       | 034°55′  | 10°       | 064°17′  | 10°       | 096°34′  | 10°       | 129°13′  | 10°       | 160°09′  |
| 12°       | 010°12′  | 12°       | 036°46′  | 12°       | 066°22′  | 12°       | 098°46′  | 12°       | 131°20′  | 12°       | 162°09′  |
| 14°       | 011°55′  | 14°       | 038°38′  | 14°       | 068°28′  | 14°       | 100°58′  | 14°       | 133°27′  | 14°       | 164°09′  |
| 16°       | 013°38′  | 16°       | 040°31′  | 16°       | 070°34′  | 16°       | 103°10′  | 16°       | 135°33′  | 16°       | 166°08′  |
| 18°       | 015°21′  | 18°       | 042°25′  | 18°       | 072°41′  | 18°       | 105°22′  | 18°       | 137°39′  | 18°       | 168°08′  |
| 20°       | 017°05′  | 20°       | 044°19′  | 20°       | 074°49′  | 20°       | 107°34′  | 20°       | 139°44′  | 20°       | 170°07′  |
| 22°       | 018°49′  | 22°       | 046°15′  | 22°       | 076°58′  | 22°       | 109°45′  | 22°       | 141°49′  | 22°       | 172°06′  |
| 24°       | 020°34′  | 24°       | 048°11′  | 24°       | 079°07′  | 24°       | 111°57′  | 24°       | 143°53′  | 24°       | 174°04′  |
| 26°       | 022°19′  | 26°       | 050°09′  | 26°       | 081°16′  | 26°       | 114°07′  | 26°       | 145°57′  | 26°       | 176°03′  |
| 28°       | 024°05′  | 28°       | 052°07′  | 28°       | 083°26′  | 28°       | 116°18′  | 28°       | 148°00′  | 28°       | 178°01′  |
| 30°       | 025°52′  | 30°       | 054°07′  | 30°       | 085°37′  | 30°       | 118°28′  | 30°       | 150°02′  | 30°       | 180°00′  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00′  | 00°       | 209°58′  | 00°         | 241°32′  | 00°       | 274°23′  | 00°       | 305°53′  | 00°       | 334°08′  |
| 02°       | 181°59′  | 02°       | 212°00′  | 02°         | 243°42′  | 02°       | 276°34′  | 02°       | 307°53′  | 02°       | 335°55′  |
| 04°       | 183°57′  | 04°       | 214°03′  | 04°         | 245°53′  | 04°       | 278°44′  | 04°       | 309°51′  | 04°       | 337°41′  |
| 06°       | 185°56′  | 06°       | 216°07′  | 06°         | 248°03′  | 06°       | 280°53′  | 06°       | 311°49′  | 06°       | 339°26′  |
| 08°       | 187°54′  | 08°       | 218°11′  | 08°         | 250°15′  | 08°       | 283°02′  | 08°       | 313°45′  | 08°       | 341°11′  |
| 10°       | 189°53′  | 10°       | 220°16′  | 10°         | 252°26′  | 10°       | 285°11′  | 10°       | 315°41′  | 10°       | 342°55′  |
| 12°       | 191°52′  | 12°       | 222°21′  | 12°         | 254°38′  | 12°       | 287°19′  | 12°       | 317°35′  | 12°       | 344°39′  |
| 14°       | 193°52′  | 14°       | 224°27′  | 14°         | 256°50′  | 14°       | 289°26′  | 14°       | 319°29′  | 14°       | 346°22′  |
| 16°       | 195°51′  | 16°       | 226°33′  | 16°         | 259°02′  | 16°       | 291°32′  | 16°       | 321°22′  | 16°       | 348°05′  |
| 18°       | 197°51′  | 18°       | 228°40′  | 18°         | 261°14′  | 18°       | 293°38′  | 18°       | 323°14′  | 18°       | 349°48′  |
| 20°       | 199°51′  | 20°       | 230°47′  | 20°         | 263°26′  | 20°       | 295°43′  | 20°       | 325°05′  | 20°       | 351°31′  |
| 22°       | 201°52′  | 22°       | 232°55′  | 22°         | 265°37′  | 22°       | 297°47′  | 22°       | 326°55′  | 22°       | 353°13′  |
| 24°       | 203°53′  | 24°       | 235°04′  | 24°         | 267°49′  | 24°       | 299°50′  | 24°       | 328°44′  | 24°       | 354°55′  |
| 26°       | 205°54′  | 26°       | 237°13′  | 26°         | 270°01′  | 26°       | 301°52′  | 26°       | 330°33′  | 26°       | 356°37′  |
| 28°       | 207°56′  | 28°       | 239°22′  | 28°         | 272°12′  | 28°       | 303°53′  | 28°       | 332°21′  | 28°       | 358°18′  |
| 30°       | 209°58′  | 30°       | 241°32′  | 30°         | 274°23′  | 30°       | 305°53′  | 30°       | 334°08′  | 30°       | 360°00′  |

Table 2.5: Oblique ascensions of the ecliptic for latitude +10°.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 000°00′  | 00°       | 023°41′  | 00°       | 050°09′  | 00°       | 080°55′  | 00°       | 114°30′  | 00°       | 147°51′  |
| 02°       | 001°33′  | 02°       | 025°20′  | 02°       | 052°04′  | 02°       | 083°07′  | 02°       | 116°46′  | 02°       | 150°02′  |
| 04°       | 003°06′  | 04°       | 026°59′  | 04°       | 054°00′  | 04°       | 085°18′  | 04°       | 119°01′  | 04°       | 152°12′  |
| 06°       | 004°38′  | 06°       | 028°40′  | 06°       | 055°57′  | 06°       | 087°31′  | 06°       | 121°16′  | 06°       | 154°22′  |
| 08°       | 006°12′  | 08°       | 030°22′  | 08°       | 057°56′  | 08°       | 089°44′  | 08°       | 123°31′  | 08°       | 156°31′  |
| 10°       | 007°45′  | 10°       | 032°04′  | 10°       | 059°56′  | 10°       | 091°58′  | 10°       | 125°46′  | 10°       | 158°40′  |
| 12°       | 009°18′  | 12°       | 033°48′  | 12°       | 061°57′  | 12°       | 094°12′  | 12°       | 128°00′  | 12°       | 160°49′  |
| 14°       | 010°52′  | 14°       | 035°32′  | 14°       | 063°59′  | 14°       | 096°27′  | 14°       | 130°14′  | 14°       | 162°58′  |
| 16°       | 012°26′  | 16°       | 037°18′  | 16°       | 066°02′  | 16°       | 098°42′  | 16°       | 132°27′  | 16°       | 165°06′  |
| 18°       | 014°01′  | 18°       | 039°04′  | 18°       | 068°07′  | 18°       | 100°57′  | 18°       | 134°40′  | 18°       | 167°14′  |
| 20°       | 015°36′  | 20°       | 040°52′  | 20°       | 070°13′  | 20°       | 103°12′  | 20°       | 136°53′  | 20°       | 169°22′  |
| 22°       | 017°12′  | 22°       | 042°41′  | 22°       | 072°19′  | 22°       | 105°28′  | 22°       | 139°06′  | 22°       | 171°30′  |
| 24°       | 018°48′  | 24°       | 044°31′  | 24°       | 074°27′  | 24°       | 107°44′  | 24°       | 141°18′  | 24°       | 173°37′  |
| 26°       | 020°25′  | 26°       | 046°23′  | 26°       | 076°35′  | 26°       | 109°59′  | 26°       | 143°29′  | 26°       | 175°45′  |
| 28°       | 022°02′  | 28°       | 048°15′  | 28°       | 078°45′  | 28°       | 112°15′  | 28°       | 145°40′  | 28°       | 177°53′  |
| 30°       | 023°41′  | 30°       | 050°09′  | 30°       | 080°55′  | 30°       | 114°30′  | 30°       | 147°51′  | 30°       | 180°00′  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00′  | 00°       | 212°09′  | 00°         | 245°30′  | 00°       | 279°05′  | 00°       | 309°51′  | 00°       | 336°19′  |
| 02°       | 182°07′  | 02°       | 214°20′  | 02°         | 247°45′  | 02°       | 281°15′  | 02°       | 311°45′  | 02°       | 337°58′  |
| 04°       | 184°15′  | 04°       | 216°31′  | 04°         | 250°01′  | 04°       | 283°25′  | 04°       | 313°37′  | 04°       | 339°35′  |
| 06°       | 186°23′  | 06°       | 218°42′  | 06°         | 252°16′  | 06°       | 285°33′  | 06°       | 315°29′  | 06°       | 341°12′  |
| 08°       | 188°30′  | 08°       | 220°54′  | 08°         | 254°32′  | 08°       | 287°41′  | 08°       | 317°19′  | 08°       | 342°48′  |
| 10°       | 190°38′  | 10°       | 223°07′  | 10°         | 256°48′  | 10°       | 289°47′  | 10°       | 319°08′  | 10°       | 344°24′  |
| 12°       | 192°46′  | 12°       | 225°20′  | 12°         | 259°03′  | 12°       | 291°53′  | 12°       | 320°56′  | 12°       | 345°59′  |
| 14°       | 194°54′  | 14°       | 227°33′  | 14°         | 261°18′  | 14°       | 293°58′  | 14°       | 322°42′  | 14°       | 347°34′  |
| 16°       | 197°02′  | 16°       | 229°46′  | 16°         | 263°33′  | 16°       | 296°01′  | 16°       | 324°28′  | 16°       | 349°08′  |
| 18°       | 199°11′  | 18°       | 232°00′  | 18°         | 265°48′  | 18°       | 298°03′  | 18°       | 326°12′  | 18°       | 350°42′  |
| 20°       | 201°20′  | 20°       | 234°14′  | 20°         | 268°02′  | 20°       | 300°04′  | 20°       | 327°56′  | 20°       | 352°15′  |
| 22°       | 203°29′  | 22°       | 236°29′  | 22°         | 270°16′  | 22°       | 302°04′  | 22°       | 329°38′  | 22°       | 353°48′  |
| 24°       | 205°38′  | 24°       | 238°44′  | 24°         | 272°29′  | 24°       | 304°03′  | 24°       | 331°20′  | 24°       | 355°22′  |
| 26°       | 207°48′  | 26°       | 240°59′  | 26°         | 274°42′  | 26°       | 306°00′  | 26°       | 333°01′  | 26°       | 356°54′  |
| 28°       | 209°58′  | 28°       | 243°14′  | 28°         | 276°53′  | 28°       | 307°56′  | 28°       | 334°40′  | 28°       | 358°27′  |
| 30°       | 212°09′  | 30°       | 245°30′  | 30°         | 279°05′  | 30°       | 309°51′  | 30°       | 336°19′  | 30°       | 360°00′  |

Table 2.6: Oblique ascensions of the ecliptic for latitude +20°.



| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 000°00'  | 00°       | 021°11'  | 00°       | 045°36'  | 00°       | 075°30'  | 00°       | 109°57'  | 00°       | 145°22'  |
| 02°       | 001°23'  | 02°       | 022°41'  | 02°       | 047°24'  | 02°       | 077°42'  | 02°       | 112°19'  | 02°       | 147°42'  |
| 04°       | 002°45'  | 04°       | 024°11'  | 04°       | 049°14'  | 04°       | 079°55'  | 04°       | 114°41'  | 04°       | 154°40'  |
| 06°       | 004°08'  | 06°       | 025°43'  | 06°       | 051°06'  | 06°       | 082°08'  | 06°       | 117°04'  | 06°       | 152°21'  |
| 08°       | 005°31'  | 08°       | 027°15'  | 08°       | 053°00'  | 08°       | 084°23'  | 08°       | 119°26'  | 08°       | 154°40'  |
| 10°       | 006°54'  | 10°       | 028°49'  | 10°       | 054°55'  | 10°       | 086°39'  | 10°       | 121°48'  | 10°       | 156°59'  |
| 12°       | 008°17'  | 12°       | 030°23'  | 12°       | 056°51'  | 12°       | 088°56'  | 12°       | 124°10'  | 12°       | 159°18'  |
| 14°       | 009°41'  | 14°       | 031°59'  | 14°       | 058°50'  | 14°       | 091°14'  | 14°       | 126°32'  | 14°       | 161°37'  |
| 16°       | 011°05'  | 16°       | 033°37'  | 16°       | 060°49'  | 16°       | 093°32'  | 16°       | 128°54'  | 16°       | 163°55'  |
| 18°       | 012°30'  | 18°       | 035°15'  | 18°       | 062°51'  | 18°       | 095°51'  | 18°       | 131°16'  | 18°       | 166°13'  |
| 20°       | 013°55'  | 20°       | 036°55'  | 20°       | 064°54'  | 20°       | 098°11'  | 20°       | 133°38'  | 20°       | 168°31'  |
| 22°       | 015°21'  | 22°       | 038°36'  | 22°       | 066°58'  | 22°       | 100°32'  | 22°       | 135°59'  | 22°       | 170°49'  |
| 24°       | 016°47'  | 24°       | 040°19'  | 24°       | 069°04'  | 24°       | 102°52'  | 24°       | 138°20'  | 24°       | 173°07'  |
| 26°       | 018°15'  | 26°       | 042°03'  | 26°       | 071°12'  | 26°       | 105°14'  | 26°       | 140°41'  | 26°       | 175°25'  |
| 28°       | 019°42'  | 28°       | 043°48'  | 28°       | 073°20'  | 28°       | 107°35'  | 28°       | 143°01'  | 28°       | 177°42'  |
| 30°       | 021°11'  | 30°       | 045°36'  | 30°       | 075°30'  | 30°       | 109°57'  | 30°       | 145°22'  | 30°       | 180°00'  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00'  | 00°       | 214°38'  | 00°         | 250°03'  | 00°       | 284°30'  | 00°       | 314°24'  | 00°       | 338°49'  |
| 02°       | 182°18'  | 02°       | 216°59'  | 02°         | 252°25'  | 02°       | 286°40'  | 02°       | 316°12'  | 02°       | 340°18'  |
| 04°       | 184°35'  | 04°       | 219°19'  | 04°         | 254°46'  | 04°       | 288°48'  | 04°       | 317°57'  | 04°       | 341°45'  |
| 06°       | 186°53'  | 06°       | 221°40'  | 06°         | 257°08'  | 06°       | 290°56'  | 06°       | 319°41'  | 06°       | 343°13'  |
| 08°       | 189°11'  | 08°       | 224°01'  | 08°         | 259°28'  | 08°       | 293°02'  | 08°       | 321°24'  | 08°       | 344°39'  |
| 10°       | 191°29'  | 10°       | 226°22'  | 10°         | 261°49'  | 10°       | 295°06'  | 10°       | 323°05'  | 10°       | 346°05'  |
| 12°       | 193°47'  | 12°       | 228°44'  | 12°         | 264°09'  | 12°       | 297°09'  | 12°       | 324°45'  | 12°       | 347°30'  |
| 14°       | 196°05'  | 14°       | 231°06'  | 14°         | 266°28'  | 14°       | 299°11'  | 14°       | 326°23'  | 14°       | 348°55'  |
| 16°       | 198°23'  | 16°       | 233°28'  | 16°         | 268°46'  | 16°       | 301°10'  | 16°       | 328°01'  | 16°       | 350°19'  |
| 18°       | 200°42'  | 18°       | 235°50'  | 18°         | 271°04'  | 18°       | 303°09'  | 18°       | 329°37'  | 18°       | 351°43'  |
| 20°       | 203°01'  | 20°       | 238°12'  | 20°         | 273°21'  | 20°       | 305°05'  | 20°       | 331°11'  | 20°       | 353°06'  |
| 22°       | 205°20'  | 22°       | 240°34'  | 22°         | 275°37'  | 22°       | 307°00'  | 22°       | 332°45'  | 22°       | 354°29'  |
| 24°       | 207°39'  | 24°       | 242°56'  | 24°         | 277°52'  | 24°       | 308°54'  | 24°       | 334°17'  | 24°       | 355°52'  |
| 26°       | 209°59'  | 26°       | 245°19'  | 26°         | 280°05'  | 26°       | 310°46'  | 26°       | 335°49'  | 26°       | 357°15'  |
| 28°       | 212°18'  | 28°       | 247°41'  | 28°         | 282°18'  | 28°       | 312°36'  | 28°       | 337°19'  | 28°       | 358°37'  |
| 30°       | 214°38'  | 30°       | 250°03'  | 30°         | 284°30'  | 30°       | 314°24'  | 30°       | 338°49'  | 30°       | 360°00'  |

Table 2.7: Oblique ascensions of the ecliptic for latitude +30°.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 000°00′  | 00°       | 018°07′  | 00°       | 039°54′  | 00°       | 068°40′  | 00°       | 104°15′  | 00°       | 142°17′  |
| 02°       | 001°10′  | 02°       | 019°24′  | 02°       | 041°34′  | 02°       | 070°52′  | 02°       | 106°46′  | 02°       | 144°49′  |
| 04°       | 002°20′  | 04°       | 020°43′  | 04°       | 043°16′  | 04°       | 073°06′  | 04°       | 109°17′  | 04°       | 147°21′  |
| 06°       | 003°30′  | 06°       | 022°03′  | 06°       | 045°01′  | 06°       | 075°21′  | 06°       | 111°48′  | 06°       | 149°52′  |
| 08°       | 004°41′  | 08°       | 023°24′  | 08°       | 046°48′  | 08°       | 077°38′  | 08°       | 114°20′  | 08°       | 152°24′  |
| 10°       | 005°52′  | 10°       | 024°46′  | 10°       | 048°36′  | 10°       | 079°57′  | 10°       | 116°53′  | 10°       | 154°55′  |
| 12°       | 007°03′  | 12°       | 026°10′  | 12°       | 050°27′  | 12°       | 082°18′  | 12°       | 119°25′  | 12°       | 157°26′  |
| 14°       | 008°14′  | 14°       | 027°35′  | 14°       | 052°20′  | 14°       | 084°39′  | 14°       | 121°57′  | 14°       | 159°57′  |
| 16°       | 009°26′  | 16°       | 029°02′  | 16°       | 054°15′  | 16°       | 087°03′  | 16°       | 124°30′  | 16°       | 162°28′  |
| 18°       | 010°38′  | 18°       | 030°30′  | 18°       | 056°12′  | 18°       | 089°27′  | 18°       | 127°03′  | 18°       | 164°58′  |
| 20°       | 011°51′  | 20°       | 031°59′  | 20°       | 058°12′  | 20°       | 091°53′  | 20°       | 129°35′  | 20°       | 167°29′  |
| 22°       | 013°05′  | 22°       | 033°31′  | 22°       | 060°13′  | 22°       | 094°19′  | 22°       | 132°08′  | 22°       | 169°59′  |
| 24°       | 014°19′  | 24°       | 035°04′  | 24°       | 062°17′  | 24°       | 096°47′  | 24°       | 134°40′  | 24°       | 172°29′  |
| 26°       | 015°34′  | 26°       | 036°38′  | 26°       | 064°23′  | 26°       | 099°16′  | 26°       | 137°13′  | 26°       | 175°00′  |
| 28°       | 016°50′  | 28°       | 038°15′  | 28°       | 066°31′  | 28°       | 101°45′  | 28°       | 139°45′  | 28°       | 177°30′  |
| 30°       | 018°07′  | 30°       | 039°54′  | 30°       | 068°40′  | 30°       | 104°15′  | 30°       | 142°17′  | 30°       | 180°00′  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00′  | 00°       | 217°43′  | 00°         | 255°45′  | 00°       | 291°20′  | 00°       | 320°06′  | 00°       | 341°53′  |
| 02°       | 182°30′  | 02°       | 220°15′  | 02°         | 258°15′  | 02°       | 293°29′  | 02°       | 321°45′  | 02°       | 343°10′  |
| 04°       | 185°00′  | 04°       | 222°47′  | 04°         | 260°44′  | 04°       | 295°37′  | 04°       | 323°22′  | 04°       | 344°26′  |
| 06°       | 187°31′  | 06°       | 225°20′  | 06°         | 263°13′  | 06°       | 297°43′  | 06°       | 324°56′  | 08°       | 346°55′  |
| 08°       | 190°01′  | 08°       | 227°52′  | 08°         | 265°41′  | 08°       | 299°17′  | 08°       | 326°29′  | 08°       | 346°55′  |
| 10°       | 192°31′  | 10°       | 230°25′  | 10°         | 268°07′  | 10°       | 301°48′  | 10°       | 328°01′  | 10°       | 348°09′  |
| 12°       | 195°02′  | 12°       | 232°57′  | 12°         | 270°33′  | 12°       | 303°48′  | 12°       | 329°30′  | 12°       | 349°22′  |
| 14°       | 197°32′  | 14°       | 235°30′  | 14°         | 272°57′  | 14°       | 305°45′  | 14°       | 330°58′  | 14°       | 350°34′  |
| 16°       | 200°03′  | 16°       | 238°03′  | 16°         | 275°21′  | 16°       | 307°40′  | 16°       | 332°25′  | 16°       | 351°46′  |
| 18°       | 202°34′  | 18°       | 240°35′  | 18°         | 277°42′  | 18°       | 309°33′  | 18°       | 333°50′  | 18°       | 352°57′  |
| 20°       | 205°05′  | 20°       | 243°07′  | 20°         | 280°03′  | 20°       | 311°24′  | 20°       | 335°14′  | 20°       | 354°08′  |
| 22°       | 207°36′  | 22°       | 245°40′  | 22°         | 282°22′  | 22°       | 313°12′  | 22°       | 336°36′  | 22°       | 355°19′  |
| 24°       | 210°08′  | 24°       | 248°12′  | 24°         | 284°39′  | 24°       | 314°59′  | 24°       | 337°57′  | 24°       | 356°30′  |
| 26°       | 212°39′  | 26°       | 250°43′  | 26°         | 286°54′  | 26°       | 316°44′  | 26°       | 339°17′  | 26°       | 357°40′  |
| 28°       | 215°11′  | 28°       | 253°14′  | 28°         | 289°08′  | 28°       | 318°26′  | 28°       | 340°36′  | 28°       | 358°50′  |
| 30°       | 217°43′  | 30°       | 255°45′  | 30°         | 291°20′  | 30°       | 320°06′  | 30°       | 341°53′  | 30°       | 360°00′  |

Table 2.8: Oblique ascensions of the ecliptic for latitude +40°.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 000°00′  | 00°       | 013°55′  | 00°       | 031°54′  | 00°       | 058°54′  | 00°       | 096°15′  | 00°       | 138°06′  |
| 02°       | 000°53′  | 02°       | 014°56′  | 02°       | 033°22′  | 02°       | 061°06′  | 02°       | 098°59′  | 02°       | 140°54′  |
| 04°       | 001°47′  | 04°       | 015°59′  | 04°       | 034°52′  | 04°       | 063°21′  | 04°       | 101°44′  | 04°       | 143°43′  |
| 06°       | 002°40′  | 06°       | 017°02′  | 06°       | 036°25′  | 06°       | 065°40′  | 06°       | 104°29′  | 06°       | 146°31′  |
| 08°       | 003°34′  | 08°       | 018°07′  | 08°       | 038°01′  | 08°       | 068°00′  | 08°       | 107°15′  | 08°       | 149°19′  |
| 10°       | 004°27′  | 10°       | 019°13′  | 10°       | 039°40′  | 10°       | 070°24′  | 10°       | 110°02′  | 10°       | 152°07′  |
| 12°       | 005°22′  | 12°       | 020°21′  | 12°       | 041°22′  | 12°       | 072°50′  | 12°       | 112°50′  | 12°       | 154°55′  |
| 14°       | 006°16′  | 14°       | 021°31′  | 14°       | 043°06′  | 14°       | 075°18′  | 14°       | 115°37′  | 14°       | 157°42′  |
| 16°       | 007°11′  | 16°       | 022°42′  | 16°       | 044°54′  | 16°       | 077°49′  | 16°       | 118°26′  | 16°       | 160°30′  |
| 18°       | 008°07′  | 18°       | 023°54′  | 18°       | 046°45′  | 18°       | 080°22′  | 18°       | 121°14′  | 18°       | 163°17′  |
| 20°       | 009°03′  | 20°       | 025°09′  | 20°       | 048°38′  | 20°       | 082°57′  | 20°       | 124°02′  | 20°       | 166°05′  |
| 22°       | 010°00′  | 22°       | 026°26′  | 22°       | 050°35′  | 22°       | 085°33′  | 22°       | 126°51′  | 22°       | 168°52′  |
| 24°       | 010°57′  | 24°       | 027°44′  | 24°       | 052°35′  | 24°       | 088°12′  | 24°       | 129°40′  | 24°       | 171°39′  |
| 26°       | 011°56′  | 26°       | 029°05′  | 26°       | 054°38′  | 26°       | 090°51′  | 26°       | 132°28′  | 26°       | 174°26′  |
| 28°       | 012°55′  | 28°       | 030°28′  | 28°       | 056°45′  | 28°       | 093°33′  | 28°       | 135°17′  | 28°       | 177°13′  |
| 30°       | 013°55′  | 30°       | 031°54′  | 30°       | 058°54′  | 30°       | 096°15′  | 30°       | 138°06′  | 30°       | 180°00′  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00′  | 00°       | 221°54′  | 00°         | 263°45′  | 00°       | 301°06′  | 00°       | 328°06′  | 00°       | 346°05′  |
| 02°       | 182°47′  | 02°       | 224°43′  | 02°         | 266°27′  | 02°       | 303°15′  | 02°       | 329°32′  | 02°       | 347°05′  |
| 04°       | 185°34′  | 04°       | 227°32′  | 04°         | 269°09′  | 04°       | 305°22′  | 04°       | 330°55′  | 04°       | 348°04′  |
| 06°       | 188°21′  | 06°       | 230°20′  | 06°         | 271°48′  | 06°       | 307°25′  | 06°       | 332°16′  | 06°       | 349°03′  |
| 08°       | 191°08′  | 08°       | 233°09′  | 08°         | 274°27′  | 08°       | 309°25′  | 08°       | 333°34′  | 08°       | 350°00′  |
| 10°       | 193°55′  | 10°       | 235°58′  | 10°         | 277°03′  | 10°       | 311°22′  | 10°       | 334°51′  | 10°       | 350°57′  |
| 12°       | 196°43′  | 12°       | 238°46′  | 12°         | 279°38′  | 12°       | 313°15′  | 12°       | 336°06′  | 12°       | 351°53′  |
| 14°       | 199°30′  | 14°       | 241°34′  | 14°         | 282°11′  | 14°       | 315°06′  | 14°       | 337°18′  | 14°       | 352°49′  |
| 16°       | 202°18′  | 16°       | 244°23′  | 16°         | 284°42′  | 16°       | 316°54′  | 16°       | 338°29′  | 16°       | 353°44′  |
| 18°       | 205°05′  | 18°       | 247°10′  | 18°         | 287°10′  | 18°       | 318°38′  | 18°       | 339°39′  | 18°       | 354°38′  |
| 20°       | 207°53′  | 20°       | 249°58′  | 20°         | 289°36′  | 20°       | 320°20′  | 20°       | 340°47′  | 20°       | 355°33′  |
| 22°       | 210°41′  | 22°       | 252°45′  | 22°         | 292°00′  | 22°       | 321°59′  | 22°       | 341°53′  | 22°       | 356°26′  |
| 24°       | 213°29′  | 24°       | 255°31′  | 24°         | 294°20′  | 24°       | 323°35′  | 24°       | 342°58′  | 24°       | 357°20′  |
| 26°       | 216°17′  | 26°       | 258°16′  | 26°         | 296°39′  | 26°       | 325°08′  | 26°       | 344°01′  | 26°       | 358°13′  |
| 28°       | 219°06′  | 28°       | 261°01′  | 28°         | 298°54′  | 28°       | 326°38′  | 28°       | 345°04′  | 28°       | 359°07′  |
| 30°       | 221°54′  | 30°       | 263°45′  | 30°         | 301°06′  | 30°       | 328°06′  | 30°       | 346°05′  | 30°       | 360°00′  |

Table 2.9: Oblique ascensions of the ecliptic for latitude +50°.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 000°00′  | 00°       | 011°04′  | 00°       | 026°14′  | 00°       | 051°45′  | 00°       | 090°35′  | 00°       | 135°15′  |
| 02°       | 000°42′  | 02°       | 011°54′  | 02°       | 027°31′  | 02°       | 053°58′  | 02°       | 093°29′  | 02°       | 138°15′  |
| 04°       | 001°24′  | 04°       | 012°44′  | 04°       | 028°52′  | 04°       | 056°15′  | 04°       | 096°24′  | 04°       | 141°15′  |
| 06°       | 002°06′  | 06°       | 013°36′  | 06°       | 030°16′  | 06°       | 058°35′  | 06°       | 099°21′  | 06°       | 144°15′  |
| 08°       | 002°48′  | 08°       | 014°30′  | 08°       | 031°44′  | 08°       | 060°59′  | 08°       | 102°18′  | 08°       | 147°14′  |
| 10°       | 003°31′  | 10°       | 015°24′  | 10°       | 033°14′  | 10°       | 063°27′  | 10°       | 105°16′  | 10°       | 150°14′  |
| 12°       | 004°14′  | 12°       | 016°21′  | 12°       | 034°48′  | 12°       | 065°57′  | 12°       | 108°14′  | 12°       | 153°13′  |
| 14°       | 004°57′  | 14°       | 017°18′  | 14°       | 036°26′  | 14°       | 068°31′  | 14°       | 111°14′  | 14°       | 156°12′  |
| 16°       | 005°41′  | 16°       | 018°18′  | 16°       | 038°07′  | 16°       | 071°08′  | 16°       | 114°13′  | 16°       | 159°11′  |
| 18°       | 006°25′  | 18°       | 019°19′  | 18°       | 039°52′  | 18°       | 073°48′  | 18°       | 117°13′  | 18°       | 162°10′  |
| 20°       | 007°10′  | 20°       | 020°23′  | 20°       | 041°41′  | 20°       | 076°31′  | 20°       | 120°13′  | 20°       | 165°08′  |
| 22°       | 007°55′  | 22°       | 021°28′  | 22°       | 043°34′  | 22°       | 079°16′  | 22°       | 123°14′  | 22°       | 168°07′  |
| 24°       | 008°41′  | 24°       | 022°36′  | 24°       | 045°31′  | 24°       | 082°03′  | 24°       | 126°14′  | 24°       | 171°05′  |
| 26°       | 009°28′  | 26°       | 023°46′  | 26°       | 047°32′  | 26°       | 084°52′  | 26°       | 129°14′  | 26°       | 174°03′  |
| 28°       | 010°15′  | 28°       | 024°58′  | 28°       | 049°37′  | 28°       | 087°43′  | 28°       | 132°14′  | 28°       | 177°02′  |
| 30°       | 011°04′  | 30°       | 026°14′  | 30°       | 051°45′  | 30°       | 090°35′  | 30°       | 135°15′  | 30°       | 180°00′  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00′  | 00°       | 224°45′  | 00°         | 269°25′  | 00°       | 308°15′  | 00°       | 333°46′  | 00°       | 348°56′  |
| 02°       | 182°58′  | 02°       | 227°46′  | 02°         | 272°17′  | 02°       | 310°23′  | 02°       | 335°02′  | 02°       | 349°45′  |
| 04°       | 185°57′  | 04°       | 230°46′  | 04°         | 275°08′  | 04°       | 312°28′  | 04°       | 336°14′  | 04°       | 350°32′  |
| 06°       | 188°55′  | 06°       | 233°46′  | 06°         | 277°57′  | 06°       | 314°29′  | 06°       | 337°24′  | 06°       | 351°19′  |
| 08°       | 191°53′  | 08°       | 236°46′  | 08°         | 280°44′  | 08°       | 316°26′  | 08°       | 338°32′  | 08°       | 352°05′  |
| 10°       | 194°52′  | 10°       | 239°47′  | 10°         | 283°29′  | 10°       | 318°19′  | 10°       | 339°37′  | 10°       | 352°50′  |
| 12°       | 197°50′  | 12°       | 242°47′  | 12°         | 286°12′  | 12°       | 320°08′  | 12°       | 340°41′  | 12°       | 353°35′  |
| 14°       | 200°49′  | 14°       | 245°47′  | 14°         | 288°52′  | 14°       | 321°53′  | 14°       | 341°42′  | 14°       | 354°19′  |
| 16°       | 203°48′  | 16°       | 248°46′  | 16°         | 291°29′  | 16°       | 323°34′  | 16°       | 342°42′  | 16°       | 355°03′  |
| 18°       | 206°47′  | 18°       | 251°46′  | 18°         | 294°03′  | 18°       | 325°12′  | 18°       | 343°39′  | 18°       | 355°46′  |
| 20°       | 209°46′  | 20°       | 254°44′  | 20°         | 296°33′  | 20°       | 326°46′  | 20°       | 344°36′  | 20°       | 356°29′  |
| 22°       | 212°46′  | 22°       | 257°42′  | 22°         | 299°01′  | 22°       | 328°16′  | 22°       | 345°30′  | 22°       | 357°12′  |
| 24°       | 215°45′  | 24°       | 260°39′  | 24°         | 301°25′  | 24°       | 329°44′  | 24°       | 346°24′  | 24°       | 357°54′  |
| 26°       | 218°45′  | 26°       | 263°36′  | 26°         | 303°45′  | 26°       | 331°08′  | 26°       | 347°16′  | 26°       | 358°36′  |
| 28°       | 221°45′  | 28°       | 266°31′  | 28°         | 306°02′  | 28°       | 332°29′  | 28°       | 348°06′  | 28°       | 359°18′  |
| 30°       | 224°45′  | 30°       | 269°25′  | 30°         | 308°15′  | 30°       | 333°46′  | 30°       | 348°56′  | 30°       | 360°00′  |

Table 2.10: Oblique ascensions of the ecliptic for latitude +55°.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 000°00′  | 00°       | 007°20′  | 00°       | 018°22′  | 00°       | 041°21′  | 00°       | 082°44′  | 00°       | 131°31′  |
| 02°       | 000°27′  | 02°       | 007°54′  | 02°       | 019°24′  | 02°       | 043°34′  | 02°       | 085°54′  | 02°       | 134°47′  |
| 04°       | 000°55′  | 04°       | 008°29′  | 04°       | 020°29′  | 04°       | 045°54′  | 04°       | 089°06′  | 04°       | 138°02′  |
| 06°       | 001°23′  | 06°       | 009°05′  | 06°       | 021°38′  | 06°       | 048°18′  | 06°       | 092°19′  | 06°       | 141°17′  |
| 08°       | 001°50′  | 08°       | 009°42′  | 08°       | 022°50′  | 08°       | 050°48′  | 08°       | 095°33′  | 08°       | 144°32′  |
| 10°       | 002°18′  | 10°       | 010°20′  | 10°       | 024°07′  | 10°       | 053°23′  | 10°       | 098°48′  | 10°       | 147°47′  |
| 12°       | 002°46′  | 12°       | 011°00′  | 12°       | 025°27′  | 12°       | 056°03′  | 12°       | 102°04′  | 12°       | 151°01′  |
| 14°       | 003°15′  | 14°       | 011°41′  | 14°       | 026°52′  | 14°       | 058°47′  | 14°       | 105°20′  | 14°       | 154°15′  |
| 16°       | 003°44′  | 16°       | 012°24′  | 16°       | 028°22′  | 16°       | 061°35′  | 16°       | 108°36′  | 16°       | 157°29′  |
| 18°       | 004°13′  | 18°       | 013°08′  | 18°       | 029°57′  | 18°       | 064°27′  | 18°       | 111°52′  | 18°       | 160°42′  |
| 20°       | 004°43′  | 20°       | 013°55′  | 20°       | 031°38′  | 20°       | 067°23′  | 20°       | 115°09′  | 20°       | 163°55′  |
| 22°       | 005°13′  | 22°       | 014°43′  | 22°       | 033°23′  | 22°       | 070°22′  | 22°       | 118°26′  | 22°       | 167°09′  |
| 24°       | 005°44′  | 24°       | 015°34′  | 24°       | 035°14′  | 24°       | 073°24′  | 24°       | 121°42′  | 24°       | 170°22′  |
| 26°       | 006°15′  | 26°       | 016°28′  | 26°       | 037°11′  | 26°       | 076°28′  | 26°       | 124°59′  | 26°       | 173°34′  |
| 28°       | 006°47′  | 28°       | 017°23′  | 28°       | 039°13′  | 28°       | 079°35′  | 28°       | 128°15′  | 28°       | 176°47′  |
| 30°       | 007°20′  | 30°       | 018°22′  | 30°       | 041°21′  | 30°       | 082°44′  | 30°       | 131°31′  | 30°       | 180°00′  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00′  | 00°       | 228°29′  | 00°         | 277°16′  | 00°       | 318°39′  | 00°       | 341°38′  | 00°       | 352°40′  |
| 02°       | 183°13′  | 02°       | 231°45′  | 02°         | 280°25′  | 02°       | 320°47′  | 02°       | 342°37′  | 02°       | 353°13′  |
| 04°       | 186°26′  | 04°       | 235°01′  | 04°         | 283°32′  | 04°       | 322°49′  | 04°       | 343°32′  | 04°       | 353°45′  |
| 06°       | 189°38′  | 06°       | 238°18′  | 06°         | 286°36′  | 06°       | 324°46′  | 06°       | 344°26′  | 06°       | 354°16′  |
| 08°       | 192°51′  | 08°       | 241°34′  | 08°         | 289°38′  | 08°       | 326°37′  | 08°       | 345°17′  | 08°       | 354°47′  |
| 10°       | 196°05′  | 10°       | 244°51′  | 10°         | 292°37′  | 10°       | 328°22′  | 10°       | 346°05′  | 10°       | 355°17′  |
| 12°       | 199°18′  | 12°       | 248°08′  | 12°         | 295°33′  | 12°       | 330°03′  | 12°       | 346°52′  | 12°       | 355°47′  |
| 14°       | 202°31′  | 14°       | 251°24′  | 14°         | 298°25′  | 14°       | 331°38′  | 14°       | 347°36′  | 14°       | 356°16′  |
| 16°       | 205°45′  | 16°       | 254°40′  | 16°         | 301°13′  | 16°       | 333°08′  | 16°       | 348°19′  | 16°       | 356°45′  |
| 18°       | 208°59′  | 18°       | 257°56′  | 18°         | 303°57′  | 18°       | 334°33′  | 18°       | 349°00′  | 18°       | 357°14′  |
| 20°       | 212°13′  | 20°       | 261°12′  | 20°         | 306°37′  | 20°       | 335°53′  | 20°       | 349°40′  | 20°       | 357°42′  |
| 22°       | 215°28′  | 22°       | 264°27′  | 22°         | 309°12′  | 22°       | 337°10′  | 22°       | 350°18′  | 22°       | 358°10′  |
| 24°       | 218°43′  | 24°       | 267°41′  | 24°         | 311°42′  | 24°       | 338°22′  | 24°       | 350°55′  | 24°       | 358°37′  |
| 26°       | 221°58′  | 26°       | 270°54′  | 26°         | 314°06′  | 26°       | 339°31′  | 26°       | 351°31′  | 26°       | 359°05′  |
| 28°       | 225°13′  | 28°       | 274°06′  | 28°         | 316°26′  | 28°       | 340°36′  | 28°       | 352°06′  | 28°       | 359°33′  |
| 30°       | 228°29′  | 30°       | 277°16′  | 30°         | 318°39′  | 30°       | 341°38′  | 30°       | 352°40′  | 30°       | 360°00′  |

Table 2.11: Oblique ascensions of the ecliptic for latitude  $+60^\circ$ .

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 000°00'  | 00°       | 002°07'  | 00°       | 005°56'  | 00°       | 021°39'  | 00°       | 070°18'  | 00°       | 126°18'  |
| 02°       | 000°08'  | 02°       | 002°17'  | 02°       | 006°22'  | 02°       | 023°56'  | 02°       | 074°04'  | 02°       | 129°57'  |
| 04°       | 000°16'  | 04°       | 002°28'  | 04°       | 006°51'  | 04°       | 026°25'  | 04°       | 077°50'  | 04°       | 133°35'  |
| 06°       | 000°23'  | 06°       | 002°39'  | 06°       | 007°22'  | 06°       | 029°06'  | 06°       | 081°36'  | 06°       | 137°12'  |
| 08°       | 000°31'  | 08°       | 002°51'  | 08°       | 007°57'  | 08°       | 031°58'  | 08°       | 085°22'  | 08°       | 140°49'  |
| 10°       | 000°39'  | 10°       | 003°03'  | 10°       | 008°36'  | 10°       | 034°59'  | 10°       | 089°08'  | 10°       | 144°25'  |
| 12°       | 000°47'  | 12°       | 003°16'  | 12°       | 009°19'  | 12°       | 038°09'  | 12°       | 092°54'  | 12°       | 148°00'  |
| 14°       | 000°55'  | 14°       | 003°29'  | 14°       | 010°07'  | 14°       | 041°26'  | 14°       | 096°39'  | 14°       | 151°35'  |
| 16°       | 001°04'  | 16°       | 003°44'  | 16°       | 011°02'  | 16°       | 044°50'  | 16°       | 100°24'  | 16°       | 155°09'  |
| 18°       | 001°12'  | 18°       | 003°59'  | 18°       | 012°04'  | 18°       | 048°19'  | 18°       | 104°08'  | 18°       | 158°43'  |
| 20°       | 001°21'  | 20°       | 004°15'  | 20°       | 013°14'  | 20°       | 051°52'  | 20°       | 107°52'  | 20°       | 162°16'  |
| 22°       | 001°29'  | 22°       | 004°32'  | 22°       | 014°33'  | 22°       | 055°29'  | 22°       | 111°35'  | 22°       | 165°50'  |
| 24°       | 001°38'  | 24°       | 004°51'  | 24°       | 016°02'  | 24°       | 059°08'  | 24°       | 115°17'  | 24°       | 169°22'  |
| 26°       | 001°48'  | 26°       | 005°11'  | 26°       | 017°42'  | 26°       | 062°50'  | 26°       | 118°58'  | 26°       | 172°55'  |
| 28°       | 001°57'  | 28°       | 005°33'  | 28°       | 019°34'  | 28°       | 066°33'  | 28°       | 122°38'  | 28°       | 176°28'  |
| 30°       | 002°07'  | 30°       | 005°56'  | 30°       | 021°39'  | 30°       | 070°18'  | 30°       | 126°18'  | 30°       | 180°00'  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00'  | 00°       | 233°42'  | 00°         | 289°42'  | 00°       | 338°21'  | 00°       | 354°04'  | 00°       | 357°53'  |
| 02°       | 183°32'  | 02°       | 237°22'  | 02°         | 293°27'  | 02°       | 340°26'  | 02°       | 354°27'  | 02°       | 358°03'  |
| 04°       | 187°05'  | 04°       | 241°02'  | 04°         | 297°10'  | 04°       | 342°18'  | 04°       | 354°49'  | 04°       | 358°12'  |
| 06°       | 190°38'  | 06°       | 244°43'  | 06°         | 300°52'  | 06°       | 343°58'  | 06°       | 355°09'  | 06°       | 358°22'  |
| 08°       | 194°10'  | 08°       | 248°25'  | 08°         | 304°31'  | 08°       | 345°27'  | 08°       | 355°28'  | 08°       | 358°31'  |
| 10°       | 197°44'  | 10°       | 252°08'  | 10°         | 308°08'  | 10°       | 346°46'  | 10°       | 355°45'  | 10°       | 358°39'  |
| 12°       | 201°17'  | 12°       | 255°52'  | 12°         | 311°41'  | 12°       | 347°56'  | 12°       | 356°01'  | 12°       | 358°48'  |
| 14°       | 204°51'  | 14°       | 259°36'  | 14°         | 315°10'  | 14°       | 348°58'  | 14°       | 356°16'  | 14°       | 358°56'  |
| 16°       | 208°25'  | 16°       | 263°21'  | 16°         | 318°34'  | 16°       | 349°53'  | 16°       | 356°31'  | 16°       | 359°05'  |
| 18°       | 212°00'  | 18°       | 267°06'  | 18°         | 321°51'  | 18°       | 350°41'  | 18°       | 356°44'  | 18°       | 359°13'  |
| 20°       | 215°35'  | 20°       | 270°52'  | 20°         | 325°01'  | 20°       | 351°24'  | 20°       | 356°57'  | 20°       | 359°21'  |
| 22°       | 219°11'  | 22°       | 274°38'  | 22°         | 328°02'  | 22°       | 352°03'  | 22°       | 357°09'  | 22°       | 359°29'  |
| 24°       | 222°48'  | 24°       | 278°24'  | 24°         | 330°54'  | 24°       | 352°38'  | 24°       | 357°21'  | 24°       | 359°37'  |
| 26°       | 226°25'  | 26°       | 282°10'  | 26°         | 333°35'  | 26°       | 353°09'  | 26°       | 357°32'  | 26°       | 359°44'  |
| 28°       | 230°03'  | 28°       | 285°56'  | 28°         | 336°04'  | 28°       | 353°38'  | 28°       | 357°43'  | 28°       | 359°52'  |
| 30°       | 233°42'  | 30°       | 289°42'  | 30°         | 338°21'  | 30°       | 354°04'  | 30°       | 357°53'  | 30°       | 360°00'  |

Table 2.12: Oblique ascensions of the ecliptic for latitude +65°.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 360°00'  | 00°       | 354°02'  | 00°       | —        | 00°       | —        | 00°41'    | 032°53'  | 00°       | 118°13'  |
| 02°       | 359°39'  | 02°       | 353°30'  | 02°       | —        | 02°       | —        | 02°       | 044°26'  | 02°       | 122°31'  |
| 04°       | 359°18'  | 04°       | 352°57'  | 04°       | —        | 04°       | —        | 04°       | 052°41'  | 04°       | 126°47'  |
| 06°       | 358°57'  | 06°       | 352°21'  | 06°       | —        | 06°       | —        | 06°       | 059°22'  | 06°       | 131°01'  |
| 08°       | 358°35'  | 08°       | 351°42'  | 08°       | —        | 08°       | —        | 08°       | 065°22'  | 08°       | 135°13'  |
| 10°       | 358°14'  | 10°       | 351°00'  | 10°       | —        | 10°       | —        | 10°       | 070°57'  | 10°       | 139°22'  |
| 12°       | 357°52'  | 12°       | 350°14'  | 12°       | —        | 12°       | —        | 12°       | 076°15'  | 12°       | 143°31'  |
| 14°       | 357°29'  | 14°       | 349°23'  | 14°       | —        | 14°       | —        | 14°       | 081°21'  | 14°       | 147°37'  |
| 16°       | 357°06'  | 16°       | 348°26'  | 16°       | —        | 16°       | —        | 16°       | 086°18'  | 16°       | 151°43'  |
| 18°       | 356°43'  | 18°       | 347°20'  | 18°       | —        | 18°       | —        | 18°       | 091°07'  | 18°       | 155°47'  |
| 20°       | 356°18'  | 20°       | 346°04'  | 20°       | —        | 20°       | —        | 20°       | 095°49'  | 20°       | 159°51'  |
| 22°       | 355°53'  | 22°       | 344°32'  | 22°       | —        | 22°       | —        | 22°       | 100°26'  | 22°       | 163°54'  |
| 24°       | 355°27'  | 24°       | 342°37'  | 24°       | —        | 24°       | —        | 24°       | 104°58'  | 24°       | 167°56'  |
| 26°       | 355°00'  | 26°       | 340°03'  | 26°       | —        | 26°       | —        | 26°       | 109°27'  | 26°       | 171°57'  |
| 28°       | 354°32'  | 28°       | 335°55'  | 28°       | —        | 28°       | —        | 28°       | 113°51'  | 28°       | 175°59'  |
| 30°       | 354°02'  | 29°19'    | 327°07'  | 30°       | —        | 30°       | —        | 30°       | 118°13'  | 30°       | 180°00'  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00'  | 00°       | 241°47'  | 00°         | —        | 00°       | —        | 00°41'    | 032°53'  | 00°       | 005°58'  |
| 02°       | 184°01'  | 02°       | 246°09'  | 02°         | —        | 02°       | —        | 02°       | 024°05'  | 02°       | 005°28'  |
| 04°       | 188°03'  | 04°       | 250°33'  | 04°         | —        | 04°       | —        | 04°       | 019°57'  | 04°       | 005°00'  |
| 06°       | 192°04'  | 06°       | 255°02'  | 06°         | —        | 06°       | —        | 06°       | 017°23'  | 06°       | 004°33'  |
| 08°       | 196°06'  | 08°       | 259°34'  | 08°         | —        | 08°       | —        | 08°       | 015°28'  | 08°       | 004°07'  |
| 10°       | 200°09'  | 10°       | 264°11'  | 10°         | —        | 10°       | —        | 10°       | 013°56'  | 10°       | 003°42'  |
| 12°       | 204°13'  | 12°       | 268°53'  | 12°         | —        | 12°       | —        | 12°       | 012°40'  | 12°       | 003°17'  |
| 14°       | 208°17'  | 14°       | 273°42'  | 14°         | —        | 14°       | —        | 14°       | 011°34'  | 14°       | 002°54'  |
| 16°       | 212°23'  | 16°       | 278°39'  | 16°         | —        | 16°       | —        | 16°       | 010°37'  | 16°       | 002°31'  |
| 18°       | 216°29'  | 18°       | 283°45'  | 18°         | —        | 18°       | —        | 18°       | 009°46'  | 18°       | 002°08'  |
| 20°       | 220°38'  | 20°       | 289°03'  | 20°         | —        | 20°       | —        | 20°       | 009°00'  | 20°       | 001°46'  |
| 22°       | 224°47'  | 22°       | 294°38'  | 22°         | —        | 22°       | —        | 22°       | 008°18'  | 22°       | 001°25'  |
| 24°       | 228°59'  | 24°       | 300°38'  | 24°         | —        | 24°       | —        | 24°       | 007°39'  | 24°       | 001°03'  |
| 26°       | 233°13'  | 26°       | 307°19'  | 26°         | —        | 26°       | —        | 26°       | 007°03'  | 26°       | 000°42'  |
| 28°       | 237°29'  | 28°       | 315°34'  | 28°         | —        | 28°       | —        | 28°       | 006°30'  | 28°       | 000°21'  |
| 30°       | 241°47'  | 29°19'    | 327°07'  | 30°         | —        | 30°       | —        | 30°       | 005°58'  | 30°       | 000°00'  |

Table 2.13: Oblique ascensions of the ecliptic for latitude +70°.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 360°00′  | 00°       | 338°42′  | 00°       | —        | 00°       | —        | 00°       | —        | 00°       | 102°52′  |
| 02°       | 358°52′  | 02°       | 336°16′  | 02°       | —        | 02°       | —        | 02°       | —        | 02°       | 108°49′  |
| 04°       | 357°44′  | 04°       | 333°24′  | 04°       | —        | 04°       | —        | 04°       | —        | 04°       | 114°32′  |
| 06°       | 356°35′  | 06°       | 329°54′  | 06°       | —        | 06°       | —        | 06°       | —        | 06°       | 120°04′  |
| 08°       | 355°25′  | 08°       | 325°10′  | 08°       | —        | 08°       | —        | 08°       | —        | 08°       | 125°27′  |
| 10°       | 354°13′  | 10°       | 316°55′  | 10°       | —        | 10°       | —        | 10°       | —        | 10°       | 130°43′  |
| 12°       | 353°00′  | 10°36′    | 308°11′  | 12°       | —        | 12°       | —        | 12°       | —        | 12°       | 135°52′  |
| 14°       | 351°44′  | 14°       | —        | 14°       | —        | 14°       | —        | 14°       | —        | 14°       | 140°57′  |
| 16°       | 350°26′  | 16°       | —        | 16°       | —        | 16°       | —        | 16°       | —        | 16°       | 145°58′  |
| 18°       | 349°05′  | 18°       | —        | 18°       | —        | 18°       | —        | 19°24′    | 051°49′  | 18°       | 150°56′  |
| 20°       | 347°39′  | 20°       | —        | 20°       | —        | 20°       | —        | 20°       | 061°44′  | 20°       | 155°50′  |
| 22°       | 346°08′  | 22°       | —        | 22°       | —        | 22°       | —        | 22°       | 073°54′  | 22°       | 160°43′  |
| 24°       | 344°30′  | 24°       | —        | 24°       | —        | 24°       | —        | 24°       | 082°31′  | 24°       | 165°34′  |
| 26°       | 342°45′  | 26°       | —        | 26°       | —        | 26°       | —        | 26°       | 089°54′  | 26°       | 170°23′  |
| 28°       | 340°50′  | 28°       | —        | 28°       | —        | 28°       | —        | 28°       | 096°36′  | 28°       | 175°12′  |
| 30°       | 338°42′  | 30°       | —        | 30°       | —        | 30°       | —        | 30°       | 102°52′  | 30°       | 180°00′  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00′  | 00°       | 257°08′  | 00°         | —        | 00°       | —        | 00°       | —        | 00°       | 021°18′  |
| 02°       | 184°48′  | 02°       | 263°24′  | 02°         | —        | 02°       | —        | 02°       | —        | 02°       | 019°10′  |
| 04°       | 189°37′  | 04°       | 270°06′  | 04°         | —        | 04°       | —        | 04°       | —        | 04°       | 017°15′  |
| 06°       | 194°26′  | 06°       | 277°29′  | 06°         | —        | 06°       | —        | 06°       | —        | 06°       | 015°30′  |
| 08°       | 199°17′  | 08°       | 286°06′  | 08°         | —        | 08°       | —        | 08°       | —        | 08°       | 013°52′  |
| 10°       | 204°10′  | 10°       | 298°16′  | 10°         | —        | 10°       | —        | 10°       | —        | 10°       | 012°21′  |
| 12°       | 209°04′  | 10°36′    | 308°11′  | 12°         | —        | 12°       | —        | 12°       | —        | 12°       | 010°55′  |
| 14°       | 214°02′  | 14°       | —        | 14°         | —        | 14°       | —        | 14°       | —        | 14°       | 009°34′  |
| 16°       | 219°03′  | 16°       | —        | 16°         | —        | 16°       | —        | 16°       | —        | 16°       | 008°16′  |
| 18°       | 224°08′  | 18°       | —        | 18°         | —        | 18°       | —        | 19°24′    | 051°49′  | 18°       | 007°00′  |
| 20°       | 229°17′  | 20°       | —        | 20°         | —        | 20°       | —        | 20°       | 043°05′  | 20°       | 005°47′  |
| 22°       | 234°33′  | 22°       | —        | 22°         | —        | 22°       | —        | 22°       | 034°50′  | 22°       | 004°35′  |
| 24°       | 239°56′  | 24°       | —        | 24°         | —        | 24°       | —        | 24°       | 030°06′  | 24°       | 003°25′  |
| 26°       | 245°28′  | 26°       | —        | 26°         | —        | 26°       | —        | 26°       | 026°36′  | 26°       | 002°16′  |
| 28°       | 251°11′  | 28°       | —        | 28°         | —        | 28°       | —        | 28°       | 023°44′  | 28°       | 001°08′  |
| 30°       | 257°08′  | 30°       | —        | 30°         | —        | 30°       | —        | 30°       | 021°18′  | 30°       | 000°00′  |

Table 2.14: Oblique ascensions of the ecliptic for latitude  $+75^\circ$ .



| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 360°00′  | 00°       | —        | 00°       | —        | 00°       | —        | 00°       | —        | 00°       | —        |
| 02°       | 357°19′  | 02°       | —        | 02°       | —        | 02°       | —        | 02°       | —        | 02°       | —        |
| 04°       | 354°37′  | 04°       | —        | 04°       | —        | 04°       | —        | 04°       | —        | 04°07′    | 066°00′  |
| 06°       | 351°52′  | 06°       | —        | 06°       | —        | 06°       | —        | 06°       | —        | 06°       | 089°25′  |
| 08°       | 349°02′  | 08°       | —        | 08°       | —        | 08°       | —        | 08°       | —        | 08°       | 100°58′  |
| 10°       | 346°04′  | 10°       | —        | 10°       | —        | 10°       | —        | 10°       | —        | 10°       | 110°24′  |
| 12°       | 342°58′  | 12°       | —        | 12°       | —        | 12°       | —        | 12°       | —        | 12°       | 118°47′  |
| 14°       | 339°39′  | 14°       | —        | 14°       | —        | 14°       | —        | 14°       | —        | 14°       | 126°33′  |
| 16°       | 336°02′  | 16°       | —        | 16°       | —        | 16°       | —        | 16°       | —        | 16°       | 133°52′  |
| 18°       | 331°59′  | 18°       | —        | 18°       | —        | 18°       | —        | 18°       | —        | 18°       | 140°54′  |
| 20°       | 327°20′  | 20°       | —        | 20°       | —        | 20°       | —        | 20°       | —        | 20°       | 147°42′  |
| 22°       | 321°39′  | 22°       | —        | 22°       | —        | 22°       | —        | 22°       | —        | 22°       | 154°20′  |
| 24°       | 313°51′  | 24°       | —        | 24°       | —        | 24°       | —        | 24°       | —        | 24°       | 160°51′  |
| 25°53′    | 294°00′  | 26°       | —        | 26°       | —        | 26°       | —        | 26°       | —        | 26°       | 167°16′  |
| 28°       | —        | 28°       | —        | 28°       | —        | 28°       | —        | 28°       | —        | 28°       | 173°39′  |
| 30°       | —        | 30°       | —        | 30°       | —        | 30°       | —        | 30°       | —        | 30°       | 180°00′  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00′  | 00°       | —        | 00°         | —        | 00°       | —        | 00°       | —        | 00°       | —        |
| 02°       | 186°21′  | 02°       | —        | 02°         | —        | 02°       | —        | 02°       | —        | 02°       | —        |
| 04°       | 192°44′  | 04°       | —        | 04°         | —        | 04°       | —        | 04°       | —        | 04°07′    | 066°00′  |
| 06°       | 199°09′  | 06°       | —        | 06°         | —        | 06°       | —        | 06°       | —        | 06°       | 046°09′  |
| 08°       | 205°40′  | 08°       | —        | 08°         | —        | 08°       | —        | 08°       | —        | 08°       | 038°21′  |
| 10°       | 212°18′  | 10°       | —        | 10°         | —        | 10°       | —        | 10°       | —        | 10°       | 032°40′  |
| 12°       | 219°06′  | 12°       | —        | 12°         | —        | 12°       | —        | 12°       | —        | 12°       | 028°01′  |
| 14°       | 226°08′  | 14°       | —        | 14°         | —        | 14°       | —        | 14°       | —        | 14°       | 023°58′  |
| 16°       | 233°27′  | 16°       | —        | 16°         | —        | 16°       | —        | 16°       | —        | 16°       | 020°21′  |
| 18°       | 241°13′  | 18°       | —        | 18°         | —        | 18°       | —        | 18°       | —        | 18°       | 017°02′  |
| 20°       | 249°36′  | 20°       | —        | 20°         | —        | 20°       | —        | 20°       | —        | 20°       | 013°56′  |
| 22°       | 259°02′  | 22°       | —        | 22°         | —        | 22°       | —        | 22°       | —        | 22°       | 010°58′  |
| 24°       | 270°35′  | 24°       | —        | 24°         | —        | 24°       | —        | 24°       | —        | 24°       | 008°08′  |
| 25°53′    | 294°00′  | 26°       | —        | 26°         | —        | 26°       | —        | 26°       | —        | 26°       | 005°23′  |
| 28°       | —        | 28°       | —        | 28°         | —        | 28°       | —        | 28°       | —        | 28°       | 002°41′  |
| 30°       | —        | 30°       | —        | 30°         | —        | 30°       | —        | 30°       | —        | 30°       | 000°00′  |

Table 2.15: Oblique ascensions of the ecliptic for latitude +80°.

| Aries     |          | Taurus    |          | Gemini    |          | Cancer    |          | Leo       |          | Virgo     |          |
|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 360°00'  | 00°       | —        | 00°       | —        | 00°       | —        | 00°       | —        | 00°       | —        |
| 02°       | 352°42'  | 02°       | —        | 02°       | —        | 02°       | —        | 02°       | —        | 02°       | —        |
| 04°       | 345°11'  | 04°       | —        | 04°       | —        | 04°       | —        | 04°       | —        | 04°       | —        |
| 06°       | 337°07'  | 06°       | —        | 06°       | —        | 06°       | —        | 06°       | —        | 06°       | —        |
| 08°       | 328°02'  | 08°       | —        | 08°       | —        | 08°       | —        | 08°       | —        | 08°       | —        |
| 10°       | 316°53'  | 10°       | —        | 10°       | —        | 10°       | —        | 10°       | —        | 10°       | —        |
| 12°       | 299°32'  | 12°       | —        | 12°       | —        | 12°       | —        | 12°       | —        | 12°       | —        |
| 12°40'    | 281°39'  | 14°       | —        | 14°       | —        | 14°       | —        | 14°       | —        | 14°       | —        |
| 16°       | —        | 16°       | —        | 16°       | —        | 16°       | —        | 16°       | —        | 17°20'    | 078°21'  |
| 18°       | —        | 18°       | —        | 18°       | —        | 18°       | —        | 18°       | —        | 18°       | 097°28'  |
| 20°       | —        | 20°       | —        | 20°       | —        | 20°       | —        | 20°       | —        | 20°       | 118°31'  |
| 22°       | —        | 22°       | —        | 22°       | —        | 22°       | —        | 22°       | —        | 22°       | 133°20'  |
| 24°       | —        | 24°       | —        | 24°       | —        | 24°       | —        | 24°       | —        | 24°       | 146°06'  |
| 26°       | —        | 26°       | —        | 26°       | —        | 26°       | —        | 26°       | —        | 26°       | 157°50'  |
| 28°       | —        | 28°       | —        | 28°       | —        | 28°       | —        | 28°       | —        | 28°       | 169°02'  |
| 30°       | —        | 30°       | —        | 30°       | —        | 30°       | —        | 30°       | —        | 30°       | 180°00'  |

| Libra     |          | Scorpio   |          | Sagittarius |          | Capricorn |          | Aquarius  |          | Pisces    |          |
|-----------|----------|-----------|----------|-------------|----------|-----------|----------|-----------|----------|-----------|----------|
| $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$   | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ | $\lambda$ | $\alpha$ |
| 00°       | 180°00'  | 00°       | —        | 00°         | —        | 00°       | —        | 00°       | —        | 00°       | —        |
| 02°       | 190°58'  | 02°       | —        | 02°         | —        | 02°       | —        | 02°       | —        | 02°       | —        |
| 04°       | 202°10'  | 04°       | —        | 04°         | —        | 04°       | —        | 04°       | —        | 04°       | —        |
| 06°       | 213°54'  | 06°       | —        | 06°         | —        | 06°       | —        | 06°       | —        | 06°       | —        |
| 08°       | 226°40'  | 08°       | —        | 08°         | —        | 08°       | —        | 08°       | —        | 08°       | —        |
| 10°       | 241°29'  | 10°       | —        | 10°         | —        | 10°       | —        | 10°       | —        | 10°       | —        |
| 12°       | 262°32'  | 12°       | —        | 12°         | —        | 12°       | —        | 12°       | —        | 12°       | —        |
| 12°39'    | 281°39'  | 14°       | —        | 14°         | —        | 14°       | —        | 14°       | —        | 14°       | —        |
| 16°       | —        | 16°       | —        | 16°         | —        | 16°       | —        | 16°       | —        | 17°20'    | 078°21'  |
| 18°       | —        | 18°       | —        | 18°         | —        | 18°       | —        | 18°       | —        | 18°       | 060°28'  |
| 20°       | —        | 20°       | —        | 20°         | —        | 20°       | —        | 20°       | —        | 20°       | 043°07'  |
| 22°       | —        | 22°       | —        | 22°         | —        | 22°       | —        | 22°       | —        | 22°       | 031°58'  |
| 24°       | —        | 24°       | —        | 24°         | —        | 24°       | —        | 24°       | —        | 24°       | 022°53'  |
| 26°       | —        | 26°       | —        | 26°         | —        | 26°       | —        | 26°       | —        | 26°       | 014°49'  |
| 28°       | —        | 28°       | —        | 28°         | —        | 28°       | —        | 28°       | —        | 28°       | 007°18'  |
| 30°       | —        | 30°       | —        | 30°         | —        | 30°       | —        | 30°       | —        | 30°       | 000°00'  |

Table 2.16: Oblique ascensions of the ecliptic for latitude  $+85^\circ$ .

| Aries  |        |         |         | Libra       |        |         |         |
|--------|--------|---------|---------|-------------|--------|---------|---------|
| 00:00  | 90°00' | 156°34' | 336°34' | 00:00       | 90°00' | 203°26' | 023°26' |
| 01:00  | 75°00' | 156°34' | 336°34' | 01:00       | 75°00' | 203°26' | 023°26' |
| 02:00  | 60°00' | 156°34' | 336°34' | 02:00       | 60°00' | 203°26' | 023°26' |
| 03:00  | 45°00' | 156°34' | 336°34' | 03:00       | 45°00' | 203°26' | 023°26' |
| 04:00  | 30°00' | 156°34' | 336°34' | 04:00       | 30°00' | 203°26' | 023°26' |
| 05:00  | 15°00' | 156°34' | 336°34' | 05:00       | 15°00' | 203°26' | 023°26' |
| 06:00  | 00°00' | 156°34' | 336°34' | 06:00       | 00°00' | 203°26' | 023°26' |
| Taurus |        |         |         | Scorpio     |        |         |         |
| 00:00  | 78°32' | 249°26' | 249°26' | 00:00       | 78°32' | 110°34' | 110°34' |
| 01:00  | 71°12' | 196°00' | 302°51' | 01:00       | 71°12' | 164°00' | 057°09' |
| 02:00  | 58°04' | 178°26' | 320°25' | 02:00       | 58°04' | 181°34' | 039°35' |
| 03:00  | 43°52' | 170°40' | 328°11' | 03:00       | 43°52' | 189°20' | 031°49' |
| 04:00  | 29°20' | 165°58' | 332°53' | 04:00       | 29°20' | 194°02' | 027°07' |
| 05:00  | 14°42' | 162°29' | 336°23' | 05:00       | 14°42' | 197°31' | 023°37' |
| 06:00  | 00°00' | 159°26' | 339°26' | 06:00       | 00°00' | 200°34' | 020°34' |
| Gemini |        |         |         | Sagittarius |        |         |         |
| 00:00  | 69°51' | 257°46' | 257°46' | 00:00       | 69°51' | 102°14' | 102°14' |
| 01:00  | 65°04' | 219°53' | 295°39' | 01:00       | 65°04' | 140°07' | 064°21' |
| 02:00  | 54°24' | 198°35' | 316°57' | 02:00       | 54°24' | 161°25' | 043°03' |
| 03:00  | 41°36' | 186°47' | 328°46' | 03:00       | 41°36' | 173°13' | 031°14' |
| 04:00  | 28°00' | 179°01' | 336°32' | 04:00       | 28°00' | 180°59' | 023°28' |
| 05:00  | 14°04' | 173°03' | 342°30' | 05:00       | 14°04' | 186°57' | 017°30' |
| 06:00  | 00°00' | 167°46' | 347°46' | 06:00       | 00°00' | 192°14' | 012°14' |
| Cancer |        |         |         | Capricorn   |        |         |         |
| 00:00  | 66°34' | 270°00' | 270°00' | 00:00       | 66°34' | 090°00' | 090°00' |
| 01:00  | 62°24' | 236°02' | 303°58' | 01:00       | 62°24' | 123°58' | 056°02' |
| 02:00  | 52°37' | 214°34' | 325°26' | 02:00       | 52°37' | 145°26' | 034°34' |
| 03:00  | 40°27' | 201°41' | 338°19' | 03:00       | 40°27' | 158°19' | 021°41' |
| 04:00  | 27°18' | 192°56' | 347°04' | 04:00       | 27°18' | 167°04' | 012°56' |
| 05:00  | 13°44' | 186°05' | 353°55' | 05:00       | 13°44' | 173°55' | 006°05' |
| 06:00  | 00°00' | 180°00' | 000°00' | 06:00       | 00°00' | 180°00' | 000°00' |
| Leo    |        |         |         | Aquarius    |        |         |         |
| 00:00  | 69°51' | 282°14' | 282°14' | 00:00       | 69°51' | 077°46' | 077°46' |
| 01:00  | 65°04' | 244°21' | 320°07' | 01:00       | 65°04' | 115°39' | 039°53' |
| 02:00  | 54°24' | 223°03' | 341°25' | 02:00       | 54°24' | 136°57' | 018°35' |
| 03:00  | 41°36' | 211°14' | 353°13' | 03:00       | 41°36' | 148°46' | 006°47' |
| 04:00  | 28°00' | 203°28' | 000°59' | 04:00       | 28°00' | 156°32' | 359°01' |
| 05:00  | 14°04' | 197°30' | 006°57' | 05:00       | 14°04' | 162°30' | 353°03' |
| 06:00  | 00°00' | 192°14' | 012°14' | 06:00       | 00°00' | 167°46' | 347°46' |
| Virgo  |        |         |         | Pisces      |        |         |         |
| 00:00  | 78°32' | 290°34' | 290°34' | 00:00       | 78°32' | 069°26' | 069°26' |
| 01:00  | 71°12' | 237°09' | 344°00' | 01:00       | 71°12' | 122°51' | 016°00' |
| 02:00  | 58°04' | 219°35' | 001°34' | 02:00       | 58°04' | 140°25' | 358°26' |
| 03:00  | 43°52' | 211°49' | 009°20' | 03:00       | 43°52' | 148°11' | 350°40' |
| 04:00  | 29°20' | 207°07' | 014°02' | 04:00       | 29°20' | 152°53' | 345°58' |
| 05:00  | 14°42' | 203°37' | 017°31' | 05:00       | 14°42' | 156°23' | 342°29' |
| 06:00  | 00°00' | 200°34' | 020°34' | 06:00       | 00°00' | 159°26' | 339°26' |

Table 2.17: Ecliptic altitude and parallactic angle for latitude 0°.

|        |        |         |         |             |        |         |         |
|--------|--------|---------|---------|-------------|--------|---------|---------|
| Aries  |        |         |         | 06:00       | 01°59' | 190°46' | 030°23' |
| 00:00  | 80°00' | 066°34' | 066°34' | 06:08       | 00°00' | 190°22' | 030°47' |
| 01:00  | 72°02' | 122°18' | 010°50' | Libra       |        |         |         |
| 02:00  | 58°32' | 137°08' | 356°00' | 00:00       | 80°00' | 113°26' | 113°26' |
| 03:00  | 44°08' | 142°34' | 350°34' | 01:00       | 72°02' | 169°10' | 057°42' |
| 04:00  | 29°30' | 145°03' | 348°05' | 02:00       | 58°32' | 184°00' | 042°52' |
| 05:00  | 14°46' | 146°13' | 346°55' | 03:00       | 44°08' | 189°26' | 037°26' |
| 06:00  | 00°00' | 146°34' | 346°34' | 04:00       | 29°30' | 191°55' | 034°57' |
| Taurus |        |         |         | 05:00       | 14°46' | 193°05' | 033°47' |
| 00:00  | 88°32' | 249°26' | 249°26' | 06:00       | 00°00' | 193°26' | 033°26' |
| 01:00  | 75°11' | 163°41' | 335°10' | Scorpio     |        |         |         |
| 02:00  | 60°30' | 159°21' | 339°30' | 00:00       | 68°32' | 110°34' | 110°34' |
| 03:00  | 45°48' | 156°49' | 342°02' | 01:00       | 63°52' | 145°55' | 075°13' |
| 04:00  | 31°08' | 154°35' | 344°16' | 02:00       | 53°15' | 165°58' | 055°11' |
| 05:00  | 16°31' | 152°16' | 346°35' | 03:00       | 40°23' | 176°40' | 044°29' |
| 06:00  | 01°59' | 149°37' | 349°14' | 04:00       | 26°37' | 183°07' | 038°01' |
| 06:08  | 00°00' | 149°13' | 349°38' | 05:00       | 12°26' | 187°30' | 033°39' |
| Gemini |        |         |         | 05:52       | 00°00' | 190°22' | 030°47' |
| 00:00  | 79°51' | 257°46' | 257°46' | Sagittarius |        |         |         |
| 01:00  | 72°20' | 200°37' | 314°55' | 00:00       | 59°51' | 102°14' | 102°14' |
| 02:00  | 59°22' | 182°38' | 332°54' | 01:00       | 56°26' | 129°41' | 074°47' |
| 03:00  | 45°32' | 174°04' | 341°29' | 02:00       | 47°48' | 149°23' | 055°05' |
| 04:00  | 31°28' | 168°13' | 347°20' | 03:00       | 36°26' | 162°11' | 042°17' |
| 05:00  | 17°24' | 163°15' | 352°18' | 04:00       | 23°44' | 170°55' | 033°32' |
| 06:00  | 03°26' | 158°22' | 357°10' | 05:00       | 10°20' | 177°27' | 027°00' |
| 06:15  | 00°00' | 157°07' | 358°26' | 05:45       | 00°00' | 181°34' | 022°53' |
| Cancer |        |         |         | Capricorn   |        |         |         |
| 00:00  | 76°34' | 270°00' | 270°00' | 00:00       | 56°34' | 090°00' | 090°00' |
| 01:00  | 70°22' | 220°40' | 319°20' | 01:00       | 53°29' | 115°22' | 064°38' |
| 02:00  | 58°23' | 200°04' | 339°56' | 02:00       | 45°31' | 134°39' | 045°21' |
| 03:00  | 45°04' | 189°35' | 350°25' | 03:00       | 34°44' | 147°56' | 032°04' |
| 04:00  | 31°23' | 182°27' | 357°33' | 04:00       | 22°30' | 157°24' | 022°36' |
| 05:00  | 17°38' | 176°31' | 003°29' | 05:00       | 09°29' | 164°40' | 015°20' |
| 06:00  | 03°58' | 170°49' | 009°11' | 05:42       | 00°00' | 169°05' | 010°55' |
| 06:18  | 00°00' | 169°05' | 010°55' | Aquarius    |        |         |         |
| Leo    |        |         |         | 00:00       | 59°51' | 077°46' | 077°46' |
| 00:00  | 79°51' | 282°14' | 282°14' | 01:00       | 56°26' | 105°13' | 050°19' |
| 01:00  | 72°20' | 225°05' | 339°23' | 02:00       | 47°48' | 124°55' | 030°37' |
| 02:00  | 59°22' | 207°06' | 357°22' | 03:00       | 36°26' | 137°43' | 017°49' |
| 03:00  | 45°32' | 198°31' | 005°56' | 04:00       | 23°44' | 146°28' | 009°05' |
| 04:00  | 31°28' | 192°40' | 011°47' | 05:00       | 10°20' | 153°00' | 002°33' |
| 05:00  | 17°24' | 187°42' | 016°45' | 05:45       | 00°00' | 157°07' | 358°26' |
| 06:00  | 03°26' | 182°50' | 021°38' | Pisces      |        |         |         |
| 06:15  | 00°00' | 181°34' | 022°53' | 00:00       | 68°32' | 069°26' | 069°26' |
| Virgo  |        |         |         | 01:00       | 63°52' | 104°47' | 034°05' |
| 00:00  | 88°32' | 290°34' | 290°34' | 02:00       | 53°15' | 124°49' | 014°02' |
| 01:00  | 75°11' | 204°50' | 016°19' | 03:00       | 40°23' | 135°31' | 003°20' |
| 02:00  | 60°30' | 200°30' | 020°39' | 04:00       | 26°37' | 141°59' | 356°53' |
| 03:00  | 45°48' | 197°58' | 023°11' | 05:00       | 12°26' | 146°21' | 352°30' |
| 04:00  | 31°08' | 195°44' | 025°25' | 05:52       | 00°00' | 149°13' | 349°38' |
| 05:00  | 16°31' | 193°25' | 027°44' |             |        |         |         |

Table 2.18: Ecliptic altitude and parallactic angle for latitude +10°.

|        |        |         |         |             |        |         |         |
|--------|--------|---------|---------|-------------|--------|---------|---------|
| Aries  |        |         |         |             |        |         |         |
| 00:00  | 70°00′ | 066°34′ | 066°34′ | 06:00       | 03°54′ | 180°57′ | 040°12′ |
| 01:00  | 65°11′ | 101°59′ | 031°09′ | 06:17       | 00°00′ | 180°09′ | 041°00′ |
| 02:00  | 54°28′ | 120°31′ | 012°37′ | Libra       |        |         |         |
| 03:00  | 41°38′ | 129°20′ | 003°48′ | 00:00       | 70°00′ | 113°26′ | 113°26′ |
| 04:00  | 28°01′ | 133°46′ | 359°22′ | 01:00       | 65°11′ | 148°51′ | 078°01′ |
| 05:00  | 14°05′ | 135°55′ | 357°13′ | 02:00       | 54°28′ | 167°23′ | 059°29′ |
| 06:00  | 00°00′ | 136°34′ | 356°34′ | 03:00       | 41°38′ | 176°12′ | 050°40′ |
| Taurus |        |         |         | 04:00       | 28°01′ | 180°38′ | 046°14′ |
| 00:00  | 81°28′ | 069°26′ | 069°26′ | 05:00       | 14°05′ | 182°47′ | 044°05′ |
| 01:00  | 73°15′ | 126°58′ | 011°53′ | 06:00       | 00°00′ | 183°26′ | 043°26′ |
| 02:00  | 59°57′ | 139°10′ | 359°41′ | Scorpio     |        |         |         |
| 03:00  | 45°59′ | 142°26′ | 356°25′ | 00:00       | 58°32′ | 110°34′ | 110°34′ |
| 04:00  | 31°54′ | 142°53′ | 355°58′ | 01:00       | 55°14′ | 135°49′ | 085°19′ |
| 05:00  | 17°50′ | 141°53′ | 356°58′ | 02:00       | 46°51′ | 153°58′ | 067°11′ |
| 06:00  | 03°54′ | 139°48′ | 359°03′ | 03:00       | 35°41′ | 165°27′ | 055°42′ |
| 06:17  | 00°00′ | 139°00′ | 359°51′ | 04:00       | 23°06′ | 172°48′ | 048°21′ |
| Gemini |        |         |         | 05:00       | 09°48′ | 177°40′ | 043°29′ |
| 00:00  | 89°51′ | 257°46′ | 257°46′ | 05:43       | 00°00′ | 180°09′ | 041°00′ |
| 01:00  | 75°55′ | 165°46′ | 349°46′ | Sagittarius |        |         |         |
| 02:00  | 61°52′ | 162°48′ | 352°44′ | 00:00       | 49°51′ | 102°14′ | 102°14′ |
| 03:00  | 47°52′ | 159°52′ | 355°41′ | 01:00       | 47°15′ | 123°13′ | 081°14′ |
| 04:00  | 33°59′ | 156°42′ | 358°51′ | 02:00       | 40°15′ | 140°14′ | 064°14′ |
| 05:00  | 20°15′ | 153°07′ | 002°26′ | 03:00       | 30°24′ | 152°37′ | 051°50′ |
| 06:00  | 06°46′ | 148°54′ | 006°38′ | 04:00       | 18°52′ | 161°33′ | 042°55′ |
| 06:31  | 00°00′ | 146°24′ | 009°08′ | 05:00       | 06°21′ | 168°11′ | 036°16′ |
| Cancer |        |         |         | 05:29       | 00°00′ | 170°52′ | 033°36′ |
| 00:00  | 86°34′ | 270°00′ | 270°00′ | Capricorn   |        |         |         |
| 01:00  | 75°39′ | 190°58′ | 202°37′ | 00:00       | 46°34′ | 090°00′ | 090°00′ |
| 02:00  | 61°58′ | 181°12′ | 204°00′ | 01:00       | 44°10′ | 109°49′ | 070°11′ |
| 03:00  | 48°13′ | 175°44′ | 150°21′ | 02:00       | 37°38′ | 126°24′ | 053°36′ |
| 04:00  | 34°33′ | 171°08′ | 143°47′ | 03:00       | 28°16′ | 138°59′ | 041°01′ |
| 05:00  | 21°03′ | 166°33′ | 137°02′ | 04:00       | 17°10′ | 148°24′ | 031°36′ |
| 06:00  | 07°49′ | 161°32′ | 130°26′ | 05:00       | 05°00′ | 155°40′ | 024°20′ |
| 06:36  | 00°00′ | 158°07′ | 126°33′ | 05:24       | 00°00′ | 158°07′ | 021°53′ |
| Leo    |        |         |         | Aquarius    |        |         |         |
| 00:00  | 89°51′ | 282°14′ | 282°14′ | 00:00       | 49°51′ | 077°46′ | 077°46′ |
| 01:00  | 75°55′ | 190°14′ | 014°14′ | 01:00       | 47°15′ | 098°46′ | 056°47′ |
| 02:00  | 61°52′ | 187°16′ | 017°12′ | 02:00       | 40°15′ | 115°46′ | 039°46′ |
| 03:00  | 47°52′ | 184°19′ | 020°08′ | 03:00       | 30°24′ | 128°10′ | 027°23′ |
| 04:00  | 33°59′ | 181°09′ | 023°18′ | 04:00       | 18°52′ | 137°05′ | 018°27′ |
| 05:00  | 20°15′ | 177°34′ | 026°53′ | 05:00       | 06°21′ | 143°44′ | 011°49′ |
| 06:00  | 06°46′ | 173°22′ | 031°06′ | 05:29       | 00°00′ | 146°24′ | 009°08′ |
| 06:31  | 00°00′ | 170°52′ | 033°36′ | Pisces      |        |         |         |
| Virgo  |        |         |         | 00:00       | 58°32′ | 069°26′ | 069°26′ |
| 00:00  | 81°28′ | 110°34′ | 110°34′ | 01:00       | 55°14′ | 094°41′ | 044°11′ |
| 01:00  | 73°15′ | 168°07′ | 053°02′ | 02:00       | 46°51′ | 112°49′ | 026°02′ |
| 02:00  | 59°57′ | 180°19′ | 040°50′ | 03:00       | 35°41′ | 124°18′ | 014°33′ |
| 03:00  | 45°59′ | 183°35′ | 037°34′ | 04:00       | 23°06′ | 131°39′ | 007°12′ |
| 04:00  | 31°54′ | 184°02′ | 037°07′ | 05:00       | 09°48′ | 136°31′ | 002°20′ |
| 05:00  | 17°50′ | 183°02′ | 038°07′ | 05:43       | 00°00′ | 139°00′ | 359°51′ |

Table 2.19: Ecliptic altitude and parallactic angle for latitude +20°.

|        |        |         |         |             |        |         |         |
|--------|--------|---------|---------|-------------|--------|---------|---------|
| Aries  |        |         |         |             |        |         |         |
| 00:00  | 60°00′ | 066°34′ | 066°34′ | 06:00       | 05°42′ | 171°04′ | 050°05′ |
| 01:00  | 56°46′ | 090°43′ | 042°25′ | 06:27       | 00°00′ | 169°54′ | 051°15′ |
| 02:00  | 48°35′ | 107°28′ | 025°40′ | Libra       |        |         |         |
| 03:00  | 37°46′ | 117°20′ | 015°48′ | 00:00       | 60°00′ | 113°26′ | 113°26′ |
| 04:00  | 25°40′ | 122°53′ | 010°15′ | 01:00       | 56°46′ | 137°35′ | 089°17′ |
| 05:00  | 12°57′ | 125°42′ | 007°26′ | 02:00       | 48°35′ | 154°20′ | 072°32′ |
| 06:00  | 00°00′ | 126°34′ | 006°34′ | 03:00       | 37°46′ | 164°12′ | 062°40′ |
| Taurus |        |         |         | 04:00       | 25°40′ | 169°45′ | 057°07′ |
| 00:00  | 71°28′ | 069°26′ | 069°26′ | 05:00       | 12°57′ | 172°34′ | 054°18′ |
| 01:00  | 66°49′ | 104°08′ | 034°43′ | 06:00       | 00°00′ | 173°26′ | 053°26′ |
| 02:00  | 56°33′ | 121°13′ | 017°38′ | Scorpio     |        |         |         |
| 03:00  | 44°24′ | 128°24′ | 010°27′ | 00:00       | 48°32′ | 110°34′ | 110°34′ |
| 04:00  | 31°35′ | 131°07′ | 007°44′ | 01:00       | 46°05′ | 129°26′ | 091°43′ |
| 05:00  | 18°36′ | 131°23′ | 007°28′ | 02:00       | 39°28′ | 144°41′ | 076°27′ |
| 06:00  | 05°42′ | 129°55′ | 008°56′ | 03:00       | 30°03′ | 155°36′ | 065°33′ |
| 06:27  | 00°00′ | 128°45′ | 010°06′ | 04:00       | 18°58′ | 163°03′ | 058°06′ |
| Gemini |        |         |         | 05:00       | 06°54′ | 168°00′ | 053°09′ |
| 00:00  | 80°09′ | 077°46′ | 077°46′ | 05:33       | 00°00′ | 169°54′ | 051°15′ |
| 01:00  | 73°15′ | 128°48′ | 026°45′ | Sagittarius |        |         |         |
| 02:00  | 61°12′ | 141°47′ | 013°46′ | 00:00       | 39°51′ | 102°14′ | 102°14′ |
| 03:00  | 48°20′ | 144°53′ | 010°40′ | 01:00       | 37°49′ | 118°43′ | 085°45′ |
| 04:00  | 35°22′ | 144°39′ | 010°54′ | 02:00       | 32°08′ | 132°59′ | 071°28′ |
| 05:00  | 22°30′ | 142°39′ | 012°54′ | 03:00       | 23°45′ | 144°13′ | 060°14′ |
| 06:00  | 09°55′ | 139°19′ | 016°14′ | 04:00       | 13°33′ | 152°43′ | 051°44′ |
| 06:49  | 00°00′ | 135°36′ | 019°57′ | 05:00       | 02°11′ | 159°04′ | 045°23′ |
| Cancer |        |         |         | 05:11       | 00°00′ | 160°03′ | 044°24′ |
| 00:00  | 83°26′ | 090°00′ | 090°00′ | Capricorn   |        |         |         |
| 01:00  | 75°06′ | 150°38′ | 029°22′ | 00:00       | 36°34′ | 090°00′ | 090°00′ |
| 02:00  | 62°30′ | 159°40′ | 020°20′ | 01:00       | 34°40′ | 105°49′ | 074°11′ |
| 03:00  | 49°32′ | 160°38′ | 019°22′ | 02:00       | 29°18′ | 119°46′ | 060°14′ |
| 04:00  | 36°36′ | 159°05′ | 020°55′ | 03:00       | 21°17′ | 131°05′ | 048°55′ |
| 05:00  | 23°52′ | 156°10′ | 023°50′ | 04:00       | 11°27′ | 139°56′ | 040°04′ |
| 06:00  | 11°28′ | 152°05′ | 027°55′ | 05:00       | 00°23′ | 146°47′ | 033°13′ |
| 06:58  | 00°00′ | 146°59′ | 033°01′ | 05:02       | 00°00′ | 146°59′ | 033°01′ |
| Leo    |        |         |         | Aquarius    |        |         |         |
| 00:00  | 80°09′ | 102°14′ | 102°14′ | 00:00       | 39°51′ | 077°46′ | 077°46′ |
| 01:00  | 73°15′ | 153°15′ | 051°12′ | 01:00       | 37°49′ | 094°15′ | 061°17′ |
| 02:00  | 61°12′ | 166°14′ | 038°13′ | 02:00       | 32°08′ | 108°32′ | 047°01′ |
| 03:00  | 48°20′ | 169°20′ | 035°07′ | 03:00       | 23°45′ | 119°46′ | 035°47′ |
| 04:00  | 35°22′ | 169°06′ | 035°21′ | 04:00       | 13°33′ | 128°16′ | 027°17′ |
| 05:00  | 22°30′ | 167°06′ | 037°21′ | 05:00       | 02°11′ | 134°37′ | 020°56′ |
| 06:00  | 09°55′ | 163°46′ | 040°41′ | 05:11       | 00°00′ | 135°36′ | 019°57′ |
| 06:49  | 00°00′ | 160°03′ | 044°24′ | Pisces      |        |         |         |
| Virgo  |        |         |         | 00:00       | 48°32′ | 069°26′ | 069°26′ |
| 00:00  | 71°28′ | 110°34′ | 110°34′ | 01:00       | 46°05′ | 088°17′ | 050°34′ |
| 01:00  | 66°49′ | 145°17′ | 075°52′ | 02:00       | 39°28′ | 103°33′ | 035°19′ |
| 02:00  | 56°33′ | 162°22′ | 058°47′ | 03:00       | 30°03′ | 114°27′ | 024°24′ |
| 03:00  | 44°24′ | 169°33′ | 051°36′ | 04:00       | 18°58′ | 121°54′ | 016°57′ |
| 04:00  | 31°35′ | 172°16′ | 048°53′ | 05:00       | 06°54′ | 126°51′ | 012°00′ |
| 05:00  | 18°36′ | 172°32′ | 048°37′ | 05:33       | 00°00′ | 128°45′ | 010°06′ |

Table 2.20: Ecliptic altitude and parallactic angle for latitude +30°.

|        |        |         |         |             |        |         |         |
|--------|--------|---------|---------|-------------|--------|---------|---------|
| Aries  |        |         |         | 03:00       | 41°12' | 156°37' | 064°32' |
| 00:00  | 50°00' | 066°34' | 066°34' | 04:00       | 30°13' | 160°43' | 060°26' |
| 01:00  | 47°44' | 083°43' | 049°25' | 05:00       | 18°47' | 161°59' | 059°10' |
| 02:00  | 41°34' | 097°21' | 035°47' | 06:00       | 07°21' | 161°09' | 060°00' |
| 03:00  | 32°48' | 106°41' | 026°27' | 06:39       | 00°00' | 159°35' | 061°34' |
| 04:00  | 22°31' | 112°28' | 020°40' | Libra       |        |         |         |
| 05:00  | 11°26' | 115°35' | 017°33' | 00:00       | 50°00' | 113°26' | 113°26' |
| 06:00  | 00°00' | 116°34' | 016°34' | 01:00       | 47°44' | 130°35' | 096°17' |
| Taurus |        |         |         | 02:00       | 41°34' | 144°13' | 082°39' |
| 00:00  | 61°28' | 069°26' | 069°26' | 03:00       | 32°48' | 153°33' | 073°19' |
| 01:00  | 58°32' | 091°45' | 047°06' | 04:00       | 22°31' | 159°20' | 067°32' |
| 02:00  | 51°05' | 106°59' | 031°52' | 05:00       | 11°26' | 162°27' | 064°25' |
| 03:00  | 41°12' | 115°28' | 023°23' | 06:00       | 00°00' | 163°26' | 063°26' |
| 04:00  | 30°13' | 119°34' | 019°17' | Scorpio     |        |         |         |
| 05:00  | 18°47' | 120°50' | 018°01' | 00:00       | 38°32' | 110°34' | 110°34' |
| 06:00  | 07°21' | 120°00' | 018°51' | 01:00       | 36°41' | 124°53' | 096°16' |
| 06:39  | 00°00' | 118°26' | 020°25' | 02:00       | 31°29' | 137°16' | 083°53' |
| Gemini |        |         |         | 03:00       | 23°46' | 146°52' | 074°17' |
| 00:00  | 70°09' | 077°46' | 077°46' | 04:00       | 14°20' | 153°47' | 067°22' |
| 01:00  | 66°21' | 107°24' | 048°09' | 05:00       | 03°49' | 158°26' | 062°42' |
| 02:00  | 57°35' | 123°23' | 032°10' | 05:21       | 00°00' | 159°35' | 061°34' |
| 03:00  | 46°53' | 130°11' | 025°21' | Sagittarius |        |         |         |
| 04:00  | 35°31' | 132°22' | 023°11' | 00:00       | 29°51' | 102°14' | 102°14' |
| 05:00  | 24°03' | 131°54' | 023°39' | 01:00       | 28°15' | 115°14' | 089°13' |
| 06:00  | 12°47' | 129°33' | 026°00' | 02:00       | 23°40' | 126°57' | 077°30' |
| 07:00  | 02°01' | 125°32' | 030°00' | 03:00       | 16°41' | 136°40' | 067°47' |
| 07:12  | 00°00' | 124°34' | 030°59' | 04:00       | 07°57' | 144°17' | 060°10' |
| Cancer |        |         |         | 04:48       | 00°00' | 149°01' | 055°26' |
| 00:00  | 73°26' | 090°00' | 090°00' | Capricorn   |        |         |         |
| 01:00  | 69°09' | 123°52' | 056°08' | 00:00       | 26°34' | 090°00' | 090°00' |
| 02:00  | 59°48' | 139°36' | 040°24' | 01:00       | 25°03' | 102°38' | 077°22' |
| 03:00  | 48°49' | 145°21' | 034°39' | 02:00       | 20°41' | 114°10' | 065°50' |
| 04:00  | 37°23' | 146°36' | 033°24' | 03:00       | 13°58' | 123°56' | 056°04' |
| 05:00  | 25°57' | 145°23' | 034°37' | 04:00       | 05°30' | 131°48' | 048°12' |
| 06:00  | 14°49' | 142°24' | 037°36' | 04:35       | 00°00' | 135°32' | 044°28' |
| 07:00  | 04°14' | 137°54' | 042°06' | Aquarius    |        |         |         |
| 07:25  | 00°00' | 135°32' | 044°28' | 00:00       | 29°51' | 077°46' | 077°46' |
| Leo    |        |         |         | 01:00       | 28°15' | 090°47' | 064°46' |
| 00:00  | 70°09' | 102°14' | 102°14' | 02:00       | 23°40' | 102°30' | 053°03' |
| 01:00  | 66°21' | 131°51' | 072°36' | 03:00       | 16°41' | 112°13' | 043°20' |
| 02:00  | 57°35' | 147°50' | 056°37' | 04:00       | 07°57' | 119°50' | 035°43' |
| 03:00  | 46°53' | 154°39' | 049°49' | 04:48       | 00°00' | 124°34' | 030°59' |
| 04:00  | 35°31' | 156°49' | 047°38' | Pisces      |        |         |         |
| 05:00  | 24°03' | 156°21' | 048°06' | 00:00       | 38°32' | 069°26' | 069°26' |
| 06:00  | 12°47' | 154°00' | 050°27' | 01:00       | 36°41' | 083°44' | 055°07' |
| 07:00  | 02°01' | 150°00' | 054°28' | 02:00       | 31°29' | 096°07' | 042°44' |
| 07:12  | 00°00' | 149°01' | 055°26' | 03:00       | 23°46' | 105°43' | 033°08' |
| Virgo  |        |         |         | 04:00       | 14°20' | 112°38' | 026°13' |
| 00:00  | 61°28' | 110°34' | 110°34' | 05:00       | 03°49' | 117°18' | 021°34' |
| 01:00  | 58°32' | 132°54' | 088°15' | 05:21       | 00°00' | 118°26' | 020°25' |
| 02:00  | 51°05' | 148°08' | 073°01' |             |        |         |         |

Table 2.21: Ecliptic altitude and parallactic angle for latitude +40°.

|        |        |         |         |             |        |         |         |
|--------|--------|---------|---------|-------------|--------|---------|---------|
| Aries  |        |         |         | 02:00       | 44°15′ | 137°14′ | 083°55′ |
| 00:00  | 40°00′ | 066°34′ | 066°34′ | 03:00       | 36°43′ | 145°07′ | 076°02′ |
| 01:00  | 38°23′ | 078°49′ | 054°19′ | 04:00       | 27°52′ | 149°36′ | 071°33′ |
| 02:00  | 33°50′ | 089°20′ | 043°48′ | 05:00       | 18°23′ | 151°26′ | 069°43′ |
| 03:00  | 27°02′ | 097°15′ | 035°53′ | 06:00       | 08°46′ | 151°09′ | 070°00′ |
| 04:00  | 18°45′ | 102°34′ | 030°34′ | 06:56       | 00°00′ | 149°10′ | 071°59′ |
| 05:00  | 09°35′ | 105°36′ | 027°32′ | Libra       |        |         |         |
| 06:00  | 00°00′ | 106°34′ | 026°34′ | 00:00       | 40°00′ | 113°26′ | 113°26′ |
| Taurus |        |         |         | 01:00       | 38°23′ | 125°41′ | 101°11′ |
| 00:00  | 51°28′ | 069°26′ | 069°26′ | 02:00       | 33°50′ | 136°12′ | 090°40′ |
| 01:00  | 49°32′ | 084°17′ | 054°34′ | 03:00       | 27°02′ | 144°07′ | 082°45′ |
| 02:00  | 44°15′ | 096°05′ | 042°46′ | 04:00       | 18°45′ | 149°26′ | 077°26′ |
| 03:00  | 36°43′ | 103°58′ | 034°53′ | 05:00       | 09°35′ | 152°28′ | 074°24′ |
| 04:00  | 27°52′ | 108°27′ | 030°24′ | 06:00       | 00°00′ | 153°26′ | 073°26′ |
| 05:00  | 18°23′ | 110°17′ | 028°34′ | 06:00       | 00°00′ | 153°26′ | 073°26′ |
| 06:00  | 08°46′ | 110°00′ | 028°51′ | Scorpio     |        |         |         |
| 06:56  | 00°00′ | 108°01′ | 030°50′ | 00:00       | 28°32′ | 110°34′ | 110°34′ |
| Gemini |        |         |         | 01:00       | 27°08′ | 121°21′ | 099°48′ |
| 00:00  | 60°09′ | 077°46′ | 077°46′ | 02:00       | 23°09′ | 131°02′ | 090°07′ |
| 01:00  | 57°51′ | 096°00′ | 059°33′ | 03:00       | 17°03′ | 138°58′ | 082°11′ |
| 02:00  | 51°51′ | 109°08′ | 046°25′ | 04:00       | 09°22′ | 144°55′ | 076°14′ |
| 03:00  | 43°40′ | 116°42′ | 038°50′ | 05:00       | 00°37′ | 148°57′ | 072°11′ |
| 04:00  | 34°26′ | 120°14′ | 035°19′ | 05:04       | 00°00′ | 149°10′ | 071°59′ |
| 05:00  | 24°50′ | 120°57′ | 034°36′ | Sagittarius |        |         |         |
| 06:00  | 15°18′ | 119°34′ | 035°59′ | 00:00       | 19°51′ | 102°14′ | 102°14′ |
| 07:00  | 06°11′ | 116°25′ | 039°08′ | 01:00       | 18°36′ | 112°20′ | 092°07′ |
| 07:44  | 00°00′ | 113°05′ | 042°27′ | 02:00       | 15°00′ | 121°40′ | 082°48′ |
| Cancer |        |         |         | 03:00       | 09°22′ | 129°39′ | 074°48′ |
| 00:00  | 63°26′ | 090°00′ | 090°00′ | 04:00       | 02°10′ | 136°05′ | 068°22′ |
| 01:00  | 60°58′ | 110°03′ | 069°57′ | 04:16       | 00°00′ | 137°33′ | 066°55′ |
| 02:00  | 54°38′ | 123°43′ | 056°17′ | Capricorn   |        |         |         |
| 03:00  | 46°12′ | 131°02′ | 048°58′ | 00:00       | 16°34′ | 090°00′ | 090°00′ |
| 04:00  | 36°50′ | 134°04′ | 045°56′ | 01:00       | 15°22′ | 099°56′ | 080°04′ |
| 05:00  | 27°13′ | 134°17′ | 045°43′ | 02:00       | 11°54′ | 109°10′ | 070°50′ |
| 06:00  | 17°44′ | 132°27′ | 047°33′ | 03:00       | 06°27′ | 117°13′ | 062°47′ |
| 07:00  | 08°45′ | 128°55′ | 051°05′ | 03:56       | 00°00′ | 123°24′ | 056°36′ |
| 08:00  | 00°34′ | 123°50′ | 056°10′ | Aquarius    |        |         |         |
| 08:04  | 00°00′ | 123°24′ | 056°36′ | 00:00       | 19°51′ | 077°46′ | 077°46′ |
| Leo    |        |         |         | 01:00       | 18°36′ | 087°53′ | 067°40′ |
| 00:00  | 60°09′ | 102°14′ | 102°14′ | 02:00       | 15°00′ | 097°12′ | 058°20′ |
| 01:00  | 57°51′ | 120°27′ | 084°00′ | 03:00       | 09°22′ | 105°12′ | 050°21′ |
| 02:00  | 51°51′ | 133°35′ | 070°52′ | 04:00       | 02°10′ | 111°38′ | 043°55′ |
| 03:00  | 43°40′ | 141°10′ | 063°18′ | 04:16       | 00°00′ | 113°05′ | 042°27′ |
| 04:00  | 34°26′ | 144°41′ | 059°46′ | Pisces      |        |         |         |
| 05:00  | 24°50′ | 145°24′ | 059°03′ | 00:00       | 28°32′ | 069°26′ | 069°26′ |
| 06:00  | 15°18′ | 144°01′ | 060°26′ | 01:00       | 27°08′ | 080°12′ | 058°39′ |
| 07:00  | 06°11′ | 140°52′ | 063°35′ | 02:00       | 23°09′ | 089°53′ | 048°58′ |
| 07:44  | 00°00′ | 137°33′ | 066°55′ | 03:00       | 17°03′ | 097°49′ | 041°02′ |
| Virgo  |        |         |         | 04:00       | 09°22′ | 103°46′ | 035°05′ |
| 00:00  | 51°28′ | 110°34′ | 110°34′ | 05:00       | 00°37′ | 107°49′ | 031°03′ |
| 01:00  | 49°32′ | 125°26′ | 095°43′ | 05:04       | 00°00′ | 108°01′ | 030°50′ |

Table 2.22: Ecliptic altitude and parallactic angle for latitude +50°.



|        |        |         |         |             |        |         |         |
|--------|--------|---------|---------|-------------|--------|---------|---------|
| Aries  |        |         |         | 08:38       | 00°00′ | 124°56′ | 079°31′ |
| 00:00  | 30°00′ | 066°34′ | 066°34′ | Virgo       |        |         |         |
| 01:00  | 28°53′ | 075°04′ | 058°04′ | 00:00       | 41°28′ | 110°34′ | 110°34′ |
| 02:00  | 25°40′ | 082°40′ | 050°28′ | 01:00       | 40°12′ | 120°20′ | 100°49′ |
| 03:00  | 20°42′ | 088°46′ | 044°22′ | 02:00       | 36°37′ | 128°43′ | 092°25′ |
| 04:00  | 14°29′ | 093°08′ | 040°00′ | 03:00       | 31°15′ | 135°00′ | 086°09′ |
| 05:00  | 07°26′ | 095°43′ | 037°25′ | 04:00       | 24°40′ | 139°02′ | 082°07′ |
| 06:00  | 00°00′ | 096°34′ | 036°34′ | 05:00       | 17°24′ | 140°59′ | 080°10′ |
| Taurus |        |         |         | 06:00       | 09°55′ | 141°05′ | 080°04′ |
| 00:00  | 41°28′ | 069°26′ | 069°26′ | 07:00       | 02°36′ | 139°29′ | 081°40′ |
| 01:00  | 40°12′ | 079°11′ | 059°40′ | 07:22       | 00°00′ | 138°29′ | 082°40′ |
| 02:00  | 36°37′ | 087°35′ | 051°17′ | Libra       |        |         |         |
| 03:00  | 31°15′ | 093°51′ | 045°00′ | 00:00       | 30°00′ | 113°26′ | 113°26′ |
| 04:00  | 24°40′ | 097°53′ | 040°58′ | 01:00       | 28°53′ | 121°56′ | 104°56′ |
| 05:00  | 17°24′ | 099°50′ | 039°01′ | 02:00       | 25°40′ | 129°32′ | 097°20′ |
| 06:00  | 09°55′ | 099°56′ | 038°55′ | 03:00       | 20°42′ | 135°38′ | 091°14′ |
| 07:00  | 02°36′ | 098°20′ | 040°31′ | 04:00       | 14°29′ | 140°00′ | 086°52′ |
| 07:22  | 00°00′ | 097°20′ | 041°31′ | 05:00       | 07°26′ | 142°35′ | 084°17′ |
| Gemini |        |         |         | 06:00       | 00°00′ | 143°26′ | 083°26′ |
| 00:00  | 50°09′ | 077°46′ | 077°46′ | Scorpio     |        |         |         |
| 01:00  | 48°44′ | 089°05′ | 066°27′ | 00:00       | 18°32′ | 110°34′ | 110°34′ |
| 02:00  | 44°49′ | 098°24′ | 057°08′ | 01:00       | 17°31′ | 118°22′ | 102°46′ |
| 03:00  | 39°04′ | 104°52′ | 050°41′ | 02:00       | 14°36′ | 125°33′ | 095°36′ |
| 04:00  | 32°12′ | 108°33′ | 046°59′ | 03:00       | 10°02′ | 131°37′ | 089°32′ |
| 05:00  | 24°49′ | 109°55′ | 045°37′ | 04:00       | 04°11′ | 136°18′ | 084°51′ |
| 06:00  | 17°21′ | 109°22′ | 046°11′ | 04:38       | 00°00′ | 138°29′ | 082°40′ |
| 07:00  | 10°11′ | 107°09′ | 048°23′ | Sagittarius |        |         |         |
| 08:00  | 03°39′ | 103°29′ | 052°03′ | 00:00       | 09°51′ | 102°14′ | 102°14′ |
| 08:38  | 00°00′ | 100°29′ | 055°04′ | 01:00       | 08°56′ | 109°45′ | 094°42′ |
| Cancer |        |         |         | 02:00       | 06°13′ | 116°48′ | 087°40′ |
| 00:00  | 53°26′ | 090°00′ | 090°00′ | 03:00       | 01°56′ | 122°57′ | 081°31′ |
| 01:00  | 51°57′ | 102°07′ | 077°53′ | 03:22       | 00°00′ | 124°56′ | 079°31′ |
| 02:00  | 47°53′ | 111°53′ | 068°07′ | Capricorn   |        |         |         |
| 03:00  | 41°58′ | 118°24′ | 061°36′ | 00:00       | 06°34′ | 090°00′ | 090°00′ |
| 04:00  | 35°01′ | 121°55′ | 058°05′ | 01:00       | 05°40′ | 097°28′ | 082°32′ |
| 05:00  | 27°35′ | 123°01′ | 056°59′ | 02:00       | 03°02′ | 104°30′ | 075°30′ |
| 06:00  | 20°09′ | 122°11′ | 057°49′ | 02:45       | 00°00′ | 109°17′ | 070°43′ |
| 07:00  | 13°03′ | 119°43′ | 060°17′ | Aquarius    |        |         |         |
| 08:00  | 06°36′ | 115°51′ | 064°09′ | 00:00       | 09°51′ | 077°46′ | 077°46′ |
| 09:00  | 01°09′ | 110°43′ | 069°17′ | 01:00       | 08°56′ | 085°18′ | 070°15′ |
| 09:15  | 00°00′ | 109°17′ | 070°43′ | 02:00       | 06°13′ | 092°20′ | 063°12′ |
| Leo    |        |         |         | 03:00       | 01°56′ | 098°29′ | 057°03′ |
| 00:00  | 50°09′ | 102°14′ | 102°14′ | 03:22       | 00°00′ | 100°29′ | 055°04′ |
| 01:00  | 48°44′ | 113°33′ | 090°55′ | Pisces      |        |         |         |
| 02:00  | 44°49′ | 122°52′ | 081°36′ | 00:00       | 18°32′ | 069°26′ | 069°26′ |
| 03:00  | 39°04′ | 129°19′ | 075°08′ | 01:00       | 17°31′ | 077°14′ | 061°38′ |
| 04:00  | 32°12′ | 133°01′ | 071°27′ | 02:00       | 14°36′ | 084°24′ | 054°27′ |
| 05:00  | 24°49′ | 134°23′ | 070°05′ | 03:00       | 10°02′ | 090°28′ | 048°23′ |
| 06:00  | 17°21′ | 133°49′ | 070°38′ | 04:00       | 04°11′ | 095°09′ | 043°42′ |
| 07:00  | 10°11′ | 131°37′ | 072°51′ | 04:38       | 00°00′ | 097°20′ | 041°31′ |
| 08:00  | 03°39′ | 127°57′ | 076°31′ |             |        |         |         |

Table 2.23: Ecliptic altitude and parallactic angle for latitude +60°.

|        |        |         |         |         |        |         |         |
|--------|--------|---------|---------|---------|--------|---------|---------|
| Aries  |        |         |         | 12:00   | 03°26′ | 090°00′ | 090°00′ |
| 00:00  | 20°00′ | 066°34′ | 066°34′ | Leo     |        |         |         |
| 01:00  | 19°17′ | 071°57′ | 061°11′ | 00:00   | 40°09′ | 102°14′ | 102°14′ |
| 02:00  | 17°14′ | 076°53′ | 056°15′ | 01:00   | 39°20′ | 108°48′ | 095°39′ |
| 03:00  | 14°00′ | 081°00′ | 052°08′ | 02:00   | 37°00′ | 114°35′ | 089°52′ |
| 04:00  | 09°51′ | 084°04′ | 049°04′ | 03:00   | 33°25′ | 119°04′ | 085°23′ |
| 05:00  | 05°05′ | 085°56′ | 047°12′ | 04:00   | 28°58′ | 122°01′ | 082°26′ |
| 06:00  | 00°00′ | 086°34′ | 046°34′ | 05:00   | 24°00′ | 123°26′ | 081°02′ |
| Taurus |        |         |         | 06:00   | 18°53′ | 123°25′ | 081°02′ |
| 00:00  | 31°28′ | 069°26′ | 069°26′ | 07:00   | 13°55′ | 122°08′ | 082°20′ |
| 01:00  | 30°42′ | 075°20′ | 063°31′ | 08:00   | 09°23′ | 119°42′ | 084°45′ |
| 02:00  | 28°30′ | 080°39′ | 058°12′ | 09:00   | 05°33′ | 116°17′ | 088°10′ |
| 03:00  | 25°05′ | 084°55′ | 053°56′ | 10:00   | 02°37′ | 112°05′ | 092°22′ |
| 04:00  | 20°46′ | 087°54′ | 050°58′ | 11:00   | 00°46′ | 107°18′ | 097°09′ |
| 05:00  | 15°53′ | 089°31′ | 049°20′ | 12:00   | 00°09′ | 102°14′ | 102°14′ |
| 06:00  | 10°46′ | 089°48′ | 049°03′ | Virgo   |        |         |         |
| 07:00  | 05°45′ | 088°49′ | 050°02′ | 00:00   | 31°28′ | 110°34′ | 110°34′ |
| 08:00  | 01°06′ | 086°40′ | 052°12′ | 01:00   | 30°42′ | 116°29′ | 104°40′ |
| 08:16  | 00°00′ | 085°55′ | 052°56′ | 02:00   | 28°30′ | 121°48′ | 099°21′ |
| Gemini |        |         |         | 03:00   | 25°05′ | 126°04′ | 095°05′ |
| 00:00  | 40°09′ | 077°46′ | 077°46′ | 04:00   | 20°46′ | 129°02′ | 092°06′ |
| 01:00  | 39°20′ | 084°21′ | 071°12′ | 05:00   | 15°53′ | 130°40′ | 090°29′ |
| 02:00  | 37°00′ | 090°08′ | 065°25′ | 06:00   | 10°46′ | 130°57′ | 090°12′ |
| 03:00  | 33°25′ | 094°37′ | 060°56′ | 07:00   | 05°45′ | 129°58′ | 091°11′ |
| 04:00  | 28°58′ | 097°34′ | 057°59′ | 08:00   | 01°06′ | 127°48′ | 093°20′ |
| 05:00  | 24°00′ | 098°58′ | 056°34′ | 08:16   | 00°00′ | 127°04′ | 094°05′ |
| 06:00  | 18°53′ | 098°58′ | 056°35′ | Libra   |        |         |         |
| 07:00  | 13°55′ | 097°40′ | 057°52′ | 00:00   | 20°00′ | 113°26′ | 113°26′ |
| 08:00  | 09°23′ | 095°15′ | 060°18′ | 01:00   | 19°17′ | 118°49′ | 108°03′ |
| 09:00  | 05°33′ | 091°50′ | 063°43′ | 02:00   | 17°14′ | 123°45′ | 103°07′ |
| 10:00  | 02°37′ | 087°38′ | 067°55′ | 03:00   | 14°00′ | 127°52′ | 099°00′ |
| 11:00  | 00°46′ | 082°51′ | 072°42′ | 04:00   | 09°51′ | 130°56′ | 095°56′ |
| 12:00  | 00°09′ | 077°46′ | 077°46′ | 05:00   | 05°05′ | 132°48′ | 094°04′ |
| Cancer |        |         |         | 06:00   | 00°00′ | 133°26′ | 093°26′ |
| 00:00  | 43°26′ | 090°00′ | 090°00′ | Scorpio |        |         |         |
| 01:00  | 42°36′ | 096°54′ | 083°06′ | 00:00   | 08°32′ | 110°34′ | 110°34′ |
| 02:00  | 40°12′ | 102°56′ | 077°04′ | 01:00   | 07°52′ | 115°42′ | 105°27′ |
| 03:00  | 36°33′ | 107°31′ | 072°29′ | 02:00   | 05°56′ | 120°28′ | 100°40′ |
| 04:00  | 32°03′ | 110°27′ | 069°33′ | 03:00   | 02°53′ | 124°35′ | 096°34′ |
| 05:00  | 27°04′ | 111°47′ | 068°13′ | 03:44   | 00°00′ | 127°04′ | 094°05′ |
| 06:00  | 21°57′ | 111°38′ | 068°22′ | Pisces  |        |         |         |
| 07:00  | 17°00′ | 110°13′ | 069°47′ | 00:00   | 08°32′ | 069°26′ | 069°26′ |
| 08:00  | 12°31′ | 107°40′ | 072°20′ | 01:00   | 07°52′ | 074°33′ | 064°18′ |
| 09:00  | 08°44′ | 104°10′ | 075°50′ | 02:00   | 05°56′ | 079°20′ | 059°32′ |
| 10:00  | 05°51′ | 099°54′ | 080°06′ | 03:00   | 02°53′ | 083°26′ | 055°25′ |
| 11:00  | 04°03′ | 095°05′ | 084°55′ | 03:44   | 00°00′ | 085°55′ | 052°56′ |

Table 2.24: Ecliptic altitude and parallactic angle for latitude +70°.

|        |        |         |         |       |        |         |         |
|--------|--------|---------|---------|-------|--------|---------|---------|
| Aries  |        |         |         |       |        |         |         |
| 00:00  | 10°00′ | 066°34′ | 066°34′ | 08:00 | 18°11′ | 099°06′ | 080°54′ |
| 01:00  | 09°39′ | 069°11′ | 063°57′ | 09:00 | 16°12′ | 097°21′ | 082°39′ |
| 02:00  | 08°39′ | 071°36′ | 061°32′ | 10:00 | 14°42′ | 095°09′ | 084°51′ |
| 03:00  | 07°03′ | 073°40′ | 059°28′ | 11:00 | 13°45′ | 092°39′ | 087°21′ |
| 04:00  | 04°59′ | 075°15′ | 057°53′ | 12:00 | 13°26′ | 090°00′ | 090°00′ |
| 05:00  | 02°35′ | 076°14′ | 056°54′ | Leo   |        |         |         |
| 06:00  | 00°00′ | 076°34′ | 056°34′ | 00:00 | 30°09′ | 102°14′ | 102°14′ |
| Taurus |        |         |         | 01:00 | 29°47′ | 105°12′ | 099°16′ |
| 00:00  | 21°28′ | 069°26′ | 069°26′ | 02:00 | 28°43′ | 107°55′ | 096°33′ |
| 01:00  | 21°07′ | 072°11′ | 066°40′ | 03:00 | 27°02′ | 110°09′ | 094°18′ |
| 02:00  | 20°04′ | 074°44′ | 064°07′ | 04:00 | 24°53′ | 111°46′ | 092°41′ |
| 03:00  | 18°26′ | 076°52′ | 061°59′ | 05:00 | 22°25′ | 112°41′ | 091°46′ |
| 04:00  | 16°19′ | 078°26′ | 060°25′ | 06:00 | 19°50′ | 112°52′ | 091°35′ |
| 05:00  | 13°53′ | 079°23′ | 059°29′ | 07:00 | 17°17′ | 112°21′ | 092°07′ |
| 06:00  | 11°18′ | 079°38′ | 059°14′ | 08:00 | 14°56′ | 111°11′ | 093°16′ |
| 07:00  | 08°44′ | 079°12′ | 059°39′ | 09:00 | 12°56′ | 109°28′ | 094°59′ |
| 08:00  | 06°21′ | 078°08′ | 060°43′ | 10:00 | 11°25′ | 107°19′ | 097°09′ |
| 09:00  | 04°20′ | 076°30′ | 062°21′ | 11:00 | 10°28′ | 104°51′ | 099°36′ |
| 10:00  | 02°47′ | 074°25′ | 064°26′ | 12:00 | 10°09′ | 102°14′ | 102°14′ |
| 11:00  | 01°48′ | 072°00′ | 066°51′ | Virgo |        |         |         |
| 12:00  | 01°28′ | 069°26′ | 069°26′ | 00:00 | 21°28′ | 110°34′ | 110°34′ |
| Gemini |        |         |         | 01:00 | 21°07′ | 113°20′ | 107°49′ |
| 00:00  | 30°09′ | 077°46′ | 077°46′ | 02:00 | 20°04′ | 115°53′ | 105°16′ |
| 01:00  | 29°47′ | 080°44′ | 074°48′ | 03:00 | 18°26′ | 118°01′ | 103°08′ |
| 02:00  | 28°43′ | 083°27′ | 072°05′ | 04:00 | 16°19′ | 119°35′ | 101°34′ |
| 03:00  | 27°02′ | 085°42′ | 069°51′ | 05:00 | 13°53′ | 120°31′ | 100°37′ |
| 04:00  | 24°53′ | 087°19′ | 068°14′ | 06:00 | 11°18′ | 120°46′ | 100°22′ |
| 05:00  | 22°25′ | 088°14′ | 067°19′ | 07:00 | 08°44′ | 120°21′ | 100°48′ |
| 06:00  | 19°50′ | 088°25′ | 067°08′ | 08:00 | 06°21′ | 119°17′ | 101°52′ |
| 07:00  | 17°17′ | 087°53′ | 067°39′ | 09:00 | 04°20′ | 117°39′ | 103°30′ |
| 08:00  | 14°56′ | 086°44′ | 068°49′ | 10:00 | 02°47′ | 115°34′ | 105°35′ |
| 09:00  | 12°56′ | 085°01′ | 070°32′ | 11:00 | 01°48′ | 113°09′ | 108°00′ |
| 10:00  | 11°25′ | 082°51′ | 072°41′ | 12:00 | 01°28′ | 110°34′ | 110°34′ |
| 11:00  | 10°28′ | 080°24′ | 075°09′ | Libra |        |         |         |
| 12:00  | 10°09′ | 077°46′ | 077°46′ | 00:00 | 10°00′ | 113°26′ | 113°26′ |
| Cancer |        |         |         | 01:00 | 09°39′ | 116°03′ | 110°49′ |
| 00:00  | 33°26′ | 090°00′ | 090°00′ | 02:00 | 08°39′ | 118°28′ | 108°24′ |
| 01:00  | 33°04′ | 093°04′ | 086°56′ | 03:00 | 07°03′ | 120°32′ | 106°20′ |
| 02:00  | 31°59′ | 095°53′ | 084°07′ | 04:00 | 04°59′ | 122°07′ | 104°45′ |
| 03:00  | 30°17′ | 098°10′ | 081°50′ | 05:00 | 02°35′ | 123°06′ | 103°46′ |
| 04:00  | 28°07′ | 099°49′ | 080°11′ | 06:00 | 00°00′ | 123°26′ | 103°26′ |
| 05:00  | 25°39′ | 100°43′ | 079°17′ |       |        |         |         |
| 06:00  | 23°03′ | 100°53′ | 079°07′ |       |        |         |         |
| 07:00  | 20°31′ | 100°19′ | 079°41′ |       |        |         |         |

Table 2.25: Ecliptic altitude and parallactic angle for latitude +80°.



### 3. Dates and times

#### 3.1 *Julian and Gregorian calendars*

The Julian calendar was instituted by Julius Caesar in 45 BC. However, it was not correctly implemented until 8 AD. The Julian calendar assumes that the length of a tropical year (that is, the time between successive passages of the Sun through the vernal equinox) is 365.25 days. Unfortunately, the true length of the tropical year is 365.2422 days. This slight error caused the date of the vernal equinox to regress through the calendar over the course of many centuries. In order to correct this problem, the Gregorian calendar was promulgated by Pope Gregory XIII in 1582. The new calendar was immediately adopted by all of the Catholic countries in Europe, and was eventually (in some cases, after a delay of hundreds of years) also adopted by all of the Protestant countries. According to the Gregorian calendar, the length of a tropical year is 365.2425 days, which is very close to the correct length.

#### 3.2 *Julian day number*

Following modern astronomical practice, we shall specify dates in this treatise by means of *Julian day numbers*. According to this scheme, days are numbered consecutively from January 1, 4713 BC (in the Julian calendar), which is designated day zero. For instance, October 14, 1066 AD (the Julian date of the battle of Hastings) is day 2 110 701. Each Julian day commences at 12:00 universal time (UT). (Universal time is virtually identical to Greenwich mean time.)

#### 3.3 *Determination of Julian day numbers*

The Julian day number of a given day can be determined from Tables 3.1–3.3. The date must be expressed in terms of the Gregorian calendar.

The procedure is as follows:

1. Enter the table of century years (Table 3.1) with the century year immediately preceding the date in question, and take out the tabular value. If the century year is marked with a †, note this fact for use in step 2.
2. Enter the table of years of the century (Table 3.2) with the last two digits of the year in question, and take out the tabular value. If the century year used in step 1 was marked with a †, diminish the tabular value by one day, unless the tabular value is zero. If the year in question is a leap year, marked with a \*, note this fact for use in step 3.
3. Enter the table of the days of the year (Table 3.3) with the day in question, and take out the tabular value. If the year in question is a leap year and the table entry falls after February 28, add one day to the tabular value. The sum of the values obtained in steps 1, 2, and 3 then gives the Julian day number of the date in question.

*Example 1: June 10, 1992 AD*

|                        |                |                 |                  |
|------------------------|----------------|-----------------|------------------|
| 1. Century year        | $\dagger 1900$ |                 | 2 415 020        |
| 2. Year of the century | $*92$          | $33\,603 - 1 =$ | 33 602           |
| 3. Day of the year     | June 10        | $161 + 1 =$     | 162              |
| Julian day number      |                |                 | <u>2 448 784</u> |

Observe that in step 2 the tabular value has been diminished by 1 because 1900 is a common year (marked with a  $\dagger$  in Table 3.1). In step 3, the tabular value has been increased by 1 because 1992 is a leap year (marked with a  $*$  in Table 3.2), and the date falls after February 28.

*Example 2: January 18, 1824 AD*

|                        |                |                |                  |
|------------------------|----------------|----------------|------------------|
| 1. Century year        | $\dagger 1800$ |                | 2 378 496        |
| 2. Year of the century | $*24$          | $8\,766 - 1 =$ | 8 765            |
| 3. Day of the year     | January 18     | $18 =$         | 18               |
| Julian day number      |                |                | <u>2 387 279</u> |

Observe that in step 2 the tabular value has been diminished by 1 because 1800 is a common year (marked with a  $\dagger$  in Table 3.1). In step 3, the tabular value has not been increased by 1, despite the fact that 1824 is a leap year (marked with an  $*$  in Table 3.2), because the date falls before February 28.

We can specify the time of day (in universal time), as well as the date, by means of fractional Julian day numbers. For instance,  $t = 2\,448\,784.0$  JD corresponds to 12:00 UT on June 10, 1992 AD, whereas  $t = 2\,448\,784.5$  JD corresponds to 24:00 UT later the same day.

### 3.4 Tables

|                |           |
|----------------|-----------|
| $\dagger 1700$ | 2 341 972 |
| $\dagger 1800$ | 2 378 496 |
| $\dagger 1900$ | 2 415 020 |
| 2000           | 2 451 544 |
| $\dagger 2100$ | 2 488 069 |
| $\dagger 2200$ | 2 524 593 |
| $\dagger 2300$ | 2 561 117 |

Table 3.1: Julian Day Number: Century Years.  $\dagger$  Common years. All years are AD. Table reproduced, with permission, from Evans (1998).

|     |       |     |        |     |        |     |        |     |        |
|-----|-------|-----|--------|-----|--------|-----|--------|-----|--------|
| §0  | 0     | *20 | 7 305  | *40 | 14 610 | *60 | 21 915 | *80 | 29 220 |
| 1   | 336   | 21  | 7 671  | 41  | 14 976 | 61  | 22 281 | 81  | 29 586 |
| 2   | 731   | 22  | 8 036  | 42  | 15 341 | 62  | 22 646 | 82  | 29 951 |
| 3   | 1 096 | 23  | 8 401  | 43  | 15 706 | 63  | 22 011 | 83  | 30 316 |
|     |       |     |        |     |        |     |        |     |        |
| *4  | 1 461 | *24 | 8 766  | *44 | 16 071 | *64 | 23 376 | *84 | 30 681 |
| 5   | 1 827 | 25  | 9 132  | 45  | 16 437 | 65  | 23 742 | 85  | 31 047 |
| 6   | 2 192 | 26  | 9 497  | 46  | 16 802 | 66  | 24 107 | 86  | 31 412 |
| 7   | 2 557 | 27  | 9 862  | 47  | 17 167 | 67  | 24 472 | 87  | 31 777 |
|     |       |     |        |     |        |     |        |     |        |
| *8  | 2 922 | *28 | 10 227 | *48 | 17 532 | *68 | 24 837 | *88 | 32 142 |
| 9   | 3 288 | 29  | 10 593 | 49  | 17 898 | 69  | 25 203 | 89  | 32 508 |
| 10  | 3 653 | 30  | 10 958 | 50  | 18 263 | 70  | 25 568 | 90  | 32 873 |
| 11  | 4 018 | 31  | 11 323 | 51  | 18 628 | 71  | 25 933 | 91  | 33 238 |
|     |       |     |        |     |        |     |        |     |        |
| *12 | 4 383 | *32 | 11 688 | *52 | 18 993 | *72 | 26 298 | *92 | 33 603 |
| 13  | 4 749 | 33  | 12 054 | 53  | 19 359 | 73  | 26 664 | 93  | 33 969 |
| 14  | 5 114 | 34  | 12 419 | 54  | 19 724 | 74  | 27 029 | 94  | 34 334 |
| 15  | 5 479 | 35  | 12 784 | 55  | 20 089 | 75  | 27 394 | 95  | 34 699 |
|     |       |     |        |     |        |     |        |     |        |
| *16 | 5 844 | *36 | 13 149 | *56 | 20 454 | *76 | 27 759 | *96 | 35 064 |
| 17  | 6 210 | 37  | 13 515 | 57  | 20 820 | 77  | 28 125 | 97  | 35 430 |
| 18  | 6 575 | 38  | 13 880 | 58  | 21 185 | 78  | 28 490 | 98  | 35 795 |
| 19  | 6 940 | 39  | 14 245 | 59  | 21 550 | 79  | 28 855 | 99  | 36 160 |

Table 3.2: Julian Day Number: Years of the Century. \* Leap year. § Leap year unless century is marked †. In centuries marked †, subtract one day from the tabulated values for the years 1 through 99. Reproduced, with permission, from Evans (1998).

| Day | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 1   | 1   | 32  | 60  | 91  | 121 | 152 | 182 | 213 | 244 | 274 | 305 | 335 |
| 2   | 2   | 33  | 61  | 92  | 122 | 153 | 183 | 214 | 245 | 275 | 306 | 336 |
| 3   | 3   | 34  | 62  | 93  | 123 | 154 | 184 | 215 | 246 | 276 | 307 | 337 |
| 4   | 4   | 35  | 63  | 94  | 124 | 155 | 185 | 216 | 247 | 277 | 308 | 338 |
| 5   | 5   | 36  | 64  | 95  | 125 | 156 | 186 | 217 | 248 | 278 | 309 | 339 |
| 6   | 6   | 37  | 65  | 96  | 126 | 157 | 187 | 218 | 249 | 279 | 310 | 340 |
| 7   | 7   | 38  | 66  | 97  | 127 | 158 | 188 | 219 | 250 | 280 | 311 | 341 |
| 8   | 8   | 39  | 67  | 98  | 128 | 159 | 189 | 220 | 251 | 281 | 312 | 342 |
| 9   | 9   | 40  | 68  | 99  | 129 | 160 | 190 | 221 | 252 | 282 | 313 | 343 |
| 10  | 10  | 41  | 69  | 100 | 130 | 161 | 191 | 222 | 253 | 283 | 314 | 344 |
| 11  | 11  | 42  | 70  | 101 | 131 | 162 | 192 | 223 | 254 | 284 | 315 | 345 |
| 12  | 12  | 43  | 71  | 102 | 132 | 163 | 193 | 224 | 255 | 285 | 316 | 346 |
| 13  | 13  | 44  | 72  | 103 | 133 | 164 | 194 | 225 | 256 | 286 | 317 | 347 |
| 14  | 14  | 45  | 73  | 104 | 134 | 165 | 195 | 226 | 257 | 285 | 318 | 348 |
| 15  | 15  | 46  | 74  | 105 | 135 | 166 | 196 | 227 | 258 | 288 | 319 | 349 |
| 16  | 16  | 47  | 75  | 106 | 136 | 167 | 197 | 228 | 259 | 289 | 320 | 350 |
| 17  | 17  | 48  | 76  | 107 | 137 | 168 | 198 | 229 | 260 | 290 | 321 | 351 |
| 18  | 18  | 49  | 77  | 108 | 138 | 169 | 199 | 230 | 261 | 291 | 322 | 352 |
| 19  | 19  | 50  | 78  | 109 | 139 | 170 | 200 | 231 | 262 | 292 | 323 | 353 |
| 20  | 20  | 51  | 79  | 110 | 140 | 171 | 201 | 232 | 263 | 293 | 324 | 354 |
| 21  | 21  | 52  | 80  | 111 | 141 | 172 | 202 | 233 | 264 | 294 | 325 | 355 |
| 22  | 22  | 53  | 81  | 112 | 142 | 173 | 203 | 234 | 265 | 295 | 326 | 356 |
| 23  | 23  | 54  | 82  | 113 | 143 | 174 | 204 | 235 | 266 | 296 | 327 | 357 |
| 24  | 24  | 55  | 83  | 114 | 144 | 175 | 205 | 236 | 267 | 297 | 328 | 358 |
| 25  | 25  | 56  | 84  | 115 | 145 | 176 | 206 | 237 | 268 | 298 | 329 | 359 |
| 26  | 26  | 57  | 85  | 116 | 146 | 177 | 207 | 238 | 269 | 299 | 330 | 360 |
| 27  | 27  | 58  | 86  | 117 | 147 | 178 | 208 | 239 | 270 | 300 | 331 | 361 |
| 28  | 28  | 59  | 87  | 118 | 148 | 179 | 209 | 240 | 271 | 301 | 332 | 362 |
| 29  | 29  | *   | 88  | 119 | 149 | 180 | 210 | 241 | 272 | 302 | 333 | 363 |
| 30  | 30  |     | 89  | 120 | 150 | 181 | 211 | 242 | 273 | 303 | 334 | 364 |
| 31  | 31  |     | 90  |     | 151 |     | 212 | 243 |     | 304 |     | 365 |

Table 3.3: Julian Day Number: Days of the Year. \* In leap year, after February 28, add 1 to the tabulated value. Reproduced, with permission, from Evans (1998).



## 4. Geometric planetary orbit models

### 4.1 The model of Kepler

In this chapter, Kepler's geometric model of a geocentric planetary orbit is examined in detail, and then compared to the less accurate geometric models of Hipparchus, Ptolemy, and Copernicus. In the following, all orbits are viewed from the northern ecliptic pole.

Kepler's geometric model of a heliocentric planetary orbit is summed up in his three well-known laws of planetary motion. According to Kepler's first law, all planetary orbits are ellipses that are confocal with the Sun and lie in a fixed plane. Moreover, according to Kepler's second law, the radius vector which connects the Sun to a given planet sweeps out equal areas in equal time intervals.

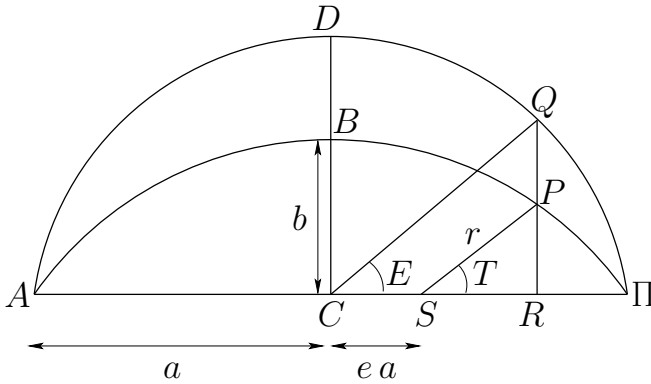


Figure 4.1: A Keplerian orbit.

Consider Figure 4.1. The curve  $\Pi PBA$  is half of an elliptical planetary orbit. Furthermore,  $C$  is the geometric center of the orbit,  $S$  the focus at which the Sun is located,  $P$  the instantaneous position of the planet,  $\Pi$  the perihelion point (that is, the planet's point of closest approach to the Sun), and  $A$  the aphelion point (that is, the point of furthest distance from the Sun). The ellipse is symmetric about  $\Pi A$ , which is termed the *major axis*, and about  $CB$ , which is termed the *minor axis*. The length  $CA \equiv a$  is called the orbital *major radius*. The length  $CS$  represents the displacement of the Sun from the geometric center of the orbit, and is generally written  $ea$ , where  $e$  is termed the orbital *eccentricity*, and  $0 \leq e \leq 1$ . The length  $CB \equiv b = a(1 - e^2)^{1/2}$  is called the orbital *minor radius*. The length  $SP \equiv r$  represents the radial distance of the planet from the Sun. Finally, the angle  $RSP \equiv T$  is the angular bearing of the planet from the Sun, relative to the major axis of the orbit, and is termed the *true anomaly*.

The curve  $\Pi QDA$  is half of a circle whose geometric center is  $C$ , and whose radius is  $a$ . Hence, the circle passes through the perihelion and aphelion points.  $R$  is the point at which the perpendicular from  $P$  meets the major axis  $\Pi A$ . The point where  $RP$  produced meets

circle  $\Pi QDA$  is denoted  $Q$ . Finally, the angle  $SCQ \equiv E$  is called the *elliptic anomaly*.

Now, the equation of the ellipse  $\Pi PBA$  is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad (4.1)$$

where  $x$  and  $y$  are the perpendicular distances from the minor and major axes, respectively. Likewise, the equation of the circle  $\Pi QDA$  is

$$\frac{x'^2}{a^2} + \frac{y'^2}{a^2} = 1. \quad (4.2)$$

Hence, if  $x = x'$  then

$$\frac{y}{y'} = \frac{b}{a}, \quad (4.3)$$

and it follows that

$$\frac{RP}{RQ} = \frac{b}{a}. \quad (4.4)$$

Now,  $CS = e a$ . Furthermore, it is easily demonstrated that  $SR = r \cos T$ ,  $RP = r \sin T$ ,  $CR = a \cos E$ , and  $RQ = a \sin E$ . Consequently, Equation (4.4) yields

$$r \sin T = b \sin E = a (1 - e^2)^{1/2} \sin E. \quad (4.5)$$

Also, because  $SR = CR - CS$ , we have

$$r \cos T = a (\cos E - e). \quad (4.6)$$

Taking the square root of the sum of the squares of the previous two equations, we obtain

$$r = a (1 - e \cos E), \quad (4.7)$$

which can be combined with Equation (4.6) to give

$$\cos T = \frac{\cos E - e}{1 - e \cos E}. \quad (4.8)$$

Now, according to Kepler's second law,

$$\frac{\text{Area } \Pi PS}{\pi a b} = \frac{t - t_*}{\tau}, \quad (4.9)$$

where  $t$  is the time at which the planet passes point  $P$ ,  $t_*$  the time at which it passes the perihelion point, and  $\tau$  the *orbital period*. However,

$$\text{Area } \Pi PS = \text{Area } SRP + \text{Area } \Pi RP = \frac{1}{2} r^2 \cos T \sin T + \text{Area } \Pi RP. \quad (4.10)$$

But,

$$\text{Area } \Pi RP = \frac{b}{a} \text{Area } \Pi RQ, \quad (4.11)$$

because  $RP/RQ = b/a$  for all values of  $T$ . In addition,

$$\text{Area } \Pi RQ = \text{Area } \Pi QC - \text{Area } RQC = \frac{1}{2} E a^2 - \frac{1}{2} a^2 \cos E \sin E. \quad (4.12)$$

Hence, we can write

$$\left( \frac{t - t_*}{\tau} \right) \pi a b = \frac{1}{2} r^2 \cos T \sin T + \frac{b}{a} \frac{a^2}{2} (E - \cos E \sin E). \quad (4.13)$$

According to Equations (4.5) and (4.6),  $r \sin T = b \sin E$ , and  $r \cos T = a (\cos E - e)$ , so the previous expression reduces to

$$M = E - e \sin E, \quad (4.14)$$

where

$$M = \left( \frac{2\pi}{\tau} \right) (t - t_*) \quad (4.15)$$

is an angle that is zero at the perihelion point, increases uniformly in time, and has a repetition period that matches the period of the planetary orbit. This angle is termed the *mean anomaly*.

In summary, the radial and angular polar coordinates,  $r$  and  $T$ , respectively, of a planet in a Keplerian orbit about the Sun are specified as implicit functions of the mean anomaly, which is a linear function of time, by the following three equations:

$$M = E - e \sin E, \quad (4.16)$$

$$r = a (1 - e \cos E), \quad (4.17)$$

$$\cos T = \frac{\cos E - e}{1 - e \cos E}. \quad (4.18)$$

It turns out that the Earth and the five visible (to the naked eye) planets all possess low eccentricity orbits characterized by  $0 < e \ll 1$ . Hence, it is a good approximation to expand the previous three equations using  $e$  as a small parameter. To second order, we get

$$E = M + e \sin M + (1/2) e^2 \sin 2M, \quad (4.19)$$

$$r = a (1 - e \cos T - e^2 \sin^2 T), \quad (4.20)$$

$$T = E + e \sin E + (1/4) e^2 \sin 2E. \quad (4.21)$$

Finally, these equations can be combined to give  $r$  and  $T$  as explicit functions of the mean anomaly:

$$\frac{r}{a} = 1 - e \cos M + e^2 \sin^2 M, \quad (4.22)$$

$$T = M + 2e \sin M + (5/4) e^2 \sin 2M. \quad (4.23)$$

### 4.2 The model of Hipparchus

Hipparchus's geometric model of the apparent orbit of the Sun around the Earth can also be used to describe a heliocentric planetary orbit. The model is illustrated in Figure 4.2. The orbit of the planet corresponds to the circle  $\Pi PDA$ , where  $\Pi$  is the perihelion point,  $P$  the planet's instantaneous position, and  $A$  the aphelion point. The diameter  $\Pi SCA$  is the effective major axis of the orbit (to be more exact, it is the line of apsides), where  $C$  is the geometric center of circle  $\Pi PDA$ , and  $S$  the fixed position of the Sun. The radius  $CP$  of circle  $\Pi PDA$  is the effective major radius,  $a$ , of the orbit. The distance  $SC$  is equal to  $2ea$ , where  $e$  is the orbit's effective eccentricity. The angle  $PC\Pi$  is identified with the mean anomaly,  $M$ , and increases linearly in time. In other words, as seen from  $C$ , the planet  $P$  moves uniformly around circle  $\Pi PDA$  in a counterclockwise direction. Finally,  $SP$  is the radial distance,  $r$ , of the planet from the Sun, and angle  $PS\Pi$  is the planet's true anomaly,  $T$ .

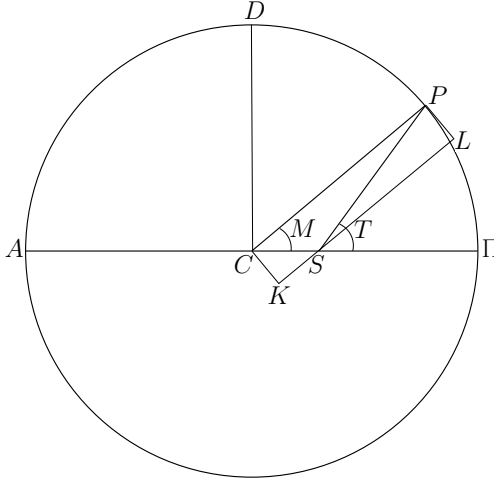


Figure 4.2: A Hipparchian orbit.

Let us draw the straight-line  $KSL$  parallel to  $CP$ , and passing through point  $S$ , and then complete the rectangle  $PCKL$ . Simple geometry reveals that  $CK = PL = 2ea \sin M$ ,  $KS = 2ea \cos M$ , and  $SL = a - 2ea \cos M$ . Moreover, according to the theorem of Pythagoras,  $SP^2 = SL^2 + PL^2$ , which implies that

$$\frac{r}{a} = (1 - 4e \cos M + 4e^2)^{1/2}. \quad (4.24)$$

Now,  $T = M + q$ , where  $q$  is angle  $PSL$ . However,

$$\sin q = \frac{PL}{SP} = \frac{2e \sin M}{(1 - 4e \cos M + 4e^2)^{1/2}}. \quad (4.25)$$

Finally, expanding the previous two equations to second order in the small parameter  $e$ , we obtain

$$\frac{r}{a} = 1 - 2e \cos M + 2e^2 \sin^2 M, \quad (4.26)$$

$$T = M + 2e \sin M + 2e^2 \sin 2M. \quad (4.27)$$

It can be seen, by comparison with Equations (4.22) and (4.23), that the relative radial distance,  $r/a$ , in the Hipparchian model deviates from that in the (correct) Keplerian model to first order in  $e$  (in fact, the variation of  $r/a$  is greater by a factor of two in the former model), whereas the true anomaly,  $T$ , only deviates to second order in  $e$ . We conclude that Hipparchus's geometric model of a heliocentric planetary orbit does a reasonably good job at predicting the angular position of the planet, relative to the Sun, but significantly exaggerates (by a factor of two) the variation in the radial distance between the two during the course of a complete orbital rotation.

### 4.3 The model of Ptolemy

Ptolemy's geometric model of the motion of the center of an epicycle around a deferent can also be used to describe a heliocentric planetary orbit. The model is illustrated in Figure 4.3. The orbit of the planet corresponds to the circle  $\Pi PDA$  (only half of which is shown), where  $\Pi$  is the perihelion point,  $P$  the planet's instantaneous position, and  $A$  the aphelion point. The diameter  $\Pi SCQA$  is the effective major axis of the orbit, where  $C$  is the geometric center of circle  $\Pi PDA$ ,  $S$  the fixed position of the Sun, and  $Q$  the location of the so-called *equant*. The radius  $CP$  of circle  $\Pi PDA$  is the effective major radius,  $a$ , of the orbit. The distances  $SC$  and  $CQ$  are both equal to  $e a$ , where  $e$  is the orbit's effective eccentricity. The angle  $PQ\Pi$  is identified with the mean anomaly,  $M$ , and increases linearly in time. In other words, as seen from  $Q$ , the planet  $P$  moves uniformly around circle  $\Pi PDA$  in a counterclockwise direction. Finally,  $SP$  is the radial distance,  $r$ , of the planet from the Sun, and angle  $PS\Pi$  is the planet's true anomaly,  $T$ .

Let us draw the straight-line  $KSL$  parallel to  $QP$ , and passing through point  $S$ , and then complete the rectangle  $PQKL$ . Simple geometry reveals that  $QK = PL = 2e a \sin M$ ,  $KS = 2e a \cos M$ , and  $SL = \rho - 2e a \cos M$ , where  $\rho = QP$ . The cosine rule applied to triangle  $CQP$  yields  $CP^2 = CQ^2 + QP^2 - 2CQQP \cos M$ , or  $\rho^2 - 2e a \cos M \rho - a^2(1 - e^2) = 0$ , which can be solved to give  $\rho/a = e \cos M + (1 - e^2 \sin^2 M)^{1/2}$ . Moreover, according to the theorem of Pythagoras,  $SP^2 = SL^2 + PL^2$ , which implies that

$$\frac{r}{a} = [1 - 2e \cos M (1 - e^2 \sin^2 M)^{1/2} + e^2 + 2e^2 \sin^2 M]^{1/2}. \quad (4.28)$$

Now,  $T = M + q$ , where  $q$  is angle  $PSL$ . However,

$$\sin q = \frac{PL}{SP} = \frac{2e \sin M}{[1 - 2e \cos M (1 - e^2 \sin^2 M)^{1/2} + e^2 + 2e^2 \sin^2 M]^{1/2}}. \quad (4.29)$$

Finally, expanding the previous two equations to second order in the small parameter  $e$ , we

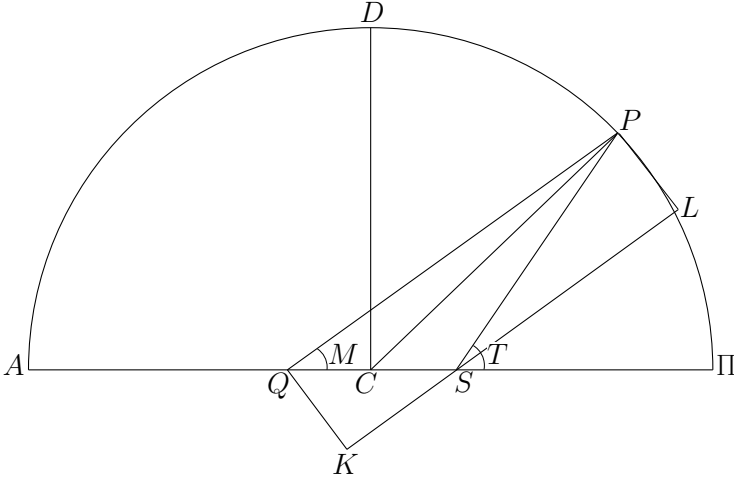


Figure 4.3: A Ptolemaic orbit.

obtain

$$\frac{r}{a} = 1 - e \cos M + (3/2) e^2 \sin^2 M, \quad (4.30)$$

$$T = M + 2 e \sin M + e^2 \sin 2M. \quad (4.31)$$

It can be seen, by comparison with Equations (4.22)–(4.23) and (4.26)–(4.27), that Ptolemy's geometric model of a heliocentric planetary orbit is significantly more accurate than Hipparchus's model, because the relative radial distance,  $r/a$ , and the true anomaly,  $T$ , in the former model both only deviate from those in the (correct) Keplerian model to second order in  $e$ .

#### 4.4 The model of Copernicus

Copernicus's geometric model of a heliocentric planetary orbit is illustrated in Figure 4.4. The planet  $P$  rotates on a circular epicycle  $YP$  whose center  $X$  moves around the Sun on the eccentric circle  $\Pi XDA$  (only half of which is shown). The diameter  $\Pi SCA$  is the effective major axis of the orbit, where  $C$  is the geometric center of circle  $\Pi XDA$ , and  $S$  the fixed position of the Sun. When  $X$  is at  $\Pi$  or  $A$  the planet is at its perihelion or aphelion points, respectively. The radius  $CX$  of circle  $\Pi XDA$  is the effective major radius,  $a$ , of the orbit. The distance  $SC$  is equal to  $(3/2) e a$ , where  $e$  is the orbit's effective eccentricity. Moreover, the radius  $XP$  of the epicycle is equal to  $(1/2) e a$ . The angle  $XC\Pi$  is identified with the mean anomaly,  $M$ , and increases linearly in time. In other words, as seen from  $C$ , the center of the epicycle  $X$  moves uniformly around circle  $\Pi XDA$  in a counterclockwise direction. The angle  $PXY$ , where  $Y$  is point at which  $CX$  produced meets the epicycle, is equal to the mean anomaly  $M$ . In other words, the planet  $P$  moves uniformly around the epicycle  $YP$ , in a counterclockwise direction, at twice the speed that point  $X$  moves around circle  $\Pi XDA$ . Finally,  $SP$  is

Let us draw the straight-line  $KSL$  parallel to  $CX$ , and passing through point  $S$ , and then complete the rectangle  $XCKL$ . Simple geometry reveals that  $CK = XL = (3/2)ea \sin M$ ,  $KS = (3/2)ea \cos M$ , and  $SL = a - (3/2)ea \cos M$ . Let  $PZ$  be drawn normal to  $XY$ , and let it meet  $KSL$  produced at point  $W$ . Simple geometry reveals that  $ZW = XL$ ,  $ZP = (1/2)ea \sin M$ , and  $XZ = LW = (1/2)ea \cos M$ . It follows that  $WP = ZW + ZP = XL + ZP = 2ea \sin M$ , and  $SW = SL + LW = SL + XZ = a - ea \cos M$ . Moreover, according to the theorem of Pythagoras,  $SP^2 = SW^2 + WP^2$ , which implies that

Now,  $T = M + q$ , where  $q$  is angle  $PSW$ . However,

Finally, expanding the previous two equations to second order in the small parameter  $e$ , we obtain

$$T = M + 2e \sin M + e^2 \sin 2M. \quad (4.35)$$

It can be seen, by comparison with Equations (4.22)–(4.23) and (4.30)–(4.31), that, as is the case for Ptolemy’s model, both the relative radial distance,  $r/a$ , and the true anomaly,  $T$ , in

Copernicus's geometric model of a heliocentric planetary orbit only deviate from those in the (correct) Keplerian model to second order in  $e$ . However, the deviation in the Ptolemaic model is slightly smaller than that in the Copernican model. To be more exact, the maximum deviation in  $r/a$  is  $(1/2) e^2$  in the former model, and  $e^2$  in the latter. On the other hand, the maximum deviation in  $T$  is  $(1/4) e^2$  in both models.



## 5. The Sun

### 5.1 Solar ecliptic longitude model

Our solar longitude model is sketched in Figure 5.1. From a geocentric point of view, the Sun,  $S$ , appears to execute a (counterclockwise) Keplerian orbit of major radius  $a$ , and eccentricity  $e$ , about the Earth,  $G$ . As has already been mentioned, the circle traced out by the Sun on the celestial sphere is known as the ecliptic circle. This circle is inclined at  $23^\circ 26'$  to the celestial equator, which is the projection of the Earth's equator onto the celestial sphere. Suppose that the angle subtended at the Earth between the vernal equinox (that is, the point at which the ecliptic crosses the celestial equator from south to north) and the Sun's perigee (that is, the point of closest approach to the Earth) is  $\varpi$ . This angle is termed the *longitude of the perigee*, and is assumed to vary linearly with time: that is,

$$\varpi = \varpi_0 + \varpi_1 (t - t_0). \quad (5.1)$$

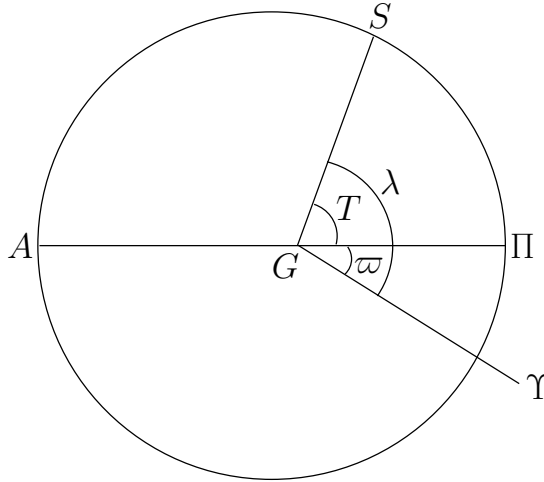


Figure 5.1: The apparent orbit of the Sun about the Earth. Here,  $S$ ,  $G$ ,  $\Pi$ ,  $A$ ,  $\varpi$ ,  $T$ ,  $\lambda$ , and  $\Upsilon$  represent the Sun, the Earth, the perigee, the apogee, the longitude of the perigee, the true anomaly, the ecliptic longitude, and The vernal equinox, respectively. the view is from northern ecliptic pole. The Sun orbits counterclockwise.

The Sun's *ecliptic longitude* is defined as the angle subtended at the Earth between the vernal equinox and the Sun. Hence, from Figure 5.1,

$$\lambda = \varpi + T, \quad (5.2)$$

where  $T$  is the true anomaly. (See Chapter 4.) By analogy, the *mean longitude* is written

$$\bar{\lambda} = \varpi + M, \quad (5.3)$$

where  $M$  is the mean anomaly. (See Chapter 4.) It follows from Equation (4.23) that

$$\lambda = \bar{\lambda} + q, \quad (5.4)$$

where

$$q = 2e \sin M + (5/4)e^2 \sin 2M, \quad (5.5)$$

is called the *equation of center*. Note that  $\lambda$ ,  $\bar{\lambda}$ ,  $T$ , and  $M$  are usually written as angles in the range  $0^\circ$  to  $360^\circ$ , whereas  $q$  is generally written as an angle in the range  $-180^\circ$  to  $+180^\circ$ .

The mean longitude increases uniformly with time (because both  $\varpi$  and  $M$  increase uniformly with time) as

$$\bar{\lambda} = \bar{\lambda}_0 + n(t - t_0), \quad (5.6)$$

where  $\bar{\lambda}_0$  is termed the *mean longitude at epoch*,  $n$  the *rate of motion in mean longitude*, and  $t_0$  the *epoch*. We can also write

$$M = M_0 + \tilde{n}(t - t_0), \quad (5.7)$$

where

$$M_0 = \bar{\lambda}_0 - \varpi_0 \quad (5.8)$$

is called the *mean anomaly at epoch*, and

$$\tilde{n} = n - \varpi_1 \quad (5.9)$$

the *rate of motion in mean anomaly*.

Our procedure for determining the ecliptic longitude of the Sun is as follows. The requisite orbital elements (that is,  $e$ ,  $n$ ,  $\tilde{n}$ ,  $\bar{\lambda}_0$ , and  $M_0$ ) for the J2000 epoch (that is, 12:00 UT on January 1, 2000 AD, which corresponds to  $t_0 = 2\,451\,545.0$  JD) are listed in Table 5.1. These elements are calculated on the assumption that the vernal equinox precesses at the uniform rate of  $-3.8246 \times 10^{-5}$  °/day. The ecliptic longitude of the Sun is specified by the following formulae:

$$\bar{\lambda} = \bar{\lambda}_0 + n(t - t_0), \quad (5.10)$$

$$M = M_0 + \tilde{n}(t - t_0), \quad (5.11)$$

$$q = 2e \sin M + (5/4)e^2 \sin 2M, \quad (5.12)$$

$$\lambda = \bar{\lambda} + q. \quad (5.13)$$

These formulae are capable of matching NASA ephemeris data<sup>1</sup> during the years 1995–2006 AD with a mean error of  $0.2'$  and a maximum error of  $0.7'$ .

The ecliptic longitude of the Sun can be calculated with the aid of Tables 5.3 and 5.4. Table 5.3 allows the mean longitude,  $\bar{\lambda}$ , and mean anomaly,  $M$ , of the Sun to be determined as functions of time. Table 5.4 specifies the equation of center,  $q$ , as a function of the mean anomaly.

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<sup>1</sup>See <http://ssd.jpl.nasa.gov/>

Note that Table 5.3 contains essentially the same information as that contained in the “Table of the Sun’s mean motion” (Κανόνιον τῆς ὁμαλῆς τοῦ ἡλίου κινήσεως) that appears in Section 2 of Book III of the Almagest. Likewise, Table 5.4 contains essentially the same information as that contained in the “Table of the solar anomaly” (Κανόνιον τῆς ἡλιακῆς ἀνωμαλίας) that appears in Section 6 of Book III of the Almagest.

## 5.2 Determination of solar ecliptic longitude

The procedure for using Tables 5.3 and 5.4 to determine solar longitude is as follows:

1. Determine the fractional Julian day number,  $t$ , corresponding to the date and time at which the Sun’s ecliptic longitude is to be calculated with the aid of Tables 3.1–3.3. Form  $\Delta t = t - t_0$ , where  $t_0 = 2\,451\,545.0$  is the epoch.
2. Enter Table 5.3 with the digit for each power of 10 in  $\Delta t$  and take out the corresponding values of  $\Delta\bar{\lambda}$  and  $\Delta M$ . If  $\Delta t$  is negative then the corresponding values are also negative. The value of the mean longitude,  $\bar{\lambda}$ , is the sum of all the  $\Delta\bar{\lambda}$  values plus the value of  $\bar{\lambda}$  at the epoch. Likewise, the value of the mean anomaly,  $M$ , is the sum of all the  $\Delta M$  values plus the value of  $M$  at the epoch. Add as many multiples of  $360^\circ$  to  $\bar{\lambda}$  and  $M$  as is required to make them both fall in the range  $0^\circ$  to  $360^\circ$ . Round  $M$  to the nearest degree.
3. Enter Table 5.4 with the value of  $M$  and take out the corresponding value of the equation of center,  $q$ , and the radial anomaly,  $\zeta$ . (The latter step is only necessary if the ecliptic longitude of the Sun is to be used to determine that of a planet.) It is necessary to interpolate if  $M$  is odd.
4. The ecliptic longitude,  $\lambda$ , is the sum of the mean longitude,  $\bar{\lambda}$ , and the equation of center,  $q$ . If necessary, convert  $\lambda$  into an angle in the range  $0^\circ$  to  $360^\circ$ . The decimal fraction can be converted into arc minutes using Table 5.2. Round to the nearest arc minute.

Two examples of the use of this procedure are given in the next section.

## 5.3 Example solar longitude calculations

*Example 1:* May 5, 2005 AD, 00:00 UT

According to Tables 3.1–3.3,  $t = 2\,453\,495.5$  JD. Hence,

$$t - t_0 = 2\,453\,495.5 - 2\,451\,545.0 = 1\,950.5 \text{ JD.}$$

Making use of Table 5.3, we find:

| $t(\text{JD})$ | $\bar{\lambda}(^{\circ})$ | $M(^{\circ})$  |
|----------------|---------------------------|----------------|
| +1 000         | 265.647                   | 265.600        |
| +900           | 167.083                   | 167.040        |
| +50            | 49.282                    | 49.280         |
| + .5           | 0.493                     | 0.493          |
| Epoch          | 280.458                   | 357.588        |
|                | <u>762.963</u>            | <u>840.001</u> |
| Modulus        | 42.963                    | 120.001        |

Rounding the mean anomaly to the nearest degree, we obtain  $M \simeq 120^{\circ}$ . It follows from Table 5.4 that

$$q(120^{\circ}) = 1.641^{\circ},$$

so

$$\lambda = \bar{\lambda} + q = 42.961 + 1.641 = 44.602 \simeq 44^{\circ}36'.$$

Here, we have converted the decimal fraction into arc minutes using Table 5.2, and then rounded the final result to the nearest arc minute.

Following the practice of the Ancient Greeks (and modern-day astrologers), we shall express ecliptic longitudes in terms of the signs of the zodiac, which are listed in Section 2.6. The ecliptic longitude  $44^{\circ}36'$  is conventionally written 14TA36; that is,  $14^{\circ}36'$  into the sign of Taurus. Thus, we conclude that the position of the Sun at 00:00 UT on May 5, 2005 AD was 14TA36.

*Example 2:* December 25, 1800 AD, 00:00 UT

According to Tables 3.1–3.3,  $t = 2\,378\,854.5$  JD. Hence,

$$t - t_0 = 2\,378\,854.5 - 2\,451\,545.0 = -72\,690.5 \text{ JD}.$$

Making use of Table 5.3, we find:

| $t(\text{JD})$ | $\bar{\lambda}(^{\circ})$ | $M(^{\circ})$   |
|----------------|---------------------------|-----------------|
| −70 000        | −235.315                  | −232.017        |
| −2 000         | −171.295                  | −171.200        |
| −600           | −231.388                  | −231.360        |
| −90            | −88.708                   | −88.704         |
| − .5           | −0.493                    | −0.493          |
| Epoch          | 280.458                   | 357.588         |
|                | <u>−446.741</u>           | <u>−366.186</u> |
| Modulus        | 273.259                   | 353.814         |

We conclude that  $M \simeq 354^\circ$ . From Table 5.4,

$$q(354^\circ) = -0.204^\circ,$$

so

$$\lambda = \bar{\lambda} + q = 273.259 - 0.204 = 273.055 \simeq 273^\circ 03'.$$

Thus, the position of the Sun at 00:00 UT on December 25, 1800 AD was 3CP03.

#### 5.4 Determination of equinox and solstice dates

We can also use Tables 5.3 and 5.4 to calculate the dates of the equinoxes and solstices, and, hence, the lengths of the seasons, in a given year. The *vernal equinox* (that is, the point on the Sun's apparent orbit at which it passes through the celestial equator from south to north) corresponds to  $\lambda = 0^\circ$ , the *summer solstice* (that is, the point at which the Sun is farthest north of the celestial equator) to  $\lambda = 90^\circ$ , the *autumnal equinox* (that is, the point at which the Sun passes through the celestial equator from north to south) to  $\lambda = 180^\circ$ , and the *winter solstice* (that is, the point at which the Sun is farthest south of the celestial equator) to  $\lambda = 270^\circ$ . See Figure 5.2. Furthermore, *spring* is defined as the period between the spring equinox and the summer solstice, *summer* as the period between the summer solstice and the autumnal equinox, *autumn* as the period between the autumnal equinox and the winter solstice, and *winter* as the period between the winter solstice and the following vernal equinox. Consider the year 2000 AD. For the case of the vernal equinox, we can first estimate the time at which this event takes place by approximating the solar longitude as the mean solar longitude; that is,

$$\lambda \simeq \bar{\lambda} = \bar{\lambda}_0 + n(t - t_0) = 280.458 + 0.98564735(t - t_0),$$

We obtain

$$t \simeq t_0 + (360 - 280.458)/0.98564735 \simeq t_0 + 81 \text{ JD}.$$

Calculating the true solar longitude at this time, using Tables 5.3 and 5.4, we get  $\lambda = 2.177^\circ$ . Now, the actual vernal equinox occurs when  $\lambda = 0^\circ$ . Thus, a much better estimate for the date of the vernal equinox is

$$t = t_0 + 81 - 2.177/0.98564735 \simeq t_0 + 78.8 \text{ JD},$$

which corresponds to 7:00 UT on March 20. Similar calculations show that the summer solstice takes place at

$$t = t_0 + 171.6 \text{ JD},$$

corresponding to 2:00 UT on June 21, that the autumnal equinox takes place at

$$t = t_0 + 265.2 \text{ JD},$$

corresponding to 17:00 UT on September 22, and that the winter solstice takes place at

$$t = t_0 + 355.1 \text{ JD},$$

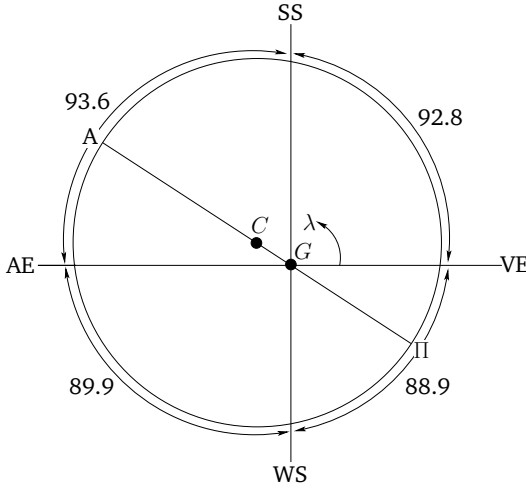


Figure 5.2: The Sun's apparent orbit around the Earth,  $G$ , showing the vernal equinox (VE), summer solstice (SS), autumnal equinox (AE), and winter solstice (WS). Here,  $\lambda$ ,  $\Pi$ ,  $A$ , and  $C$  are the ecliptic longitude, the perigee, the apogee, and the geometric center of the orbit, respectively. The lengths of the seasons (in days) are indicated.

corresponding to 14:00 UT on December 21. Thus, the length of spring is 92.8 days, the length of summer 93.6 days, and the length of autumn 89.9 days. Finally, the length of winter is the length of the tropical year (that is, the time period between successive vernal equinoxes), which is  $360/0.98564735 = 325.24$  days, minus the sum of the lengths of the other three seasons. This gives 88.9 days.

Figure 5.2 illustrates the relationship between the equinox and solstice points, and the lengths of the seasons. The Earth is displaced from the geometric center of the Sun's apparent orbit in the direction of the solar perigee, which presently lies between the winter solstice and the vernal equinox. This displacement (which is greatly exaggerated in the figure) has two effects. Firstly, it causes the arc of the Sun's apparent orbit between the summer solstice and autumnal equinox to be longer than that between the winter solstice and the vernal equinox. Secondly, it causes the Sun to appear to move faster in winter than in summer, in accordance with Kepler's second law, because the Sun is closer to the Earth in the former season. Both of these effects tend to lengthen summer, and shorten winter. Hence, summer is presently the longest season, and winter the shortest.

In Section 4 of Book III of the *Almagest*, Ptolemy effectively performs the calculation described in this section in reverse, in order to determine the eccentricity of the Sun's apparent orbit about the Earth, as well as the location of the solar perigee, from the observed differences in the lengths of the seasons. However, Ptolemy's figures for the lengths of spring, summer, autumn, and winter, which he inherited from Hipparchus, were 94.5, 92.5, 88.125, and 90.125 days, respectively. The lengths of the seasons have changed since the time of Hipparchus because the solar perigee has rotated by about  $38^\circ$  (in the direction of the Sun's apparent motion)

over the last 2 200 years.

### 5.5 The equation of time

At any particular observation site on the Earth's surface, *local noon* is defined as the instant in time when the Sun culminates at the meridian. However, as a consequence of the inclination of the ecliptic to the celestial equator, as well as the uneven motion of the Sun around the ecliptic, the time interval between successive local noons, which is known as a *solar day*, is not constant, but varies throughout the year. Hence, if we were to define a second as  $1/86\,400$  of a solar day then the length of a second would also vary throughout the year, which is clearly undesirable. In order to avoid this problem, astronomers have invented a fictitious body called the *mean Sun*. The mean Sun travels around the celestial equator (from west to east) at a constant rate that is such that it completes one orbit every tropical year. Moreover, the mean Sun and the true Sun coincide at the spring equinox. *Local mean noon* at a particular observation site is defined as the instance in time when the mean Sun culminates at the meridian. Because the orbit of the mean Sun is not inclined to the celestial equator, and the mean Sun travels around the celestial equator at a uniform rate, the time interval between successive mean noons, which is known as a *mean solar day*, takes the constant value of 24 hours, or 86 400 seconds, throughout the year. *Universal time* (UT) is defined such that 12:00 UT coincides with mean noon every day at an observation site of terrestrial longitude  $0^\circ$ . If we define *local time* (LT) as  $LT = UT - \phi(^{\circ})/15^{\circ}$  hrs., where  $\phi$  is the terrestrial longitude of the observation site, then 12:00 LT coincides with mean noon every day at a general observation site on the Earth's surface.

According to the previous definition, the right ascension,  $\bar{\alpha}$ , of the mean Sun satisfies

$$\bar{\alpha} = \bar{\lambda}, \quad (5.14)$$

where  $\bar{\lambda}$  is the Sun's mean ecliptic longitude. Moreover, it follows from Equations (2.16) and (5.4) that the right ascension of the true Sun is given by

$$\tan \alpha = \cos \epsilon \tan(\bar{\lambda} + q), \quad (5.15)$$

where  $\epsilon$  is the inclination of the ecliptic to the celestial equator,  $q(M)$  the Sun's equation of center, and  $M$  its mean anomaly. Now, neglecting the small time variation of the longitude of the Sun's perigee [that is, setting  $\varpi_1 = 0$  in Equation (5.1)], we can write [see Equations (5.6), (5.7), and (5.9), as well as Table 5.1]

$$M = \bar{\lambda} + M_0 - \bar{\lambda}_0 = \bar{\lambda} + 77.213^\circ. \quad (5.16)$$

It follows that, to first order in the solar eccentricity,  $e$ , we have

$$\Delta\alpha = \bar{\alpha} - \alpha = \lambda - \tan^{-1}(\cos \epsilon \tan \lambda) - 2e \sin M, \quad (5.17)$$

where

$$M = \lambda + 77.213^\circ. \quad (5.18)$$

Now,

$$\Delta t = \Delta\alpha(^{\circ})/15^{\circ} \quad (5.19)$$

represents the time difference (in hours) between local noon and mean local noon (because right ascension crosses the meridian at the uniform rate of  $15^\circ$  an hour), and is known as the *equation of time*. If  $\Delta t$  is positive then local noon occurs before mean local noon, and vice versa.

The equation of time specifies the difference between time calculated using a sundial or sextant—which is known as *solar time*—and time obtained from an accurate clock—which is known as *mean solar time*. Table 5.5 shows the equation of time as a function of the Sun's ecliptic longitude. It can be seen that the difference between solar time and mean solar time can be as much as 16 minutes, and attains its maximum value between the autumnal equinox and the winter solstice, and its minimum value between the winter solstice and vernal equinox.

The equation of time is discussed in Section 9 of Book III of the *Almagest*.

### 5.6 Solar distance model

The distance,  $r$ , of the Sun from the Earth varies slightly over the course of a year because the Earth is not quite at the geometric center of the Sun's apparent orbit. It follows that the apparent angular size of the solar disk in the Earth's sky also exhibits an annual variation. We need to understand this variation in order to accurately predict lunar and solar eclipses. According to Equation (4.22),

$$\frac{r}{a} = 1 - \zeta \quad (5.20)$$

where

$$\zeta = e \cos M - e^2 \sin^2 M. \quad (5.21)$$

Here,  $a = 1.498 \times 10^8$  km is the Sun's (apparent) orbital major radius, and  $\zeta$  is known as a *radial anomaly*. The solar radial anomaly is tabulated as a function of its argument ( $M$ ) in Table 5.4.

The apparent radius of the solar disk is

$$\rho_S = \frac{R}{r}, \quad (5.22)$$

where  $R = 6.957 \times 10^5$  km is the mean physical radius of the Sun. Here, use has been made of the small angle approximation. It follows from Equations (5.20) and (5.22) that

$$\rho_S = \frac{15.987'}{1 - \zeta}. \quad (5.23)$$

Clearly, the mean diameter of the solar disk is about half a degree.

As an example of a solar apparent radius calculation, we have already seen that at 00:00 UT on May 5, 2005 AD the mean anomaly of the Sun was  $M \simeq 120^\circ$ . It follows from Table 5.4 that the solar radial anomaly was  $\zeta = -0.856 \times 10^{-2}$ . Hence, the apparent radius of the solar disk was

$$\rho_S = \frac{15.987}{1 + 0.856 \times 10^{-2}} \simeq 15.85'. \quad (5.24)$$

As a second example of a solar radius calculation, we have also seen that at 00:00 UT on December 25, 1800 AD the mean anomaly of the Sun was  $M \simeq 354^\circ$ . It follows from Table 5.4



that the solar radial anomaly was  $\zeta = 1.662 \times 10^{-2}$ . Hence, the apparent radius of the solar disk was

$$\rho_S = \frac{15.987}{1 - 1.662 \times 10^{-2}} \simeq 16.26'. \quad (5.25)$$

### 5.7 Tables

| Object  | $a$ (AU) | $e$      | $n$ ( $^\circ$ /day) | $\tilde{n}$ ( $^\circ$ /day) | $\bar{\lambda}_0$ ( $^\circ$ ) | $M_0$ ( $^\circ$ ) |
|---------|----------|----------|----------------------|------------------------------|--------------------------------|--------------------|
| Mercury | 0.387098 | 0.205636 | 4.09237703           | 4.09233439                   | 252.087                        | 174.693            |
| Venus   | 0.723334 | 0.006777 | 1.60216872           | 1.60213040                   | 181.973                        | 49.237             |
| Sun     | 1.000000 | 0.016711 | 0.98564735           | 0.98560025                   | 280.458                        | 357.588            |
| Mars    | 1.523706 | 0.093394 | 0.52407118           | 0.52402076                   | 355.460                        | 19.388             |
| Jupiter | 5.202873 | 0.048386 | 0.08312507           | 0.08308100                   | 34.365                         | 19.348             |
| Saturn  | 9.536651 | 0.053862 | 0.03350830           | 0.03348152                   | 50.059                         | 317.857            |

Table 5.1: Keplerian orbital elements for the Sun and the five visible planets at the J2000 epoch (that is, 12:00 UT, January 1, 2000 AD, which corresponds to  $t_0 = 2\,451\,545.0$  JD). The elements are optimized for use in the time period 1800 AD to 2050 AD. Source: Jet Propulsion Laboratory (NASA), <http://ssd.jpl.nasa.gov/>. The motion rates have been converted into tropical motion rates assuming a uniform precession of the equinoxes of  $3.8246 \times 10^{-5}$   $^\circ$  per day. An astronomical unit (AU) is  $1.496 \times 10^8$  km.

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|       |      |       |      |       |      |       |      |       |      |       |      |
|-------|------|-------|------|-------|------|-------|------|-------|------|-------|------|
| 00.0' | .000 | 10.0' | .167 | 20.0' | .333 | 30.0' | .500 | 40.0' | .667 | 50.0' | .833 |
| 00.2' | .003 | 10.2' | .170 | 20.2' | .337 | 30.2' | .503 | 40.2' | .670 | 50.2' | .837 |
| 00.4' | .007 | 10.4' | .173 | 20.4' | .340 | 30.4' | .507 | 40.4' | .673 | 50.4' | .840 |
| 00.6' | .010 | 10.6' | .177 | 20.6' | .343 | 30.6' | .510 | 40.6' | .677 | 50.6' | .843 |
| 00.8' | .013 | 10.8' | .180 | 20.8' | .347 | 30.8' | .513 | 40.8' | .680 | 50.8' | .847 |
| 01.0' | .017 | 11.0' | .183 | 21.0' | .350 | 31.0' | .517 | 41.0' | .683 | 51.0' | .850 |
| 01.2' | .020 | 11.2' | .187 | 21.2' | .353 | 31.2' | .520 | 41.2' | .687 | 51.2' | .853 |
| 01.4' | .023 | 11.4' | .190 | 21.4' | .357 | 31.4' | .523 | 41.4' | .690 | 51.4' | .857 |
| 01.6' | .027 | 11.6' | .193 | 21.6' | .360 | 31.6' | .527 | 41.6' | .693 | 51.6' | .860 |
| 01.8' | .030 | 11.8' | .197 | 21.8' | .363 | 31.8' | .530 | 41.8' | .697 | 51.8' | .863 |
| 02.0' | .033 | 12.0' | .200 | 22.0' | .367 | 32.0' | .533 | 42.0' | .700 | 52.0' | .867 |
| 02.2' | .037 | 12.2' | .203 | 22.2' | .370 | 32.2' | .537 | 42.2' | .703 | 52.2' | .870 |
| 02.4' | .040 | 12.4' | .207 | 22.4' | .373 | 32.4' | .540 | 42.4' | .707 | 52.4' | .873 |
| 02.6' | .043 | 12.6' | .210 | 22.6' | .377 | 32.6' | .543 | 42.6' | .710 | 52.6' | .877 |
| 02.8' | .047 | 12.8' | .213 | 22.8' | .380 | 32.8' | .547 | 42.8' | .713 | 52.8' | .880 |
| 03.0' | .050 | 13.0' | .217 | 23.0' | .383 | 33.0' | .550 | 43.0' | .717 | 53.0' | .883 |
| 03.2' | .053 | 13.2' | .220 | 23.2' | .387 | 33.2' | .553 | 43.2' | .720 | 53.2' | .887 |
| 03.4' | .057 | 13.4' | .223 | 23.4' | .390 | 33.4' | .557 | 43.4' | .723 | 53.4' | .890 |
| 03.6' | .060 | 13.6' | .227 | 23.6' | .393 | 33.6' | .560 | 43.6' | .727 | 53.6' | .893 |
| 03.8' | .063 | 13.8' | .230 | 23.8' | .397 | 33.8' | .563 | 43.8' | .730 | 53.8' | .897 |
| 04.0' | .067 | 14.0' | .233 | 24.0' | .400 | 34.0' | .567 | 44.0' | .733 | 54.0' | .900 |
| 04.2' | .070 | 14.2' | .237 | 24.2' | .403 | 34.2' | .570 | 44.2' | .737 | 54.2' | .903 |
| 04.4' | .073 | 14.4' | .240 | 24.4' | .407 | 34.4' | .573 | 44.4' | .740 | 54.4' | .907 |
| 04.6' | .077 | 14.6' | .243 | 24.6' | .410 | 34.6' | .577 | 44.6' | .743 | 54.6' | .910 |
| 04.8' | .080 | 14.8' | .247 | 24.8' | .413 | 34.8' | .580 | 44.8' | .747 | 54.8' | .913 |

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|       |      |       |      |       |      |       |      |       |      |       |      |
|-------|------|-------|------|-------|------|-------|------|-------|------|-------|------|
| 05.0' | .083 | 15.0' | .250 | 25.0' | .417 | 35.0' | .583 | 45.0' | .750 | 55.0' | .917 |
| 05.2' | .087 | 15.2' | .253 | 25.2' | .420 | 35.2' | .587 | 45.2' | .753 | 55.2' | .920 |
| 05.4' | .090 | 15.4' | .257 | 25.4' | .423 | 35.4' | .590 | 45.4' | .757 | 55.4' | .923 |
| 05.6' | .093 | 15.6' | .260 | 25.6' | .427 | 35.6' | .593 | 45.6' | .760 | 55.6' | .927 |
| 05.8' | .097 | 15.8' | .263 | 25.8' | .430 | 35.8' | .597 | 45.8' | .763 | 55.8' | .930 |
| 06.0' | .100 | 16.0' | .267 | 26.0' | .433 | 36.0' | .600 | 46.0' | .767 | 56.0' | .933 |
| 06.2' | .103 | 16.2' | .270 | 26.2' | .437 | 36.2' | .603 | 46.2' | .770 | 56.2' | .937 |
| 06.4' | .107 | 16.4' | .273 | 26.4' | .440 | 36.4' | .607 | 46.4' | .773 | 56.4' | .940 |
| 06.6' | .110 | 16.6' | .277 | 26.6' | .443 | 36.6' | .610 | 46.6' | .777 | 56.6' | .943 |
| 06.8' | .113 | 16.8' | .280 | 26.8' | .447 | 36.8' | .613 | 46.8' | .780 | 56.8' | .947 |
| 07.0' | .117 | 17.0' | .283 | 27.0' | .450 | 37.0' | .617 | 47.0' | .783 | 57.0' | .950 |
| 07.2' | .120 | 17.2' | .287 | 27.2' | .453 | 37.2' | .620 | 47.2' | .787 | 57.2' | .953 |
| 07.4' | .123 | 17.4' | .290 | 27.4' | .457 | 37.4' | .623 | 47.4' | .790 | 57.4' | .957 |
| 07.6' | .127 | 17.6' | .293 | 27.6' | .460 | 37.6' | .627 | 47.6' | .793 | 57.6' | .960 |
| 07.8' | .130 | 17.8' | .297 | 27.8' | .463 | 37.8' | .630 | 47.8' | .797 | 57.8' | .963 |
| 08.0' | .133 | 18.0' | .300 | 28.0' | .467 | 38.0' | .633 | 48.0' | .800 | 58.0' | .967 |
| 08.2' | .137 | 18.2' | .303 | 28.2' | .470 | 38.2' | .637 | 48.2' | .803 | 58.2' | .970 |
| 08.4' | .140 | 18.4' | .307 | 28.4' | .473 | 38.4' | .640 | 48.4' | .807 | 58.4' | .973 |
| 08.6' | .143 | 18.6' | .310 | 28.6' | .477 | 38.6' | .643 | 48.6' | .810 | 58.6' | .977 |
| 08.8' | .147 | 18.8' | .313 | 28.8' | .480 | 38.8' | .647 | 48.8' | .813 | 58.8' | .980 |
| 09.0' | .150 | 19.0' | .317 | 29.0' | .483 | 39.0' | .650 | 49.0' | .817 | 59.0' | .983 |
| 09.2' | .153 | 19.2' | .320 | 29.2' | .487 | 39.2' | .653 | 49.2' | .820 | 59.2' | .987 |
| 09.4' | .157 | 19.4' | .323 | 29.4' | .490 | 39.4' | .657 | 49.4' | .823 | 59.4' | .990 |
| 09.6' | .160 | 19.6' | .327 | 29.6' | .493 | 39.6' | .660 | 49.6' | .827 | 59.6' | .993 |
| 09.8' | .163 | 19.8' | .330 | 29.8' | .497 | 39.8' | .663 | 49.8' | .830 | 59.8' | .997 |

Table 5.2: Arc minute to decimal fraction conversion table.

| $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(^{\circ})$ | $\Delta M(^{\circ})$ | $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(^{\circ})$ | $\Delta M(^{\circ})$ | $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(^{\circ})$ | $\Delta M(^{\circ})$ |
|-----------------------|----------------------------------|----------------------|-----------------------|----------------------------------|----------------------|-----------------------|----------------------------------|----------------------|
| 10 000                | 136.474                          | 136.002              | 1 000                 | 265.647                          | 265.600              | 100                   | 98.565                           | 98.560               |
| 20 000                | 272.947                          | 272.005              | 2 000                 | 171.295                          | 171.200              | 200                   | 197.129                          | 197.120              |
| 30 000                | 49.421                           | 48.007               | 3 000                 | 76.942                           | 76.801               | 300                   | 295.694                          | 295.680              |
| 40 000                | 185.894                          | 184.010              | 4 000                 | 342.589                          | 342.401              | 400                   | 34.259                           | 34.240               |
| 50 000                | 322.367                          | 320.012              | 5 000                 | 248.237                          | 248.001              | 500                   | 132.824                          | 132.800              |
| 60 000                | 98.841                           | 96.015               | 6 000                 | 153.884                          | 153.601              | 600                   | 231.388                          | 231.360              |
| 70 000                | 235.315                          | 232.017              | 7 000                 | 59.531                           | 59.202               | 700                   | 329.953                          | 329.920              |
| 80 000                | 11.788                           | 8.020                | 8 000                 | 325.179                          | 324.802              | 800                   | 68.518                           | 68.480               |
| 90 000                | 148.262                          | 144.022              | 9 000                 | 230.826                          | 230.402              | 900                   | 167.083                          | 167.040              |
| 10                    | 9.856                            | 9.856                | 1                     | 0.986                            | 0.986                | 0.1                   | 0.099                            | 0.099                |
| 20                    | 19.713                           | 19.712               | 2                     | 1.971                            | 1.971                | 0.2                   | 0.197                            | 0.197                |
| 30                    | 29.569                           | 29.568               | 3                     | 2.957                            | 2.957                | 0.3                   | 0.296                            | 0.296                |
| 40                    | 39.426                           | 39.424               | 4                     | 3.943                            | 3.942                | 0.4                   | 0.394                            | 0.394                |
| 50                    | 49.282                           | 49.280               | 5                     | 4.928                            | 4.928                | 0.5                   | 0.493                            | 0.493                |
| 60                    | 59.139                           | 59.136               | 6                     | 5.914                            | 5.914                | 0.6                   | 0.591                            | 0.591                |
| 70                    | 68.995                           | 68.992               | 7                     | 6.900                            | 6.899                | 0.7                   | 0.690                            | 0.690                |
| 80                    | 78.852                           | 78.848               | 8                     | 7.885                            | 7.885                | 0.8                   | 0.789                            | 0.788                |
| 90                    | 88.708                           | 88.704               | 9                     | 8.871                            | 8.870                | 0.9                   | 0.887                            | 0.887                |

Table 5.3: Mean motion of the Sun. Here,  $\Delta t = t - t_0$ ,  $\Delta \bar{\lambda} = \bar{\lambda} - \bar{\lambda}_0$ , and  $\Delta M = M - M_0$ . At epoch ( $t_0 = 2\,451\,545.0$  JD),  $\bar{\lambda}_0 = 280.458^{\circ}$ , and  $M_0 = 357.588^{\circ}$ .

| $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ |
|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|
| 0             | 0.000         | 1.671       | 90            | 1.915         | -0.028      | 180           | 0.000         | -1.671      | 270           | -1.915        | -0.028      |
| 2             | 0.068         | 1.670       | 92            | 1.912         | -0.086      | 182           | -0.065        | -1.670      | 272           | -1.915        | 0.030       |
| 4             | 0.136         | 1.667       | 94            | 1.907         | -0.144      | 184           | -0.131        | -1.667      | 274           | -1.913        | 0.089       |
| 6             | 0.204         | 1.662       | 96            | 1.900         | -0.202      | 186           | -0.196        | -1.662      | 276           | -1.909        | 0.147       |
| 8             | 0.272         | 1.654       | 98            | 1.891         | -0.260      | 188           | -0.261        | -1.655      | 278           | -1.902        | 0.205       |
| 10            | 0.339         | 1.645       | 100           | 1.879         | -0.317      | 190           | -0.326        | -1.647      | 280           | -1.893        | 0.263       |
| 12            | 0.406         | 1.633       | 102           | 1.865         | -0.374      | 192           | -0.390        | -1.636      | 282           | -1.881        | 0.321       |
| 14            | 0.473         | 1.620       | 104           | 1.849         | -0.431      | 194           | -0.454        | -1.623      | 284           | -1.867        | 0.378       |
| 16            | 0.538         | 1.604       | 106           | 1.830         | -0.486      | 196           | -0.517        | -1.608      | 286           | -1.851        | 0.435       |
| 18            | 0.604         | 1.587       | 108           | 1.809         | -0.542      | 198           | -0.580        | -1.592      | 288           | -1.833        | 0.491       |
| 20            | 0.668         | 1.567       | 110           | 1.787         | -0.596      | 200           | -0.642        | -1.574      | 290           | -1.812        | 0.547       |
| 22            | 0.731         | 1.545       | 112           | 1.762         | -0.650      | 202           | -0.703        | -1.553      | 292           | -1.789        | 0.602       |
| 24            | 0.794         | 1.522       | 114           | 1.735         | -0.703      | 204           | -0.764        | -1.531      | 294           | -1.764        | 0.656       |
| 26            | 0.855         | 1.497       | 116           | 1.705         | -0.755      | 206           | -0.824        | -1.507      | 296           | -1.737        | 0.710       |
| 28            | 0.916         | 1.469       | 118           | 1.674         | -0.806      | 208           | -0.882        | -1.482      | 298           | -1.707        | 0.763       |
| 30            | 0.975         | 1.440       | 120           | 1.641         | -0.856      | 210           | -0.940        | -1.454      | 300           | -1.676        | 0.815       |
| 32            | 1.033         | 1.409       | 122           | 1.606         | -0.906      | 212           | -0.997        | -1.425      | 302           | -1.642        | 0.865       |
| 34            | 1.089         | 1.377       | 124           | 1.569         | -0.954      | 214           | -1.052        | -1.394      | 304           | -1.606        | 0.915       |
| 36            | 1.145         | 1.342       | 126           | 1.530         | -1.001      | 216           | -1.107        | -1.362      | 306           | -1.568        | 0.964       |
| 38            | 1.198         | 1.306       | 128           | 1.490         | -1.046      | 218           | -1.160        | -1.327      | 308           | -1.528        | 1.011       |
| 40            | 1.251         | 1.269       | 130           | 1.447         | -1.091      | 220           | -1.211        | -1.292      | 310           | -1.487        | 1.058       |
| 42            | 1.301         | 1.229       | 132           | 1.403         | -1.134      | 222           | -1.261        | -1.254      | 312           | -1.443        | 1.103       |
| 44            | 1.350         | 1.189       | 134           | 1.358         | -1.175      | 224           | -1.310        | -1.216      | 314           | -1.397        | 1.146       |
| 46            | 1.397         | 1.146       | 136           | 1.310         | -1.216      | 226           | -1.358        | -1.175      | 316           | -1.350        | 1.189       |
| 48            | 1.443         | 1.103       | 138           | 1.261         | -1.254      | 228           | -1.403        | -1.134      | 318           | -1.301        | 1.229       |
| 50            | 1.487         | 1.058       | 140           | 1.211         | -1.292      | 230           | -1.447        | -1.091      | 320           | -1.251        | 1.269       |
| 52            | 1.528         | 1.011       | 142           | 1.160         | -1.327      | 232           | -1.490        | -1.046      | 322           | -1.198        | 1.306       |
| 54            | 1.568         | 0.964       | 144           | 1.107         | -1.362      | 234           | -1.530        | -1.001      | 324           | -1.145        | 1.342       |
| 56            | 1.606         | 0.915       | 146           | 1.052         | -1.394      | 236           | -1.569        | -0.954      | 326           | -1.089        | 1.377       |
| 58            | 1.642         | 0.865       | 148           | 0.997         | -1.425      | 238           | -1.606        | -0.906      | 328           | -1.033        | 1.409       |
| 60            | 1.676         | 0.815       | 150           | 0.940         | -1.454      | 240           | -1.641        | -0.856      | 330           | -0.975        | 1.440       |
| 62            | 1.707         | 0.763       | 152           | 0.882         | -1.482      | 242           | -1.674        | -0.806      | 332           | -0.916        | 1.469       |
| 64            | 1.737         | 0.710       | 154           | 0.824         | -1.507      | 244           | -1.705        | -0.755      | 334           | -0.855        | 1.497       |
| 66            | 1.764         | 0.656       | 156           | 0.764         | -1.531      | 246           | -1.735        | -0.703      | 336           | -0.794        | 1.522       |
| 68            | 1.789         | 0.602       | 158           | 0.703         | -1.553      | 248           | -1.762        | -0.650      | 338           | -0.731        | 1.545       |
| 70            | 1.812         | 0.547       | 160           | 0.642         | -1.574      | 250           | -1.787        | -0.596      | 340           | -0.668        | 1.567       |
| 72            | 1.833         | 0.491       | 162           | 0.580         | -1.592      | 252           | -1.809        | -0.542      | 342           | -0.604        | 1.587       |
| 74            | 1.851         | 0.435       | 164           | 0.517         | -1.608      | 254           | -1.830        | -0.486      | 344           | -0.538        | 1.604       |
| 76            | 1.867         | 0.378       | 166           | 0.454         | -1.623      | 256           | -1.849        | -0.431      | 346           | -0.473        | 1.620       |
| 78            | 1.881         | 0.321       | 168           | 0.390         | -1.636      | 258           | -1.865        | -0.374      | 348           | -0.406        | 1.633       |
| 80            | 1.893         | 0.263       | 170           | 0.326         | -1.647      | 260           | -1.879        | -0.317      | 350           | -0.339        | 1.645       |
| 82            | 1.902         | 0.205       | 172           | 0.261         | -1.655      | 262           | -1.891        | -0.260      | 352           | -0.272        | 1.654       |
| 84            | 1.909         | 0.147       | 174           | 0.196         | -1.662      | 264           | -1.900        | -0.202      | 354           | -0.204        | 1.662       |
| 86            | 1.913         | 0.089       | 176           | 0.131         | -1.667      | 266           | -1.907        | -0.144      | 356           | -0.136        | 1.667       |
| 88            | 1.915         | 0.030       | 178           | 0.065         | -1.670      | 268           | -1.912        | -0.086      | 358           | -0.068        | 1.670       |
| 90            | 1.915         | -0.028      | 180           | 0.000         | -1.671      | 270           | -1.915        | -0.028      | 360           | -0.000        | 1.671       |

Table 5.4: Anomalies of the Sun.

| Aries     |                                  | Taurus    |                                  | Gemini      |                                  | Cancer    |                                  | Leo       |                                  | Virgo     |                                  |
|-----------|----------------------------------|-----------|----------------------------------|-------------|----------------------------------|-----------|----------------------------------|-----------|----------------------------------|-----------|----------------------------------|
| $\lambda$ | $\Delta t$                       | $\lambda$ | $\Delta t$                       | $\lambda$   | $\Delta t$                       | $\lambda$ | $\Delta t$                       | $\lambda$ | $\Delta t$                       | $\lambda$ | $\Delta t$                       |
| 00°       | -07 <sup>m</sup> 28 <sup>s</sup> | 00°       | +01 <sup>m</sup> 02 <sup>s</sup> | 00°         | +03 <sup>m</sup> 31 <sup>s</sup> | 00°       | -01 <sup>m</sup> 42 <sup>s</sup> | 00°       | -06 <sup>m</sup> 27 <sup>s</sup> | 00°       | -02 <sup>m</sup> 44 <sup>s</sup> |
| 02°       | -06 <sup>m</sup> 52 <sup>s</sup> | 02°       | +01 <sup>m</sup> 28 <sup>s</sup> | 02°         | +03 <sup>m</sup> 22 <sup>s</sup> | 02°       | -02 <sup>m</sup> 09 <sup>s</sup> | 02°       | -06 <sup>m</sup> 30 <sup>s</sup> | 02°       | -02 <sup>m</sup> 11 <sup>s</sup> |
| 04°       | -06 <sup>m</sup> 15 <sup>s</sup> | 04°       | +01 <sup>m</sup> 51 <sup>s</sup> | 04°         | +03 <sup>m</sup> 11 <sup>s</sup> | 04°       | -02 <sup>m</sup> 36 <sup>s</sup> | 04°       | -06 <sup>m</sup> 31 <sup>s</sup> | 04°       | -01 <sup>m</sup> 36 <sup>s</sup> |
| 06°       | -05 <sup>m</sup> 38 <sup>s</sup> | 06°       | +02 <sup>m</sup> 12 <sup>s</sup> | 06°         | +02 <sup>m</sup> 57 <sup>s</sup> | 06°       | -03 <sup>m</sup> 03 <sup>s</sup> | 06°       | -06 <sup>m</sup> 29 <sup>s</sup> | 06°       | +00 <sup>m</sup> 59 <sup>s</sup> |
| 08°       | -05 <sup>m</sup> 01 <sup>s</sup> | 08°       | +02 <sup>m</sup> 32 <sup>s</sup> | 08°         | +02 <sup>m</sup> 42 <sup>s</sup> | 08°       | -03 <sup>m</sup> 28 <sup>s</sup> | 08°       | -06 <sup>m</sup> 24 <sup>s</sup> | 08°       | +00 <sup>m</sup> 21 <sup>s</sup> |
| 10°       | -04 <sup>m</sup> 25 <sup>s</sup> | 10°       | +02 <sup>m</sup> 49 <sup>s</sup> | 10°         | +02 <sup>m</sup> 24 <sup>s</sup> | 10°       | -03 <sup>m</sup> 53 <sup>s</sup> | 10°       | -06 <sup>m</sup> 16 <sup>s</sup> | 10°       | +00 <sup>m</sup> 18 <sup>s</sup> |
| 12°       | -03 <sup>m</sup> 48 <sup>s</sup> | 12°       | +03 <sup>m</sup> 04 <sup>s</sup> | 12°         | +02 <sup>m</sup> 05 <sup>s</sup> | 12°       | -04 <sup>m</sup> 17 <sup>s</sup> | 12°       | -06 <sup>m</sup> 06 <sup>s</sup> | 12°       | +00 <sup>m</sup> 59 <sup>s</sup> |
| 14°       | -03 <sup>m</sup> 12 <sup>s</sup> | 14°       | +03 <sup>m</sup> 17 <sup>s</sup> | 14°         | +01 <sup>m</sup> 44 <sup>s</sup> | 14°       | -04 <sup>m</sup> 39 <sup>s</sup> | 14°       | -05 <sup>m</sup> 53 <sup>s</sup> | 14°       | +01 <sup>m</sup> 40 <sup>s</sup> |
| 16°       | -02 <sup>m</sup> 37 <sup>s</sup> | 16°       | +03 <sup>m</sup> 27 <sup>s</sup> | 16°         | +01 <sup>m</sup> 21 <sup>s</sup> | 16°       | -04 <sup>m</sup> 59 <sup>s</sup> | 16°       | -05 <sup>m</sup> 38 <sup>s</sup> | 16°       | +02 <sup>m</sup> 23 <sup>s</sup> |
| 18°       | -02 <sup>m</sup> 02 <sup>s</sup> | 18°       | +03 <sup>m</sup> 35 <sup>s</sup> | 18°         | +00 <sup>m</sup> 58 <sup>s</sup> | 18°       | -05 <sup>m</sup> 18 <sup>s</sup> | 18°       | -05 <sup>m</sup> 20 <sup>s</sup> | 18°       | +03 <sup>m</sup> 06 <sup>s</sup> |
| 20°       | -01 <sup>m</sup> 28 <sup>s</sup> | 20°       | +03 <sup>m</sup> 40 <sup>s</sup> | 20°         | +00 <sup>m</sup> 33 <sup>s</sup> | 20°       | -05 <sup>m</sup> 35 <sup>s</sup> | 20°       | -05 <sup>m</sup> 00 <sup>s</sup> | 20°       | +03 <sup>m</sup> 49 <sup>s</sup> |
| 22°       | +00 <sup>m</sup> 55 <sup>s</sup> | 22°       | +03 <sup>m</sup> 43 <sup>s</sup> | 22°         | +00 <sup>m</sup> 07 <sup>s</sup> | 22°       | -05 <sup>m</sup> 50 <sup>s</sup> | 22°       | -04 <sup>m</sup> 37 <sup>s</sup> | 22°       | +04 <sup>m</sup> 33 <sup>s</sup> |
| 24°       | +00 <sup>m</sup> 24 <sup>s</sup> | 24°       | +03 <sup>m</sup> 44 <sup>s</sup> | 24°         | +00 <sup>m</sup> 20 <sup>s</sup> | 24°       | -06 <sup>m</sup> 03 <sup>s</sup> | 24°       | -04 <sup>m</sup> 12 <sup>s</sup> | 24°       | +05 <sup>m</sup> 17 <sup>s</sup> |
| 26°       | +00 <sup>m</sup> 06 <sup>s</sup> | 26°       | +03 <sup>m</sup> 42 <sup>s</sup> | 26°         | +00 <sup>m</sup> 47 <sup>s</sup> | 26°       | -06 <sup>m</sup> 14 <sup>s</sup> | 26°       | -03 <sup>m</sup> 45 <sup>s</sup> | 26°       | +06 <sup>m</sup> 01 <sup>s</sup> |
| 28°       | +00 <sup>m</sup> 35 <sup>s</sup> | 28°       | +03 <sup>m</sup> 38 <sup>s</sup> | 28°         | -01 <sup>m</sup> 14 <sup>s</sup> | 28°       | -06 <sup>m</sup> 22 <sup>s</sup> | 28°       | -03 <sup>m</sup> 15 <sup>s</sup> | 28°       | +06 <sup>m</sup> 45 <sup>s</sup> |
| 30°       | +01 <sup>m</sup> 02 <sup>s</sup> | 30°       | +03 <sup>m</sup> 31 <sup>s</sup> | 30°         | -01 <sup>m</sup> 42 <sup>s</sup> | 30°       | -06 <sup>m</sup> 27 <sup>s</sup> | 30°       | -02 <sup>m</sup> 44 <sup>s</sup> | 30°       | +07 <sup>m</sup> 28 <sup>s</sup> |
| Libra     |                                  | Scorpio   |                                  | Sagittarius |                                  | Capricorn |                                  | Aquarius  |                                  | Pisces    |                                  |
| $\lambda$ | $\Delta t$                       | $\lambda$ | $\Delta t$                       | $\lambda$   | $\Delta t$                       | $\lambda$ | $\Delta t$                       | $\lambda$ | $\Delta t$                       | $\lambda$ | $\Delta t$                       |
| 00°       | +07 <sup>m</sup> 28 <sup>s</sup> | 00°       | +15 <sup>m</sup> 40 <sup>s</sup> | 00°         | +13 <sup>m</sup> 55 <sup>s</sup> | 00°       | +01 <sup>m</sup> 42 <sup>s</sup> | 00°       | -10 <sup>m</sup> 58 <sup>s</sup> | 00°       | -13 <sup>m</sup> 58 <sup>s</sup> |
| 02°       | +08 <sup>m</sup> 11 <sup>s</sup> | 02°       | +15 <sup>m</sup> 55 <sup>s</sup> | 02°         | +13 <sup>m</sup> 22 <sup>s</sup> | 02°       | +00 <sup>m</sup> 43 <sup>s</sup> | 02°       | -11 <sup>m</sup> 33 <sup>s</sup> | 02°       | -13 <sup>m</sup> 47 <sup>s</sup> |
| 04°       | +08 <sup>m</sup> 53 <sup>s</sup> | 04°       | +16 <sup>m</sup> 08 <sup>s</sup> | 04°         | +12 <sup>m</sup> 46 <sup>s</sup> | 04°       | +00 <sup>m</sup> 16 <sup>s</sup> | 04°       | -12 <sup>m</sup> 03 <sup>s</sup> | 04°       | -13 <sup>m</sup> 32 <sup>s</sup> |
| 06°       | +09 <sup>m</sup> 34 <sup>s</sup> | 06°       | +16 <sup>m</sup> 17 <sup>s</sup> | 06°         | +12 <sup>m</sup> 08 <sup>s</sup> | 06°       | -01 <sup>m</sup> 14 <sup>s</sup> | 06°       | -12 <sup>m</sup> 31 <sup>s</sup> | 06°       | -13 <sup>m</sup> 15 <sup>s</sup> |
| 08°       | +10 <sup>m</sup> 15 <sup>s</sup> | 08°       | +16 <sup>m</sup> 23 <sup>s</sup> | 08°         | +11 <sup>m</sup> 26 <sup>s</sup> | 08°       | -02 <sup>m</sup> 12 <sup>s</sup> | 08°       | -12 <sup>m</sup> 55 <sup>s</sup> | 08°       | -12 <sup>m</sup> 56 <sup>s</sup> |
| 10°       | +10 <sup>m</sup> 53 <sup>s</sup> | 10°       | +16 <sup>m</sup> 27 <sup>s</sup> | 10°         | +10 <sup>m</sup> 42 <sup>s</sup> | 10°       | -03 <sup>m</sup> 08 <sup>s</sup> | 10°       | -13 <sup>m</sup> 17 <sup>s</sup> | 10°       | -12 <sup>m</sup> 34 <sup>s</sup> |
| 12°       | +11 <sup>m</sup> 31 <sup>s</sup> | 12°       | +16 <sup>m</sup> 26 <sup>s</sup> | 12°         | +09 <sup>m</sup> 55 <sup>s</sup> | 12°       | -04 <sup>m</sup> 04 <sup>s</sup> | 12°       | -13 <sup>m</sup> 35 <sup>s</sup> | 12°       | -12 <sup>m</sup> 11 <sup>s</sup> |
| 14°       | +12 <sup>m</sup> 07 <sup>s</sup> | 14°       | +16 <sup>m</sup> 23 <sup>s</sup> | 14°         | +09 <sup>m</sup> 07 <sup>s</sup> | 14°       | -04 <sup>m</sup> 58 <sup>s</sup> | 14°       | -13 <sup>m</sup> 50 <sup>s</sup> | 14°       | -11 <sup>m</sup> 45 <sup>s</sup> |
| 16°       | +12 <sup>m</sup> 41 <sup>s</sup> | 16°       | +16 <sup>m</sup> 16 <sup>s</sup> | 16°         | +08 <sup>m</sup> 16 <sup>s</sup> | 16°       | -05 <sup>m</sup> 51 <sup>s</sup> | 16°       | -14 <sup>m</sup> 02 <sup>s</sup> | 16°       | -11 <sup>m</sup> 18 <sup>s</sup> |
| 18°       | +13 <sup>m</sup> 14 <sup>s</sup> | 18°       | +16 <sup>m</sup> 06 <sup>s</sup> | 18°         | +07 <sup>m</sup> 23 <sup>s</sup> | 18°       | -06 <sup>m</sup> 42 <sup>s</sup> | 18°       | -14 <sup>m</sup> 10 <sup>s</sup> | 18°       | -10 <sup>m</sup> 49 <sup>s</sup> |
| 20°       | +13 <sup>m</sup> 44 <sup>s</sup> | 20°       | +15 <sup>m</sup> 53 <sup>s</sup> | 20°         | +06 <sup>m</sup> 29 <sup>s</sup> | 20°       | -07 <sup>m</sup> 31 <sup>s</sup> | 20°       | -14 <sup>m</sup> 16 <sup>s</sup> | 20°       | -10 <sup>m</sup> 18 <sup>s</sup> |
| 22°       | +14 <sup>m</sup> 12 <sup>s</sup> | 22°       | +15 <sup>m</sup> 36 <sup>s</sup> | 22°         | +05 <sup>m</sup> 33 <sup>s</sup> | 22°       | -08 <sup>m</sup> 17 <sup>s</sup> | 22°       | -14 <sup>m</sup> 18 <sup>s</sup> | 22°       | -09 <sup>m</sup> 46 <sup>s</sup> |
| 24°       | +14 <sup>m</sup> 38 <sup>s</sup> | 24°       | +15 <sup>m</sup> 15 <sup>s</sup> | 24°         | +04 <sup>m</sup> 36 <sup>s</sup> | 24°       | -09 <sup>m</sup> 02 <sup>s</sup> | 24°       | -14 <sup>m</sup> 18 <sup>s</sup> | 24°       | -09 <sup>m</sup> 13 <sup>s</sup> |
| 26°       | +15 <sup>m</sup> 01 <sup>s</sup> | 26°       | +14 <sup>m</sup> 52 <sup>s</sup> | 26°         | +03 <sup>m</sup> 39 <sup>s</sup> | 26°       | -09 <sup>m</sup> 44 <sup>s</sup> | 26°       | -14 <sup>m</sup> 14 <sup>s</sup> | 26°       | -08 <sup>m</sup> 39 <sup>s</sup> |
| 28°       | +15 <sup>m</sup> 22 <sup>s</sup> | 28°       | +14 <sup>m</sup> 25 <sup>s</sup> | 28°         | +02 <sup>m</sup> 40 <sup>s</sup> | 28°       | -10 <sup>m</sup> 23 <sup>s</sup> | 28°       | -14 <sup>m</sup> 08 <sup>s</sup> | 28°       | -08 <sup>m</sup> 04 <sup>s</sup> |
| 30°       | +15 <sup>m</sup> 40 <sup>s</sup> | 30°       | +13 <sup>m</sup> 55 <sup>s</sup> | 30°         | +01 <sup>m</sup> 42 <sup>s</sup> | 30°       | -10 <sup>m</sup> 58 <sup>s</sup> | 30°       | -13 <sup>m</sup> 58 <sup>s</sup> | 30°       | -07 <sup>m</sup> 28 <sup>s</sup> |

Table 5.5: The equation of time. The superscripts  $m$  and  $s$  denote minutes and seconds.

## 6. The Moon

### 6.1 Lunar ecliptic longitude model

The orbit of the Moon around the Earth is strongly perturbed by the gravitational influence of the Sun. It follows that we cannot derive an accurate lunar longitude model from Keplerian orbit theory alone. Instead, we shall employ a greatly simplified version of modern lunar theory. According to such theory, the time variation of the ecliptic longitude of the Moon is fairly well represented by the following formulae:<sup>1</sup>

$$\bar{\lambda} = \bar{\lambda}_0 + n(t - t_0), \quad (6.1)$$

$$M = M_0 + \tilde{n}(t - t_0), \quad (6.2)$$

$$\bar{F} = \bar{F}_0 + \tilde{n}(t - t_0), \quad (6.3)$$

$$\tilde{D} = \bar{\lambda} - \lambda_S, \quad (6.4)$$

$$q_1 = 2e \sin M + 1.2379 e^2 \sin 2M, \quad (6.5)$$

$$q_2 = 0.4052 e \sin(2\tilde{D} - M), \quad (6.6)$$

$$q_3 = 0.2094 e (\sin 2\tilde{D} - 0.0527 \sin \tilde{D}), \quad (6.7)$$

$$q_4 = -0.0589 e \sin M_S, \quad (6.8)$$

$$q_5 = -0.0364 e \sin 2\bar{F}, \quad (6.9)$$

$$\lambda = \bar{\lambda} + q_1 + q_2 + q_3 + q_4 + q_5. \quad (6.10)$$

Here,  $\lambda_S$  and  $M_S$  are the longitude and mean anomaly of the Sun, respectively. Moreover,  $e$ ,  $\lambda$ ,  $\bar{\lambda}$ ,  $\bar{F}$ , and  $q_i$  are the eccentricity, longitude, mean longitude, mean argument of latitude, and  $i$ th anomaly of the Moon, respectively. The Moon's first anomaly is due to the eccentricity of its orbit, and is very similar in form to that obtained from Keplerian orbit theory. (See Chapter 4.) The Moon's second, third, and fourth anomalies are known as *evection*, *variation*, and the *annual inequality*, respectively, and originate from the perturbing influence of the Sun. Finally, the Moon's fifth anomaly is called the *reduction to the ecliptic*, and is a consequence of the fact that the Moon's orbit is slightly tilted with respect to the plane of the ecliptic. Note that Ptolemy's lunar theory only takes the first two lunar anomalies into account. (Evection was discovered by Ptolemy. However, variation was only discovered in the 1580s by Tycho Brahe, who also discovered the annual inequality about a decade later.)

The Moon's orbital elements— $e$ ,  $n$ ,  $\tilde{n}$ ,  $\bar{\lambda}_0$ ,  $M_0$ , and  $F_0$ —for the J2000 epoch are listed in Table 6.1. Note that the lunar perigee precesses in the direction of the Moon's orbital motion at the rate of  $n - \tilde{n} = 0.11140^\circ$  per day, or  $360^\circ$  in 8.85 years. This very large precession rate (more than 2000 times the corresponding precession rate for the Sun's apparent orbit) is another consequence of the strong perturbing influence of the Sun on the Moon's orbit. The

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<sup>1</sup> See Meeus (1998) and Fitzpatrick (2012).

previous formulae are capable of matching NASA ephemeris data during the years 1995–2006 AD with a mean error of  $5'$  and a maximum error of  $14'$ .

The ecliptic longitude of the Moon can be calculated with the aid of Tables 6.2 and 6.3. Table 6.2 allows the lunar mean longitude,  $\bar{\lambda}$ , the mean anomaly,  $M$ , and the mean argument of latitude,  $\bar{F}$ , to be determined as functions of time. Table 6.3 specifies the lunar anomalies,  $q_1$ – $q_5$ , as functions of their various arguments.

Incidentally, Table 6.2 contains essentially the same information as is contained in the “Table of the Moon’s mean motion” (Κανόνες τῶν τῆς σελήνης μέσων κινήσεων) that appears in Section 4 of Book IV of the *Almagest*. Moreover, Table 6.3 contains equivalent information to that contained in the “Table of the whole lunar anomaly” (Κανόνιον τῆς καθόλου σεληνακῆς ἀνωμαλίας) that appears in Section 8 of Book V of the *Almagest*.

## 6.2 Determination of lunar ecliptic longitude

The procedure for using Tables 6.1 and 6.3 to determine lunar longitude is as follows:

1. Determine the fractional Julian day number,  $t$ , corresponding to the date and time at which the Moon’s ecliptic longitude is to be calculated with the aid of Tables 3.1–3.3. Form  $\Delta t = t - t_0$ , where  $t_0 = 2\,451\,545.0$  is the epoch.
2. Calculate the ecliptic longitude,  $\lambda_S$ , and the mean anomaly,  $M_S$ , of the Sun using the procedure set out in Section 5.2.
3. Enter Table 6.2 with the digit for each power of 10 in  $\Delta t$  and take out the corresponding values of  $\Delta\bar{\lambda}$ ,  $\Delta M$ , and  $\Delta\bar{F}$ . If  $\Delta t$  is negative then the values are minus those shown in the table. The value of the mean longitude,  $\bar{\lambda}$ , is the sum of all the  $\Delta\bar{\lambda}$  values plus the value of  $\bar{\lambda}$  at the epoch. Likewise, the value of the mean anomaly,  $M$ , is the sum of all the  $\Delta M$  values plus the value of  $M$  at the epoch. Finally, the value of the mean argument of latitude,  $\bar{F}$ , is the sum of all the  $\Delta\bar{F}$  values plus the value of  $\bar{F}$  at the epoch. Add as many multiples of  $360^\circ$  to  $\bar{\lambda}$ ,  $M$ , and  $\bar{F}$  as is required to make them all fall in the range  $0^\circ$  to  $360^\circ$ .
4. Form  $\tilde{D} = \bar{\lambda} - \lambda_S$ .
5. Form the five arguments  $a_1 = M$ ,  $a_2 = 2\tilde{D} - M$ ,  $a_3 = \tilde{D}$ ,  $a_4 = M_S$ ,  $a_5 = 2\bar{F}$ . Add as many multiples of  $360^\circ$  to the arguments as is required to make them all fall in the range  $0^\circ$  to  $360^\circ$ . Round each argument to the nearest degree.
6. Enter Table 6.3 with the value of each of the five arguments  $a_1$ – $a_5$  and take out the value of each of the five corresponding anomalies  $q_1$ – $q_5$ . It is necessary to interpolate if the arguments are odd.
7. The Moon’s ecliptic longitude is given by  $\lambda = \bar{\lambda} + q_1 + q_2 + q_3 + q_4 + q_5$ . If necessary, convert  $\lambda$  into an angle in the range  $0^\circ$  to  $360^\circ$ . The decimal fraction can be converted into arc minutes using Table 5.2. Round to the nearest arc minute.

Two examples of the use of the procedure that has just been described are given in the following section.



### 6.3 Example lunar longitude calculations

*Example 1:* May 5, 2005 AD, 00:00 UT

From Section 5.2,  $t - t_0 = 1950.5$  JD,  $\lambda_S = 44.602^\circ$ , and  $M_S = 120.001^\circ$ . Making use of Table 6.2, we find:

| $t(\text{JD})$ | $\bar{\lambda}(\circ)$ | $M(\circ)$     | $\bar{F}(\circ)$ |
|----------------|------------------------|----------------|------------------|
| +1 000         | 216.396                | 104.993        | 269.350          |
| +900           | 338.757                | 238.494        | 26.415           |
| +50            | 298.820                | 293.250        | 301.468          |
| +.5            | 6.588                  | 6.532          | 6.615            |
| Epoch          | 218.322                | 134.916        | 93.284           |
|                | <u>1078.883</u>        | <u>778.185</u> | <u>697.132</u>   |
| Modulus        | 358.883                | 58.185         | 337.132          |

It follows that

$$\tilde{D} = \bar{\lambda} - \lambda_S = 358.883 - 44.602 = 314.281^\circ.$$

Thus,

$$a_1 = M \simeq 58^\circ, \quad a_2 = 2\tilde{D} - M = 2 \times 314.281 - 58.185 = 570.377 \simeq 210^\circ,$$

$$a_3 = \tilde{D} \simeq 314^\circ, \quad a_4 = M_S \simeq 120^\circ,$$

$$a_5 = 2\bar{F} = 2 \times 337.132 = 674.264 \simeq 314^\circ.$$

Table 6.3 yields

$$q_1(a_1) = 5.525^\circ, \quad q_2(a_2) = -0.637^\circ, \quad q_3(a_3) = -0.633^\circ,$$

$$q_4(a_4) = -0.160^\circ, \quad q_5(a_5) = 0.082^\circ.$$

Hence,

$$\begin{aligned} \lambda &= \bar{\lambda} + q_1 + q_2 + q_3 + q_4 + q_5 \\ &= 358.883 + 5.525 - 0.637 - 0.633 - 0.160 + 0.082 \\ &= 363.060^\circ, \end{aligned}$$

or

$$\lambda = 3.060 \simeq 3^\circ 04'.$$

Thus, the ecliptic longitude of the Moon at 00:00 UT on May 5, 2005 AD was 3AR04.

*Example 2:* December 25, 1800 AD, 00:00 UT

From Section 5.2,  $t - t_0 = -72\,690.5$  JD,  $\lambda_S = 273.055^\circ$ , and  $M_S = 353.814^\circ$ . Making use of Table 6.2, we find:

| $t(\text{JD})$ | $\bar{\lambda}(^\circ)$ | $M(^\circ)$     | $\bar{F}(^\circ)$ |
|----------------|-------------------------|-----------------|-------------------|
| -70 000        | -27.752                 | -149.506        | -134.519          |
| -2 000         | -72.793                 | -209.986        | -178.701          |
| -600           | -345.838                | -278.996        | -17.610           |
| -90            | -105.876                | -95.849         | -110.642          |
| -.5            | -6.588                  | -6.532          | -6.615            |
| Epoch          | 218.322                 | 134.916         | 93.284            |
|                | <u>-340.525</u>         | <u>-605.953</u> | <u>-354.803</u>   |
| Modulus        | 19.475                  | 114.047         | 5.197             |

It follows that

$$\tilde{D} = \bar{\lambda} - \lambda_S = 19.475 - 273.055 = -253.580^\circ.$$

Thus,

$$a_1 = M \simeq 114^\circ, \quad a_2 = 2\tilde{D} - M = -2 \times 253.580 - 114.047 = -621.207 \simeq 99^\circ,$$

$$a_3 = \tilde{D} \simeq 106^\circ, \quad a_4 = M_S \simeq 354^\circ,$$

$$a_5 = 2\bar{F} = 2 \times 5.197 = 10.394 \simeq 10^\circ.$$

Table 6.3 yields

$$q_1(a_1) = 5.586^\circ, \quad q_2(a_2) = 1.259^\circ, \quad q_3(a_3) = -0.382^\circ,$$

$$q_4(a_4) = 0.019^\circ, \quad q_5(a_5) = -0.020^\circ.$$

Hence,

$$\begin{aligned} \lambda &= \bar{\lambda} + q_1 + q_2 + q_3 + q_4 + q_5 \\ &= 19.475 + 5.586 + 1.259 - 0.382 + 0.019 - 0.020 \\ &= 25.937^\circ, \end{aligned}$$

or

$$\lambda = 25.937 \simeq 25^\circ 56'.$$

Thus, the ecliptic longitude of the Moon at 00:00 UT on December 25, 1800 AD was 25AR56.

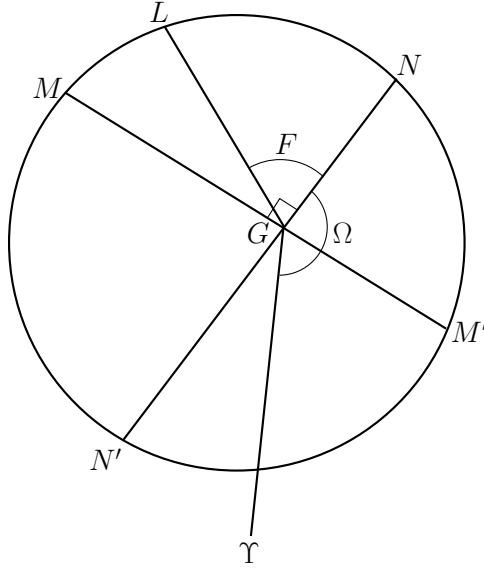


Figure 6.1: The orbit of the Moon about the Earth. Here,  $G$ ,  $L$ ,  $N$ ,  $N'$ ,  $\Omega$ ,  $F$ , and  $\Upsilon$  represent the Earth, the Moon, the ascending node, the descending node, the longitude of the ascending node, the argument of latitude, and the vernal equinox, respectively. The view is from northern ecliptic pole. The Moon orbits counterclockwise.

#### 6.4 Lunar ecliptic latitude model

A model of the Moon's ecliptic latitude is needed in order to predict the occurrence of solar and lunar eclipses. Figure 6.1 shows the Moon's orbit about the Earth. The plane of this orbit is fixed, but slightly tilted with respect to the plane of the ecliptic (that is, the plane of the Sun's apparent orbit about the Earth). Let the two planes intersect along the line of nodes,  $NGN'$ . Here,  $N$  is the point at which the orbit crosses the ecliptic plane from south to north (in the direction of the Moon's orbital motion), and is termed the *ascending node*. Likewise,  $N'$  is the point at which the orbit crosses the ecliptic plane from north to south, and is called the *descending node*. Incidentally, the line of nodes must pass through point  $G$ , because the Earth is common to the ecliptic plane and the plane of the lunar orbit. The angle,  $\Omega$ , subtended between the radius vector  $G\Upsilon$ , connecting the Earth to the vernal equinox, and the line  $GN$ , is known as the *longitude of the ascending node*. Note, incidentally, that the ascending node precesses in the opposite direction to the Moon's orbital motion at the rate  $\dot{\Omega} - n = 5.2954 \times 10^{-2}^\circ$  per day, or  $360^\circ$  in 18.6 years. This unusually large precession rate is another consequence of the Sun's strong perturbing influence on the Moon's orbit. Let the line  $MGM'$  lie in the plane of the Moon's orbit such that it is perpendicular to  $NGN'$ . The inclination,  $i$ , of the Moon's orbital plane is the angle that  $GM$  subtends with its projection onto the ecliptic plane. Likewise, the Moon's ecliptic longitude,  $\beta$ , is the angle that  $GL$  subtends with its projection onto the ecliptic

plane. Simple geometry yields  $\sin \beta = \sin i \sin F$ , where  $F$  is the angle between  $GN$  and  $GL$ . This angle is termed the *argument of latitude*. Now, it is easily seen that  $F \simeq \lambda - \Omega$ , where  $\lambda$  is the Moon's ecliptic longitude (that is, the angle subtended between  $G\Upsilon$  and  $GL$ ). Here, we are assuming that the orbital inclination  $i$  is relatively small. The *mean argument of latitude* is defined  $\bar{F} = \bar{\lambda} - \Omega$ . Hence, our model for the Moon's ecliptic latitude becomes

$$F = \bar{F} + q_1 + q_2 + q_3 + q_4 + q_5, \quad (6.11)$$

$$\sin \beta = \sin i \sin F. \quad (6.12)$$

The value of the lunar orbital inclination,  $i$ , for the J2000 epoch is specified in Table 6.1. The previous model is capable of matching NASA ephemeris data during the years 1995-2006 AD with a mean error of  $6'$ , and a maximum error of  $11'$ .

### 6.5 Determination of lunar ecliptic latitude

The ecliptic latitude of the Moon can be calculated with the aid of Table 6.4. The procedure for using this table is as follows:

1. Determine the fractional Julian day number,  $t$ , corresponding to the date and time at which the Moon's ecliptic latitude is to be calculated with the aid of Tables 3.1–3.3. Form  $\Delta t = t - t_0$ , where  $t_0 = 2\,451\,545.0$  is the epoch.
2. Calculate the lunar mean argument of latitude,  $\bar{F}$ , and the five lunar anomalies,  $q_1$ – $q_5$ , using the procedure outlined earlier in this section.
3. Form the argument  $F = \bar{F} + q_1 + q_2 + q_3 + q_4 + q_5$ . Add as many multiples of  $360^\circ$  to  $F$  as is required to make it fall in the range  $0^\circ$  to  $360^\circ$ . Round  $F$  to the nearest degree.
4. Enter Table 6.4 with the value of  $F$  and take out the lunar ecliptic latitude,  $\beta$ . It is necessary to interpolate if  $F$  is odd.

Two examples of the use of the procedure that has just been described are given in the following section.

### 6.6 Example lunar latitude calculations

*Example 1:* May 5, 2005 AD, 00:00 UT

We have already seen in Section 6.3 that at 00:00 UT on May 5, 2005 AD the lunar mean argument of latitude, and the lunar anomalies, were  $\bar{F} = 337.132^\circ$ , and  $q_1 = 5.525^\circ$ ,  $q_2 = -0.637^\circ$ ,  $q_3 = -0.633^\circ$ ,  $q_4 = -0.160^\circ$ , and  $q_5 = 0.082^\circ$ , respectively. Hence,

$$\begin{aligned} F &= \bar{F} + q_1 + q_2 + q_3 + q_4 + q_5 \\ &= 337.132 + 5.525 - 0.637 - 0.633 - 0.160 + 0.082 \\ &\simeq 341^\circ. \end{aligned}$$

Thus, according to Table 6.4, the ecliptic latitude of the Moon at 00:00 UT on May 5, 2005 AD was  $-1.668^\circ \simeq -1^\circ 40'$ .

*Example 2: December 25, 1800 AD, 00:00 UT*

We have already seen in Section 6.3 that at 00:00 UT on December 25, 1800 AD the lunar mean argument of latitude, and the lunar anomalies, were  $\bar{F} = 5.197^\circ$ , and  $q_1 = 5.586^\circ$ ,  $q_2 = 1.259^\circ$ ,  $q_3 = -0.382^\circ$ ,  $q_4 = -0.019^\circ$ , and  $q_5 = -0.020^\circ$ , respectively. Hence,

$$\begin{aligned} F &= \bar{F} + q_1 + q_2 + q_3 + q_4 + q_5 \\ &= 5.197 + 5.586 + 1.259 - 0.382 + 0.019 - 0.020 \\ &\simeq 12^\circ. \end{aligned}$$

Thus, according to Table 6.4, the ecliptic latitude of the Moon at 00:00 UT on December 25, 1800 AD was  $+1.065^\circ \simeq +1^\circ 04'$ .

### 6.7 The length of a month

A *sidereal month* is the mean period of time between successive passages of the Moon through the vernal equinox, and is  $360/n = 360/13.17639646 = 27.322$  days in length, where  $n$  comes from Table 6.1. A *synodic month* is the mean period of time between successive new moons, or successive full moons, and is  $360/(n - n_S) = 360/(13.17639646 - 0.98564735) = 29.531$  days in length, where  $n_S$  (which is the mean motion of the Sun) comes from Table 5.1. An *anomalistic month* is the mean period of time between successive passages of the Moon through its perigee, and is  $360/\tilde{n} = 360/13.06499295 = 27.555$  days in length, where  $\tilde{n}$  comes from Table 6.1. Finally, a *draconic month* is the mean period of time between successive passages of the Moon through its ascending node, and is  $360/\check{n} = 360/13.22935027 = 27.212$  days in length, where  $\check{n}$  comes from Table 6.1. Incidentally, the lengths of the various months are discussed in Section 3 of Book IV of the Almagest. Impressively, Ptolemy's estimates of the various lengths match those just given to three decimal places.

### 6.8 Lunar distance model

According to a simplified version of modern lunar theory, the variation of the distance,  $r$ , of the Moon from the Earth is fairly well represented by the following formulae:<sup>2</sup>

$$\frac{r}{a_M} = 1 - \zeta_1 - \zeta_2 - \zeta_3 - \zeta_4 - \zeta_5, \quad (6.13)$$

$$\zeta_1 = 0.9894 e \cos M + 0.4915 e^2 \cos 2M, \quad (6.14)$$

$$\zeta_2 = 0.1751 e \cos(2\tilde{D} - M), \quad (6.15)$$

$$\zeta_3 = 0.1399 e (\cos 2\tilde{D} - 0.0368 \cos \tilde{D}), \quad (6.16)$$

$$\zeta_4 = -0.0023 e \cos M_S, \quad (6.17)$$

$$\zeta_5 = 0.0001 e \cos 2\bar{F}, \quad (6.18)$$

where  $a_M = 3.850 \times 10^5$  km is the mean Earth-Moon distance, and  $\zeta_i$  is termed the  $i$ th radial anomaly of the Moon. As before, the first anomaly is due to the eccentricity of the lunar orbit, and is very similar to that obtained from Keplerian orbit theory. The second, third, and fourth anomalies are due to the perturbing action of the Sun. Finally, the fifth anomaly is due to the slight inclination of the lunar orbit to the ecliptic plane. The radial anomalies of the Moon are specified as functions of their arguments in Table 6.5.

Our lunar distance model can be used to calculate the apparent radius of the Moon's disk (were it fully illuminated) in the Earth's sky. In fact, the apparent radius is

$$\rho_M = \frac{R_M}{r}, \quad (6.19)$$

where  $R_M = 1737$  km is the mean physical radius of the Moon, and use has been made of the small angle approximation. It follows that

$$\rho_M = \frac{15.510'}{1 - \zeta_1 - \zeta_2 - \zeta_3 - \zeta_4 - \zeta_5}. \quad (6.20)$$

Clearly, the mean diameter of the (illuminated) lunar disk is about half a degree.

As an example of an apparent lunar radius calculation, we have seen that at 00:00 UT on May 5, 2005 AD the arguments of the lunar radial anomalies took the values

$$\begin{aligned} a_1 = M &\simeq 58^\circ, & a_2 = 2\tilde{D} - M &\simeq 210^\circ, \\ a_3 = \tilde{D} &\simeq 314^\circ, & a_4 = M_S &\simeq 120^\circ & a_5 = 2\bar{F} &\simeq 314^\circ. \end{aligned}$$

Table 6.5 yields

$$100 \zeta_1(a_1) = 2.813, \quad 100 \zeta_2(a_2) = -0.832, \quad 100 \zeta_3(a_3) = 0.046,$$

<sup>2</sup>See Meeus (1998) and Fitzpatrick (2012).

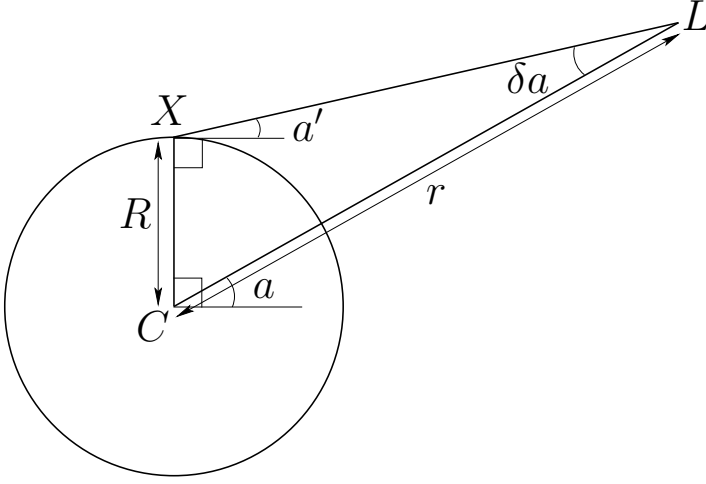


Figure 6.2: The Moon,  $L$ , as viewed by a hypothetical observer,  $C$ , at the center of the Earth, and a real observer,  $X$ , on the surface of the Earth.

$$100 \zeta_4(a_4) = 0.006, \quad 100 \zeta_5(a_5) = -0.001.$$

Hence, we deduce that the apparent lunar radius was

$$\rho_M = \frac{15.510'}{1 - (2.813 - 0.832 + 0.046 + 0.006 - 0.001)/100} = 15.83'.$$

As a second example of a lunar radius calculation, we have seen that at 00:00 UT on December 25, 1800 AD the arguments of the lunar radial anomalies took the values

$$a_1 = M \simeq 114^\circ, \quad a_2 = 2\tilde{D} - M \simeq 99^\circ,$$

$$a_3 = \tilde{D} \simeq 106^\circ, \quad a_4 = M_S \simeq 354^\circ, \quad a_5 = 2\bar{F} \simeq 10^\circ.$$

Table 6.5 yields

$$100 \zeta_1(a_1) = -2.308, \quad 100 \zeta_2(a_2) = 0.151, \quad 100 \zeta_3(a_3) = -0.643,$$

$$100 \zeta_4(a_4) = 0.013, \quad 100 \zeta_5(a_5) = 0.001.$$

Hence, we deduce that the apparent lunar radius was

$$\rho_M = \frac{15.510'}{1 - (-2.308 + 0.151 - 0.643 + 0.013 + 0.001)/100} = 15.09'.$$

### 6.9 Lunar parallax

Now, it turns out that the Moon is sufficiently close to the Earth that its position in the sky is significantly modified by *parallax*. All of our previous analysis applies to a hypothetical observer situated at the center of the Earth. Consider a real observer situated on the Earth's surface. It can be seen from Figure 6.2 that the altitude of the Moon is  $a'$  for the real observer, and  $a$  for the hypothetical observer. Simple trigonometry reveals that  $a' = a - \delta a$ , which implies that the real observer sees the Moon at a lower altitude than the hypothetical observer. Let  $R_E = 6371$  km be the mean radius of the Earth, and  $r$  the distance from the center of the Earth to the Moon. More simple trigonometry yields

$$\sin \delta a = \frac{R_E}{r} \cos a'. \quad (6.21)$$

Making use of Equation (6.13), as well as the small angle approximation, we deduce that

$$\delta a \simeq \delta_M \cos a, \quad (6.22)$$

where

$$\delta_M = \frac{56.888'}{1 - \zeta_1 - \zeta_2 - \zeta_3 - \zeta_4 - \zeta_5}. \quad (6.23)$$

Note that lunar parallax increases with decreasing lunar altitude, reaching a maximum value of about  $57'$  when the Moon is close to the horizon.

We have seen that at 00:00 UT on May 5, 2005 AD the lunar radial anomalies were  $100 \zeta_1 = 2.813$ ,  $100 \zeta_2 = -0.832$ ,  $100 \zeta_3 = 0.046$ ,  $100 \zeta_4 = 0.006$ , and  $100 \zeta_5 = -0.001$ . Thus, the maximum lunar parallax was

$$\delta_M = \frac{56.888'}{1 - (2.813 - 0.832 + 0.046 + 0.006 - 0.001)/100} = 58.08'.$$

Likewise, we have also seen that at 00:00 UT on December 15, 1800 AD the lunar radial anomalies were  $100 \zeta_1 = -2.308$ ,  $100 \zeta_2 = 0.151$ ,  $100 \zeta_3 = -0.0643$ ,  $100 \zeta_4 = 0.013$ , and  $100 \zeta_5 = 0.001$ . Thus, the maximum lunar parallax was

$$\delta_M = \frac{56.888'}{1 - (-2.308 + 0.151 - 0.643 + 0.013 + 0.001)/100} = 55.35'.$$

Incidentally, the information contained in this section is equivalent to that contained in the "Parallax table" (Κανὼν παραλλακτικός) that appears in Section 18 of Book V of the *Almagest*. Moreover, in Section 15 of the same book, Ptolemy uses the measured lunar parallax to determine the distance of the Moon from the Earth. Ptolemy finds that the Moon is 59 Earth radii from the center of the Earth. (The correct Earth-Moon distance is 60.33 Earth radii.) Furthermore, in Section 16 of the same book, Ptolemy uses his Earth-Moon distance, in combination with the apparent size of the Moon in the sky, to estimate that the Earth's radius is 3.4 times that of the Moon. (The correct answer is that the Earth's radius is 3.67 times that of the Moon.)



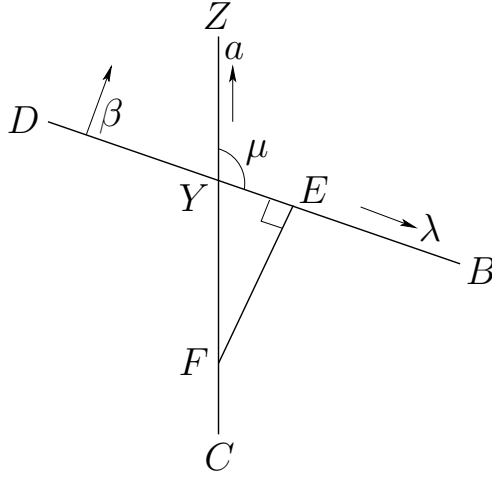


Figure 6.3: Parallaxic shifts in the Moon's ecliptic longitude and latitude.

It now remains to investigate how parallax affects the Moon's ecliptic longitude and latitude. Figure 6.3 shows a detail of Figure 2.12. Point  $Y$  is the Moon's geocentric position on the celestial sphere.  $DB$  is a line passing through this point that is parallel to the local ecliptic circle, whereas  $ZC$  is a small section of an altitude circle passing through  $Y$ . The angle subtended between the ecliptic and the altitude circle is the parallactic angle,  $\mu$ . Let  $F$  be the true position of the Moon. It follows that  $\delta a = YF$ . The changes in the Moon's ecliptic longitude and latitude are  $\delta\lambda = YE$  and  $-\delta\beta = EF$ , respectively. Here, we are considering the case where increasing altitude corresponds to increasing ecliptic latitude. Assuming that the arcs  $\delta a$ ,  $\delta\lambda$ , and  $\delta\beta$  are all fairly small, the triangle  $YEF$  can be treated as a plane triangle. Hence, we obtain

$$\delta\lambda = -\delta a \cos \mu, \quad (6.24)$$

$$\delta\beta = -\delta a \sin \mu. \quad (6.25)$$

As is easily demonstrated, the previous formulae also apply to the case in which increasing altitude corresponds to decreasing ecliptic latitude.

For example, consider a day on which the geocentric ecliptic longitude of the Moon is  $\lambda = 210^\circ$  (that is, 00SC00). Let the maximum lunar parallax be  $\delta_M = 56.89'$ . Suppose that the Moon is viewed from an observation site located at terrestrial latitude  $+10^\circ$ . The "Scorpio" entry in Table 2.18 gives the Moon's geocentric altitude,  $a$ , as a function of time, as well as the value of the parallactic angle  $\mu$ . Making use of this data, in combination with Equations (6.22), (6.24), and (6.25), we can calculate the parallax-induced changes in the Moon's ecliptic longitude and latitude as it transits the sky. Data from such a calculation is given in the following table. The first column specifies time since the Moon's upper transit (thus,  $t = +1$  hrs. means one hour after the upper transit), the second column gives the Moon's geocentric altitude, the

third column the parallactic angle, the fourth column the decrease in the Moon's real altitude due to parallax, and the fifth and sixth columns the parallax-induced changes in its ecliptic longitude and latitude, respectively. It can be seen that parallax causes the Moon's apparent location to shift by almost  $2^\circ$  relative to the fixed stars as it transits the sky. Note that the previous calculation is somewhat inaccurate because it does not take into account the Moon's motion along the ecliptic (which can easily amount to  $6^\circ$  during the course of a night). However, the calculation does illustrate how the data contained in Tables 2.17–2.25, in combination with the data in Table 6.5, permits the parallax-induced shift in the Moon's ecliptic position to be calculated for a wide range of different lunar phases, observation sites, and observation times.

| $t$ (hrs.) | $a$    | $\mu$   | $\delta a$ | $\delta \lambda$ | $\delta \beta$ |
|------------|--------|---------|------------|------------------|----------------|
| −5.52      | 00°00′ | 190°22′ | 57′        | +56′             | +10′           |
| −5.00      | 12°26′ | 187°30′ | 56′        | +55′             | +07′           |
| −4.00      | 26°37′ | 183°07′ | 51′        | +51′             | +03′           |
| −3.00      | 40°23′ | 176°40′ | 43′        | +43′             | −03′           |
| −2.00      | 53°15′ | 165°58′ | 34′        | +33′             | −08′           |
| −1.00      | 63°52′ | 145°55′ | 25′        | +21′             | −14′           |
| +0.00      | 68°32′ | 110°34′ | 21′        | +07′             | −20′           |
| +1.00      | 63°52′ | 075°13′ | 25′        | −06′             | −24′           |
| +2.00      | 53°15′ | 055°11′ | 34′        | −19′             | −28′           |
| +3.00      | 40°23′ | 044°29′ | 43′        | −31′             | −30′           |
| +4.00      | 26°37′ | 038°01′ | 51′        | −40′             | −31′           |
| +5.00      | 12°26′ | 033°39′ | 56′        | −46′             | −31′           |
| +5.52      | 00°00′ | 030°47′ | 57′        | −49′             | −29′           |

### 6.10 Tables

| $e$      | $n(^{\circ}/\text{day})$ | $\tilde{n}(^{\circ}/\text{day})$ | $\tilde{\bar{n}}(^{\circ}/\text{day})$ | $\bar{\lambda}_0(^{\circ})$ | $M_0(^{\circ})$ | $F_0(^{\circ})$ | $i(^{\circ})$ |
|----------|--------------------------|----------------------------------|----------------------------------------|-----------------------------|-----------------|-----------------|---------------|
| 0.054881 | 13.17639646              | 13.06499295                      | 13.22935027                            | 218.322                     | 134.916         | 93.284          | 5.128         |

Table 6.1: Orbital elements of the Moon for the J2000 epoch (that is, 12:00 UT, January 1, 2000 AD, which corresponds to  $t_0 = 2\,451\,545.0$  JD).

| $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(^{\circ})$ | $\Delta M(^{\circ})$ | $\Delta \bar{F}(^{\circ})$ | $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(^{\circ})$ | $\Delta M(^{\circ})$ | $\Delta \bar{F}(^{\circ})$ |
|-----------------------|----------------------------------|----------------------|----------------------------|-----------------------|----------------------------------|----------------------|----------------------------|
| 10 000                | 3.965                            | 329.930              | 173.503                    | 1 000                 | 216.396                          | 104.993              | 269.350                    |
| 20 000                | 7.929                            | 299.859              | 347.005                    | 2 000                 | 72.793                           | 209.986              | 178.701                    |
| 30 000                | 11.894                           | 269.788              | 160.508                    | 3 000                 | 289.189                          | 314.979              | 88.051                     |
| 40 000                | 15.858                           | 239.718              | 334.011                    | 4 000                 | 145.586                          | 59.972               | 357.401                    |
| 50 000                | 19.823                           | 209.648              | 147.513                    | 5 000                 | 1.982                            | 164.965              | 266.751                    |
| 60 000                | 23.788                           | 179.577              | 321.016                    | 6 000                 | 218.379                          | 269.958              | 176.102                    |
| 70 000                | 27.752                           | 149.506              | 134.519                    | 7 000                 | 74.775                           | 14.951               | 85.452                     |
| 80 000                | 31.717                           | 119.436              | 308.022                    | 8 000                 | 291.172                          | 119.944              | 354.802                    |
| 90 000                | 35.681                           | 89.366               | 121.524                    | 9 000                 | 147.568                          | 224.937              | 264.152                    |
|                       |                                  |                      |                            |                       |                                  |                      |                            |
| 100                   | 237.640                          | 226.499              | 242.935                    | 10                    | 131.764                          | 130.650              | 132.294                    |
| 200                   | 115.279                          | 92.999               | 125.870                    | 20                    | 263.528                          | 261.300              | 264.587                    |
| 300                   | 352.919                          | 319.498              | 8.805                      | 30                    | 35.292                           | 31.950               | 36.881                     |
| 400                   | 230.559                          | 185.997              | 251.740                    | 40                    | 167.056                          | 162.600              | 169.174                    |
| 500                   | 108.198                          | 52.496               | 134.675                    | 50                    | 298.820                          | 293.250              | 301.468                    |
| 600                   | 345.838                          | 278.996              | 17.610                     | 60                    | 70.584                           | 63.900               | 73.761                     |
| 700                   | 223.478                          | 145.495              | 260.545                    | 70                    | 202.348                          | 194.550              | 206.055                    |
| 800                   | 101.117                          | 11.994               | 143.480                    | 80                    | 334.112                          | 325.199              | 338.348                    |
| 900                   | 338.757                          | 238.494              | 26.415                     | 90                    | 105.876                          | 95.849               | 110.642                    |
|                       |                                  |                      |                            |                       |                                  |                      |                            |
| 1                     | 13.176                           | 13.065               | 13.229                     | 0.1                   | 1.318                            | 1.306                | 1.323                      |
| 2                     | 26.353                           | 26.130               | 26.459                     | 0.2                   | 2.635                            | 2.613                | 2.646                      |
| 3                     | 39.529                           | 39.195               | 39.688                     | 0.3                   | 3.953                            | 3.919                | 3.969                      |
| 4                     | 52.706                           | 52.260               | 52.917                     | 0.4                   | 5.271                            | 5.226                | 5.292                      |
| 5                     | 65.882                           | 65.325               | 66.147                     | 0.5                   | 6.588                            | 6.532                | 6.615                      |
| 6                     | 79.058                           | 78.390               | 79.376                     | 0.6                   | 7.906                            | 7.839                | 7.938                      |
| 7                     | 92.235                           | 91.455               | 92.605                     | 0.7                   | 9.223                            | 9.145                | 9.261                      |
| 8                     | 105.411                          | 104.520              | 105.835                    | 0.8                   | 10.541                           | 10.452               | 10.583                     |
| 9                     | 118.588                          | 117.585              | 119.064                    | 0.9                   | 11.859                           | 11.758               | 11.906                     |

Table 6.2: Mean motion of the Moon. Here,  $\Delta t = t - t_0$ ,  $\Delta \bar{\lambda} = \bar{\lambda} - \bar{\lambda}_0$ ,  $\Delta M = M - M_0$ , and  $\Delta \bar{F} = \bar{F} - \bar{F}_0$ . At epoch ( $t_0 = 2\,451\,545.0$  JD),  $\bar{\lambda}_0 = 218.322^{\circ}$ ,  $M_0 = 134.916^{\circ}$ , and  $\bar{F}_0 = 93.284^{\circ}$ .

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| Arg. ( $^{\circ}$ ) | $q_1(^{\circ})$ | $q_2(^{\circ})$ | $q_3(^{\circ})$ | $q_4(^{\circ})$ | $q_5(^{\circ})$ |
|---------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| 000(360)            | +0.000          | +0.000          | +0.000          | -0.000          | -0.000          |
| 002(358)            | +0.234          | +0.044          | +0.045          | -0.006          | -0.004          |
| 004(356)            | +0.468          | +0.089          | +0.089          | -0.013          | -0.008          |
| 006(354)            | +0.702          | +0.133          | +0.133          | -0.019          | -0.012          |
| 008(352)            | +0.934          | +0.177          | +0.177          | -0.026          | -0.016          |
| 010(350)            | +1.165          | +0.221          | +0.219          | -0.032          | -0.020          |
| 012(348)            | +1.394          | +0.265          | +0.261          | -0.039          | -0.024          |
| 014(346)            | +1.622          | +0.308          | +0.301          | -0.045          | -0.028          |
| 016(344)            | +1.847          | +0.351          | +0.339          | -0.051          | -0.032          |
| 018(342)            | +2.069          | +0.394          | +0.376          | -0.057          | -0.035          |
| 020(340)            | +2.288          | +0.436          | +0.411          | -0.063          | -0.039          |
| 022(338)            | +2.504          | +0.477          | +0.444          | -0.069          | -0.043          |
| 024(336)            | +2.717          | +0.518          | +0.475          | -0.075          | -0.047          |
| 026(334)            | +2.925          | +0.559          | +0.504          | -0.081          | -0.050          |
| 028(332)            | +3.130          | +0.598          | +0.530          | -0.087          | -0.054          |
| 030(330)            | +3.329          | +0.637          | +0.553          | -0.093          | -0.057          |
| 032(328)            | +3.525          | +0.675          | +0.573          | -0.098          | -0.061          |
| 034(326)            | +3.715          | +0.712          | +0.591          | -0.104          | -0.064          |
| 036(324)            | +3.900          | +0.749          | +0.606          | -0.109          | -0.067          |
| 038(322)            | +4.079          | +0.784          | +0.618          | -0.114          | -0.070          |
| 040(320)            | +4.253          | +0.819          | +0.626          | -0.119          | -0.074          |
| 042(318)            | +4.421          | +0.853          | +0.632          | -0.124          | -0.077          |
| 044(316)            | +4.582          | +0.885          | +0.634          | -0.129          | -0.080          |
| 046(314)            | +4.737          | +0.917          | +0.633          | -0.133          | -0.082          |
| 048(312)            | +4.886          | +0.947          | +0.629          | -0.138          | -0.085          |
| 050(310)            | +5.028          | +0.976          | +0.622          | -0.142          | -0.088          |
| 052(308)            | +5.163          | +1.004          | +0.612          | -0.146          | -0.090          |
| 054(306)            | +5.291          | +1.031          | +0.598          | -0.150          | -0.093          |
| 056(304)            | +5.412          | +1.056          | +0.582          | -0.154          | -0.095          |
| 058(302)            | +5.525          | +1.081          | +0.562          | -0.157          | -0.097          |
| 060(300)            | +5.631          | +1.103          | +0.540          | -0.160          | -0.099          |

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| Arg. (°) | $q_1$ (°) | $q_2$ (°) | $q_3$ (°) | $q_4$ (°) | $q_5$ (°) |
|----------|-----------|-----------|-----------|-----------|-----------|
| 060(300) | +5.631    | +1.103    | +0.540    | -0.160    | -0.099    |
| 062(298) | +5.730    | +1.125    | +0.515    | -0.164    | -0.101    |
| 064(296) | +5.821    | +1.145    | +0.488    | -0.166    | -0.103    |
| 066(294) | +5.904    | +1.164    | +0.458    | -0.169    | -0.105    |
| 068(292) | +5.979    | +1.181    | +0.425    | -0.172    | -0.106    |
| 070(290) | +6.047    | +1.197    | +0.391    | -0.174    | -0.108    |
| 072(288) | +6.107    | +1.212    | +0.354    | -0.176    | -0.109    |
| 074(286) | +6.158    | +1.225    | +0.316    | -0.178    | -0.110    |
| 076(284) | +6.202    | +1.236    | +0.275    | -0.180    | -0.111    |
| 078(282) | +6.238    | +1.246    | +0.234    | -0.181    | -0.112    |
| 080(280) | +6.266    | +1.255    | +0.191    | -0.182    | -0.113    |
| 082(278) | +6.287    | +1.262    | +0.147    | -0.183    | -0.113    |
| 084(276) | +6.299    | +1.267    | +0.102    | -0.184    | -0.114    |
| 086(274) | +6.303    | +1.271    | +0.057    | -0.185    | -0.114    |
| 088(272) | +6.300    | +1.273    | +0.011    | -0.185    | -0.114    |
| 090(270) | +6.289    | +1.274    | -0.035    | -0.185    | -0.114    |
| 092(268) | +6.270    | +1.273    | -0.081    | -0.185    | -0.114    |
| 094(266) | +6.244    | +1.271    | -0.126    | -0.185    | -0.114    |
| 096(264) | +6.210    | +1.267    | -0.171    | -0.184    | -0.114    |
| 098(262) | +6.169    | +1.262    | -0.216    | -0.183    | -0.113    |
| 100(260) | +6.120    | +1.255    | -0.259    | -0.182    | -0.113    |
| 102(258) | +6.065    | +1.246    | -0.302    | -0.181    | -0.112    |
| 104(256) | +6.002    | +1.236    | -0.343    | -0.180    | -0.111    |
| 106(254) | +5.932    | +1.225    | -0.382    | -0.178    | -0.110    |
| 108(252) | +5.856    | +1.212    | -0.420    | -0.176    | -0.109    |
| 110(250) | +5.772    | +1.197    | -0.456    | -0.174    | -0.108    |
| 112(248) | +5.683    | +1.181    | -0.490    | -0.172    | -0.106    |
| 114(246) | +5.586    | +1.164    | -0.521    | -0.169    | -0.105    |
| 116(244) | +5.484    | +1.145    | -0.550    | -0.166    | -0.103    |
| 118(242) | +5.376    | +1.125    | -0.577    | -0.164    | -0.101    |
| 120(240) | +5.261    | +1.103    | -0.600    | -0.160    | -0.099    |

| Arg. ( $^{\circ}$ ) | $q_1(^{\circ})$ | $q_2(^{\circ})$ | $q_3(^{\circ})$ | $q_4(^{\circ})$ | $q_5(^{\circ})$ |
|---------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| 120(240)            | +5.261          | +1.103          | -0.600          | -0.160          | -0.099          |
| 122(238)            | +5.141          | +1.081          | -0.621          | -0.157          | -0.097          |
| 124(236)            | +5.016          | +1.056          | -0.639          | -0.154          | -0.095          |
| 126(234)            | +4.885          | +1.031          | -0.654          | -0.150          | -0.093          |
| 128(232)            | +4.748          | +1.004          | -0.666          | -0.146          | -0.090          |
| 130(230)            | +4.607          | +0.976          | -0.675          | -0.142          | -0.088          |
| 132(228)            | +4.461          | +0.947          | -0.681          | -0.138          | -0.085          |
| 134(226)            | +4.310          | +0.917          | -0.683          | -0.133          | -0.082          |
| 136(224)            | +4.155          | +0.885          | -0.682          | -0.129          | -0.080          |
| 138(222)            | +3.996          | +0.853          | -0.678          | -0.124          | -0.077          |
| 140(220)            | +3.832          | +0.819          | -0.671          | -0.119          | -0.074          |
| 142(218)            | +3.665          | +0.784          | -0.660          | -0.114          | -0.070          |
| 144(216)            | +3.493          | +0.749          | -0.647          | -0.109          | -0.067          |
| 146(214)            | +3.319          | +0.712          | -0.630          | -0.104          | -0.064          |
| 148(212)            | +3.141          | +0.675          | -0.610          | -0.098          | -0.061          |
| 150(210)            | +2.959          | +0.637          | -0.588          | -0.093          | -0.057          |
| 152(208)            | +2.775          | +0.598          | -0.562          | -0.087          | -0.054          |
| 154(206)            | +2.589          | +0.559          | -0.534          | -0.081          | -0.050          |
| 156(204)            | +2.399          | +0.518          | -0.503          | -0.075          | -0.047          |
| 158(202)            | +2.207          | +0.477          | -0.470          | -0.069          | -0.043          |
| 160(200)            | +2.014          | +0.436          | -0.435          | -0.063          | -0.039          |
| 162(198)            | +1.818          | +0.394          | -0.398          | -0.057          | -0.035          |
| 164(196)            | +1.620          | +0.351          | -0.358          | -0.051          | -0.032          |
| 166(194)            | +1.421          | +0.308          | -0.318          | -0.045          | -0.028          |
| 168(192)            | +1.221          | +0.265          | -0.275          | -0.039          | -0.024          |
| 170(190)            | +1.019          | +0.221          | -0.231          | -0.032          | -0.020          |
| 172(188)            | +0.816          | +0.177          | -0.186          | -0.026          | -0.016          |
| 174(186)            | +0.613          | +0.133          | -0.141          | -0.019          | -0.012          |
| 176(184)            | +0.409          | +0.089          | -0.094          | -0.013          | -0.008          |
| 178(182)            | +0.205          | +0.044          | -0.047          | -0.006          | -0.004          |
| 180(180)            | +0.000          | +0.000          | -0.000          | -0.000          | -0.000          |

Table 6.3: Longitudinal anomalies of the Moon. The common argument corresponds to  $M$ ,  $2\tilde{D} - M$ ,  $\tilde{D}$ ,  $M_S$ , and  $2\bar{F}$  for the case of  $q_1$ ,  $q_2$ ,  $q_3$ ,  $q_4$ , and  $q_5$ , respectively. If the argument is in parentheses then the anomalies are minus the values shown in the table.

| $F(^{\circ})$ | $\beta(^{\circ})$ | $F(^{\circ})$ | $F(^{\circ})$ | $\beta(^{\circ})$ | $F(^{\circ})$ |
|---------------|-------------------|---------------|---------------|-------------------|---------------|
| 000/180       | 0.000             | (180)/(360)   | 046/134       | 3.686             | (226)/(314)   |
| 002/178       | 0.179             | (182)/(358)   | 048/132       | 3.809             | (228)/(312)   |
| 004/176       | 0.357             | (184)/(356)   | 050/130       | 3.926             | (230)/(310)   |
| 006/174       | 0.535             | (186)/(354)   | 052/128       | 4.039             | (232)/(308)   |
| 008/172       | 0.713             | (188)/(352)   | 054/126       | 4.147             | (234)/(306)   |
| 010/170       | 0.889             | (190)/(350)   | 056/124       | 4.250             | (236)/(304)   |
| 012/168       | 1.065             | (192)/(348)   | 058/122       | 4.347             | (238)/(302)   |
| 014/166       | 1.239             | (194)/(346)   | 060/120       | 4.440             | (240)/(300)   |
| 016/164       | 1.412             | (196)/(344)   | 062/118       | 4.527             | (242)/(298)   |
| 018/162       | 1.583             | (198)/(342)   | 064/116       | 4.608             | (244)/(296)   |
| 020/160       | 1.752             | (200)/(340)   | 066/114       | 4.684             | (246)/(294)   |
| 022/158       | 1.919             | (202)/(338)   | 068/112       | 4.754             | (248)/(292)   |
| 024/156       | 2.083             | (204)/(336)   | 070/110       | 4.818             | (250)/(290)   |
| 026/154       | 2.246             | (206)/(334)   | 072/108       | 4.877             | (252)/(288)   |
| 028/152       | 2.405             | (208)/(332)   | 074/106       | 4.929             | (254)/(286)   |
| 030/150       | 2.561             | (210)/(330)   | 076/104       | 4.975             | (256)/(284)   |
| 032/148       | 2.715             | (212)/(328)   | 078/102       | 5.016             | (258)/(282)   |
| 034/146       | 2.865             | (214)/(326)   | 080/100       | 5.050             | (260)/(280)   |
| 036/144       | 3.012             | (216)/(324)   | 082/098       | 5.078             | (262)/(278)   |
| 038/142       | 3.155             | (218)/(322)   | 084/096       | 5.100             | (264)/(276)   |
| 040/140       | 3.294             | (220)/(320)   | 086/094       | 5.116             | (266)/(274)   |
| 042/138       | 3.429             | (222)/(318)   | 088/092       | 5.125             | (268)/(272)   |
| 044/136       | 3.560             | (224)/(316)   | 090/090       | 5.128             | (270)/(270)   |

Table 6.4: Ecliptic latitude of the Moon. The latitude is minus the value shown in the table if the argument is in parentheses.

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| Arg. (°) | 100 $\zeta_1$ | 100 $\zeta_2$ | 100 $\zeta_3$ | 100 $\zeta_4$ | 100 $\zeta_5$ |
|----------|---------------|---------------|---------------|---------------|---------------|
| 000(360) | +5.578        | -0.961        | +0.740        | -0.013        | +0.001        |
| 002(358) | +5.574        | -0.960        | +0.738        | -0.013        | +0.001        |
| 004(356) | +5.563        | -0.959        | +0.732        | -0.013        | +0.001        |
| 006(354) | +5.545        | -0.956        | +0.723        | -0.013        | +0.001        |
| 008(352) | +5.519        | -0.952        | +0.710        | -0.012        | +0.001        |
| 010(350) | +5.487        | -0.946        | +0.694        | -0.012        | +0.001        |
| 012(348) | +5.447        | -0.940        | +0.674        | -0.012        | +0.001        |
| 014(346) | +5.399        | -0.932        | +0.650        | -0.012        | +0.001        |
| 016(344) | +5.345        | -0.924        | +0.624        | -0.012        | +0.001        |
| 018(342) | +5.284        | -0.914        | +0.594        | -0.012        | +0.001        |
| 020(340) | +5.216        | -0.903        | +0.562        | -0.012        | +0.001        |
| 022(338) | +5.141        | -0.891        | +0.526        | -0.012        | +0.001        |
| 024(336) | +5.060        | -0.878        | +0.488        | -0.012        | +0.001        |
| 026(334) | +4.972        | -0.864        | +0.447        | -0.011        | +0.001        |
| 028(332) | +4.877        | -0.848        | +0.404        | -0.011        | +0.001        |
| 030(330) | +4.776        | -0.832        | +0.359        | -0.011        | +0.001        |
| 032(328) | +4.670        | -0.815        | +0.313        | -0.011        | +0.001        |
| 034(326) | +4.557        | -0.797        | +0.264        | -0.010        | +0.001        |
| 036(324) | +4.439        | -0.777        | +0.214        | -0.010        | +0.001        |
| 038(322) | +4.315        | -0.757        | +0.163        | -0.010        | +0.001        |
| 040(320) | +4.185        | -0.736        | +0.112        | -0.010        | +0.001        |
| 042(318) | +4.051        | -0.714        | +0.059        | -0.009        | +0.001        |
| 044(316) | +3.911        | -0.691        | +0.006        | -0.009        | +0.001        |
| 046(314) | +3.767        | -0.668        | -0.046        | -0.009        | +0.001        |
| 048(312) | +3.618        | -0.643        | -0.099        | -0.008        | +0.001        |
| 050(310) | +3.465        | -0.618        | -0.151        | -0.008        | +0.001        |
| 052(308) | +3.307        | -0.592        | -0.203        | -0.008        | +0.001        |
| 054(306) | +3.146        | -0.565        | -0.254        | -0.007        | +0.000        |
| 056(304) | +2.981        | -0.537        | -0.303        | -0.007        | +0.000        |
| 058(302) | +2.813        | -0.509        | -0.352        | -0.007        | +0.000        |
| 060(300) | +2.641        | -0.480        | -0.398        | -0.006        | +0.000        |



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| Arg. (°) | 100 $\zeta_1$ | 100 $\zeta_2$ | 100 $\zeta_3$ | 100 $\zeta_4$ | 100 $\zeta_5$ |
|----------|---------------|---------------|---------------|---------------|---------------|
| 060(300) | +2.641        | -0.480        | -0.398        | -0.006        | +0.000        |
| 062(298) | +2.466        | -0.451        | -0.443        | -0.006        | +0.000        |
| 064(296) | +2.289        | -0.421        | -0.485        | -0.006        | +0.000        |
| 066(294) | +2.109        | -0.391        | -0.525        | -0.005        | +0.000        |
| 068(292) | +1.928        | -0.360        | -0.563        | -0.005        | +0.000        |
| 070(290) | +1.744        | -0.329        | -0.598        | -0.004        | +0.000        |
| 072(288) | +1.558        | -0.297        | -0.630        | -0.004        | +0.000        |
| 074(286) | +1.371        | -0.265        | -0.659        | -0.003        | +0.000        |
| 076(284) | +1.183        | -0.232        | -0.685        | -0.003        | +0.000        |
| 078(282) | +0.994        | -0.200        | -0.707        | -0.003        | +0.000        |
| 080(280) | +0.804        | -0.167        | -0.726        | -0.002        | +0.000        |
| 082(278) | +0.613        | -0.134        | -0.742        | -0.002        | +0.000        |
| 084(276) | +0.423        | -0.100        | -0.754        | -0.001        | +0.000        |
| 086(274) | +0.232        | -0.067        | -0.762        | -0.001        | +0.000        |
| 088(272) | +0.042        | -0.034        | -0.767        | -0.000        | +0.000        |
| 090(270) | -0.148        | -0.000        | -0.768        | -0.000        | +0.000        |
| 092(268) | -0.337        | +0.034        | -0.765        | +0.000        | -0.000        |
| 094(266) | -0.525        | +0.067        | -0.758        | +0.001        | -0.000        |
| 096(264) | -0.712        | +0.100        | -0.748        | +0.001        | -0.000        |
| 098(262) | -0.898        | +0.134        | -0.734        | +0.002        | -0.000        |
| 100(260) | -1.082        | +0.167        | -0.717        | +0.002        | -0.000        |
| 102(258) | -1.264        | +0.200        | -0.696        | +0.003        | -0.000        |
| 104(256) | -1.444        | +0.232        | -0.671        | +0.003        | -0.000        |
| 106(254) | -1.622        | +0.265        | -0.643        | +0.003        | -0.000        |
| 108(252) | -1.798        | +0.297        | -0.612        | +0.004        | -0.000        |
| 110(250) | -1.971        | +0.329        | -0.578        | +0.004        | -0.000        |
| 112(248) | -2.141        | +0.360        | -0.542        | +0.005        | -0.000        |
| 114(246) | -2.308        | +0.391        | -0.502        | +0.005        | -0.000        |
| 116(244) | -2.471        | +0.421        | -0.460        | +0.006        | -0.000        |
| 118(242) | -2.632        | +0.451        | -0.416        | +0.006        | -0.000        |
| 120(240) | -2.789        | +0.480        | -0.370        | +0.006        | -0.000        |

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| Arg. (°) | $100 \zeta_1$ | $100 \zeta_2$ | $100 \zeta_3$ | $100 \zeta_4$ | $100 \zeta_5$ |
|----------|---------------|---------------|---------------|---------------|---------------|
| 120(240) | -2.789        | +0.480        | -0.370        | +0.006        | -0.000        |
| 122(238) | -2.942        | +0.509        | -0.322        | +0.007        | -0.000        |
| 124(236) | -3.092        | +0.537        | -0.272        | +0.007        | -0.000        |
| 126(234) | -3.237        | +0.565        | -0.221        | +0.007        | -0.000        |
| 128(232) | -3.379        | +0.592        | -0.168        | +0.008        | -0.001        |
| 130(230) | -3.516        | +0.618        | -0.115        | +0.008        | -0.001        |
| 132(228) | -3.649        | +0.643        | -0.061        | +0.008        | -0.001        |
| 134(226) | -3.777        | +0.668        | -0.007        | +0.009        | -0.001        |
| 136(224) | -3.901        | +0.691        | +0.047        | +0.009        | -0.001        |
| 138(222) | -4.020        | +0.714        | +0.101        | +0.009        | -0.001        |
| 140(220) | -4.134        | +0.736        | +0.155        | +0.010        | -0.001        |
| 142(218) | -4.243        | +0.757        | +0.208        | +0.010        | -0.001        |
| 144(216) | -4.347        | +0.777        | +0.260        | +0.010        | -0.001        |
| 146(214) | -4.446        | +0.797        | +0.311        | +0.010        | -0.001        |
| 148(212) | -4.540        | +0.815        | +0.361        | +0.011        | -0.001        |
| 150(210) | -4.628        | +0.832        | +0.408        | +0.011        | -0.001        |
| 152(208) | -4.712        | +0.848        | +0.454        | +0.011        | -0.001        |
| 154(206) | -4.789        | +0.864        | +0.498        | +0.011        | -0.001        |
| 156(204) | -4.861        | +0.878        | +0.540        | +0.012        | -0.001        |
| 158(202) | -4.928        | +0.891        | +0.578        | +0.012        | -0.001        |
| 160(200) | -4.989        | +0.903        | +0.615        | +0.012        | -0.001        |
| 162(198) | -5.044        | +0.914        | +0.648        | +0.012        | -0.001        |
| 164(196) | -5.094        | +0.924        | +0.678        | +0.012        | -0.001        |
| 166(194) | -5.138        | +0.932        | +0.705        | +0.012        | -0.001        |
| 168(192) | -5.176        | +0.940        | +0.729        | +0.012        | -0.001        |
| 170(190) | -5.208        | +0.946        | +0.749        | +0.012        | -0.001        |
| 172(188) | -5.235        | +0.952        | +0.766        | +0.012        | -0.001        |
| 174(186) | -5.255        | +0.956        | +0.779        | +0.013        | -0.001        |
| 176(184) | -5.270        | +0.959        | +0.788        | +0.013        | -0.001        |
| 178(182) | -5.279        | +0.960        | +0.794        | +0.013        | -0.001        |
| 180(180) | -5.282        | +0.961        | +0.796        | +0.013        | -0.001        |

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Table 6.5: Radial anomalies of the Moon. The common argument corresponds to  $M$ ,  $2\tilde{D} - M$ ,  $\tilde{D}$ ,  $M_S$ , and  $2\bar{F}$  for the case of  $\zeta_1$ ,  $\zeta_2$ ,  $\zeta_3$ ,  $\zeta_4$ , and  $\zeta_5$ , respectively. If the argument is in parentheses then the anomalies are minus the values shown in the table.

## 7. Lunar-Solar Syzygies and Eclipses

### 7.1 Syzygies

Let  $\lambda_S$  and  $\lambda_M$  represent the ecliptic longitudes of the Sun and the Moon, respectively. The lunar-solar *elongation* is defined

$$D = \lambda_M - \lambda_S. \quad (7.1)$$

Because the Moon is only visible because of light reflected from the Sun,<sup>1</sup> there is a fairly obvious relationship between lunar-solar elongation and lunar phase. See Figure 7.1. For instance, a *new moon* corresponds to  $D = 0^\circ$ , a *quarter moon* to  $D = 90^\circ$  or  $270^\circ$ , and a *full moon* to  $D = 180^\circ$ . New moons and full moons are collectively known as lunar-solar *syzygies*. (From the Greek *συζυγία*, meaning “yoking together” or conjunction.)

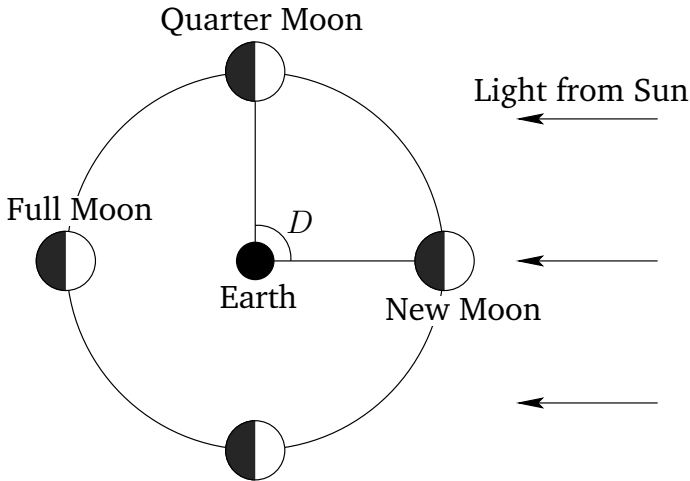


Figure 7.1: The phases of the Moon.

<sup>1</sup>Anaxagoras of Clazomenae (Ἀναξαγόρας ὁ Κλαζομενίας), who lived in the 5th century BC, is generally credited with this discovery.

## 7.2 Lunar-solar elongation model

We can determine the lunar-solar elongation by combining the solar and lunar models described in the previous two chapters. Our elongation model is as follows:

$$\bar{D} = \bar{\lambda}_M - \bar{\lambda}_S, \quad (7.2)$$

$$q_1 = 2 e_M \sin M_M + 1.2379 e_M^2 \sin 2M_M, \quad (7.3)$$

$$q_2 = 0.4052 e_M \sin(2\bar{D} - M_M), \quad (7.4)$$

$$q_3 = 0.2094 e_M (\sin 2\bar{D} - 0.0527 \sin \bar{D}), \quad (7.5)$$

$$q_4 = -(0.0589 e_M + 2 e_S) \sin M_S - (5/4) e_S^2 \sin 2M_S, \quad (7.6)$$

$$q_5 = -0.0364 e_M \sin 2\bar{F}_M, \quad (7.7)$$

$$D = \bar{D} + q_1 + q_2 + q_3 + q_4 + q_5. \quad (7.8)$$

Here,  $e_S$ ,  $M_S$ , and  $\bar{\lambda}_S$  are the eccentricity, mean anomaly, and mean longitude of the Sun's apparent orbit about the Earth, respectively. Moreover,  $e_M$ ,  $M_M$ ,  $\bar{\lambda}_M$ , and  $\bar{F}_M$  are the eccentricity, mean anomaly, mean longitude, and mean argument of latitude of the Moon's orbit, respectively.

The lunar-solar elongation can be calculated with the aid of Tables 7.1 and 7.2. Table 7.1 allows the mean lunar-solar elongation,  $\bar{D}$ , the mean lunar argument of latitude,  $\bar{F}_M$ , the mean anomaly of the Sun,  $M_S$ , and the mean anomaly of the Moon,  $M_M$ , to be determined as functions of time. Table 7.2 specifies the anomalies  $q_1$ – $q_5$  as functions of their various arguments.

Tables 7.1 and 7.2 contain equivalent information to that contained in the “Table of conjunctions” (Συνόδων κανόνιον) and the “Table of full moons” (Πανσελήνων κανόνιον) that appear in Section 3 of Book VI of the *Almagest*.

## 7.3 Determination of lunar-solar elongation

The procedure for using Tables 7.1 and 7.2 to determine the lunar-solar elongation is as follows:

1. Determine the fractional Julian day number,  $t$ , corresponding to the date and time at which the lunar-solar elongation is to be calculated with the aid of Tables 3.1–3.3. Form  $\Delta t = t - t_0$ , where  $t_0 = 2\,451\,545.0$  is the epoch.
2. Enter Table 7.1 with the digit for each power of 10 in  $\Delta t$  and take out the corresponding values of  $\Delta\bar{D}$ ,  $\Delta\bar{F}_M$ ,  $\Delta M_S$ , and  $\Delta M_M$ . If  $\Delta t$  is negative then the values are minus those shown in the table. The value of the mean lunar-solar elongation,  $\bar{D}$ , is the sum of all the  $\Delta\bar{D}$  values plus the value of  $\bar{D}$  at the epoch. Likewise, the value of the mean lunar argument of latitude,  $\bar{F}_M$ , is the sum of all the  $\Delta\bar{F}_M$  values plus the value of  $\bar{F}_M$  at the epoch. Moreover, the value of the solar mean anomaly,  $M_S$ , is the sum of all the  $\Delta M_S$  values plus the value of  $M_S$  at the epoch. Finally, the value of the lunar mean anomaly,  $M_M$ , is the sum of all the  $\Delta M_M$  values plus the value of  $M_M$  at the epoch. Add as many multiples of  $360^\circ$  to  $\bar{D}$ ,  $\bar{F}_M$ ,  $M_S$ , and  $M_M$  as is required to make them all fall in the range  $0^\circ$  to  $360^\circ$ .

3. Form the five arguments  $a_1 = M_M$ ,  $a_2 = 2\bar{D} - M_M$ ,  $a_3 = \bar{D}$ ,  $a_4 = M_S$ ,  $a_5 = 2\bar{F}_M$ . Add as many multiples of  $360^\circ$  to the arguments as is required to make them all fall in the range  $0^\circ$  to  $360^\circ$ . Round each argument to the nearest degree.
4. Enter Table 7.2 with the value of each of the five arguments  $a_1$ – $a_5$  and take out the value of each of the five corresponding anomalies  $q_1$ – $q_5$ . It is necessary to interpolate if the arguments are odd.
5. The lunar-solar elongation is given by  $D = \bar{D} + q_1 + q_2 + q_3 + q_4 + q_5$ . If necessary, convert  $D$  into an angle in the range  $0^\circ$  to  $360^\circ$ . The decimal fraction can be converted into arc minutes using Table 5.2.

In order to facilitate the calculation of syzygies, the previous model has been used to construct Table 7.3, which lists the dates and fractional Julian day numbers of the first new moons of the years 1900–2099 CE. Two examples of syzygy calculations are given in the following section. Incidentally, our model is capable of predicting the times of new moons and full moons to within 10 minutes.

## 7.4 Example syzygy calculations

### Example 1: Sixth new moon in 2004 AD

From Table 7.3, the date of first new moon in 2004 AD is 2453026.4 JD. Now, the lunar-solar elongation increases at the mean rate  $n_M - n_S = 13.17639646 - 0.98564735 = 12.1907491^\circ$  per day, or  $360^\circ$  in 29.53 days; the latter time period is known as a synodic month. Hence, a rough estimate for the date of the sixth new moon in 2004 AD is five synodic months after that of the first: that is,  $2453026.4 + 5 \times 29.53 \simeq 2453174.1$  JD. It follows that  $\Delta t = 2453174.1 - 2451545.0 = 1629.1$  JD. Let us calculate the lunar-solar elongation at this date. From Table 7.1:

| $t(\text{JD})$ | $\bar{D}(^\circ)$ | $\bar{F}_M(^\circ)$ | $M_S(^\circ)$ | $M_M(^\circ)$ |
|----------------|-------------------|---------------------|---------------|---------------|
| +1 000         | 310.749           | 269.350             | 265.600       | 104.993       |
| +600           | 114.449           | 17.610              | 231.360       | 278.996       |
| +20            | 243.815           | 264.587             | 19.712        | 261.300       |
| +9             | 109.717           | 119.064             | 8.870         | 117.585       |
| +1             | 1.219             | 1.323               | 0.099         | 1.306         |
| Epoch          | 297.864           | 93.284              | 357.588       | 134.916       |
|                | 1077.813          | 765.218             | 883.229       | 899.096       |
| Modulus        | 357.813           | 45.218              | 163.229       | 179.096       |

Thus,

$$a_1 = M_M \simeq 179^\circ, \quad a_2 = 2\bar{D} - M_M = 2 \times 357.813 - 179.082 \simeq 177^\circ,$$

$$a_3 = \bar{D} \simeq 358^\circ, \quad a_4 = M_S \simeq 163^\circ,$$

$$a_5 = 2\bar{F}_M = 2 \times 45.218 \simeq 90^\circ.$$

Table 7.2 yields

$$q_1(a_1) = 0.103^\circ, \quad q_2(a_2) = 0.067^\circ, \quad q_3(a_3) = -0.045^\circ,$$

$$q_4(a_4) = -0.603^\circ, \quad q_5(a_5) = -0.114^\circ.$$

Hence,

$$D = \bar{D} + q_1 + q_2 + q_3 + q_4 + q_5$$

$$= 357.813 + 0.103 + 0.067 - 0.045 - 0.603 - 0.114$$

$$\simeq 357.22^\circ.$$

Now, the actual new moon takes place when  $D = 360.00^\circ$ . Thus, a far better estimate for the date of the sixth new moon in 2004 AD is  $2453174.10 + (360.00 - 357.22)/12.1907491 = 2453174.33$  JD. This corresponds to 19:55 UT on June 17th.

*Example 2: Third full moon in 1982 AD*

From Table 7.3, the fractional Julian day number of first new moon in 1982 AD is 2444994.7 JD, which corresponds to January 25th. Because there is more than half a synodic month between this event and the start of year, we conclude that the first full moon in 1982 AD took place before January 25th. Hence, a rough estimate for the date of the third full moon in 1982 AD is one and a half synodic months after that of the first new moon; that is,  $2444994.7 + 1.5 \times 29.53 \simeq 2445039.0$  JD. It follows that  $\Delta t = 2445039.0 - 2451545.0 = -6506.0$  JD. Let us calculate the lunar-solar elongation at this date. From Table 7.1:

| $t(\text{JD})$ | $\bar{D}(\circ)$ | $\bar{F}_M(\circ)$ | $M_S(\circ)$ | $M_M(\circ)$ |
|----------------|------------------|--------------------|--------------|--------------|
| -6 000         | -64.495          | -176.102           | -153.601     | -269.958     |
| -500           | -335.375         | -134.675           | -132.800     | -52.496      |
| -6             | -73.144          | -79.376            | -5.914       | -78.390      |
| Epoch          | 297.864          | 93.284             | 357.588      | 134.916      |
|                | -175.150         | -296.869           | 65.273       | -265.928     |
| Modulus        | 184.131          | 63.062             | 65.273       | 94.072       |

Thus,

$$a_1 = M_M \simeq 94^\circ, \quad a_2 = 2\bar{D} - M_M = 2 \times 184.850 - 94.072 \simeq 276^\circ,$$

$$a_3 = \bar{D} \simeq 185^\circ, \quad a_4 = M_S \simeq 65^\circ,$$

$$a_5 = 2\bar{F}_M = 2 \times 63.062 \simeq 126^\circ.$$

Table 7.2 yields

$$\begin{aligned} q_1(a_1) &= 6.244^\circ, \quad q_2(a_2) = -1.267^\circ, \quad q_3(a_3) = 0.118^\circ, \\ q_4(a_4) &= -1.918^\circ, \quad q_5(a_5) = -0.093^\circ. \end{aligned}$$

Hence,

$$\begin{aligned} D &= \bar{D} + q_1 + q_2 + q_3 + q_4 + q_5 \\ &= 184.850 + 6.244 - 1.267 + 0.118 - 1.918 - 0.093 \\ &\simeq 187.93^\circ. \end{aligned}$$

Now, the actual full moon takes place when  $D = 180.00^\circ$ . Thus, a far better estimate for the date of the third full moon in 1982 AD is  $2445039.0 + (180.00 - 187.93)/12.1907491 = 2445038.35$  JD. This corresponds to 20:04 UT on March 9th.

### 7.5 Solar and lunar eclipses

Eclipses are discussed in Book VI of the *Almagest*. A *solar eclipse*—or, more accurately, a lunar-solar occultation—occurs when the Moon blocks the light of the Sun. Clearly, this is only possible at a new moon. See Figure 7.1. On the other hand, an umbral *lunar eclipse* occurs when the Moon falls into the umbra of the Earth. Of course, this is only possible at a full moon. It follows that eclipses can only take place at lunar-solar syzygies.

In order to determine whether a particular lunar-solar syzygy coincides with an eclipse, we first need to calculate the angular radii of the Sun, the Moon, and the Earth's umbra in the sky. We have already seen that the angular radius of the Sun,  $\rho_S$ , is given by formula (5.23). Likewise, the angular radius of the Moon,  $\rho_M$ , is given by formula (6.20). Let  $R_S$  and  $R_E$  be the physical radii of the Sun and the Earth, respectively. Moreover, let  $r_S$  and  $r_M$  be the Earth-Sun and the Earth-Moon distances, respectively. The ratio  $\delta_M = R_E/r_M$  is the maximum parallax of the Moon, and is specified in formula (6.23). Simple trigonometry reveals that the angular size of the Earth's umbra at the radius of the Moon's orbit is

$$\rho_U = \delta_M - \rho_S. \quad (7.9)$$

This can be seen from Figure 7.2. The radius of the umbra at the position of the Moon is  $R_U = R_E - x = R_E - r_M \rho_S$ . Hence, the angular radius of the umbra is  $\rho_U = R_U/r_M = \delta_M - \rho_S$ . Incidentally, the identification of two of the angles in the figure with  $\rho_S = R_S/r_S$  follows because  $R_S \gg R_E$ .

A solar eclipse does not take place every new moon, nor a lunar eclipse every full moon, because of the inclination of the Moon's orbit to the ecliptic plane, which causes the Moon to pass either above or below the Sun, or the Earth's umbra, respectively, in the majority of cases. It follows that the critical parameter that determines the occurrence of eclipses is the ecliptic latitude of the Moon at syzygy,  $\beta_{syz}$ . Of course, once the date and time of a syzygy has been

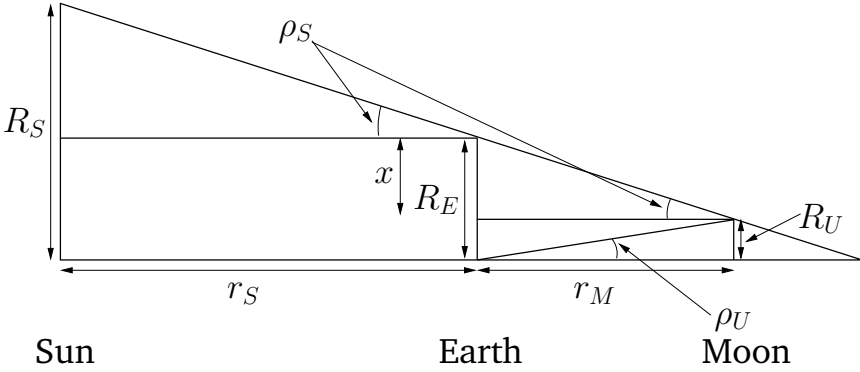


Figure 7.2: The Earth's umbra. Here,  $R_S$ ,  $R_E$ ,  $R_U$ ,  $r_S$ , and  $r_M$  are the physical radius of the Sun, the physical radius of the Earth, the physical radius of the Earth's umbra at the Moon, the Earth-Sun distance, and the Earth-Moon distances, respectively.

established,  $\beta_{syx}$  can be calculated from Table 6.4. However, the lunar argument of latitude,  $F$ , must first be determined using

$$F = \bar{F}_M + q_1 + q_2 + q_3 + q_{4'} + q_5, \quad (7.10)$$

where  $\bar{F}_M$  comes from Table 7.1,  $q_1$ ,  $q_2$ ,  $q_3$ , and  $q_5$  are obtained from Table 7.2, and  $q_{4'}$  is the  $q_4$  from Table 6.3. For instance, we have seen that for the third full moon of 1982 AD,  $\bar{F}_M = 63.062$ ,  $M_S \simeq 65^\circ$ ,  $q_1 = 6.244^\circ$ ,  $q_2 = -1.267^\circ$ ,  $q_3 = 0.118^\circ$ , and  $q_5 = -0.093^\circ$ . According to Table 6.3,  $q_{4'}(M_S) = -0.168^\circ$ . Hence,

$$\begin{aligned} F &= \bar{F}_M + q_1 + q_2 + q_3 + q_{4'} + q_5 \\ &= 63.062 + 6.244 - 1.267 + 0.118 - 0.168 - 0.093 = 67.926 \\ &\simeq 68^\circ. \end{aligned}$$

It follows from Table 6.4 that  $\beta_{syx} = 4.784^\circ \simeq 4^\circ 47'$ .

The criterion for an umbral lunar eclipse is particularly simple, because it is not complicated by lunar parallax. A *total lunar eclipse*, in which the Moon is completely immersed in the Earth's umbra, must take place at a full moon if  $|\beta_{syx}| < \rho_U - \rho_M$  (see Figure 7.3), or equivalently

$$|\beta_{syx}| < \beta_{Mt} \equiv \delta_M - \rho_M - \rho_S, \quad (7.11)$$

and either a total or a *partial lunar eclipse*, in which the Moon is only partially immersed in the Earth's umbra, must take place if  $|\beta_{syx}| < \rho_U + \rho_M$  (see Figure 7.3), or equivalently

$$|\beta_{syx}| < \beta_M \equiv \delta_M + \rho_M - \rho_S. \quad (7.12)$$

Note that lunar eclipses are simultaneously visible at all observation sites on the Earth for which the Moon is above the horizon, because the Earth's umbra is larger than the Moon, and the



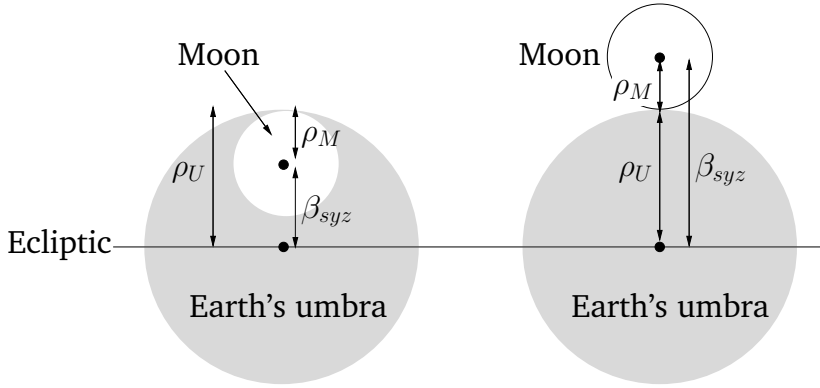


Figure 7.3: The limiting cases for a total lunar eclipse (left) and a partial lunar eclipse (right).

relative position of the Moon and the Earth's umbra is not affected by parallax (because both the Moon and the umbra are the same distance from the Earth). The *magnitude* of a lunar eclipse is defined as the linear fraction of the Moon's diameter that lies in the Earth's umbra at maximum eclipse. It follows that

$$m = \frac{\rho_U + \rho_M - |\beta_{syz}|}{2\rho_M} = \frac{\beta_{Mt} - |\beta_{syz}|}{\beta_M - \beta_{Mt}}. \quad (7.13)$$

The criterion for a solar eclipse is modified by lunar parallax, which causes the angular position of the Moon relative to the Sun to shift by up to  $\delta_M$  from its geocentric position. The amount of the shift depends on the observation site. However, a site can always be found at which the shift takes its maximum value in any particular direction. Note that the Sun has negligible parallax, because it is much farther away from the Earth than the Moon. Taking parallactic shifts into account, a *total solar eclipse*, in which the Sun is totally obscured by the Moon, must take place if  $\rho_M > \rho_S$  and

$$|\beta_{syz}| < \beta_{St} \equiv \delta_M + \rho_M - \rho_S, \quad (7.14)$$

an *annular solar eclipse*, in which all of the Sun apart from a thin outer ring is obscured by the Moon, must take place if  $\rho_S > \rho_M$  and

$$|\beta_{syz}| < \beta_{Sa} \equiv \delta_M - \rho_M + \rho_S, \quad (7.15)$$

and either a total, an annular, or a *partial solar eclipse*, in which the Sun is only partially obscured by the Moon, must take place if

$$|\beta_{syz}| < \beta_S \equiv \delta_M + \rho_M + \rho_S. \quad (7.16)$$

As a consequence of lunar parallax, and the fact that the angular sizes of the Sun and Moon in the sky are very similar, solar eclipses are only visible in very localized regions of the Earth.

The magnitude of a solar eclipse is defined as the linear fraction of the Sun's diameter that is obscured by the Moon at maximum eclipse. It follows that

$$m = \frac{\rho_M}{\rho_S} = \frac{\beta_S - \beta_{Sa}}{\beta_S - \beta_{St}} \quad (7.17)$$

for a total or annular eclipse, and

$$m = \frac{\delta_M + \rho_S + \rho_M - |\beta_{syzy}|}{2\rho_S} = \frac{\beta_S - |\beta_{syzy}|}{\beta_S - \beta_{St}}. \quad (7.18)$$

for a partial eclipse.

Note that the previous criteria represent necessary, but not sufficient, conditions for the occurrence of the various eclipses with which they are associated. This is the case because the point of closest approach of the Moon and the Earth's umbra, in the case of a lunar eclipse, and the Moon and Sun, in the case of a solar eclipse, does not necessarily occur exactly at the syzygy, due to the inclination of the Moon's orbit to the ecliptic. However, because the said inclination is fairly gentle, the previous criteria turn out to be very accurate predictors of eclipses.

## 7.6 Example eclipse calculations

Let us use our model to examine the lunar-solar syzygies of the year 1992 AD in order to see whether any of them were associated with solar or lunar eclipses. The following table shows the dates and times of the new moons of 1992 AD, calculated using the method described in Section 7.3. Also shown is the magnitude of the Moon's ecliptic latitude at each syzygy,  $|\beta_{syzy}|$ , calculated from Equation (7.10) and Table 6.4, as well as the critical values of this parameter for a general, total, and annular solar eclipse. The latter are calculated from Equations (7.14)–(7.16), with the aid of Table 6.5. The eclipse magnitude is also shown. It can be seen that the criterion for a total solar eclipse (that is,  $|\beta_{syzy}| < \beta_{St}$  and  $\beta_{St} > \beta_{Sa}$ ) is satisfied for the syzygy marked with a T, the criterion for an annular solar eclipse (that is,  $|\beta_{syzy}| < \beta_{Sa}$  and  $\beta_{Sa} > \beta_{St}$ ) for the syzygy marked with an A, and the criterion for a partial solar eclipse (that is,  $\beta_{St}, \beta_{Sa} < |\beta_{syzy}| < \beta_S$ ) for the syzygy marked with a P. It is easily verified that a total solar eclipse, an annular solar eclipse, and a partial solar eclipse did indeed take place in 1992 AD at the dates and times indicated.

| Date       | Time (UT) | $\beta_S(^{\circ})$ | $\beta_{St}(^{\circ})$ | $\beta_{Sa}(^{\circ})$ | $ \beta_{syz} (^{\circ})$ | Type | $m$  |
|------------|-----------|---------------------|------------------------|------------------------|---------------------------|------|------|
| 04/01/1992 | 23:15     | 85.0                | 52.5                   | 55.5                   | 022.9                     | A    | 0.91 |
| 03/02/1992 | 18:58     | 84.9                | 52.5                   | 55.5                   | 181.5                     |      |      |
| 04/03/1992 | 13:14     | 85.7                | 53.5                   | 55.9                   | 284.8                     |      |      |
| 03/04/1992 | 04:52     | 87.2                | 55.2                   | 56.7                   | 304.0                     |      |      |
| 02/05/1992 | 17:39     | 89.1                | 57.3                   | 57.7                   | 238.1                     |      |      |
| 01/06/1992 | 03:55     | 91.0                | 59.5                   | 58.8                   | 109.3                     |      |      |
| 30/06/1992 | 12:18     | 92.7                | 61.2                   | 59.7                   | 046.8                     |      |      |
| 29/07/1992 | 19:35     | 93.7                | 62.2                   | 60.3                   | 190.4                     |      |      |
| 28/08/1992 | 02:43     | 93.8                | 62.2                   | 60.4                   | 285.2                     |      |      |
| 26/09/1992 | 10:44     | 93.1                | 61.2                   | 60.0                   | 304.6                     |      |      |
| 25/10/1992 | 20:38     | 91.5                | 59.3                   | 59.2                   | 240.0                     | T    | 1.05 |
| 24/11/1992 | 09:14     | 89.5                | 57.1                   | 58.1                   | 105.4                     |      |      |
| 24/12/1992 | 00:49     | 87.4                | 54.9                   | 56.9                   | 061.9                     |      |      |
|            |           |                     |                        |                        |                           | P    | 0.78 |

The following table shows the dates and times of the full moons of 1992 AD. Also shown is the magnitude of the Moon's ecliptic latitude at each syzygy, as well as the critical values of this parameter for a general and a total lunar eclipse. The latter are calculated from Equations (7.11) and (7.12), with the aid of Table 6.5. The eclipse magnitude is also shown.. It can be seen that the criterion for a total lunar eclipse (that is,  $|\beta_{syz}| < \beta_{Mt}$ ) is satisfied for the syzygy marked with a T, whereas the criterion for a partial lunar eclipse (that is,  $\beta_{Mt} < |\beta_{syz}| < \beta_M$ ) is satisfied for the syzygy marked with a P. It is easily verified that a total lunar eclipse, and a partial lunar eclipse did indeed take place in 1992 AD at the dates and times indicated.

| Date       | Time (UT) | $\beta_M(^{\circ})$ | $\beta_{Mt}(^{\circ})$ | $ \beta_{syz} (^{\circ})$ | Type | $m$  |
|------------|-----------|---------------------|------------------------|---------------------------|------|------|
| 19/01/1992 | 21:25     | 61.9                | 28.4                   | 106.4                     | P    | 0.63 |
| 18/02/1992 | 07:51     | 61.4                | 28.2                   | 241.2                     |      |      |
| 18/03/1992 | 18:07     | 60.1                | 27.5                   | 305.2                     |      |      |
| 17/04/1992 | 04:43     | 58.2                | 26.5                   | 282.5                     |      |      |
| 16/05/1992 | 16:14     | 56.3                | 25.4                   | 183.0                     |      |      |
| 15/06/1992 | 05:06     | 54.6                | 24.5                   | 035.6                     |      |      |
| 14/07/1992 | 19:18     | 53.4                | 23.8                   | 120.8                     |      |      |
| 13/08/1992 | 10:26     | 52.9                | 23.5                   | 245.8                     |      |      |
| 12/09/1992 | 02:02     | 53.1                | 23.6                   | 305.6                     |      |      |
| 11/10/1992 | 17:49     | 54.1                | 24.1                   | 281.2                     |      |      |
| 10/11/1992 | 09:17     | 55.8                | 25.0                   | 176.0                     | T    | 1.25 |
| 09/12/1992 | 23:47     | 57.9                | 26.1                   | 018.3                     |      |      |
| 08/01/1993 | 12:43     | 59.9                | 27.3                   | 145.4                     |      |      |

### 7.7 Eclipse statistics

Consider a very large collection of lunar-solar syzygies. For such a collection, we expect the lunar argument of latitude,  $F$ , the lunar mean anomaly,  $M_M$ , and the solar mean anomaly,  $M_S$ , to be statistically independent of one another, and randomly distributed in the range  $0^{\circ}$  to  $360^{\circ}$ .

Using this insight, we can easily calculate the probability that a new moon is coincident with a solar eclipse, or a full moon with a lunar eclipse, using Equation (6.12) and the criteria (7.11), (7.12), and (7.14)–(7.16).

For a new moon we find:

|                                       |       |
|---------------------------------------|-------|
| Probability of total solar eclipse:   | 4.2%  |
| Probability of annular solar eclipse: | 7.7%  |
| Probability of partial solar eclipse: | 6.6%  |
| Probability of any solar eclipse:     | 18.5% |

For a full moon we get:

|                                       |       |
|---------------------------------------|-------|
| Probability of total lunar eclipse:   | 5.2%  |
| Probability of partial lunar eclipse: | 6.5%  |
| Probability of any lunar eclipse:     | 11.7% |

Thus, we can see that, over a long period of time, the ratio of the number of total/annular solar eclipses to the number of partial solar eclipses is about 9/5, whereas the ratio of the number of partial lunar eclipses to the number of total lunar eclipses is approximately 5/4. Furthermore, the ratio of the number of solar eclipses to the number of lunar eclipses is about 11/7. Because there are 12.37 synodic months in a year, the mean number of solar eclipses per year is approximately  $12.37 \times 0.185 \simeq 2.3$ , whereas the mean number of lunar eclipses per year is about  $12.37 \times 0.117 \simeq 1.4$ . Clearly, solar eclipses are more common than lunar eclipses. On the other hand, at a given observation site on the Earth, lunar eclipses are much more common than solar eclipses, because the former are visible over all regions of the Earth for which the Moon is above the horizon, whereas the latter are only visible in a very localized region.

### 7.8 *Eclipse cycles*

Now, 223 synodic months corresponds to 6585.413 days, which corresponds to 238.99 anomalistic months and 242.00 draconic months. In other words, 223 synodic months is almost exactly equal to 239 anomalistic months and 242 draconic months. This implies that if a solar eclipse takes place at a given new moon then, 223 new moons later, the Moon's perigee and ascending node will be found in almost the exact same positions, which suggests that the Moon's ecliptic latitude, as well as  $\beta_S$ ,  $\beta_{St}$ , and  $\beta_{Sa}$ , will take almost exactly the same values. Hence, another solar eclipse is almost certain to happen 223 new moons after a given eclipse. The same reasoning leads to the conclusion that if a lunar eclipse takes place at a given full moon then another eclipse is almost certain to occur 223 full moons later. It can easily be demonstrated that there is no number of synodic months less than 233 that corresponds to near integer numbers of both anomalistic and draconic months. Hence, the time period of 223 synodic months, which is (mistakenly) called the *saros*, is the minimum (almost) guaranteed period on which solar and lunar eclipses repeat. Incidentally, Ptolemy, in Section 2 of Book IV of the *Almagest*, states that the near equality of 223 synodic months to 239 anomalistic months, as well as to 245 draconic months, was known to very ancient astronomers, which probably implies that the equality was discovered by Babylonian astronomers in the 7th or 6th century BC.

The following table gives details of a series of solar eclipses that take place at regular intervals of 223 new moons. This particular series, which is known as saros cycle 146, started in 1541 AD and will end in 2893 AD.

| Date       | Time (UT) | $\beta_S(^{\circ})$ | $\beta_{St}(^{\circ})$ | $\beta_{Sa}(^{\circ})$ | $ \beta_{syz} (^{\circ})$ | Type | $m$  |
|------------|-----------|---------------------|------------------------|------------------------|---------------------------|------|------|
| 03/04/1848 | 22:53     | 93.9                | 61.9                   | 60.5                   | 76.3                      | P    | 0.55 |
| 15/04/1866 | 06:56     | 93.8                | 61.9                   | 60.4                   | 73.6                      | P    | 0.63 |
| 25/04/1884 | 14:51     | 93.6                | 61.9                   | 60.3                   | 70.6                      | P    | 0.73 |
| 07/05/1902 | 22:39     | 93.5                | 61.8                   | 60.2                   | 67.2                      | P    | 0.83 |
| 18/05/1920 | 06:20     | 93.3                | 61.7                   | 60.1                   | 63.5                      | P    | 0.94 |
| 29/05/1938 | 13:56     | 93.2                | 61.6                   | 60.0                   | 59.6                      | T    | 1.05 |
| 08/06/1956 | 21:27     | 93.0                | 61.5                   | 59.9                   | 55.5                      | T    | 1.05 |
| 20/06/1974 | 04:53     | 92.8                | 61.4                   | 59.8                   | 51.2                      | T    | 1.05 |
| 30/06/1992 | 12:18     | 92.7                | 61.2                   | 59.7                   | 46.8                      | T    | 1.05 |
| 11/07/2010 | 19:40     | 92.5                | 61.0                   | 59.6                   | 42.3                      | T    | 1.05 |
| 22/07/2028 | 03:03     | 92.3                | 60.8                   | 59.5                   | 37.9                      | T    | 1.04 |
| 02/08/2046 | 10:26     | 92.1                | 60.6                   | 59.4                   | 33.6                      | T    | 1.04 |
| 12/08/2064 | 17:52     | 91.9                | 60.4                   | 59.3                   | 29.5                      | T    | 1.03 |
| 24/08/2082 | 01:21     | 91.7                | 60.1                   | 59.2                   | 25.5                      | T    | 1.03 |
| 04/09/2100 | 08:54     | 91.5                | 59.8                   | 59.1                   | 21.8                      | T    | 1.02 |
| 15/09/2118 | 16:32     | 91.4                | 59.6                   | 59.0                   | 18.4                      | T    | 1.02 |
| 26/09/2136 | 00:17     | 91.2                | 59.3                   | 58.9                   | 15.3                      | T    | 1.01 |
| 07/10/2154 | 08:08     | 91.0                | 59.0                   | 58.9                   | 12.6                      | T    | 1.01 |

Likewise, the following table gives details of a series of lunar eclipses that take place at regular intervals of 223 full moons. This particular series, which is known as saros cycle 125, started in 1163 AD and will end in 2443 AD.

| Date       | Time (UT) | $\beta_M(^{\circ})$ | $\beta_{Mt}(^{\circ})$ | $ \beta_{syz} (^{\circ})$ | Type | $m$  |
|------------|-----------|---------------------|------------------------|---------------------------|------|------|
| 27/10/1920 | 14:17     | 58.9                | 26.7                   | 13.9                      | T    | 1.40 |
| 07/11/1938 | 22:32     | 58.6                | 26.6                   | 15.4                      | T    | 1.35 |
| 18/11/1956 | 06:52     | 58.4                | 26.4                   | 16.6                      | T    | 1.31 |
| 29/11/1974 | 15:18     | 58.1                | 26.3                   | 17.6                      | T    | 1.27 |
| 09/12/1992 | 23:47     | 57.9                | 26.1                   | 18.3                      | T    | 1.25 |
| 21/12/2010 | 08:19     | 57.7                | 26.0                   | 18.8                      | T    | 1.23 |
| 31/12/2028 | 16:52     | 57.5                | 25.9                   | 19.3                      | T    | 1.21 |
| 12/01/2047 | 01:25     | 57.3                | 25.8                   | 19.7                      | T    | 1.19 |
| 22/01/2065 | 09:55     | 57.1                | 25.7                   | 20.2                      | T    | 1.17 |
| 02/02/2083 | 18:22     | 56.9                | 25.6                   | 20.9                      | T    | 1.15 |
| 14/02/2101 | 02:43     | 56.7                | 25.5                   | 21.7                      | T    | 1.12 |
| 25/02/2119 | 10:57     | 56.6                | 25.4                   | 22.8                      | T    | 1.08 |
| 07/03/2137 | 19:04     | 56.4                | 25.3                   | 24.2                      | T    | 1.04 |
| 19/03/2155 | 03:01     | 56.2                | 25.2                   | 26.0                      | P    | 0.98 |
| 29/03/2173 | 10:49     | 56.1                | 25.2                   | 28.1                      | P    | 0.91 |
| 09/04/2191 | 18:28     | 55.9                | 25.1                   | 30.6                      | P    | 0.82 |

## 7.9 Tables

| $\Delta t(\text{JD})$ | $\Delta \bar{D}(^{\circ})$ | $\Delta \bar{F}_M(^{\circ})$ | $\Delta M_S(^{\circ})$ | $\Delta M_M(^{\circ})$ | $\Delta t(\text{JD})$ | $\Delta \bar{D}(^{\circ})$ | $\Delta \bar{F}_M(^{\circ})$ | $\Delta M_S(^{\circ})$ | $\Delta M_M(^{\circ})$ |
|-----------------------|----------------------------|------------------------------|------------------------|------------------------|-----------------------|----------------------------|------------------------------|------------------------|------------------------|
| 10 000                | 227.491                    | 173.503                      | 136.002                | 329.930                | 1 000                 | 310.749                    | 269.350                      | 265.600                | 104.993                |
| 20 000                | 94.982                     | 347.005                      | 272.005                | 299.859                | 2 000                 | 261.498                    | 178.701                      | 171.200                | 209.986                |
| 30 000                | 322.473                    | 160.508                      | 48.007                 | 269.788                | 3 000                 | 212.247                    | 88.051                       | 76.801                 | 314.979                |
| 40 000                | 189.964                    | 334.011                      | 184.010                | 239.718                | 4 000                 | 162.996                    | 357.401                      | 342.401                | 59.972                 |
| 50 000                | 57.455                     | 147.513                      | 320.012                | 209.648                | 5 000                 | 113.746                    | 266.751                      | 248.001                | 164.965                |
| 60 000                | 284.947                    | 321.016                      | 96.015                 | 179.577                | 6 000                 | 64.495                     | 176.102                      | 153.601                | 269.958                |
| 70 000                | 152.438                    | 134.519                      | 232.017                | 149.506                | 7 000                 | 15.244                     | 85.452                       | 59.202                 | 14.951                 |
| 80 000                | 19.929                     | 308.022                      | 8.020                  | 119.436                | 8 000                 | 325.993                    | 354.802                      | 324.802                | 119.944                |
| 90 000                | 247.420                    | 121.524                      | 144.022                | 89.366                 | 9 000                 | 276.742                    | 264.152                      | 230.402                | 224.937                |
|                       |                            |                              |                        |                        |                       |                            |                              |                        |                        |
| 100                   | 139.075                    | 242.935                      | 98.560                 | 226.499                | 10                    | 121.907                    | 132.294                      | 9.856                  | 130.650                |
| 200                   | 278.150                    | 125.870                      | 197.120                | 92.999                 | 20                    | 243.815                    | 264.587                      | 19.712                 | 261.300                |
| 300                   | 57.225                     | 8.805                        | 295.680                | 319.498                | 30                    | 5.722                      | 36.881                       | 29.568                 | 31.950                 |
| 400                   | 196.300                    | 251.740                      | 34.240                 | 185.997                | 40                    | 127.630                    | 169.174                      | 39.424                 | 162.600                |
| 500                   | 335.375                    | 134.675                      | 132.800                | 52.496                 | 50                    | 249.537                    | 301.468                      | 49.280                 | 293.250                |
| 600                   | 114.449                    | 17.610                       | 231.360                | 278.996                | 60                    | 11.445                     | 73.761                       | 59.136                 | 63.900                 |
| 700                   | 253.524                    | 260.545                      | 329.920                | 145.495                | 70                    | 133.352                    | 206.055                      | 68.992                 | 194.550                |
| 800                   | 32.599                     | 143.480                      | 68.480                 | 11.994                 | 80                    | 255.260                    | 338.348                      | 78.848                 | 325.199                |
| 900                   | 171.674                    | 26.415                       | 167.040                | 238.494                | 90                    | 17.167                     | 110.642                      | 88.704                 | 95.849                 |
|                       |                            |                              |                        |                        |                       |                            |                              |                        |                        |
| 1                     | 12.191                     | 13.229                       | 0.986                  | 13.065                 | 0.1                   | 1.219                      | 1.323                        | 0.099                  | 1.306                  |
| 2                     | 24.381                     | 26.459                       | 1.971                  | 26.130                 | 0.2                   | 2.438                      | 2.646                        | 0.197                  | 2.613                  |
| 3                     | 36.572                     | 39.688                       | 2.957                  | 39.195                 | 0.3                   | 3.657                      | 3.969                        | 0.296                  | 3.919                  |
| 4                     | 48.763                     | 52.917                       | 3.942                  | 52.260                 | 0.4                   | 4.876                      | 5.292                        | 0.394                  | 5.226                  |
| 5                     | 60.954                     | 66.147                       | 4.928                  | 65.325                 | 0.5                   | 6.095                      | 6.615                        | 0.493                  | 6.532                  |
| 6                     | 73.144                     | 79.376                       | 5.914                  | 78.390                 | 0.6                   | 7.314                      | 7.938                        | 0.591                  | 7.839                  |
| 7                     | 85.335                     | 92.605                       | 6.899                  | 91.455                 | 0.7                   | 8.534                      | 9.261                        | 0.690                  | 9.145                  |
| 8                     | 97.526                     | 105.835                      | 7.885                  | 104.520                | 0.8                   | 9.753                      | 10.583                       | 0.788                  | 10.452                 |
| 9                     | 109.717                    | 119.064                      | 8.870                  | 117.585                | 0.9                   | 10.972                     | 11.906                       | 0.887                  | 11.758                 |

Table 7.1: Mean motion of the lunar-solar elongation. Here,  $\Delta t = t - t_0$ ,  $\Delta \bar{D} = \bar{D} - \bar{D}_0$ ,  $\Delta \bar{F}_M = \bar{F}_M - \bar{F}_{M0}$ ,  $\Delta M_S = M_S - M_{S0}$ , and  $\Delta M_M = M_M - M_{M0}$ . At epoch ( $t_0 = 2\,451\,545.0$  JD),  $\bar{D}_0 = 297.864^{\circ}$ ,  $\bar{F}_{M0} = 93.284^{\circ}$ ,  $M_{S0} = 357.588^{\circ}$ , and  $M_{M0} = 134.916^{\circ}$ .

| Arg. ( $^{\circ}$ ) | $q_1(^{\circ})$ | $q_2(^{\circ})$ | $q_3(^{\circ})$ | $q_4(^{\circ})$ | $q_5(^{\circ})$ |
|---------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| 000(360)            | +0.000          | +0.000          | +0.000          | -0.000          | -0.000          |
| 002(358)            | +0.234          | +0.044          | +0.045          | -0.075          | -0.004          |
| 004(356)            | +0.468          | +0.089          | +0.089          | -0.149          | -0.008          |
| 006(354)            | +0.702          | +0.133          | +0.133          | -0.224          | -0.012          |
| 008(352)            | +0.934          | +0.177          | +0.177          | -0.298          | -0.016          |
| 010(350)            | +1.165          | +0.221          | +0.219          | -0.372          | -0.020          |
| 012(348)            | +1.394          | +0.265          | +0.261          | -0.445          | -0.024          |
| 014(346)            | +1.622          | +0.308          | +0.301          | -0.517          | -0.028          |
| 016(344)            | +1.847          | +0.351          | +0.339          | -0.589          | -0.032          |
| 018(342)            | +2.069          | +0.394          | +0.376          | -0.661          | -0.035          |
| 020(340)            | +2.288          | +0.436          | +0.411          | -0.731          | -0.039          |
| 022(338)            | +2.504          | +0.477          | +0.444          | -0.801          | -0.043          |
| 024(336)            | +2.717          | +0.518          | +0.475          | -0.869          | -0.047          |
| 026(334)            | +2.925          | +0.559          | +0.504          | -0.936          | -0.050          |
| 028(332)            | +3.130          | +0.598          | +0.530          | -1.003          | -0.054          |
| 030(330)            | +3.329          | +0.637          | +0.553          | -1.067          | -0.057          |
| 032(328)            | +3.525          | +0.675          | +0.573          | -1.131          | -0.061          |
| 034(326)            | +3.715          | +0.712          | +0.591          | -1.193          | -0.064          |
| 036(324)            | +3.900          | +0.749          | +0.606          | -1.253          | -0.067          |
| 038(322)            | +4.079          | +0.784          | +0.618          | -1.312          | -0.070          |
| 040(320)            | +4.253          | +0.819          | +0.626          | -1.370          | -0.074          |
| 042(318)            | +4.421          | +0.853          | +0.632          | -1.425          | -0.077          |
| 044(316)            | +4.582          | +0.885          | +0.634          | -1.479          | -0.080          |
| 046(314)            | +4.737          | +0.917          | +0.633          | -1.531          | -0.082          |
| 048(312)            | +4.886          | +0.947          | +0.629          | -1.581          | -0.085          |
| 050(310)            | +5.028          | +0.976          | +0.622          | -1.629          | -0.088          |
| 052(308)            | +5.163          | +1.004          | +0.612          | -1.674          | -0.090          |
| 054(306)            | +5.291          | +1.031          | +0.598          | -1.718          | -0.093          |
| 056(304)            | +5.412          | +1.056          | +0.582          | -1.760          | -0.095          |
| 058(302)            | +5.525          | +1.081          | +0.562          | -1.799          | -0.097          |
| 060(300)            | +5.631          | +1.103          | +0.540          | -1.836          | -0.099          |

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| Arg. ( $^{\circ}$ ) | $q_1(^{\circ})$ | $q_2(^{\circ})$ | $q_3(^{\circ})$ | $q_4(^{\circ})$ | $q_5(^{\circ})$ |
|---------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| 060(300)            | +5.631          | +1.103          | +0.540          | -1.836          | -0.099          |
| 062(298)            | +5.730          | +1.125          | +0.515          | -1.871          | -0.101          |
| 064(296)            | +5.821          | +1.145          | +0.488          | -1.903          | -0.103          |
| 066(294)            | +5.904          | +1.164          | +0.458          | -1.933          | -0.105          |
| 068(292)            | +5.979          | +1.181          | +0.425          | -1.961          | -0.106          |
| 070(290)            | +6.047          | +1.197          | +0.391          | -1.986          | -0.108          |
| 072(288)            | +6.107          | +1.212          | +0.354          | -2.009          | -0.109          |
| 074(286)            | +6.158          | +1.225          | +0.316          | -2.029          | -0.110          |
| 076(284)            | +6.202          | +1.236          | +0.275          | -2.047          | -0.111          |
| 078(282)            | +6.238          | +1.246          | +0.234          | -2.062          | -0.112          |
| 080(280)            | +6.266          | +1.255          | +0.191          | -2.075          | -0.113          |
| 082(278)            | +6.287          | +1.262          | +0.147          | -2.085          | -0.113          |
| 084(276)            | +6.299          | +1.267          | +0.102          | -2.093          | -0.114          |
| 086(274)            | +6.303          | +1.271          | +0.057          | -2.098          | -0.114          |
| 088(272)            | +6.300          | +1.273          | +0.011          | -2.100          | -0.114          |
| 090(270)            | +6.289          | +1.274          | -0.035          | -2.100          | -0.114          |
| 092(268)            | +6.270          | +1.273          | -0.081          | -2.097          | -0.114          |
| 094(266)            | +6.244          | +1.271          | -0.126          | -2.092          | -0.114          |
| 096(264)            | +6.210          | +1.267          | -0.171          | -2.084          | -0.114          |
| 098(262)            | +6.169          | +1.262          | -0.216          | -2.074          | -0.113          |
| 100(260)            | +6.120          | +1.255          | -0.259          | -2.061          | -0.113          |
| 102(258)            | +6.065          | +1.246          | -0.302          | -2.046          | -0.112          |
| 104(256)            | +6.002          | +1.236          | -0.343          | -2.028          | -0.111          |
| 106(254)            | +5.932          | +1.225          | -0.382          | -2.008          | -0.110          |
| 108(252)            | +5.856          | +1.212          | -0.420          | -1.986          | -0.109          |
| 110(250)            | +5.772          | +1.197          | -0.456          | -1.961          | -0.108          |
| 112(248)            | +5.683          | +1.181          | -0.490          | -1.933          | -0.106          |
| 114(246)            | +5.586          | +1.164          | -0.521          | -1.904          | -0.105          |
| 116(244)            | +5.484          | +1.145          | -0.550          | -1.872          | -0.103          |
| 118(242)            | +5.376          | +1.125          | -0.577          | -1.838          | -0.101          |
| 120(240)            | +5.261          | +1.103          | -0.600          | -1.801          | -0.099          |



| Arg. (°) | $q_1$ (°) | $q_2$ (°) | $q_3$ (°) | $q_4$ (°) | $q_5$ (°) |
|----------|-----------|-----------|-----------|-----------|-----------|
| 120(240) | +5.261    | +1.103    | -0.600    | -1.801    | -0.099    |
| 122(238) | +5.141    | +1.081    | -0.621    | -1.763    | -0.097    |
| 124(236) | +5.016    | +1.056    | -0.639    | -1.723    | -0.095    |
| 126(234) | +4.885    | +1.031    | -0.654    | -1.680    | -0.093    |
| 128(232) | +4.748    | +1.004    | -0.666    | -1.636    | -0.090    |
| 130(230) | +4.607    | +0.976    | -0.675    | -1.589    | -0.088    |
| 132(228) | +4.461    | +0.947    | -0.681    | -1.541    | -0.085    |
| 134(226) | +4.310    | +0.917    | -0.683    | -1.491    | -0.082    |
| 136(224) | +4.155    | +0.885    | -0.682    | -1.439    | -0.080    |
| 138(222) | +3.996    | +0.853    | -0.678    | -1.385    | -0.077    |
| 140(220) | +3.832    | +0.819    | -0.671    | -1.330    | -0.074    |
| 142(218) | +3.665    | +0.784    | -0.660    | -1.274    | -0.070    |
| 144(216) | +3.493    | +0.749    | -0.647    | -1.215    | -0.067    |
| 146(214) | +3.319    | +0.712    | -0.630    | -1.156    | -0.064    |
| 148(212) | +3.141    | +0.675    | -0.610    | -1.095    | -0.061    |
| 150(210) | +2.959    | +0.637    | -0.588    | -1.033    | -0.057    |
| 152(208) | +2.775    | +0.598    | -0.562    | -0.969    | -0.054    |
| 154(206) | +2.589    | +0.559    | -0.534    | -0.905    | -0.050    |
| 156(204) | +2.399    | +0.518    | -0.503    | -0.839    | -0.047    |
| 158(202) | +2.207    | +0.477    | -0.470    | -0.773    | -0.043    |
| 160(200) | +2.014    | +0.436    | -0.435    | -0.705    | -0.039    |
| 162(198) | +1.818    | +0.394    | -0.398    | -0.637    | -0.035    |
| 164(196) | +1.620    | +0.351    | -0.358    | -0.568    | -0.032    |
| 166(194) | +1.421    | +0.308    | -0.318    | -0.499    | -0.028    |
| 168(192) | +1.221    | +0.265    | -0.275    | -0.429    | -0.024    |
| 170(190) | +1.019    | +0.221    | -0.231    | -0.358    | -0.020    |
| 172(188) | +0.816    | +0.177    | -0.186    | -0.287    | -0.016    |
| 174(186) | +0.613    | +0.133    | -0.141    | -0.215    | -0.012    |
| 176(184) | +0.409    | +0.089    | -0.094    | -0.144    | -0.008    |
| 178(182) | +0.205    | +0.044    | -0.047    | -0.072    | -0.004    |
| 180(180) | +0.000    | +0.000    | -0.000    | -0.000    | -0.000    |

Table 7.2: Anomalies of the lunar-solar elongation. The common argument corresponds to  $M_M$ ,  $2\bar{D} - M_M$ ,  $\bar{D}$ ,  $M_S$ , and  $2\bar{F}_M$  for the case of  $q_1$ ,  $q_2$ ,  $q_3$ ,  $q_4$ , and  $q_5$ , respectively. If the argument is in parentheses then the anomalies are minus the values shown in the table.

| Date       | JD         | Date       | JD         | Date       | JD         | Date       | JD         |
|------------|------------|------------|------------|------------|------------|------------|------------|
| 01/01/1900 | 2415021.08 | 18/01/1950 | 2433299.83 | 06/01/2000 | 2451550.25 | 23/01/2050 | 2469829.71 |
| 20/01/1901 | 2415405.11 | 07/01/1951 | 2433654.34 | 24/01/2001 | 2451934.05 | 12/01/2051 | 2470184.29 |
| 09/01/1902 | 2415759.38 | 26/01/1952 | 2434038.43 | 13/01/2002 | 2452288.07 | 02/01/2052 | 2470538.62 |
| 28/01/1903 | 2416143.19 | 15/01/1953 | 2434393.09 | 02/01/2003 | 2452642.35 | 19/01/2053 | 2470922.46 |
| 17/01/1904 | 2416497.16 | 05/01/1954 | 2434747.60 | 21/01/2004 | 2453026.37 | 08/01/2054 | 2471276.44 |
| 05/01/1905 | 2416851.27 | 24/01/1955 | 2435131.54 | 10/01/2005 | 2453381.00 | 27/01/2055 | 2471660.24 |
| 24/01/1906 | 2417235.21 | 13/01/1956 | 2435485.62 | 29/01/2006 | 2453765.10 | 16/01/2056 | 2472014.43 |
| 14/01/1907 | 2417589.74 | 30/01/1957 | 2435869.40 | 19/01/2007 | 2454119.67 | 05/01/2057 | 2472368.91 |
| 03/01/1908 | 2417944.41 | 19/01/1958 | 2436223.43 | 08/01/2008 | 2454473.98 | 24/01/2058 | 2472753.00 |
| 22/01/1909 | 2418328.51 | 09/01/1959 | 2436577.73 | 26/01/2009 | 2454857.82 | 14/01/2059 | 2473107.67 |
| 11/01/1910 | 2418682.99 | 28/01/1960 | 2436961.76 | 15/01/2010 | 2455211.81 | 03/01/2060 | 2473462.20 |
| 30/01/1911 | 2419066.90 | 16/01/1961 | 2437316.40 | 04/01/2011 | 2455565.88 | 21/01/2061 | 2473846.13 |
| 19/01/1912 | 2419420.96 | 06/01/1962 | 2437671.03 | 23/01/2012 | 2455949.82 | 10/01/2062 | 2474200.24 |
| 07/01/1913 | 2419774.94 | 25/01/1963 | 2438055.07 | 11/01/2013 | 2456304.31 | 29/01/2063 | 2474584.02 |
| 26/01/1914 | 2420158.78 | 14/01/1964 | 2438409.36 | 30/01/2014 | 2456688.40 | 18/01/2064 | 2474938.03 |
| 15/01/1915 | 2420513.11 | 02/01/1965 | 2438763.38 | 20/01/2015 | 2457043.06 | 06/01/2065 | 2475292.30 |
| 05/01/1916 | 2420867.70 | 21/01/1966 | 2439147.16 | 10/01/2016 | 2457397.57 | 25/01/2066 | 2475676.34 |
| 23/01/1917 | 2421251.82 | 10/01/1967 | 2439501.26 | 28/01/2017 | 2457781.50 | 15/01/2067 | 2476030.97 |
| 12/01/1918 | 2421606.44 | 29/01/1968 | 2439885.19 | 17/01/2018 | 2458135.59 | 05/01/2068 | 2476385.61 |
| 02/01/1919 | 2421960.85 | 18/01/1969 | 2440239.70 | 06/01/2019 | 2458489.57 | 23/01/2069 | 2476769.65 |
| 21/01/1920 | 2422344.72 | 07/01/1970 | 2440594.36 | 24/01/2020 | 2458873.41 | 12/01/2070 | 2477123.97 |
| 09/01/1921 | 2422698.73 | 26/01/1971 | 2440978.46 | 13/01/2021 | 2459227.70 | 01/01/2071 | 2477478.01 |
| 27/01/1922 | 2423082.50 | 16/01/1972 | 2441332.95 | 02/01/2022 | 2459582.27 | 20/01/2072 | 2477861.78 |
| 17/01/1923 | 2423436.61 | 04/01/1973 | 2441687.15 | 21/01/2023 | 2459966.37 | 08/01/2073 | 2478215.85 |
| 06/01/1924 | 2423791.03 | 23/01/1974 | 2442070.96 | 11/01/2024 | 2460321.00 | 27/01/2074 | 2478599.78 |
| 24/01/1925 | 2424175.11 | 12/01/1975 | 2442424.94 | 29/01/2025 | 2460705.03 | 16/01/2075 | 2478954.27 |
| 14/01/1926 | 2424529.78 | 01/01/1976 | 2442779.11 | 18/01/2026 | 2461059.32 | 06/01/2076 | 2479308.93 |
| 03/01/1927 | 2424884.36 | 19/01/1977 | 2443163.09 | 07/01/2027 | 2461413.35 | 24/01/2077 | 2479693.03 |
| 22/01/1928 | 2425268.34 | 09/01/1978 | 2443517.66 | 26/01/2028 | 2461797.14 | 14/01/2078 | 2480047.55 |
| 11/01/1929 | 2425622.51 | 28/01/1979 | 2443901.77 | 14/01/2029 | 2462151.23 | 03/01/2079 | 2480401.78 |
| 29/01/1930 | 2426006.29 | 17/01/1980 | 2444256.40 | 04/01/2030 | 2462505.61 | 22/01/2080 | 2480785.58 |
| 18/01/1931 | 2426360.28 | 06/01/1981 | 2444610.81 | 23/01/2031 | 2462889.68 | 10/01/2081 | 2481139.55 |
| 07/01/1932 | 2426714.48 | 25/01/1982 | 2444994.70 | 12/01/2032 | 2463244.34 | 28/01/2082 | 2481523.37 |
| 25/01/1933 | 2427098.47 | 14/01/1983 | 2445348.72 | 30/01/2033 | 2463628.42 | 18/01/2083 | 2481877.66 |
| 15/01/1934 | 2427453.07 | 03/01/1984 | 2445702.73 | 20/01/2034 | 2463982.92 | 07/01/2084 | 2482232.22 |
| 05/01/1935 | 2427807.73 | 21/01/1985 | 2446086.61 | 09/01/2035 | 2464337.12 | 25/01/2085 | 2482616.34 |
| 24/01/1936 | 2428191.81 | 10/01/1986 | 2446441.01 | 28/01/2036 | 2464720.93 | 15/01/2086 | 2482970.98 |
| 12/01/1937 | 2428546.19 | 29/01/1987 | 2446825.07 | 16/01/2037 | 2465074.91 | 04/01/2087 | 2483325.42 |
| 01/01/1938 | 2428900.29 | 19/01/1988 | 2447179.73 | 05/01/2038 | 2465429.07 | 23/01/2088 | 2483709.31 |
| 20/01/1939 | 2429284.06 | 07/01/1989 | 2447534.31 | 24/01/2039 | 2465813.06 | 11/01/2089 | 2484063.34 |
| 09/01/1940 | 2429638.08 | 26/01/1990 | 2447918.31 | 14/01/2040 | 2466167.64 | 30/01/2090 | 2484447.11 |
| 27/01/1941 | 2430021.96 | 15/01/1991 | 2448272.49 | 02/01/2041 | 2466522.30 | 19/01/2091 | 2484801.19 |
| 16/01/1942 | 2430376.39 | 04/01/1992 | 2448626.47 | 21/01/2042 | 2466906.37 | 09/01/2092 | 2485155.57 |
| 06/01/1943 | 2430731.03 | 22/01/1993 | 2449010.28 | 11/01/2043 | 2467260.78 | 27/01/2093 | 2485539.64 |
| 25/01/1944 | 2431115.14 | 11/01/1994 | 2449364.46 | 30/01/2044 | 2467644.66 | 16/01/2094 | 2485894.30 |
| 14/01/1945 | 2431469.72 | 30/01/1995 | 2449748.45 | 18/01/2045 | 2467998.68 | 06/01/2095 | 2486248.90 |
| 03/01/1946 | 2431824.01 | 20/01/1996 | 2450103.03 | 07/01/2046 | 2468352.69 | 25/01/2096 | 2486632.91 |
| 22/01/1947 | 2432207.85 | 09/01/1997 | 2450457.69 | 26/01/2047 | 2468736.58 | 13/01/2097 | 2486987.12 |
| 11/01/1948 | 2432561.83 | 28/01/1998 | 2450841.76 | 15/01/2048 | 2469090.97 | 02/01/2098 | 2487341.11 |
| 29/01/1949 | 2432945.62 | 17/01/1999 | 2451196.16 | 04/01/2049 | 2469445.60 | 21/01/2099 | 2487724.89 |

Table 7.3: Dates and fractional Julian day numbers of the first new moons of the years 1900–2099 AD.

## 8. The superior planets

### 8.1 Planetary ecliptic longitude model

Figure 8.1 compares and contrasts heliocentric and geocentric models of the motion of a superior planet (that is, a planet that is farther from the Sun than the Earth),  $P$ , as seen from the Earth,  $G$ . The Sun is at  $S$ . In the heliocentric model, we can write the Earth-planet displacement vector,  $\mathbf{P}$ , as the sum of the Earth-Sun displacement vector,  $\mathbf{S}$ , and the Sun-planet displacement vector,  $\mathbf{P}'$ . The geocentric model, which is entirely equivalent to the heliocentric model as far as the relative motion, of the planet with respect to the Earth is concerned, and is much more convenient, relies on the simple vector identity

$$\mathbf{P} = \mathbf{S} + \mathbf{P}' \equiv \mathbf{P}' + \mathbf{S}. \quad (8.1)$$

In other words, we can get from the Earth to the planet by one of two different routes. The first route corresponds to the heliocentric model, and the second to the geocentric model. In the latter model,  $\mathbf{P}'$  gives the displacement of the so-called *guide-point*,  $G'$ , from the Earth. Because  $\mathbf{P}'$  is also the displacement of the planet,  $P$ , from the Sun,  $S$ , it is clear that  $G'$  executes a Keplerian orbit about the Earth whose elements are the same as those of the orbit of the planet about the Sun. The ellipse traced out by  $G'$  is termed the *deferent*. The vector  $\mathbf{S}$  gives the displacement of the planet from the guide-point. However,  $\mathbf{S}$  is also the displacement of the Sun from the Earth. Hence, it is clear that the planet,  $P$ , executes a Keplerian orbit about the guide-point,  $G'$ , whose elements are the same as the Sun's apparent orbit about the Earth. The ellipse traced out by  $P$  about  $G'$  is termed the *epicycle*.

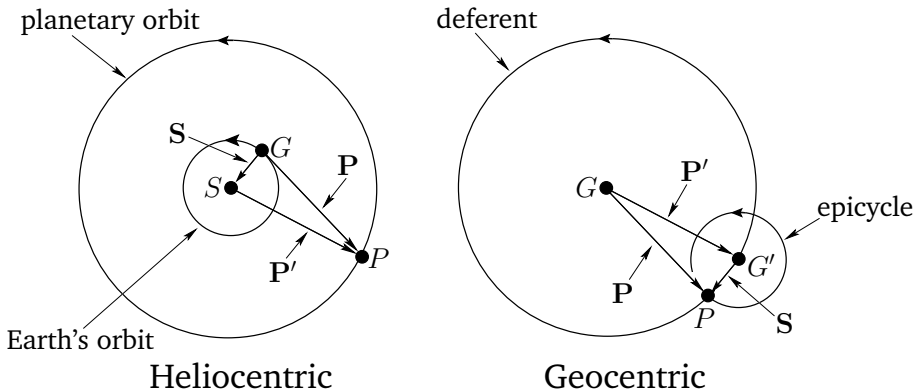


Figure 8.1: Heliocentric and geocentric models of the motion of a superior planet. Here,  $S$  is the Sun,  $G$  the Earth, and  $P$  the planet. The view is from the northern ecliptic pole.

Figure 8.2 illustrates in more detail how the deferent-epicycle model is used to determine the ecliptic longitude of a superior planet. The planet  $P$  orbits (counterclockwise) on a small

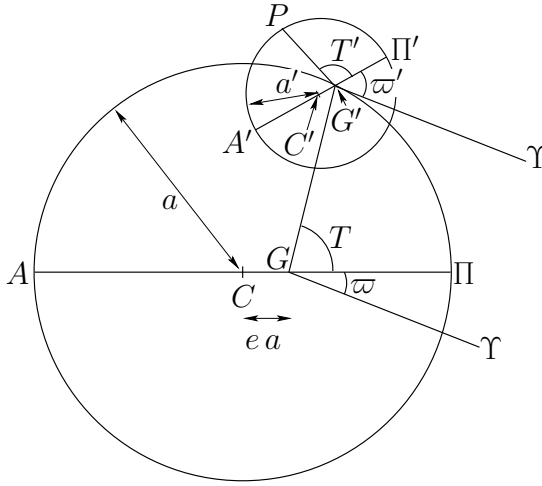


Figure 8.2: Planetary longitude model. View is from northern ecliptic pole.

Keplerian orbit  $\Pi' P A'$  about guide-point  $G'$ , which, in turn, orbits the Earth,  $G$ , (counter-clockwise) on a large Keplerian orbit  $\Pi G' A$ . As has already been mentioned, the small orbit is termed the *epicycle*, and the large orbit the *deferent*. Both orbits are assumed to lie in the plane of the ecliptic. This approximation does not introduce a large error into our calculations because the orbital inclinations of the visible planets to the ecliptic plane are all fairly small. Let  $C$ ,  $A$ ,  $\Pi$ ,  $a$ ,  $e$ ,  $\varpi$ , and  $T$  denote the geometric center, apocenter (that is, the point of farthest distance from the central object), pericenter (that is, the point of closest approach to the central object), major radius, eccentricity, longitude of the pericenter, and true anomaly of the deferent, respectively. Let  $C'$ ,  $A'$ ,  $\Pi'$ ,  $a'$ ,  $e'$ ,  $\varpi'$ , and  $T'$  denote the corresponding quantities for the epicycle.

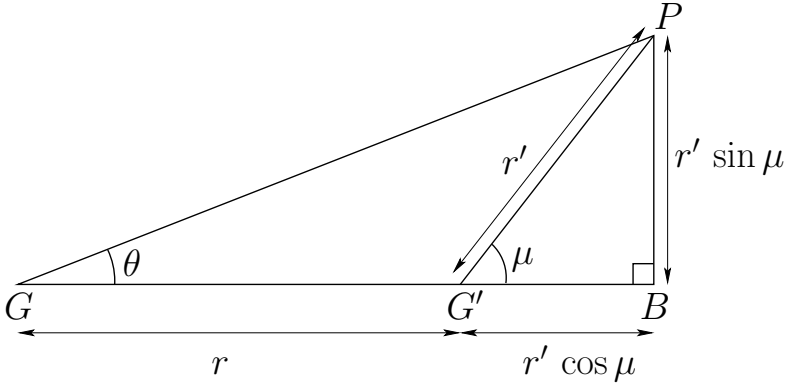
Let the line  $GG'$  be produced, and let the perpendicular  $PB$  be dropped to it from  $P$ , as shown in Figure 8.3. The angle  $\mu \equiv PG'B$  is termed the *epicyclic anomaly* (see Figures 8.4), and takes the form

$$\mu = T' + \varpi' - T - \varpi = \bar{\lambda}' + q' - \bar{\lambda} - q, \quad (8.2)$$

where  $\bar{\lambda}$  and  $q$  are the mean longitude and equation of center for the deferent, whereas  $\bar{\lambda}'$  and  $q'$  are the corresponding quantities for the epicycle. See Chapter 5. The epicyclic anomaly is generally written in the range  $0^\circ$  to  $360^\circ$ . The angle  $\theta \equiv PGG'$  is termed the *equation of the epicycle*, and is usually written in the range  $-180^\circ$  to  $+180^\circ$ . It is clear from the figure that

$$\tan \theta = \frac{\sin \mu}{r/r' + \cos \mu}, \quad (8.3)$$

where  $r \equiv GG'$  and  $r' \equiv G'P$  are the radial polar coordinates for the deferent and epicycle,

Figure 8.3: The triangle  $GBP$ .

respectively. Moreover, according to Equation (4.22),  $r/r' = (a/a') z$ , where

$$z = \frac{1 - \zeta}{1 - \zeta'}, \quad (8.4)$$

and

$$\zeta = e \cos M - e^2 \sin^2 M, \quad (8.5)$$

$$\zeta' = e' \cos M' - e'^2 \sin^2 M' \quad (8.6)$$

are termed *radial anomalies*. Finally, the ecliptic longitude of the planet is given by (see Figure 8.4)

$$\lambda = \bar{\lambda} + q + \theta. \quad (8.7)$$

Now,

$$\theta(\mu, z) \equiv \tan^{-1} \left[ \frac{\sin \mu}{(a/a') z + \cos \mu} \right] \quad (8.8)$$

is a function of two variables,  $\mu$  and  $z$ . It is impractical to tabulate such a function directly. Fortunately, while  $\theta(\mu, z)$  has a strong dependence on  $\mu$ , it only has a fairly weak dependence on  $z$ . In fact, it is easily seen that  $z$  varies between  $z_{\min} = \bar{z} - \delta z$  and  $z_{\max} = \bar{z} + \delta z$ , where

$$\bar{z} = \frac{1 + e e'}{1 - e'^2}, \quad (8.9)$$

$$\delta z = \frac{e + e'}{1 - e'^2}. \quad (8.10)$$

Let us define

$$\xi = \frac{\bar{z} - z}{\delta z}. \quad (8.11)$$

This variable takes the value  $-1$  when  $z = z_{\max}$ , the value  $0$  when  $z = \bar{z}$ , and the value  $+1$  when  $z = z_{\min}$ . Thus, using quadratic interpolation, we can write

$$\theta(\mu, z) \simeq \Theta_-(\xi) \delta\theta_-(\mu) + \bar{\theta}(\mu) + \Theta_+(\xi) \delta\theta_+(\mu), \quad (8.12)$$

where

$$\bar{\theta}(\mu) = \theta(\mu, \bar{z}), \quad (8.13)$$

$$\delta\theta_-(\mu) = \theta(\mu, \bar{z}) - \theta(\mu, z_{\max}), \quad (8.14)$$

$$\delta\theta_+(\mu) = \theta(\mu, z_{\min}) - \theta(\mu, \bar{z}), \quad (8.15)$$

and

$$\Theta_-(\xi) = -(1/2) \xi (\xi - 1), \quad (8.16)$$

$$\Theta_+(\xi) = +(1/2) \xi (\xi + 1). \quad (8.17)$$

This scheme allows us to avoid having to tabulate a two-dimensional function, while ensuring that the exact value of  $\theta(\mu, z)$  is obtained when  $z = \bar{z}$ ,  $z_{\min}$ , or  $z_{\max}$ . The previous interpolation scheme is very similar to that adopted by Ptolemy in the *Almagest*.

Our procedure for determining the ecliptic longitude of a superior planet is described in the following. It is assumed that the ecliptic longitude,  $\lambda_S$ , and the radial anomaly,  $\zeta_S$ , of the Sun have already been calculated. The latter quantity is tabulated as a function of the solar mean anomaly in Table 5.4. In the following,  $a$ ,  $e$ ,  $n$ ,  $\bar{n}$ ,  $\bar{\lambda}_0$ , and  $M_0$  represent elements of the orbit of the planet in question about the Sun, and  $e_S$  represents the eccentricity of the Sun's apparent orbit about the Earth. (In general, the subscript  $S$  denotes the Sun.) In particular,  $a$  is the major radius of the planetary orbit in units in which the major radius of the Sun's apparent orbit about the Earth is unity. The requisite elements for all of the superior planets at the J2000 epoch ( $t_0 = 2\,451\,545.0$  JD) are listed in Table 5.1. The ecliptic longitude of a superior planet

is specified by the following formulae:

$$\bar{\lambda} = \bar{\lambda}_0 + n(t - t_0), \quad (8.18)$$

$$M = M_0 + \tilde{n}(t - t_0), \quad (8.19)$$

$$q = 2e \sin M + (5/4)e^2 \sin 2M, \quad (8.20)$$

$$\zeta = e \cos M - e^2 \sin^2 M, \quad (8.21)$$

$$\mu = \lambda_S - \bar{\lambda} - q, \quad (8.22)$$

$$\bar{\theta} = \theta(\mu, \bar{z}) \equiv \tan^{-1} \left( \frac{\sin \mu}{a \bar{z} + \cos \mu} \right), \quad (8.23)$$

$$\delta\theta_- = \theta(\mu, \bar{z}) - \theta(\mu, z_{\max}), \quad (8.24)$$

$$\delta\theta_+ = \theta(\mu, z_{\min}) - \theta(\mu, \bar{z}), \quad (8.25)$$

$$z = \frac{1 - \zeta}{1 - \zeta_S}, \quad (8.26)$$

$$\xi = \frac{\bar{z} - z}{\delta z}, \quad (8.27)$$

$$\theta = \Theta_-(\xi) \delta\theta_- + \bar{\theta} + \Theta_+(\xi) \delta\theta_+, \quad (8.28)$$

$$\lambda = \bar{\lambda} + q + \theta. \quad (8.29)$$

Here,  $\bar{z} = (1 + e e_S)/(1 - e_S^2)$ ,  $\delta z = (e + e_S)/(1 - e_S^2)$ ,  $z_{\min} = \bar{z} - \delta z$ , and  $z_{\max} = \bar{z} + \delta z$ . The constants  $\bar{z}$ ,  $\delta z$ ,  $z_{\min}$ , and  $z_{\max}$  for each of the superior planets are listed in Table 8.1. Finally, the functions  $\Theta_{\pm}$  are tabulated in Table 8.2.

For the case of Mars, the previous formulae are capable of matching NASA ephemeris data during the years 1995–2006 AD with a mean error of  $3'$  and a maximum error of  $14'$ . For the case of Jupiter, the mean error is  $1.6'$  and the maximum error  $4'$ . Finally, for the case of Saturn, the mean error is  $0.5'$  and the maximum error  $1'$ .

## 8.2 Determination of the ecliptic longitude of Mars

The ecliptic longitude of Mars can be determined with the aid of Tables 8.3–8.5. Table 8.3 allows the mean longitude,  $\bar{\lambda}$ , and the mean anomaly,  $M$ , of Mars to be calculated as functions of time. Next, Table 8.4 permits the equation of center,  $q$ , and the radial anomaly,  $\zeta$ , to be determined as functions of the mean anomaly. Finally, Table 8.5 allows the quantities  $\delta\theta_-$ ,  $\bar{\theta}$ , and  $\delta\theta_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ .

The procedure for using the tables is as follows:

1. Determine the fractional Julian day number,  $t$ , corresponding to the date and time at which the ecliptic longitude is to be calculated with the aid of Tables 3.1–3.3. Form  $\Delta t = t - t_0$ , where  $t_0 = 2\,451\,545.0$  is the epoch.
2. Calculate the ecliptic longitude,  $\lambda_S$ , and radial anomaly,  $\zeta_S$ , of the Sun using the procedure set out in Section 5.2.

3. Enter Table 8.3 with the digit for each power of 10 in  $\Delta t$  and take out the corresponding values of  $\Delta\bar{\lambda}$  and  $\Delta M$ . If  $\Delta t$  is negative then the corresponding values are also negative. The value of the mean longitude,  $\bar{\lambda}$ , is the sum of all the  $\Delta\bar{\lambda}$  values plus the value of  $\bar{\lambda}$  at the epoch. Likewise, the value of the mean anomaly,  $M$ , is the sum of all the  $\Delta M$  values plus the value of  $M$  at the epoch. Add as many multiples of  $360^\circ$  to  $\bar{\lambda}$  and  $M$  as is required to make them both fall in the range  $0^\circ$  to  $360^\circ$ . Round  $M$  to the nearest degree.
4. Enter Table 8.4 with the value of  $M$  and take out the corresponding value of the equation of center,  $q$ , and the radial anomaly,  $\zeta$ . It is necessary to interpolate if  $M$  is odd.
5. Form the epicyclic anomaly,  $\mu = \lambda_S - \bar{\lambda} - q$ . Add as many multiples of  $360^\circ$  to  $\mu$  as is required to make it fall in the range  $0^\circ$  to  $360^\circ$ . Round  $\mu$  to the nearest degree.
6. Enter Table 8.5 with the value of  $\mu$  and take out the corresponding values of  $\delta\theta_-$ ,  $\bar{\theta}$ , and  $\delta\theta_+$ . If  $\mu > 180^\circ$  then it is necessary to make use of the identities  $\delta\theta_\pm(360^\circ - \mu) = -\delta\theta_\pm(\mu)$  and  $\bar{\theta}(360^\circ - \mu) = -\bar{\theta}(\mu)$ .
7. Form  $z = (1 - \zeta)/(1 - \zeta_S)$ .
8. Obtain the values of  $\bar{z}$  and  $\delta z$  from Table 8.1. Form  $\xi = (\bar{z} - z)/\delta z$ .
9. Enter Table 8.2 with the value of  $\xi$  and take out the corresponding values of  $\Theta_-$  and  $\Theta_+$ . If  $\xi < 0$  then it is necessary to use the identities  $\Theta_+(\xi) = -\Theta_-(-\xi)$  and  $\Theta_-(\xi) = -\Theta_+(-\xi)$ .
10. Form the equation of the epicycle,  $\theta = \Theta_- \delta\theta_- + \bar{\theta} + \Theta_+ \delta\theta_+$ .
11. The ecliptic longitude,  $\lambda$ , is the sum of the mean longitude,  $\bar{\lambda}$ , the equation of center,  $q$ , and the equation of the epicycle,  $\theta$ . If necessary convert  $\lambda$  into an angle in the range  $0^\circ$  to  $360^\circ$ . The decimal fraction can be converted into arc minutes using Table 5.2. Round to the nearest arc minute. The final result can be written in terms of the signs of the zodiac using the table in Section 2.6.

Two examples of this procedure are given in the following section

### 8.3 Example martian ecliptic longitude calculations

*Example 1:* May 5, 2005 AD, 00:00 UT

From Section 5.2,  $t - t_0 = 1\,950.5$  JD,  $\lambda_S = 44.602^\circ$ ,  $M_S \simeq 120^\circ$ . Hence, it follows from Table 5.4 that  $\zeta_S(M_S) = -8.56 \times 10^{-3}$ . Making use of Table 8.3, we find:



| $t(\text{JD})$ | $\bar{\lambda}(^\circ)$ | $M(^\circ)$    |
|----------------|-------------------------|----------------|
| +1000          | 164.071                 | 164.021        |
| +900           | 111.664                 | 111.619        |
| +50            | 26.204                  | 26.201         |
| +5             | 0.262                   | 0.262          |
| Epoch          | 355.460                 | 19.388         |
|                | <u>657.661</u>          | <u>321.491</u> |
| Modulus        | 297.661                 | 321.491        |

Given that  $M \simeq 321^\circ$ , Table 8.4 yields

$$q(321^\circ) = -7.345^\circ, \quad \zeta(321^\circ) = 6.912 \times 10^{-2}.$$

Thus,

$$\mu = \lambda_S - \bar{\lambda} - q = 44.602 - 297.661 + 7.345 = 114.286 \simeq 114^\circ,$$

where we have rounded the epicyclic anomaly to the nearest degree. It follows from Table 8.5 that

$$\delta\theta_-(114^\circ) = 3.853^\circ, \quad \bar{\theta}(114^\circ) = 39.209^\circ, \quad \delta\theta_+(114^\circ) = 4.612^\circ.$$

Now,

$$z = (1 - \zeta)/(1 - \zeta_S) = (1 - 6.912 \times 10^{-2})/(1 + 8.56 \times 10^{-3}) = 0.9230.$$

However, from Table 8.1,  $\bar{z} = 1.00184$  and  $\delta z = 0.11014$ , so

$$\xi = (\bar{z} - z)/\delta z = (1.00184 - 0.9230)/0.11014 \simeq 0.72.$$

According to Table 8.2,

$$\Theta_-(0.72) = 0.101, \quad \Theta_+(0.72) = 0.619,$$

so

$$\begin{aligned} \theta &= \Theta_- \delta\theta_- + \bar{\theta} + \Theta_+ \delta\theta_+ \\ &= 0.101 \times 3.853 + 39.209 + 0.619 \times 4.612 \\ &= 42.453^\circ. \end{aligned}$$

Finally,

$$\lambda = \bar{\lambda} + q + \theta = 297.661 - 7.345 + 42.453 = 332.769 \simeq 332^\circ 46'.$$

Thus, the ecliptic longitude of Mars at 00:00 UT on May 5, 2005 AD was 2PI46.

*Example 2:* December 25, 1800 AD, 00:00 UT

From Section 5.2,  $t - t_0 = -72,690.5$  JD,  $\lambda_S = 273.055^\circ$ ,  $M_S \simeq 354^\circ$ . Hence, it follows from Table 5.4 that  $\zeta_S(M_S) = 1.662 \times 10^{-2}$ . Making use of Table 8.3, we find:

| $t(\text{JD})$ | $\bar{\lambda}(^\circ)$ | $M(^\circ)$    |
|----------------|-------------------------|----------------|
| -70,000        | -324.983                | -321.453       |
| -2,000         | -328.142                | -328.042       |
| -600           | -314.443                | -314.412       |
| -90            | -47.166                 | -47.162        |
| -.5            | -0.262                  | -0.262         |
| Epoch          | 355.460                 | 19.388         |
|                | <hr/> -659.536          | <hr/> -991.943 |
| Modulus        | 60.464                  | 88.057         |

Given that  $M \simeq 88^\circ$ , Table 8.4 yields

$$q(88^\circ) = 10.739^\circ, \quad \zeta(88^\circ) = -5.45 \times 10^{-3},$$

so

$$\mu = \lambda_S - \bar{\lambda} - q = 273.055 - 60.464 - 10.739 = 201.852 \simeq 202^\circ.$$

It follows from Table 8.5 that

$$\delta\theta_-(202^\circ) = -5.980^\circ, \quad \bar{\theta}(202^\circ) = -32.007^\circ, \quad \delta\theta_+(202^\circ) = -8.955^\circ.$$

Now,

$$z = (1 - \zeta)/(1 - \zeta_S) = (1 + 5.45 \times 10^{-3})/(1 - 1.662 \times 10^{-2}) = 1.02244,$$

so

$$\xi = (\bar{z} - z)/\delta z = (1.00184 - 1.02244)/0.11014 \simeq -0.19.$$

According to Table 8.2,

$$\Theta_-(-0.19) = -0.113, \quad \Theta_+(-0.19) = -0.077,$$

so

$$\begin{aligned} \theta &= \Theta_- \delta\theta_- + \bar{\theta} + \Theta_+ \delta\theta_+ \\ &= -0.113 \times 5.980 - 32.007 - 0.077 \times 8.955 \\ &= -30.642^\circ. \end{aligned}$$

Finally,

$$\lambda = \bar{\lambda} + q + \theta = 60.464 + 10.739 - 30.642 = 40.561 \simeq 40^\circ 34'.$$

Thus, the ecliptic longitude of Mars at 00:00 UT on December 25, 1800 AD was 10TA34.

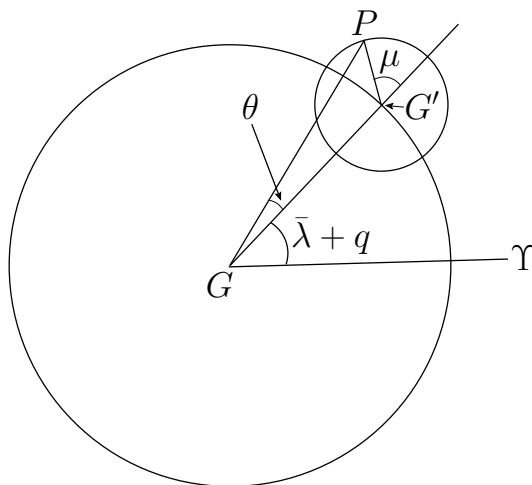


Figure 8.4: The geocentric orbit of a superior planet. Here,  $G$ ,  $G'$ ,  $P$ ,  $\mu$ ,  $\theta$ ,  $\bar{\lambda}$ ,  $q$ , and  $\Upsilon$  represent the Earth, guide-point, planet, epicyclic anomaly, equation of the epicycle, mean longitude, equation of center, and spring equinox, respectively. The view is from northern ecliptic pole. Both  $G'$  and  $P$  orbit counterclockwise.

#### 8.4 Conjunction, opposition, and station dates

Figure 8.4 shows the geocentric orbit of a superior planet. Recall that the vector  $G'P$  is always parallel to the vector connecting the Earth to the Sun. It follows that a so-called *conjunction*, at which the Sun lies directly between the planet and the Earth, occurs whenever the epicyclic anomaly,  $\mu$ , takes the value  $0^\circ$ . At a conjunction, the planet is farthest from the Earth, and has the same ecliptic longitude as the Sun, and is, therefore, invisible. Conversely, a so-called *opposition*, at which the Earth lies directly between the planet and the Sun, occurs whenever  $\mu = 180^\circ$ . At an opposition, the planet is closest to the Earth, and also directly opposite the Sun in the sky, and, therefore, at its brightest. Now, a superior planet rotates around the epicycle at a faster angular velocity than its guide-point rotates around the deferent. Moreover, both the planet and guide-point rotate in the same direction. It follows that the planet is traveling backward in the sky (relative to the direction of its mean motion) at opposition. This phenomenon is called *retrograde motion*. The period of retrograde motion begins and ends at *stations*; so-called because when the planet reaches them it appears to stand still in the sky (relative to the fixed stars) for a few days while it reverses direction.

Tables 8.3–8.5 can be used to determine the dates of the conjunctions, oppositions, and stations of Mars. Consider the first conjunction after the epoch (January 1, 2000 AD). We can estimate the time at which this event occurs by approximating the epicyclic anomaly as the

so-called *mean epicyclic anomaly*:

$$\begin{aligned}\mu &\simeq \bar{\mu} = \bar{\lambda}_S - \bar{\lambda} = \bar{\lambda}_{0S} - \bar{\lambda}_0 + (n_S - n)(t - t_0) \\ &= 284.998 + 0.46157617(t - t_0).\end{aligned}$$

We obtain

$$t \simeq t_0 + (360 - 284.998)/0.46157617 \simeq t_0 + 162 \text{ JD}.$$

A calculation of the epicyclic anomaly at this time, using Tables 8.3–8.5, yields  $\mu = -9.583^\circ$ . Now, the actual conjunction occurs when  $\mu = 0^\circ$ . Hence, our next estimate is

$$t \simeq t_0 + 162 + 9.583/0.46157617 \simeq t_0 + 183 \text{ JD}.$$

A calculation of the epicyclic anomaly at this time gives  $0.294^\circ$ . Thus, our final estimate is

$$t = t_0 + 183 - 0.294/0.46157617 = t_0 + 182.4 \text{ JD},$$

which corresponds to July 1, 2000 AD.

Consider the first opposition of Mars after the epoch. Our first estimate of the time at which this event takes place is

$$t \simeq t_0 + (540 - 284.998)/0.46157617 \simeq t_0 + 552 \text{ JD}.$$

A calculation of the epicyclic anomaly at this time yields  $\mu = 188.649^\circ$ . Now, the actual opposition occurs when  $\mu = 180^\circ$ . Hence, our second estimate is

$$t \simeq t_0 + 552 - 8.649/0.46157617 \simeq t_0 + 533 \text{ JD}.$$

A calculation of the epicyclic anomaly at this time gives  $181.455^\circ$ . Thus, our third estimate is

$$t \simeq t_0 + 533 - 1.455/0.46157617 \simeq t_0 + 530 \text{ JD}.$$

A calculation of the epicyclic anomaly at this time yields  $180.244^\circ$ . Hence, our final estimate is

$$t = t_0 + 530 - 0.244/0.46157617 = t_0 + 529.5 \text{ JD},$$

which corresponds to June 13, 2001 AD. Incidentally, it is clear from the previous analysis that the mean time period between successive conjunctions, or oppositions, of Mars is

$$360/0.46157617 = 779.9 \text{ JD},$$

which is equivalent to 2.14 years.

Let us now consider the stations of Mars. We can approximate the ecliptic longitude of a superior planet as

$$\lambda \simeq \bar{\lambda} + \bar{\theta}, \quad (8.30)$$

where

$$\bar{\theta} = \tan^{-1} \left( \frac{\sin \bar{\mu}}{\bar{a} + \cos \bar{\mu}} \right), \quad (8.31)$$

and  $\bar{a} = a \bar{z}$ . Note that  $d\bar{\lambda}/dt = n$  and  $d\bar{\mu}/dt = n_S - n$ . It follows that

$$\frac{d\lambda}{dt} \simeq n + \left( \frac{\bar{a} \cos \bar{\mu} + 1}{1 + 2 \bar{a} \cos \bar{\mu} + \bar{a}^2} \right) (n_S - n). \quad (8.32)$$

Now, a station corresponds to  $d\lambda/dt = 0$  (that is, a local maximum or minimum of  $\lambda$ ), which gives

$$\cos \bar{\mu} \simeq -\frac{(\bar{a}^2 + n_S/n)}{\bar{a} (1 + n_S/n)}. \quad (8.33)$$

For the case of Mars, we find that  $\bar{\mu} = 163.3^\circ$  or  $196.7^\circ$ . The first solution corresponds to the so-called *retrograde station*, at which the planet switches from direct to retrograde motion. The second solution corresponds to the so-called *direct station*, at which the planet switches from retrograde to direct motion. The mean time interval between a retrograde station and the following opposition, or between an opposition and the following direct station, is  $(180 - 163.3)/0.46157617 \simeq 36$  JD. Unfortunately, the only option for accurately determining the dates at which the stations occur is to calculate the ecliptic longitude of Mars over a range of days centered 36 days before and after its opposition.

Table 8.6 shows the conjunctions, oppositions, and stations of Mars for the years 2000–2020 AD, calculated using the previously described techniques.

### 8.5 Determination of ecliptic longitude of Jupiter

The ecliptic longitude of Jupiter can be determined with the aid of Tables 8.7–8.9. Table 8.7 allows the mean longitude,  $\lambda$ , and the mean anomaly,  $M$ , of Jupiter to be calculated as functions of time. Next, Table 8.8 permits the equation of center,  $q$ , and the radial anomaly,  $\zeta$ , to be determined as functions of the mean anomaly. Finally, Table 8.9 allows the quantities  $\delta\theta_-$ ,  $\bar{\theta}$ , and  $\delta\theta_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ . The procedure for using the tables is analogous to the previously described procedure for using the Mars tables. One example of this procedure is given in the following.

*Example:* May 5, 2005 AD, 00:00 UT

From before,  $t - t_0 = 1\,950.5$  JD,  $\lambda_S = 44.602^\circ$ ,  $M_S \simeq 120^\circ$ , and  $\zeta_S = -8.56 \times 10^{-3}$ . Making use of Table 8.7, we find:

| $t(\text{JD})$ | $\bar{\lambda}(^\circ)$ | $M(^\circ)$    |
|----------------|-------------------------|----------------|
| +1000          | 83.125                  | 83.081         |
| +900           | 74.813                  | 74.773         |
| +50            | 4.156                   | 4.154          |
| +5             | 0.042                   | 0.042          |
| Epoch          | 34.365                  | 19.348         |
|                | <u>196.501</u>          | <u>181.398</u> |
| Modulus        | 196.501                 | 181.398        |

Given that  $M \simeq 181^\circ$ , Table 8.8 yields

$$q(181^\circ) = -0.091^\circ, \quad \zeta(181^\circ) = -4.838 \times 10^{-2}.$$

Thus,

$$\mu = \lambda_S - \bar{\lambda} - q = 44.602 - 196.501 + 0.091 = -151.808 \simeq 208^\circ,$$

where we have rounded the epicyclic anomaly to the nearest degree. It follows from Table 8.9 that

$$\delta\theta_-(208^\circ) = -0.447^\circ, \quad \bar{\theta}(208^\circ) = -6.194^\circ, \quad \delta\theta_+(208^\circ) = -0.522^\circ.$$

Now,

$$z = (1 - \zeta)/(1 - \zeta_S) = (1 + 4.838 \times 10^{-2})/(1 + 8.56 \times 10^{-3}) = 1.0395.$$

However, from Table 8.1,  $\bar{z} = 1.00109$  and  $\delta z = 0.06512$ , so

$$\xi = (\bar{z} - z)/\delta z = (1.00109 - 1.0395)/0.06512 \simeq -0.59.$$

According to Table 8.2,

$$\Theta_-(-0.59) = -0.469, \quad \Theta_+(-0.59) = -0.121,$$

so

$$\begin{aligned} \theta &= \Theta_- \delta\theta_- + \bar{\theta} + \Theta_+ \delta\theta_+ \\ &= 0.469 \times 0.447 - 6.194 + 0.121 \times 0.522 \\ &= -5.921^\circ. \end{aligned}$$

Finally,

$$\lambda = \bar{\lambda} + q + \theta = 196.501 - 0.091 - 5.921 = 190.489 \simeq 190^\circ 29'.$$

Thus, the ecliptic longitude of Jupiter at 00:00 UT on May 5, 2005 AD was 10LI29.

The conjunctions, oppositions, and stations of Jupiter can be investigated using analogous methods to those employed earlier to examine the conjunctions, oppositions, and stations of Mars. We find that the mean time period between successive oppositions or conjunctions of Jupiter is 1.09 yr. Furthermore, on average, the retrograde and direct stations of Jupiter occur when the epicyclic anomaly takes the values  $\mu = 125.6^\circ$  and  $234.4^\circ$ , respectively. Finally, the mean time period between a retrograde station and the following opposition, or between the opposition and the following direct station, is 60 JD. The conjunctions, oppositions, and stations of Jupiter during the years 2000–2010 AD are shown in Table 8.10.

### 8.6 Determination of ecliptic longitude of Saturn

The ecliptic longitude of Saturn can be determined with the aid of Tables 8.11–8.13. Table 8.11 allows the mean longitude,  $\bar{\lambda}$ , and the mean anomaly,  $M$ , of Saturn to be calculated as functions of time. Next, Table 8.12 permits the equation of center,  $q$ , and the radial anomaly,  $\zeta$ , to be determined as functions of the mean anomaly. Finally, Table 8.13 allows the quantities  $\delta\theta_-$ ,  $\bar{\theta}$ , and  $\delta\theta_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ . The procedure for using the tables is analogous to the previously described procedure for using the Mars tables. One example of this procedure is given in the following.

*Example:* May 5, 2005 AD, 00:00 UT

From before,  $t - t_0 = 1\,950.5$  JD,  $\lambda_S = 44.602^\circ$ ,  $M_S \simeq 120^\circ$ , and  $\zeta_S = -8.56 \times 10^{-3}$ . Making use of Table 8.11, we find:

| $t(\text{JD})$ | $\bar{\lambda}(^\circ)$ | $M(^\circ)$    |
|----------------|-------------------------|----------------|
| +1000          | 33.508                  | 33.482         |
| +900           | 30.157                  | 30.133         |
| +50            | 1.675                   | 1.674          |
| +5             | 0.017                   | 0.017          |
| Epoch          | 50.059                  | 317.857        |
|                | <u>115.416</u>          | <u>383.163</u> |
| Modulus        | 115.416                 | 23.163         |

Given that  $M \simeq 23^\circ$ , Table 8.12 yields

$$q(23^\circ) = 2.561^\circ, \quad \zeta(23^\circ) = 4.913 \times 10^{-2}.$$

Thus,

$$\mu = \lambda_S - \bar{\lambda} - q = 44.602 - 115.416 - 2.561 = -73.375 \simeq 287^\circ,$$

where we have rounded the epicyclic anomaly to the nearest degree. It follows from Table 8.13 that

$$\delta\theta_-(287^\circ) = -0.353^\circ, \quad \bar{\theta}(287^\circ) = -5.551^\circ, \quad \delta\theta_+(287^\circ) = -0.405^\circ.$$

Now,

$$z = (1 - \zeta)/(1 - \zeta_S) = (1 - 4.913 \times 10^{-2})/(1 + 8.56 \times 10^{-3}) = 0.9428.$$

However, from Table 8.1,  $\bar{z} = 1.00118$  and  $\delta z = 0.07059$ , so

$$\xi = (\bar{z} - z)/\delta z = (1.00118 - 0.9428)/0.07059 \simeq 0.83.$$

According to Table 8.2,

$$\Theta_-(0.83) = 0.071, \quad \Theta_+(0.83) = 0.759,$$

so

$$\begin{aligned} \theta &= \Theta_- \delta\theta_- + \bar{\theta} + \Theta_+ \delta\theta_+ \\ &= -0.071 \times 0.353 - 5.551 - 0.759 \times 0.405 \\ &= -5.883^\circ. \end{aligned}$$

Finally,

$$\lambda = \bar{\lambda} + q + \theta = 115.416 + 2.561 - 5.883 = 112.094 \simeq 112^\circ 06'.$$

Thus, the ecliptic longitude of Saturn at 00:00 UT on May 5, 2005 AD was 22CN06.

The conjunctions, oppositions, and stations of Saturn can be investigated using analogous methods to those employed earlier to examine the conjunctions, oppositions, and stations of Mars. We find that the mean time period between successive oppositions or conjunctions of Saturn is 1.035 yr. Furthermore, on average, the retrograde and direct stations of Saturn occur when the epicyclic anomaly takes the values  $\mu = 114.5^\circ$  and  $245.5^\circ$ , respectively. Finally, the mean time period between a retrograde station and the following opposition, or between the opposition and the following direct station, is 69 JD. The conjunctions, oppositions, and stations of Saturn during the years 2000–2010 AD are shown in Table 8.14.

Incidentally, the information contained in the mean motion tables, 8.3, 8.7, and 8.11, as well as the deferential anomaly tables, 8.4, 8.8, and 8.12, is essentially equivalent to that contained in the “Tables of the mean longitudinal motion and anomalies of the five stars” (Κανόνες μέσων κινήσεων μήκους τε καὶ ἀνωμαλίας τῶν πέντε ἀστέρων) that appear in Section 4 of Book IX of the *Almagest*. Likewise, the information contained in the epicyclic anomaly tables, 8.5, 8.9, and 8.13, is equivalent to that contained in the “Tables of the longitudinal corrections of the five planets” (Κανόνες τῆς κατὰ μήκος τῶν πέντε πλανωμένων διευκρινήσεως) that appear in Section 11 of Book XI of the *Almagest*. The computation of the stations of the superior planets is discussed in Book XII of the *Almagest*.



8.7 *Tables*

| Planet  | $\bar{z}$ | $\delta z$ | $z_{\min}$ | $z_{\max}$ |
|---------|-----------|------------|------------|------------|
| Mercury | 1.04774   | 0.23216    | 0.81558    | 1.27990    |
| Venus   | 1.00016   | 0.02349    | 0.97667    | 1.02365    |
| Mars    | 1.00184   | 0.11014    | 0.89170    | 1.11198    |
| Jupiter | 1.00109   | 0.06512    | 0.93597    | 1.06602    |
| Saturn  | 1.00118   | 0.07059    | 0.93059    | 1.07177    |

Table 8.1: Constants associated with the epicycles of the inferior and superior planets.

| $\xi$ | $\Theta_-$ | $\Theta_+$ | $\xi$ | $\Theta_-$ | $\Theta_+$ | $\xi$ | $\Theta_-$ | $\Theta_+$ | $\xi$ | $\Theta_-$ | $\Theta_+$ |
|-------|------------|------------|-------|------------|------------|-------|------------|------------|-------|------------|------------|
| 0.00  | 0.000      | 0.000      | 0.25  | 0.094      | 0.156      | 0.50  | 0.125      | 0.375      | 0.75  | 0.094      | 0.656      |
| 0.01  | 0.005      | 0.005      | 0.26  | 0.096      | 0.164      | 0.51  | 0.125      | 0.385      | 0.76  | 0.091      | 0.669      |
| 0.02  | 0.010      | 0.010      | 0.27  | 0.099      | 0.171      | 0.52  | 0.125      | 0.395      | 0.77  | 0.089      | 0.681      |
| 0.03  | 0.015      | 0.015      | 0.28  | 0.101      | 0.179      | 0.53  | 0.125      | 0.405      | 0.78  | 0.086      | 0.694      |
| 0.04  | 0.019      | 0.021      | 0.29  | 0.103      | 0.187      | 0.54  | 0.124      | 0.416      | 0.79  | 0.083      | 0.707      |
| 0.05  | 0.024      | 0.026      | 0.30  | 0.105      | 0.195      | 0.55  | 0.124      | 0.426      | 0.80  | 0.080      | 0.720      |
| 0.06  | 0.028      | 0.032      | 0.31  | 0.107      | 0.203      | 0.56  | 0.123      | 0.437      | 0.81  | 0.077      | 0.733      |
| 0.07  | 0.033      | 0.037      | 0.32  | 0.109      | 0.211      | 0.57  | 0.123      | 0.447      | 0.82  | 0.074      | 0.746      |
| 0.08  | 0.037      | 0.043      | 0.33  | 0.111      | 0.219      | 0.58  | 0.122      | 0.458      | 0.83  | 0.071      | 0.759      |
| 0.09  | 0.041      | 0.049      | 0.34  | 0.112      | 0.228      | 0.59  | 0.121      | 0.469      | 0.84  | 0.067      | 0.773      |
| 0.10  | 0.045      | 0.055      | 0.35  | 0.114      | 0.236      | 0.60  | 0.120      | 0.480      | 0.85  | 0.064      | 0.786      |
| 0.11  | 0.049      | 0.061      | 0.36  | 0.115      | 0.245      | 0.61  | 0.119      | 0.491      | 0.86  | 0.060      | 0.800      |
| 0.12  | 0.053      | 0.067      | 0.37  | 0.117      | 0.253      | 0.62  | 0.118      | 0.502      | 0.87  | 0.057      | 0.813      |
| 0.13  | 0.057      | 0.073      | 0.38  | 0.118      | 0.262      | 0.63  | 0.117      | 0.513      | 0.88  | 0.053      | 0.827      |
| 0.14  | 0.060      | 0.080      | 0.39  | 0.119      | 0.271      | 0.64  | 0.115      | 0.525      | 0.89  | 0.049      | 0.841      |
| 0.15  | 0.064      | 0.086      | 0.40  | 0.120      | 0.280      | 0.65  | 0.114      | 0.536      | 0.90  | 0.045      | 0.855      |
| 0.16  | 0.067      | 0.093      | 0.41  | 0.121      | 0.289      | 0.66  | 0.112      | 0.548      | 0.91  | 0.041      | 0.869      |
| 0.17  | 0.071      | 0.099      | 0.42  | 0.122      | 0.298      | 0.67  | 0.111      | 0.559      | 0.92  | 0.037      | 0.883      |
| 0.18  | 0.074      | 0.106      | 0.43  | 0.123      | 0.307      | 0.68  | 0.109      | 0.571      | 0.93  | 0.033      | 0.897      |
| 0.19  | 0.077      | 0.113      | 0.44  | 0.123      | 0.317      | 0.69  | 0.107      | 0.583      | 0.94  | 0.028      | 0.912      |
| 0.20  | 0.080      | 0.120      | 0.45  | 0.124      | 0.326      | 0.70  | 0.105      | 0.595      | 0.95  | 0.024      | 0.926      |
| 0.21  | 0.083      | 0.127      | 0.46  | 0.124      | 0.336      | 0.71  | 0.103      | 0.607      | 0.96  | 0.019      | 0.941      |
| 0.22  | 0.086      | 0.134      | 0.47  | 0.125      | 0.345      | 0.72  | 0.101      | 0.619      | 0.97  | 0.015      | 0.955      |
| 0.23  | 0.089      | 0.141      | 0.48  | 0.125      | 0.355      | 0.73  | 0.099      | 0.631      | 0.98  | 0.010      | 0.970      |
| 0.24  | 0.091      | 0.149      | 0.49  | 0.125      | 0.365      | 0.74  | 0.096      | 0.644      | 0.99  | 0.005      | 0.985      |
| 0.25  | 0.094      | 0.156      | 0.50  | 0.125      | 0.375      | 0.75  | 0.094      | 0.656      | 1.00  | 0.000      | 1.000      |

Table 8.2: Epicyclic interpolation coefficients. Note that  $\Theta_{\pm}(\xi) = -\Theta_{\mp}(-\xi)$ .

| $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(^{\circ})$ | $\Delta M(^{\circ})$ | $\Delta \bar{F}(^{\circ})$ | $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(^{\circ})$ | $\Delta M(^{\circ})$ | $\Delta \bar{F}(^{\circ})$ |
|-----------------------|----------------------------------|----------------------|----------------------------|-----------------------|----------------------------------|----------------------|----------------------------|
| 10 000                | 200.712                          | 200.208              | 200.409                    | 1 000                 | 164.071                          | 164.021              | 164.041                    |
| 20 000                | 41.424                           | 40.415               | 40.819                     | 2 000                 | 328.142                          | 328.042              | 328.082                    |
| 30 000                | 242.135                          | 240.623              | 241.228                    | 3 000                 | 132.214                          | 132.062              | 132.123                    |
| 40 000                | 82.847                           | 80.830               | 81.638                     | 4 000                 | 296.285                          | 296.083              | 296.164                    |
| 50 000                | 283.559                          | 281.038              | 282.047                    | 5 000                 | 100.356                          | 100.104              | 100.205                    |
| 60 000                | 124.271                          | 121.246              | 122.456                    | 6 000                 | 264.427                          | 264.125              | 264.246                    |
| 70 000                | 324.983                          | 321.453              | 322.866                    | 7 000                 | 68.498                           | 68.145               | 68.287                     |
| 80 000                | 165.694                          | 161.661              | 163.275                    | 8 000                 | 232.569                          | 232.166              | 232.328                    |
| 90 000                | 6.406                            | 1.868                | 3.685                      | 9 000                 | 36.641                           | 36.187               | 36.368                     |
|                       |                                  |                      |                            |                       |                                  |                      |                            |
| 100                   | 52.407                           | 52.402               | 52.404                     | 10                    | 5.241                            | 5.240                | 5.240                      |
| 200                   | 104.814                          | 104.804              | 104.808                    | 20                    | 10.481                           | 10.480               | 10.481                     |
| 300                   | 157.221                          | 157.206              | 157.212                    | 30                    | 15.722                           | 15.721               | 15.721                     |
| 400                   | 209.628                          | 209.608              | 209.616                    | 40                    | 20.963                           | 20.961               | 20.962                     |
| 500                   | 262.036                          | 262.010              | 262.020                    | 50                    | 26.204                           | 26.201               | 26.202                     |
| 600                   | 314.443                          | 314.412              | 314.425                    | 60                    | 31.444                           | 31.441               | 31.442                     |
| 700                   | 6.850                            | 6.815                | 6.829                      | 70                    | 36.685                           | 36.681               | 36.683                     |
| 800                   | 59.257                           | 59.217               | 59.233                     | 80                    | 41.926                           | 41.922               | 41.923                     |
| 900                   | 111.664                          | 111.619              | 111.637                    | 90                    | 47.166                           | 47.162               | 47.164                     |
|                       |                                  |                      |                            |                       |                                  |                      |                            |
| 1                     | 0.524                            | 0.524                | 0.524                      | 0.1                   | 0.052                            | 0.052                | 0.052                      |
| 2                     | 1.048                            | 1.048                | 1.048                      | 0.2                   | 0.105                            | 0.105                | 0.105                      |
| 3                     | 1.572                            | 1.572                | 1.572                      | 0.3                   | 0.157                            | 0.157                | 0.157                      |
| 4                     | 2.096                            | 2.096                | 2.096                      | 0.4                   | 0.210                            | 0.210                | 0.210                      |
| 5                     | 2.620                            | 2.620                | 2.620                      | 0.5                   | 0.262                            | 0.262                | 0.262                      |
| 6                     | 3.144                            | 3.144                | 3.144                      | 0.6                   | 0.314                            | 0.314                | 0.314                      |
| 7                     | 3.668                            | 3.668                | 3.668                      | 0.7                   | 0.367                            | 0.367                | 0.367                      |
| 8                     | 4.193                            | 4.192                | 4.192                      | 0.8                   | 0.419                            | 0.419                | 0.419                      |
| 9                     | 4.717                            | 4.716                | 4.716                      | 0.9                   | 0.472                            | 0.472                | 0.472                      |

Table 8.3: Mean motion of Mars. Here,  $\Delta t = t - t_0$ ,  $\Delta \bar{\lambda} = \bar{\lambda} - \bar{\lambda}_0$ ,  $\Delta M = M - M_0$ , and  $\Delta \bar{F} = \bar{F} - \bar{F}_0$ . At epoch ( $t_0 = 2\,451\,545.0$  JD),  $\bar{\lambda}_0 = 355.460^{\circ}$ ,  $M_0 = 19.388^{\circ}$ , and  $\bar{F}_0 = 305.796^{\circ}$ .

| $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ |
|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|
| 0             | 0.000         | 9.339       | 90            | 10.702        | -0.872      | 180           | 0.000         | -9.339      | 270           | -10.702       | -0.872      |
| 2             | 0.417         | 9.333       | 92            | 10.652        | -1.197      | 182           | -0.330        | -9.335      | 272           | -10.739       | -0.545      |
| 4             | 0.833         | 9.312       | 94            | 10.589        | -1.519      | 184           | -0.660        | -9.321      | 274           | -10.763       | -0.217      |
| 6             | 1.249         | 9.279       | 96            | 10.514        | -1.839      | 186           | -0.989        | -9.298      | 276           | -10.773       | 0.114       |
| 8             | 1.662         | 9.232       | 98            | 10.426        | -2.155      | 188           | -1.317        | -9.265      | 278           | -10.770       | 0.444       |
| 10            | 2.072         | 9.171       | 100           | 10.326        | -2.468      | 190           | -1.645        | -9.224      | 280           | -10.753       | 0.776       |
| 12            | 2.479         | 9.098       | 102           | 10.214        | -2.776      | 192           | -1.971        | -9.173      | 282           | -10.722       | 1.107       |
| 14            | 2.882         | 9.011       | 104           | 10.091        | -3.081      | 194           | -2.296        | -9.113      | 284           | -10.678       | 1.438       |
| 16            | 3.281         | 8.911       | 106           | 9.957         | -3.380      | 196           | -2.619        | -9.044      | 286           | -10.619       | 1.768       |
| 18            | 3.674         | 8.799       | 108           | 9.811         | -3.675      | 198           | -2.940        | -8.966      | 288           | -10.546       | 2.097       |
| 20            | 4.062         | 8.674       | 110           | 9.655         | -3.964      | 200           | -3.259        | -8.878      | 290           | -10.458       | 2.424       |
| 22            | 4.443         | 8.537       | 112           | 9.489         | -4.248      | 202           | -3.575        | -8.782      | 292           | -10.357       | 2.749       |
| 24            | 4.817         | 8.388       | 114           | 9.313         | -4.527      | 204           | -3.889        | -8.676      | 294           | -10.241       | 3.071       |
| 26            | 5.184         | 8.227       | 116           | 9.127         | -4.799      | 206           | -4.199        | -8.562      | 296           | -10.111       | 3.389       |
| 28            | 5.542         | 8.054       | 118           | 8.932         | -5.065      | 208           | -4.506        | -8.438      | 298           | -9.967        | 3.705       |
| 30            | 5.892         | 7.870       | 120           | 8.727         | -5.324      | 210           | -4.810        | -8.306      | 300           | -9.809        | 4.016       |
| 32            | 6.233         | 7.675       | 122           | 8.514         | -5.576      | 212           | -5.110        | -8.165      | 302           | -9.637        | 4.322       |
| 34            | 6.564         | 7.470       | 124           | 8.293         | -5.822      | 214           | -5.405        | -8.015      | 304           | -9.452        | 4.623       |
| 36            | 6.885         | 7.254       | 126           | 8.064         | -6.060      | 216           | -5.696        | -7.857      | 306           | -9.252        | 4.919       |
| 38            | 7.195         | 7.029       | 128           | 7.827         | -6.292      | 218           | -5.983        | -7.690      | 308           | -9.040        | 5.208       |
| 40            | 7.494         | 6.794       | 130           | 7.583         | -6.515      | 220           | -6.264        | -7.515      | 310           | -8.814        | 5.491       |
| 42            | 7.782         | 6.550       | 132           | 7.332         | -6.731      | 222           | -6.540        | -7.331      | 312           | -8.575        | 5.768       |
| 44            | 8.059         | 6.297       | 134           | 7.074         | -6.939      | 224           | -6.810        | -7.139      | 314           | -8.323        | 6.036       |
| 46            | 8.323         | 6.036       | 136           | 6.810         | -7.139      | 226           | -7.074        | -6.939      | 316           | -8.059        | 6.297       |
| 48            | 8.575         | 5.768       | 138           | 6.540         | -7.331      | 228           | -7.332        | -6.731      | 318           | -7.782        | 6.550       |
| 50            | 8.814         | 5.491       | 140           | 6.264         | -7.515      | 230           | -7.583        | -6.515      | 320           | -7.494        | 6.794       |
| 52            | 9.040         | 5.208       | 142           | 5.983         | -7.690      | 232           | -7.827        | -6.292      | 322           | -7.195        | 7.029       |
| 54            | 9.252         | 4.919       | 144           | 5.696         | -7.857      | 234           | -8.064        | -6.060      | 324           | -6.885        | 7.254       |
| 56            | 9.452         | 4.623       | 146           | 5.405         | -8.015      | 236           | -8.293        | -5.822      | 326           | -6.564        | 7.470       |
| 58            | 9.637         | 4.322       | 148           | 5.110         | -8.165      | 238           | -8.514        | -5.576      | 328           | -6.233        | 7.675       |
| 60            | 9.809         | 4.016       | 150           | 4.810         | -8.306      | 240           | -8.727        | -5.324      | 330           | -5.892        | 7.870       |
| 62            | 9.967         | 3.705       | 152           | 4.506         | -8.438      | 242           | -8.932        | -5.065      | 332           | -5.542        | 8.054       |
| 64            | 10.111        | 3.389       | 154           | 4.199         | -8.562      | 244           | -9.127        | -4.799      | 334           | -5.184        | 8.227       |
| 66            | 10.241        | 3.071       | 156           | 3.889         | -8.676      | 246           | -9.313        | -4.527      | 336           | -4.817        | 8.388       |
| 68            | 10.357        | 2.749       | 158           | 3.575         | -8.782      | 248           | -9.489        | -4.248      | 338           | -4.443        | 8.537       |
| 70            | 10.458        | 2.424       | 160           | 3.259         | -8.878      | 250           | -9.655        | -3.964      | 340           | -4.062        | 8.674       |
| 72            | 10.546        | 2.097       | 162           | 2.940         | -8.966      | 252           | -9.811        | -3.675      | 342           | -3.674        | 8.799       |
| 74            | 10.619        | 1.768       | 164           | 2.619         | -9.044      | 254           | -9.957        | -3.380      | 344           | -3.281        | 8.911       |
| 76            | 10.678        | 1.438       | 166           | 2.296         | -9.113      | 256           | -10.091       | -3.081      | 346           | -2.882        | 9.011       |
| 78            | 10.722        | 1.107       | 168           | 1.971         | -9.173      | 258           | -10.214       | -2.776      | 348           | -2.479        | 9.098       |
| 80            | 10.753        | 0.776       | 170           | 1.645         | -9.224      | 260           | -10.326       | -2.468      | 350           | -2.072        | 9.171       |
| 82            | 10.770        | 0.444       | 172           | 1.317         | -9.265      | 262           | -10.426       | -2.155      | 352           | -1.662        | 9.232       |
| 84            | 10.773        | 0.114       | 174           | 0.989         | -9.298      | 264           | -10.514       | -1.839      | 354           | -1.249        | 9.279       |
| 86            | 10.763        | -0.217      | 176           | 0.660         | -9.321      | 266           | -10.589       | -1.519      | 356           | -0.833        | 9.312       |
| 88            | 10.739        | -0.545      | 178           | 0.330         | -9.335      | 268           | -10.652       | -1.197      | 358           | -0.417        | 9.333       |
| 90            | 10.702        | -0.872      | 180           | 0.000         | -9.339      | 270           | -10.702       | -0.872      | 360           | -0.000        | 9.339       |

Table 8.4: Differential anomalies of Mars.

| $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ |
|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|
| 0     | 0.000            | 0.000          | 0.000            | 45    | 1.159            | 17.566         | 1.329            | 90    | 2.679            | 33.228         | 3.125            | 135   | 5.180            | 40.793         | 6.547            |
| 1     | 0.025            | 0.396          | 0.028            | 46    | 1.187            | 17.945         | 1.362            | 91    | 2.721            | 33.527         | 3.176            | 136   | 5.246            | 40.716         | 6.658            |
| 2     | 0.049            | 0.792          | 0.056            | 47    | 1.216            | 18.322         | 1.394            | 92    | 2.764            | 33.822         | 3.228            | 137   | 5.312            | 40.619         | 6.771            |
| 3     | 0.074            | 1.187          | 0.084            | 48    | 1.244            | 18.699         | 1.427            | 93    | 2.807            | 34.114         | 3.281            | 138   | 5.378            | 40.503         | 6.885            |
| 4     | 0.099            | 1.583          | 0.113            | 49    | 1.273            | 19.075         | 1.461            | 94    | 2.851            | 34.403         | 3.335            | 139   | 5.442            | 40.366         | 7.001            |
| 5     | 0.123            | 1.979          | 0.141            | 50    | 1.302            | 19.450         | 1.494            | 95    | 2.895            | 34.688         | 3.390            | 140   | 5.506            | 40.206         | 7.118            |
| 6     | 0.148            | 2.374          | 0.169            | 51    | 1.331            | 19.824         | 1.528            | 96    | 2.940            | 34.969         | 3.445            | 141   | 5.568            | 40.024         | 7.235            |
| 7     | 0.173            | 2.770          | 0.197            | 52    | 1.360            | 20.196         | 1.562            | 97    | 2.985            | 35.246         | 3.501            | 142   | 5.628            | 39.817         | 7.354            |
| 8     | 0.197            | 3.165          | 0.226            | 53    | 1.390            | 20.568         | 1.596            | 98    | 3.031            | 35.519         | 3.558            | 143   | 5.687            | 39.584         | 7.474            |
| 9     | 0.222            | 3.560          | 0.254            | 54    | 1.419            | 20.939         | 1.630            | 99    | 3.078            | 35.788         | 3.616            | 144   | 5.744            | 39.325         | 7.594            |
| 10    | 0.247            | 3.955          | 0.282            | 55    | 1.449            | 21.309         | 1.665            | 100   | 3.125            | 36.053         | 3.675            | 145   | 5.797            | 39.038         | 7.714            |
| 11    | 0.272            | 4.350          | 0.311            | 56    | 1.479            | 21.677         | 1.700            | 101   | 3.173            | 36.313         | 3.735            | 146   | 5.848            | 38.721         | 7.833            |
| 12    | 0.297            | 4.745          | 0.339            | 57    | 1.510            | 22.045         | 1.735            | 102   | 3.221            | 36.568         | 3.796            | 147   | 5.895            | 38.373         | 7.952            |
| 13    | 0.322            | 5.140          | 0.368            | 58    | 1.540            | 22.411         | 1.771            | 103   | 3.270            | 36.819         | 3.857            | 148   | 5.938            | 37.992         | 8.069            |
| 14    | 0.347            | 5.534          | 0.396            | 59    | 1.571            | 22.776         | 1.807            | 104   | 3.320            | 37.065         | 3.920            | 149   | 5.976            | 37.577         | 8.184            |
| 15    | 0.372            | 5.928          | 0.425            | 60    | 1.602            | 23.139         | 1.843            | 105   | 3.370            | 37.306         | 3.984            | 150   | 6.009            | 37.126         | 8.297            |
| 16    | 0.397            | 6.322          | 0.453            | 61    | 1.633            | 23.502         | 1.879            | 106   | 3.421            | 37.541         | 4.049            | 151   | 6.036            | 36.638         | 8.405            |
| 17    | 0.422            | 6.716          | 0.482            | 62    | 1.665            | 23.863         | 1.916            | 107   | 3.472            | 37.771         | 4.115            | 152   | 6.056            | 36.110         | 8.509            |
| 18    | 0.447            | 7.110          | 0.511            | 63    | 1.696            | 24.222         | 1.953            | 108   | 3.525            | 37.996         | 4.182            | 153   | 6.069            | 35.541         | 8.607            |
| 19    | 0.472            | 7.503          | 0.540            | 64    | 1.728            | 24.581         | 1.991            | 109   | 3.578            | 38.214         | 4.251            | 154   | 6.072            | 34.929         | 8.698            |
| 20    | 0.497            | 7.896          | 0.568            | 65    | 1.761            | 24.938         | 2.029            | 110   | 3.631            | 38.426         | 4.321            | 155   | 6.066            | 34.271         | 8.780            |
| 21    | 0.523            | 8.288          | 0.597            | 66    | 1.793            | 25.293         | 2.067            | 111   | 3.686            | 38.632         | 4.391            | 156   | 6.050            | 33.567         | 8.852            |
| 22    | 0.548            | 8.680          | 0.626            | 67    | 1.826            | 25.647         | 2.106            | 112   | 3.741            | 38.831         | 4.464            | 157   | 6.022            | 32.813         | 8.911            |
| 23    | 0.573            | 9.072          | 0.656            | 68    | 1.859            | 25.999         | 2.145            | 113   | 3.796            | 39.023         | 4.537            | 158   | 5.980            | 32.007         | 8.955            |
| 24    | 0.599            | 9.464          | 0.685            | 69    | 1.893            | 26.349         | 2.184            | 114   | 3.853            | 39.209         | 4.612            | 159   | 5.925            | 31.149         | 8.982            |
| 25    | 0.625            | 9.855          | 0.714            | 70    | 1.927            | 26.698         | 2.224            | 115   | 3.910            | 39.386         | 4.688            | 160   | 5.854            | 30.235         | 8.988            |
| 26    | 0.650            | 10.246         | 0.744            | 71    | 1.961            | 27.045         | 2.264            | 116   | 3.968            | 39.556         | 4.765            | 161   | 5.766            | 29.265         | 8.972            |
| 27    | 0.676            | 10.636         | 0.773            | 72    | 1.995            | 27.390         | 2.305            | 117   | 4.026            | 39.718         | 4.844            | 162   | 5.660            | 28.236         | 8.929            |
| 28    | 0.702            | 11.026         | 0.803            | 73    | 2.030            | 27.734         | 2.346            | 118   | 4.086            | 39.872         | 4.925            | 163   | 5.535            | 27.146         | 8.855            |
| 29    | 0.728            | 11.415         | 0.833            | 74    | 2.065            | 28.075         | 2.387            | 119   | 4.146            | 40.017         | 5.007            | 164   | 5.389            | 25.996         | 8.747            |
| 30    | 0.754            | 11.804         | 0.863            | 75    | 2.100            | 28.415         | 2.429            | 120   | 4.206            | 40.153         | 5.091            | 165   | 5.221            | 24.783         | 8.601            |
| 31    | 0.780            | 12.192         | 0.893            | 76    | 2.136            | 28.753         | 2.472            | 121   | 4.268            | 40.279         | 5.176            | 166   | 5.030            | 23.506         | 8.411            |
| 32    | 0.806            | 12.580         | 0.923            | 77    | 2.172            | 29.088         | 2.515            | 122   | 4.330            | 40.396         | 5.262            | 167   | 4.815            | 22.167         | 8.174            |
| 33    | 0.833            | 12.968         | 0.953            | 78    | 2.209            | 29.421         | 2.558            | 123   | 4.393            | 40.502         | 5.351            | 168   | 4.576            | 20.764         | 7.886            |
| 34    | 0.859            | 13.354         | 0.984            | 79    | 2.246            | 29.752         | 2.602            | 124   | 4.456            | 40.598         | 5.441            | 169   | 4.311            | 19.299         | 7.541            |
| 35    | 0.886            | 13.741         | 1.014            | 80    | 2.283            | 30.081         | 2.647            | 125   | 4.520            | 40.683         | 5.533            | 170   | 4.021            | 17.774         | 7.138            |
| 36    | 0.913            | 14.126         | 1.045            | 81    | 2.321            | 30.408         | 2.692            | 126   | 4.584            | 40.756         | 5.626            | 171   | 3.707            | 16.189         | 6.673            |
| 37    | 0.939            | 14.511         | 1.076            | 82    | 2.359            | 30.732         | 2.737            | 127   | 4.649            | 40.816         | 5.721            | 172   | 3.368            | 14.549         | 6.145            |
| 38    | 0.966            | 14.896         | 1.107            | 83    | 2.397            | 31.054         | 2.784            | 128   | 4.715            | 40.864         | 5.818            | 173   | 3.005            | 12.857         | 5.553            |
| 39    | 0.994            | 15.279         | 1.138            | 84    | 2.436            | 31.373         | 2.830            | 129   | 4.780            | 40.899         | 5.917            | 174   | 2.621            | 11.116         | 4.899            |
| 40    | 1.021            | 15.662         | 1.169            | 85    | 2.475            | 31.689         | 2.878            | 130   | 4.847            | 40.920         | 6.018            | 175   | 2.217            | 9.333          | 4.186            |
| 41    | 1.048            | 16.045         | 1.201            | 86    | 2.515            | 32.003         | 2.926            | 131   | 4.913            | 40.926         | 6.120            | 176   | 1.796            | 7.513          | 3.420            |
| 42    | 1.076            | 16.426         | 1.233            | 87    | 2.555            | 32.314         | 2.975            | 132   | 4.980            | 40.918         | 6.224            | 177   | 1.360            | 5.662          | 2.608            |
| 43    | 1.103            | 16.807         | 1.265            | 88    | 2.596            | 32.622         | 3.024            | 133   | 5.047            | 40.893         | 6.330            | 178   | 0.913            | 3.788          | 1.760            |
| 44    | 1.131            | 17.187         | 1.297            | 89    | 2.637            | 32.927         | 3.074            | 134   | 5.113            | 40.851         | 6.438            | 179   | 0.458            | 1.898          | 0.886            |
| 45    | 1.159            | 17.566         | 1.329            | 90    | 2.679            | 33.228         | 3.125            | 135   | 5.180            | 40.793         | 6.547            | 180   | 0.000            | 0.000          | 0.000            |

Table 8.5: Epicyclic anomalies of Mars. All quantities are in degrees. Note that  $\bar{\theta}(360^\circ - \mu) = -\bar{\theta}(\mu)$ , and  $\delta\theta_\pm(360^\circ - \mu) = -\delta\theta_\pm(\mu)$ .

| Event       | Date       | $\lambda$ | Event       | Date       | $\lambda$ |
|-------------|------------|-----------|-------------|------------|-----------|
| Conjunction | 01/07/2000 | 10CN13    | Conjunction | 04/02/2011 | 15AQ42    |
| Station (R) | 12/05/2001 | 29SG00    | Station (R) | 24/01/2012 | 23VI01    |
| Opposition  | 13/06/2001 | 22SG44    | Opposition  | 03/03/2012 | 13VI42    |
| Station (D) | 19/07/2001 | 15SG02    | Station (D) | 14/04/2012 | 03VI51    |
| Conjunction | 10/08/2002 | 18LE05    | Conjunction | 17/04/2013 | 28AR06    |
| Station (R) | 29/07/2003 | 10PI20    | Station (R) | 01/03/2014 | 27LI31    |
| Opposition  | 28/08/2003 | 05PI03    | Opposition  | 08/04/2014 | 19LI00    |
| Station (D) | 27/09/2003 | 29AQ55    | Station (D) | 20/05/2014 | 09LI04    |
| Conjunction | 15/09/2004 | 23VI06    | Conjunction | 14/06/2015 | 23GE28    |
| Station (R) | 02/10/2005 | 23TA31    | Station (R) | 17/04/2016 | 08SG45    |
| Opposition  | 07/11/2005 | 15TA06    | Opposition  | 22/05/2016 | 01SG43    |
| Station (D) | 09/12/2005 | 08TA24    | Station (D) | 29/06/2016 | 22SC55    |
| Conjunction | 23/10/2006 | 29LI44    | Conjunction | 27/07/2017 | 04LE11    |
| Station (R) | 15/11/2007 | 12CN36    | Station (R) | 27/06/2018 | 09AQ35    |
| Opposition  | 24/12/2007 | 02CN45    | Opposition  | 27/07/2018 | 04AQ22    |
| Station (D) | 31/01/2008 | 24GE15    | Station (D) | 27/08/2018 | 28CP43    |
| Conjunction | 05/12/2008 | 14SG08    | Conjunction | 02/09/2019 | 09VI43    |
| Station (R) | 20/12/2009 | 19LE35    | Station (R) | 10/09/2020 | 28AR08    |
| Opposition  | 29/01/2010 | 09LE45    | Opposition  | 13/10/2020 | 20AR59    |
| Station (D) | 10/03/2010 | 00LE20    | Station (D) | 13/11/2020 | 15AR05    |

Table 8.6: The conjunctions, oppositions, and stations of Mars during the years 2000–2020 AD. (R) indicates a retrograde station, and (D) a direct station.

| $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(\circ)$ | $\Delta M(\circ)$ | $\Delta \bar{F}(\circ)$ | $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(\circ)$ | $\Delta M(\circ)$ | $\Delta \bar{F}(\circ)$ |
|-----------------------|-------------------------------|-------------------|-------------------------|-----------------------|-------------------------------|-------------------|-------------------------|
| 10 000                | 111.251                       | 110.810           | 110.812                 | 1 000                 | 83.125                        | 83.081            | 83.081                  |
| 20 000                | 222.501                       | 221.620           | 221.624                 | 2 000                 | 166.250                       | 166.162           | 166.162                 |
| 30 000                | 333.752                       | 332.430           | 332.437                 | 3 000                 | 249.375                       | 249.243           | 249.244                 |
| 40 000                | 85.003                        | 83.240            | 83.249                  | 4 000                 | 332.500                       | 332.324           | 332.325                 |
| 50 000                | 196.253                       | 194.050           | 194.061                 | 5 000                 | 55.625                        | 55.405            | 55.406                  |
| 60 000                | 307.504                       | 304.860           | 304.873                 | 6 000                 | 138.750                       | 138.486           | 138.487                 |
| 70 000                | 58.755                        | 55.670            | 55.685                  | 7 000                 | 221.875                       | 221.567           | 221.569                 |
| 80 000                | 170.006                       | 166.480           | 166.498                 | 8 000                 | 305.001                       | 304.648           | 304.650                 |
| 90 000                | 281.256                       | 277.290           | 277.310                 | 9 000                 | 28.126                        | 27.729            | 27.731                  |
|                       |                               |                   |                         |                       |                               |                   |                         |
| 100                   | 8.313                         | 8.308             | 8.308                   | 10                    | 0.831                         | 0.831             | 0.831                   |
| 200                   | 16.625                        | 16.616            | 16.616                  | 20                    | 1.663                         | 1.662             | 1.662                   |
| 300                   | 24.938                        | 24.924            | 24.924                  | 30                    | 2.494                         | 2.492             | 2.492                   |
| 400                   | 33.250                        | 33.232            | 33.232                  | 40                    | 3.325                         | 3.323             | 3.323                   |
| 500                   | 41.563                        | 41.541            | 41.541                  | 50                    | 4.156                         | 4.154             | 4.154                   |
| 600                   | 49.875                        | 49.849            | 49.849                  | 60                    | 4.988                         | 4.985             | 4.985                   |
| 700                   | 58.188                        | 58.157            | 58.157                  | 70                    | 5.819                         | 5.816             | 5.816                   |
| 800                   | 66.500                        | 66.465            | 66.465                  | 80                    | 6.650                         | 6.646             | 6.646                   |
| 900                   | 74.813                        | 74.773            | 74.773                  | 90                    | 7.481                         | 7.477             | 7.477                   |
|                       |                               |                   |                         |                       |                               |                   |                         |
| 1                     | 0.083                         | 0.083             | 0.083                   | 0.1                   | 0.008                         | 0.008             | 0.008                   |
| 2                     | 0.166                         | 0.166             | 0.166                   | 0.2                   | 0.017                         | 0.017             | 0.017                   |
| 3                     | 0.249                         | 0.249             | 0.249                   | 0.3                   | 0.025                         | 0.025             | 0.025                   |
| 4                     | 0.333                         | 0.332             | 0.332                   | 0.4                   | 0.033                         | 0.033             | 0.033                   |
| 5                     | 0.416                         | 0.415             | 0.415                   | 0.5                   | 0.042                         | 0.042             | 0.042                   |
| 6                     | 0.499                         | 0.498             | 0.498                   | 0.6                   | 0.050                         | 0.050             | 0.050                   |
| 7                     | 0.582                         | 0.582             | 0.582                   | 0.7                   | 0.058                         | 0.058             | 0.058                   |
| 8                     | 0.665                         | 0.665             | 0.665                   | 0.8                   | 0.067                         | 0.066             | 0.066                   |
| 9                     | 0.748                         | 0.748             | 0.748                   | 0.9                   | 0.075                         | 0.075             | 0.075                   |

Table 8.7: Mean motion of Jupiter. Here,  $\Delta t = t - t_0$ ,  $\Delta \bar{\lambda} = \bar{\lambda} - \bar{\lambda}_0$ ,  $\Delta M = M - M_0$ , and  $\Delta \bar{F} = \bar{F} - \bar{F}_0$ . At epoch ( $t_0 = 2\,451\,545.0$  JD),  $\bar{\lambda}_0 = 34.365^\circ$ ,  $M_0 = 19.348^\circ$ , and  $\bar{F}_0 = 293.660^\circ$ .

| $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ |
|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|
| 0             | 0.000         | 4.839       | 90            | 5.545         | -0.234      | 180           | 0.000         | -4.839      | 270           | -5.545        | -0.234      |
| 2             | 0.205         | 4.835       | 92            | 5.530         | -0.403      | 182           | -0.182        | -4.836      | 272           | -5.553        | -0.065      |
| 4             | 0.410         | 4.826       | 94            | 5.508         | -0.571      | 184           | -0.363        | -4.828      | 274           | -5.554        | 0.105       |
| 6             | 0.614         | 4.810       | 96            | 5.479         | -0.737      | 186           | -0.545        | -4.815      | 276           | -5.549        | 0.274       |
| 8             | 0.818         | 4.787       | 98            | 5.444         | -0.903      | 188           | -0.725        | -4.796      | 278           | -5.537        | 0.444       |
| 10            | 1.020         | 4.758       | 100           | 5.403         | -1.067      | 190           | -0.905        | -4.772      | 280           | -5.518        | 0.613       |
| 12            | 1.221         | 4.723       | 102           | 5.355         | -1.230      | 192           | -1.085        | -4.743      | 282           | -5.492        | 0.782       |
| 14            | 1.420         | 4.681       | 104           | 5.301         | -1.391      | 194           | -1.263        | -4.709      | 284           | -5.459        | 0.950       |
| 16            | 1.617         | 4.633       | 106           | 5.241         | -1.550      | 196           | -1.439        | -4.669      | 286           | -5.419        | 1.117       |
| 18            | 1.812         | 4.579       | 108           | 5.175         | -1.707      | 198           | -1.615        | -4.624      | 288           | -5.372        | 1.283       |
| 20            | 2.004         | 4.519       | 110           | 5.102         | -1.862      | 200           | -1.789        | -4.574      | 290           | -5.318        | 1.448       |
| 22            | 2.194         | 4.453       | 112           | 5.024         | -2.014      | 202           | -1.961        | -4.519      | 292           | -5.257        | 1.611       |
| 24            | 2.380         | 4.382       | 114           | 4.941         | -2.163      | 204           | -2.131        | -4.459      | 294           | -5.190        | 1.773       |
| 26            | 2.563         | 4.304       | 116           | 4.851         | -2.310      | 206           | -2.298        | -4.394      | 296           | -5.116        | 1.932       |
| 28            | 2.742         | 4.221       | 118           | 4.757         | -2.454      | 208           | -2.464        | -4.324      | 298           | -5.035        | 2.089       |
| 30            | 2.918         | 4.132       | 120           | 4.657         | -2.595      | 210           | -2.627        | -4.249      | 300           | -4.947        | 2.244       |
| 32            | 3.089         | 4.038       | 122           | 4.551         | -2.732      | 212           | -2.787        | -4.169      | 302           | -4.853        | 2.396       |
| 34            | 3.256         | 3.938       | 124           | 4.441         | -2.867      | 214           | -2.945        | -4.085      | 304           | -4.752        | 2.545       |
| 36            | 3.419         | 3.834       | 126           | 4.326         | -2.997      | 216           | -3.100        | -3.995      | 306           | -4.645        | 2.691       |
| 38            | 3.576         | 3.724       | 128           | 4.207         | -3.124      | 218           | -3.251        | -3.902      | 308           | -4.532        | 2.834       |
| 40            | 3.729         | 3.610       | 130           | 4.082         | -3.248      | 220           | -3.399        | -3.803      | 310           | -4.413        | 2.973       |
| 42            | 3.877         | 3.491       | 132           | 3.954         | -3.367      | 222           | -3.543        | -3.701      | 312           | -4.287        | 3.108       |
| 44            | 4.019         | 3.368       | 134           | 3.821         | -3.482      | 224           | -3.684        | -3.594      | 314           | -4.156        | 3.240       |
| 46            | 4.156         | 3.240       | 136           | 3.684         | -3.594      | 226           | -3.821        | -3.482      | 316           | -4.019        | 3.368       |
| 48            | 4.287         | 3.108       | 138           | 3.543         | -3.701      | 228           | -3.954        | -3.367      | 318           | -3.877        | 3.491       |
| 50            | 4.413         | 2.973       | 140           | 3.399         | -3.803      | 230           | -4.082        | -3.248      | 320           | -3.729        | 3.610       |
| 52            | 4.532         | 2.834       | 142           | 3.251         | -3.902      | 232           | -4.207        | -3.124      | 322           | -3.576        | 3.724       |
| 54            | 4.645         | 2.691       | 144           | 3.100         | -3.995      | 234           | -4.326        | -2.997      | 324           | -3.419        | 3.834       |
| 56            | 4.752         | 2.545       | 146           | 2.945         | -4.085      | 236           | -4.441        | -2.867      | 326           | -3.256        | 3.938       |
| 58            | 4.853         | 2.396       | 148           | 2.787         | -4.169      | 238           | -4.551        | -2.732      | 328           | -3.089        | 4.038       |
| 60            | 4.947         | 2.244       | 150           | 2.627         | -4.249      | 240           | -4.657        | -2.595      | 330           | -2.918        | 4.132       |
| 62            | 5.035         | 2.089       | 152           | 2.464         | -4.324      | 242           | -4.757        | -2.454      | 332           | -2.742        | 4.221       |
| 64            | 5.116         | 1.932       | 154           | 2.298         | -4.394      | 244           | -4.851        | -2.310      | 334           | -2.563        | 4.304       |
| 66            | 5.190         | 1.773       | 156           | 2.131         | -4.459      | 246           | -4.941        | -2.163      | 336           | -2.380        | 4.382       |
| 68            | 5.257         | 1.611       | 158           | 1.961         | -4.519      | 248           | -5.024        | -2.014      | 338           | -2.194        | 4.453       |
| 70            | 5.318         | 1.448       | 160           | 1.789         | -4.574      | 250           | -5.102        | -1.862      | 340           | -2.004        | 4.519       |
| 72            | 5.372         | 1.283       | 162           | 1.615         | -4.624      | 252           | -5.175        | -1.707      | 342           | -1.812        | 4.579       |
| 74            | 5.419         | 1.117       | 164           | 1.439         | -4.669      | 254           | -5.241        | -1.550      | 344           | -1.617        | 4.633       |
| 76            | 5.459         | 0.950       | 166           | 1.263         | -4.709      | 256           | -5.301        | -1.391      | 346           | -1.420        | 4.681       |
| 78            | 5.492         | 0.782       | 168           | 1.085         | -4.743      | 258           | -5.355        | -1.230      | 348           | -1.221        | 4.723       |
| 80            | 5.518         | 0.613       | 170           | 0.905         | -4.772      | 260           | -5.403        | -1.067      | 350           | -1.020        | 4.758       |
| 82            | 5.537         | 0.444       | 172           | 0.725         | -4.796      | 262           | -5.444        | -0.903      | 352           | -0.818        | 4.787       |
| 84            | 5.549         | 0.274       | 174           | 0.545         | -4.815      | 264           | -5.479        | -0.737      | 354           | -0.614        | 4.810       |
| 86            | 5.554         | 0.105       | 176           | 0.363         | -4.828      | 266           | -5.508        | -0.571      | 356           | -0.410        | 4.826       |
| 88            | 5.553         | -0.065      | 178           | 0.182         | -4.836      | 268           | -5.530        | -0.403      | 358           | -0.205        | 4.835       |
| 90            | 5.545         | -0.234      | 180           | 0.000         | -4.839      | 270           | -5.545        | -0.234      | 360           | -0.000        | 4.839       |

Table 8.8: Differential anomalies of Jupiter.



| $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ |
|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|
| 0     | 0.000            | 0.000          | 0.000            | 45    | 0.366            | 6.816          | 0.410            | 90    | 0.649            | 10.868         | 0.736            | 135   | 0.616            | 8.927          | 0.713            |
| 1     | 0.008            | 0.161          | 0.009            | 46    | 0.374            | 6.948          | 0.418            | 91    | 0.653            | 10.902         | 0.741            | 136   | 0.609            | 8.796          | 0.705            |
| 2     | 0.017            | 0.322          | 0.019            | 47    | 0.381            | 7.077          | 0.427            | 92    | 0.657            | 10.933         | 0.746            | 137   | 0.601            | 8.661          | 0.697            |
| 3     | 0.025            | 0.483          | 0.028            | 48    | 0.389            | 7.206          | 0.436            | 93    | 0.661            | 10.961         | 0.750            | 138   | 0.593            | 8.522          | 0.688            |
| 4     | 0.033            | 0.644          | 0.037            | 49    | 0.396            | 7.333          | 0.444            | 94    | 0.664            | 10.986         | 0.755            | 139   | 0.585            | 8.380          | 0.679            |
| 5     | 0.042            | 0.805          | 0.046            | 50    | 0.404            | 7.459          | 0.453            | 95    | 0.668            | 11.008         | 0.759            | 140   | 0.576            | 8.233          | 0.669            |
| 6     | 0.050            | 0.965          | 0.056            | 51    | 0.411            | 7.583          | 0.461            | 96    | 0.671            | 11.026         | 0.763            | 141   | 0.567            | 8.083          | 0.659            |
| 7     | 0.058            | 1.126          | 0.065            | 52    | 0.419            | 7.705          | 0.470            | 97    | 0.674            | 11.041         | 0.766            | 142   | 0.558            | 7.929          | 0.649            |
| 8     | 0.067            | 1.286          | 0.074            | 53    | 0.426            | 7.826          | 0.478            | 98    | 0.677            | 11.053         | 0.770            | 143   | 0.548            | 7.771          | 0.638            |
| 9     | 0.075            | 1.446          | 0.084            | 54    | 0.434            | 7.946          | 0.486            | 99    | 0.680            | 11.062         | 0.773            | 144   | 0.538            | 7.610          | 0.627            |
| 10    | 0.083            | 1.606          | 0.093            | 55    | 0.441            | 8.063          | 0.495            | 100   | 0.682            | 11.067         | 0.777            | 145   | 0.528            | 7.445          | 0.615            |
| 11    | 0.092            | 1.766          | 0.102            | 56    | 0.448            | 8.180          | 0.503            | 101   | 0.684            | 11.069         | 0.780            | 146   | 0.517            | 7.276          | 0.603            |
| 12    | 0.100            | 1.925          | 0.111            | 57    | 0.455            | 8.294          | 0.511            | 102   | 0.686            | 11.068         | 0.782            | 147   | 0.506            | 7.104          | 0.590            |
| 13    | 0.108            | 2.084          | 0.121            | 58    | 0.462            | 8.407          | 0.519            | 103   | 0.688            | 11.063         | 0.785            | 148   | 0.495            | 6.929          | 0.577            |
| 14    | 0.116            | 2.242          | 0.130            | 59    | 0.470            | 8.517          | 0.527            | 104   | 0.690            | 11.054         | 0.787            | 149   | 0.484            | 6.750          | 0.564            |
| 15    | 0.125            | 2.400          | 0.139            | 60    | 0.477            | 8.626          | 0.535            | 105   | 0.692            | 11.042         | 0.789            | 150   | 0.472            | 6.568          | 0.550            |
| 16    | 0.133            | 2.558          | 0.148            | 61    | 0.484            | 8.734          | 0.543            | 106   | 0.693            | 11.027         | 0.791            | 151   | 0.459            | 6.383          | 0.536            |
| 17    | 0.141            | 2.715          | 0.158            | 62    | 0.490            | 8.839          | 0.551            | 107   | 0.694            | 11.008         | 0.793            | 152   | 0.447            | 6.194          | 0.522            |
| 18    | 0.149            | 2.872          | 0.167            | 63    | 0.497            | 8.942          | 0.559            | 108   | 0.695            | 10.985         | 0.794            | 153   | 0.434            | 6.003          | 0.507            |
| 19    | 0.158            | 3.028          | 0.176            | 64    | 0.504            | 9.044          | 0.567            | 109   | 0.695            | 10.959         | 0.795            | 154   | 0.421            | 5.808          | 0.492            |
| 20    | 0.166            | 3.184          | 0.185            | 65    | 0.511            | 9.143          | 0.574            | 110   | 0.696            | 10.929         | 0.796            | 155   | 0.407            | 5.610          | 0.476            |
| 21    | 0.174            | 3.339          | 0.194            | 66    | 0.517            | 9.240          | 0.582            | 111   | 0.696            | 10.895         | 0.796            | 156   | 0.393            | 5.410          | 0.460            |
| 22    | 0.182            | 3.494          | 0.204            | 67    | 0.524            | 9.336          | 0.590            | 112   | 0.696            | 10.858         | 0.797            | 157   | 0.379            | 5.206          | 0.444            |
| 23    | 0.191            | 3.648          | 0.213            | 68    | 0.531            | 9.429          | 0.597            | 113   | 0.695            | 10.817         | 0.797            | 158   | 0.365            | 5.000          | 0.427            |
| 24    | 0.199            | 3.801          | 0.222            | 69    | 0.537            | 9.520          | 0.605            | 114   | 0.695            | 10.772         | 0.796            | 159   | 0.350            | 4.792          | 0.410            |
| 25    | 0.207            | 3.954          | 0.231            | 70    | 0.543            | 9.609          | 0.612            | 115   | 0.694            | 10.723         | 0.796            | 160   | 0.336            | 4.581          | 0.393            |
| 26    | 0.215            | 4.106          | 0.240            | 71    | 0.550            | 9.696          | 0.619            | 116   | 0.693            | 10.671         | 0.795            | 161   | 0.320            | 4.367          | 0.375            |
| 27    | 0.223            | 4.257          | 0.249            | 72    | 0.556            | 9.780          | 0.626            | 117   | 0.691            | 10.614         | 0.794            | 162   | 0.305            | 4.151          | 0.357            |
| 28    | 0.231            | 4.407          | 0.258            | 73    | 0.562            | 9.862          | 0.633            | 118   | 0.690            | 10.554         | 0.792            | 163   | 0.289            | 3.933          | 0.339            |
| 29    | 0.239            | 4.557          | 0.268            | 74    | 0.568            | 9.942          | 0.640            | 119   | 0.688            | 10.490         | 0.790            | 164   | 0.274            | 3.713          | 0.321            |
| 30    | 0.248            | 4.705          | 0.277            | 75    | 0.574            | 10.019         | 0.647            | 120   | 0.686            | 10.422         | 0.788            | 165   | 0.258            | 3.491          | 0.302            |
| 31    | 0.256            | 4.853          | 0.286            | 76    | 0.580            | 10.094         | 0.654            | 121   | 0.683            | 10.350         | 0.786            | 166   | 0.241            | 3.267          | 0.283            |
| 32    | 0.264            | 5.000          | 0.295            | 77    | 0.585            | 10.167         | 0.661            | 122   | 0.680            | 10.274         | 0.783            | 167   | 0.225            | 3.041          | 0.264            |
| 33    | 0.272            | 5.146          | 0.304            | 78    | 0.591            | 10.237         | 0.667            | 123   | 0.677            | 10.194         | 0.780            | 168   | 0.208            | 2.814          | 0.244            |
| 34    | 0.280            | 5.292          | 0.313            | 79    | 0.596            | 10.304         | 0.674            | 124   | 0.674            | 10.110         | 0.776            | 169   | 0.192            | 2.585          | 0.225            |
| 35    | 0.288            | 5.436          | 0.322            | 80    | 0.602            | 10.369         | 0.680            | 125   | 0.670            | 10.023         | 0.772            | 170   | 0.175            | 2.354          | 0.205            |
| 36    | 0.296            | 5.579          | 0.331            | 81    | 0.607            | 10.431         | 0.686            | 126   | 0.666            | 9.931          | 0.768            | 171   | 0.158            | 2.123          | 0.185            |
| 37    | 0.304            | 5.721          | 0.339            | 82    | 0.612            | 10.491         | 0.692            | 127   | 0.662            | 9.835          | 0.764            | 172   | 0.140            | 1.890          | 0.165            |
| 38    | 0.311            | 5.862          | 0.348            | 83    | 0.617            | 10.548         | 0.698            | 128   | 0.657            | 9.736          | 0.759            | 173   | 0.123            | 1.656          | 0.145            |
| 39    | 0.319            | 6.002          | 0.357            | 84    | 0.622            | 10.602         | 0.704            | 129   | 0.652            | 9.632          | 0.753            | 174   | 0.106            | 1.421          | 0.124            |
| 40    | 0.327            | 6.141          | 0.366            | 85    | 0.627            | 10.654         | 0.710            | 130   | 0.647            | 9.524          | 0.748            | 175   | 0.088            | 1.185          | 0.104            |
| 41    | 0.335            | 6.278          | 0.375            | 86    | 0.632            | 10.702         | 0.715            | 131   | 0.641            | 9.413          | 0.742            | 176   | 0.071            | 0.949          | 0.083            |
| 42    | 0.343            | 6.415          | 0.384            | 87    | 0.636            | 10.748         | 0.721            | 132   | 0.636            | 9.297          | 0.735            | 177   | 0.053            | 0.712          | 0.062            |
| 43    | 0.351            | 6.550          | 0.392            | 88    | 0.641            | 10.791         | 0.726            | 133   | 0.629            | 9.178          | 0.728            | 178   | 0.035            | 0.475          | 0.042            |
| 44    | 0.358            | 6.684          | 0.401            | 89    | 0.645            | 10.831         | 0.731            | 134   | 0.623            | 9.055          | 0.721            | 179   | 0.018            | 0.238          | 0.021            |
| 45    | 0.366            | 6.816          | 0.410            | 90    | 0.649            | 10.868         | 0.736            | 135   | 0.616            | 8.927          | 0.713            | 180   | 0.000            | 0.000          | 0.000            |

Table 8.9: Epicyclic anomalies of Jupiter. All quantities are in degrees. Note that  $\bar{\theta}(360^\circ - \mu) = -\bar{\theta}(\mu)$ , and  $\delta\theta_\pm(360^\circ - \mu) = -\delta\theta_\pm(\mu)$ .

| Event       | Date       | $\lambda$ | Event       | Date       | $\lambda$ |
|-------------|------------|-----------|-------------|------------|-----------|
| Conjunction | 08/05/2000 | 17TA53    | Conjunction | 22/10/2005 | 29LI16    |
| Station (R) | 29/09/2000 | 11GE13    | Station (R) | 04/03/2006 | 18SC54    |
| Opposition  | 28/11/2000 | 06GE08    | Opposition  | 04/05/2006 | 14SC03    |
| Station (D) | 25/01/2001 | 01GE10    | Station (D) | 06/07/2006 | 09SC02    |
| Conjunction | 14/06/2001 | 23GE30    | Conjunction | 22/11/2006 | 29SC34    |
| Station (R) | 02/11/2001 | 15CN41    | Station (R) | 06/04/2007 | 19SG49    |
| Opposition  | 01/01/2002 | 10CN37    | Opposition  | 06/06/2007 | 14SG57    |
| Station (D) | 01/03/2002 | 05CN37    | Station (D) | 07/08/2007 | 09SG58    |
| Conjunction | 20/07/2002 | 27CN11    | Conjunction | 23/12/2007 | 01CP03    |
| Station (R) | 04/12/2002 | 18LE06    | Station (R) | 09/05/2008 | 22CP23    |
| Opposition  | 02/02/2003 | 13LE06    | Opposition  | 09/07/2008 | 17CP30    |
| Station (D) | 04/04/2003 | 08LE03    | Station (D) | 08/09/2008 | 12CP33    |
| Conjunction | 22/08/2003 | 28LE55    | Conjunction | 24/01/2009 | 04AQ23    |
| Station (R) | 04/01/2004 | 18VI54    | Station (R) | 15/06/2009 | 27AQ01    |
| Opposition  | 04/03/2004 | 13VI58    | Opposition  | 14/08/2009 | 22AQ04    |
| Station (D) | 05/05/2004 | 08VI55    | Station (D) | 13/10/2009 | 17AQ10    |
| Conjunction | 22/09/2004 | 29VI21    | Conjunction | 28/02/2010 | 09PI43    |
| Station (R) | 02/02/2005 | 18LI53    | Station (R) | 23/07/2010 | 03AR20    |
| Opposition  | 03/04/2005 | 14LI00    | Opposition  | 21/09/2010 | 28PI19    |
| Station (D) | 05/06/2005 | 08LI58    | Station (D) | 18/11/2010 | 23PI26    |

Table 8.10: The conjunctions, oppositions, and stations of Jupiter during the years 2000–2010 AD. (R) indicates a retrograde station, and (D) a direct station.

| $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(\circ)$ | $\Delta M(\circ)$ | $\Delta \bar{F}(\circ)$ | $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(\circ)$ | $\Delta M(\circ)$ | $\Delta \bar{F}(\circ)$ |
|-----------------------|-------------------------------|-------------------|-------------------------|-----------------------|-------------------------------|-------------------|-------------------------|
| 10 000                | 335.083                       | 334.815           | 334.779                 | 1 000                 | 33.508                        | 33.482            | 33.478                  |
| 20 000                | 310.166                       | 309.630           | 309.559                 | 2 000                 | 67.017                        | 66.963            | 66.956                  |
| 30 000                | 285.249                       | 284.446           | 284.338                 | 3 000                 | 100.525                       | 100.445           | 100.434                 |
| 40 000                | 260.332                       | 259.261           | 259.118                 | 4 000                 | 134.033                       | 133.926           | 133.912                 |
| 50 000                | 235.415                       | 234.076           | 233.897                 | 5 000                 | 167.541                       | 167.408           | 167.390                 |
| 60 000                | 210.498                       | 208.891           | 208.677                 | 6 000                 | 201.050                       | 200.889           | 200.868                 |
| 70 000                | 185.581                       | 183.706           | 183.456                 | 7 000                 | 234.558                       | 234.371           | 234.346                 |
| 80 000                | 160.664                       | 158.522           | 158.236                 | 8 000                 | 268.066                       | 267.852           | 267.824                 |
| 90 000                | 135.747                       | 133.337           | 133.015                 | 9 000                 | 301.575                       | 301.334           | 301.302                 |
|                       |                               |                   |                         |                       |                               |                   |                         |
| 100                   | 3.351                         | 3.348             | 3.348                   | 10                    | 0.335                         | 0.335             | 0.335                   |
| 200                   | 6.702                         | 6.696             | 6.696                   | 20                    | 0.670                         | 0.670             | 0.670                   |
| 300                   | 10.052                        | 10.044            | 10.043                  | 30                    | 1.005                         | 1.004             | 1.004                   |
| 400                   | 13.403                        | 13.393            | 13.391                  | 40                    | 1.340                         | 1.339             | 1.339                   |
| 500                   | 16.754                        | 16.741            | 16.739                  | 50                    | 1.675                         | 1.674             | 1.674                   |
| 600                   | 20.105                        | 20.089            | 20.087                  | 60                    | 2.010                         | 2.009             | 2.009                   |
| 700                   | 23.456                        | 23.437            | 23.435                  | 70                    | 2.346                         | 2.344             | 2.343                   |
| 800                   | 26.807                        | 26.785            | 26.782                  | 80                    | 2.681                         | 2.679             | 2.678                   |
| 900                   | 30.157                        | 30.133            | 30.130                  | 90                    | 3.016                         | 3.013             | 3.013                   |
|                       |                               |                   |                         |                       |                               |                   |                         |
| 1                     | 0.034                         | 0.033             | 0.033                   | 0.1                   | 0.003                         | 0.003             | 0.003                   |
| 2                     | 0.067                         | 0.067             | 0.067                   | 0.2                   | 0.007                         | 0.007             | 0.007                   |
| 3                     | 0.101                         | 0.100             | 0.100                   | 0.3                   | 0.010                         | 0.010             | 0.010                   |
| 4                     | 0.134                         | 0.134             | 0.134                   | 0.4                   | 0.013                         | 0.013             | 0.013                   |
| 5                     | 0.168                         | 0.167             | 0.167                   | 0.5                   | 0.017                         | 0.017             | 0.017                   |
| 6                     | 0.201                         | 0.201             | 0.201                   | 0.6                   | 0.020                         | 0.020             | 0.020                   |
| 7                     | 0.235                         | 0.234             | 0.234                   | 0.7                   | 0.023                         | 0.023             | 0.023                   |
| 8                     | 0.268                         | 0.268             | 0.268                   | 0.8                   | 0.027                         | 0.027             | 0.027                   |
| 9                     | 0.302                         | 0.301             | 0.301                   | 0.9                   | 0.030                         | 0.030             | 0.030                   |

Table 8.11: Mean motion of Saturn. Here,  $\Delta t = t - t_0$ ,  $\Delta \bar{\lambda} = \bar{\lambda} - \bar{\lambda}_0$ ,  $\Delta M = M - M_0$ , and  $\Delta \bar{F} = \bar{F} - \bar{F}_0$ . At epoch ( $t_0 = 2\,451\,545.0$  JD),  $\bar{\lambda}_0 = 50.059^\circ$ ,  $M_0 = 317.857^\circ$ , and  $\bar{F}_0 = 296.482^\circ$ .

| $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ |
|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|
| 0             | 0.000         | 5.386       | 90            | 6.172         | -0.290      | 180           | 0.000         | -5.386      | 270           | -6.172        | -0.290      |
| 2             | 0.230         | 5.383       | 92            | 6.154         | -0.478      | 182           | -0.201        | -5.383      | 272           | -6.183        | -0.102      |
| 4             | 0.459         | 5.372       | 94            | 6.128         | -0.664      | 184           | -0.402        | -5.374      | 274           | -6.186        | 0.087       |
| 6             | 0.688         | 5.354       | 96            | 6.095         | -0.850      | 186           | -0.602        | -5.360      | 276           | -6.182        | 0.276       |
| 8             | 0.916         | 5.328       | 98            | 6.055         | -1.034      | 188           | -0.802        | -5.339      | 278           | -6.169        | 0.465       |
| 10            | 1.143         | 5.296       | 100           | 6.007         | -1.217      | 190           | -1.001        | -5.313      | 280           | -6.149        | 0.654       |
| 12            | 1.368         | 5.256       | 102           | 5.953         | -1.397      | 192           | -1.199        | -5.281      | 282           | -6.122        | 0.842       |
| 14            | 1.591         | 5.209       | 104           | 5.891         | -1.576      | 194           | -1.396        | -5.243      | 284           | -6.086        | 1.030       |
| 16            | 1.811         | 5.156       | 106           | 5.823         | -1.753      | 196           | -1.591        | -5.200      | 286           | -6.043        | 1.217       |
| 18            | 2.029         | 5.095       | 108           | 5.748         | -1.927      | 198           | -1.785        | -5.150      | 288           | -5.992        | 1.402       |
| 20            | 2.245         | 5.027       | 110           | 5.666         | -2.098      | 200           | -1.977        | -5.095      | 290           | -5.933        | 1.586       |
| 22            | 2.456         | 4.953       | 112           | 5.578         | -2.267      | 202           | -2.168        | -5.035      | 292           | -5.867        | 1.768       |
| 24            | 2.665         | 4.873       | 114           | 5.484         | -2.433      | 204           | -2.356        | -4.969      | 294           | -5.793        | 1.949       |
| 26            | 2.869         | 4.785       | 116           | 5.384         | -2.596      | 206           | -2.542        | -4.897      | 296           | -5.711        | 2.127       |
| 28            | 3.070         | 4.692       | 118           | 5.277         | -2.755      | 208           | -2.725        | -4.820      | 298           | -5.622        | 2.302       |
| 30            | 3.266         | 4.592       | 120           | 5.165         | -2.911      | 210           | -2.906        | -4.737      | 300           | -5.525        | 2.476       |
| 32            | 3.457         | 4.486       | 122           | 5.048         | -3.063      | 212           | -3.084        | -4.649      | 302           | -5.421        | 2.646       |
| 34            | 3.644         | 4.375       | 124           | 4.924         | -3.211      | 214           | -3.259        | -4.556      | 304           | -5.310        | 2.813       |
| 36            | 3.825         | 4.257       | 126           | 4.796         | -3.356      | 216           | -3.430        | -4.458      | 306           | -5.191        | 2.976       |
| 38            | 4.002         | 4.134       | 128           | 4.662         | -3.496      | 218           | -3.598        | -4.354      | 308           | -5.065        | 3.136       |
| 40            | 4.172         | 4.006       | 130           | 4.524         | -3.632      | 220           | -3.763        | -4.246      | 310           | -4.933        | 3.292       |
| 42            | 4.337         | 3.873       | 132           | 4.380         | -3.764      | 222           | -3.923        | -4.133      | 312           | -4.793        | 3.444       |
| 44            | 4.495         | 3.735       | 134           | 4.232         | -3.892      | 224           | -4.080        | -4.015      | 314           | -4.648        | 3.591       |
| 46            | 4.648         | 3.591       | 136           | 4.080         | -4.015      | 226           | -4.232        | -3.892      | 316           | -4.495        | 3.735       |
| 48            | 4.793         | 3.444       | 138           | 3.923         | -4.133      | 228           | -4.380        | -3.764      | 318           | -4.337        | 3.873       |
| 50            | 4.933         | 3.292       | 140           | 3.763         | -4.246      | 230           | -4.524        | -3.632      | 320           | -4.172        | 4.006       |
| 52            | 5.065         | 3.136       | 142           | 3.598         | -4.354      | 232           | -4.662        | -3.496      | 322           | -4.002        | 4.134       |
| 54            | 5.191         | 2.976       | 144           | 3.430         | -4.458      | 234           | -4.796        | -3.356      | 324           | -3.825        | 4.257       |
| 56            | 5.310         | 2.813       | 146           | 3.259         | -4.556      | 236           | -4.924        | -3.211      | 326           | -3.644        | 4.375       |
| 58            | 5.421         | 2.646       | 148           | 3.084         | -4.649      | 238           | -5.048        | -3.063      | 328           | -3.457        | 4.486       |
| 60            | 5.525         | 2.476       | 150           | 2.906         | -4.737      | 240           | -5.165        | -2.911      | 330           | -3.266        | 4.592       |
| 62            | 5.622         | 2.302       | 152           | 2.725         | -4.820      | 242           | -5.277        | -2.755      | 332           | -3.070        | 4.692       |
| 64            | 5.711         | 2.127       | 154           | 2.542         | -4.897      | 244           | -5.384        | -2.596      | 334           | -2.869        | 4.785       |
| 66            | 5.793         | 1.949       | 156           | 2.356         | -4.969      | 246           | -5.484        | -2.433      | 336           | -2.665        | 4.873       |
| 68            | 5.867         | 1.768       | 158           | 2.168         | -5.035      | 248           | -5.578        | -2.267      | 338           | -2.456        | 4.953       |
| 70            | 5.933         | 1.586       | 160           | 1.977         | -5.095      | 250           | -5.666        | -2.098      | 340           | -2.245        | 5.027       |
| 72            | 5.992         | 1.402       | 162           | 1.785         | -5.150      | 252           | -5.748        | -1.927      | 342           | -2.029        | 5.095       |
| 74            | 6.043         | 1.217       | 164           | 1.591         | -5.200      | 254           | -5.823        | -1.753      | 344           | -1.811        | 5.156       |
| 76            | 6.086         | 1.030       | 166           | 1.396         | -5.243      | 256           | -5.891        | -1.576      | 346           | -1.591        | 5.209       |
| 78            | 6.122         | 0.842       | 168           | 1.199         | -5.281      | 258           | -5.953        | -1.397      | 348           | -1.368        | 5.256       |
| 80            | 6.149         | 0.654       | 170           | 1.001         | -5.313      | 260           | -6.007        | -1.217      | 350           | -1.143        | 5.296       |
| 82            | 6.169         | 0.465       | 172           | 0.802         | -5.339      | 262           | -6.055        | -1.034      | 352           | -0.916        | 5.328       |
| 84            | 6.182         | 0.276       | 174           | 0.602         | -5.360      | 264           | -6.095        | -0.850      | 354           | -0.688        | 5.354       |
| 86            | 6.186         | 0.087       | 176           | 0.402         | -5.374      | 266           | -6.128        | -0.664      | 356           | -0.459        | 5.372       |
| 88            | 6.183         | -0.102      | 178           | 0.201         | -5.383      | 268           | -6.154        | -0.478      | 358           | -0.230        | 5.383       |
| 90            | 6.172         | -0.290      | 180           | 0.000         | -5.386      | 270           | -6.172        | -0.290      | 360           | -0.000        | 5.386       |

Table 8.12: Differential anomalies of Saturn.

| $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ |
|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|
| 0     | 0.000            | 0.000          | 0.000            | 45    | 0.242            | 3.944          | 0.276            | 90    | 0.391            | 5.979          | 0.450            | 135   | 0.322            | 4.573          | 0.375            |
| 1     | 0.006            | 0.095          | 0.006            | 46    | 0.247            | 4.017          | 0.282            | 91    | 0.393            | 5.989          | 0.452            | 136   | 0.318            | 4.499          | 0.370            |
| 2     | 0.011            | 0.190          | 0.013            | 47    | 0.252            | 4.089          | 0.287            | 92    | 0.394            | 5.997          | 0.453            | 137   | 0.313            | 4.423          | 0.364            |
| 3     | 0.017            | 0.284          | 0.019            | 48    | 0.256            | 4.160          | 0.292            | 93    | 0.395            | 6.004          | 0.454            | 138   | 0.308            | 4.346          | 0.358            |
| 4     | 0.023            | 0.379          | 0.026            | 49    | 0.261            | 4.230          | 0.298            | 94    | 0.396            | 6.008          | 0.456            | 139   | 0.302            | 4.267          | 0.352            |
| 5     | 0.028            | 0.474          | 0.032            | 50    | 0.265            | 4.299          | 0.303            | 95    | 0.397            | 6.011          | 0.457            | 140   | 0.297            | 4.186          | 0.346            |
| 6     | 0.034            | 0.568          | 0.039            | 51    | 0.270            | 4.367          | 0.308            | 96    | 0.397            | 6.012          | 0.458            | 141   | 0.292            | 4.104          | 0.340            |
| 7     | 0.040            | 0.662          | 0.045            | 52    | 0.274            | 4.433          | 0.313            | 97    | 0.398            | 6.011          | 0.459            | 142   | 0.286            | 4.020          | 0.333            |
| 8     | 0.045            | 0.757          | 0.052            | 53    | 0.279            | 4.499          | 0.318            | 98    | 0.399            | 6.008          | 0.459            | 143   | 0.280            | 3.935          | 0.327            |
| 9     | 0.051            | 0.851          | 0.058            | 54    | 0.283            | 4.564          | 0.323            | 99    | 0.399            | 6.004          | 0.460            | 144   | 0.274            | 3.848          | 0.320            |
| 10    | 0.057            | 0.945          | 0.064            | 55    | 0.287            | 4.627          | 0.328            | 100   | 0.399            | 5.997          | 0.460            | 145   | 0.269            | 3.760          | 0.313            |
| 11    | 0.062            | 1.038          | 0.071            | 56    | 0.292            | 4.689          | 0.333            | 101   | 0.399            | 5.989          | 0.461            | 146   | 0.262            | 3.670          | 0.306            |
| 12    | 0.068            | 1.132          | 0.077            | 57    | 0.296            | 4.750          | 0.338            | 102   | 0.399            | 5.979          | 0.461            | 147   | 0.256            | 3.578          | 0.299            |
| 13    | 0.074            | 1.225          | 0.084            | 58    | 0.300            | 4.810          | 0.343            | 103   | 0.399            | 5.966          | 0.461            | 148   | 0.250            | 3.486          | 0.292            |
| 14    | 0.079            | 1.318          | 0.090            | 59    | 0.304            | 4.869          | 0.347            | 104   | 0.399            | 5.952          | 0.460            | 149   | 0.243            | 3.392          | 0.284            |
| 15    | 0.085            | 1.410          | 0.096            | 60    | 0.308            | 4.926          | 0.352            | 105   | 0.399            | 5.937          | 0.460            | 150   | 0.237            | 3.296          | 0.276            |
| 16    | 0.090            | 1.502          | 0.103            | 61    | 0.312            | 4.982          | 0.356            | 106   | 0.398            | 5.919          | 0.460            | 151   | 0.230            | 3.199          | 0.269            |
| 17    | 0.096            | 1.594          | 0.109            | 62    | 0.316            | 5.037          | 0.361            | 107   | 0.397            | 5.899          | 0.459            | 152   | 0.223            | 3.101          | 0.261            |
| 18    | 0.102            | 1.686          | 0.115            | 63    | 0.320            | 5.091          | 0.365            | 108   | 0.397            | 5.877          | 0.458            | 153   | 0.216            | 3.002          | 0.253            |
| 19    | 0.107            | 1.777          | 0.122            | 64    | 0.323            | 5.143          | 0.370            | 109   | 0.396            | 5.854          | 0.457            | 154   | 0.209            | 2.901          | 0.244            |
| 20    | 0.113            | 1.868          | 0.128            | 65    | 0.327            | 5.194          | 0.374            | 110   | 0.395            | 5.828          | 0.456            | 155   | 0.202            | 2.800          | 0.236            |
| 21    | 0.118            | 1.958          | 0.134            | 66    | 0.330            | 5.243          | 0.378            | 111   | 0.394            | 5.801          | 0.455            | 156   | 0.195            | 2.697          | 0.228            |
| 22    | 0.124            | 2.048          | 0.141            | 67    | 0.334            | 5.292          | 0.382            | 112   | 0.392            | 5.772          | 0.454            | 157   | 0.187            | 2.593          | 0.219            |
| 23    | 0.129            | 2.138          | 0.147            | 68    | 0.337            | 5.338          | 0.386            | 113   | 0.391            | 5.740          | 0.452            | 158   | 0.180            | 2.488          | 0.210            |
| 24    | 0.134            | 2.227          | 0.153            | 69    | 0.341            | 5.384          | 0.390            | 114   | 0.389            | 5.707          | 0.450            | 159   | 0.172            | 2.382          | 0.202            |
| 25    | 0.140            | 2.315          | 0.159            | 70    | 0.344            | 5.428          | 0.394            | 115   | 0.387            | 5.672          | 0.448            | 160   | 0.165            | 2.275          | 0.193            |
| 26    | 0.145            | 2.403          | 0.165            | 71    | 0.347            | 5.470          | 0.398            | 116   | 0.386            | 5.635          | 0.446            | 161   | 0.157            | 2.167          | 0.184            |
| 27    | 0.151            | 2.490          | 0.171            | 72    | 0.350            | 5.511          | 0.401            | 117   | 0.384            | 5.596          | 0.444            | 162   | 0.149            | 2.059          | 0.175            |
| 28    | 0.156            | 2.577          | 0.178            | 73    | 0.353            | 5.551          | 0.405            | 118   | 0.381            | 5.555          | 0.442            | 163   | 0.142            | 1.949          | 0.166            |
| 29    | 0.161            | 2.663          | 0.184            | 74    | 0.356            | 5.589          | 0.408            | 119   | 0.379            | 5.512          | 0.439            | 164   | 0.134            | 1.839          | 0.156            |
| 30    | 0.167            | 2.749          | 0.190            | 75    | 0.359            | 5.625          | 0.412            | 120   | 0.377            | 5.467          | 0.437            | 165   | 0.126            | 1.727          | 0.147            |
| 31    | 0.172            | 2.834          | 0.196            | 76    | 0.362            | 5.660          | 0.415            | 121   | 0.374            | 5.421          | 0.434            | 166   | 0.118            | 1.616          | 0.138            |
| 32    | 0.177            | 2.918          | 0.202            | 77    | 0.365            | 5.694          | 0.418            | 122   | 0.371            | 5.372          | 0.431            | 167   | 0.109            | 1.503          | 0.128            |
| 33    | 0.182            | 3.002          | 0.208            | 78    | 0.367            | 5.726          | 0.421            | 123   | 0.368            | 5.322          | 0.427            | 168   | 0.101            | 1.390          | 0.118            |
| 34    | 0.188            | 3.085          | 0.214            | 79    | 0.370            | 5.756          | 0.424            | 124   | 0.365            | 5.270          | 0.424            | 169   | 0.093            | 1.276          | 0.109            |
| 35    | 0.193            | 3.167          | 0.219            | 80    | 0.372            | 5.784          | 0.427            | 125   | 0.362            | 5.215          | 0.420            | 170   | 0.085            | 1.162          | 0.099            |
| 36    | 0.198            | 3.248          | 0.225            | 81    | 0.375            | 5.811          | 0.430            | 126   | 0.359            | 5.159          | 0.417            | 171   | 0.076            | 1.047          | 0.089            |
| 37    | 0.203            | 3.329          | 0.231            | 82    | 0.377            | 5.837          | 0.433            | 127   | 0.355            | 5.101          | 0.413            | 172   | 0.068            | 0.932          | 0.080            |
| 38    | 0.208            | 3.409          | 0.237            | 83    | 0.379            | 5.861          | 0.435            | 128   | 0.352            | 5.042          | 0.409            | 173   | 0.060            | 0.816          | 0.070            |
| 39    | 0.213            | 3.488          | 0.243            | 84    | 0.381            | 5.883          | 0.438            | 129   | 0.348            | 4.980          | 0.404            | 174   | 0.051            | 0.700          | 0.060            |
| 40    | 0.218            | 3.566          | 0.248            | 85    | 0.383            | 5.903          | 0.440            | 130   | 0.344            | 4.917          | 0.400            | 175   | 0.043            | 0.584          | 0.050            |
| 41    | 0.223            | 3.644          | 0.254            | 86    | 0.385            | 5.922          | 0.442            | 131   | 0.340            | 4.851          | 0.395            | 176   | 0.034            | 0.467          | 0.040            |
| 42    | 0.228            | 3.720          | 0.260            | 87    | 0.387            | 5.939          | 0.444            | 132   | 0.336            | 4.784          | 0.390            | 177   | 0.026            | 0.351          | 0.030            |
| 43    | 0.233            | 3.796          | 0.265            | 88    | 0.388            | 5.954          | 0.446            | 133   | 0.331            | 4.716          | 0.386            | 178   | 0.017            | 0.234          | 0.020            |
| 44    | 0.238            | 3.871          | 0.271            | 89    | 0.390            | 5.967          | 0.448            | 134   | 0.327            | 4.645          | 0.380            | 179   | 0.009            | 0.117          | 0.010            |
| 45    | 0.242            | 3.944          | 0.276            | 90    | 0.391            | 5.979          | 0.450            | 135   | 0.322            | 4.573          | 0.375            | 180   | 0.000            | 0.000          | 0.000            |

Table 8.13: Epicyclic anomalies of Saturn. All quantities are in degrees. Note that  $\bar{\theta}(360^\circ - \mu) = -\bar{\theta}(\mu)$ , and  $\delta\theta_\pm(360^\circ - \mu) = -\delta\theta_\pm(\mu)$ .

| Event       | Date       | $\lambda$ | Event       | Date       | $\lambda$ |
|-------------|------------|-----------|-------------|------------|-----------|
| Conjunction | 10/05/2000 | 20TA26    | Station (R) | 22/11/2005 | 11LE19    |
| Station (R) | 12/09/2000 | 00GE59    | Opposition  | 27/01/2006 | 07LE51    |
| Opposition  | 19/11/2000 | 27TA29    | Station (D) | 05/04/2006 | 04LE22    |
| Station (D) | 24/01/2001 | 24TA03    | Conjunction | 07/08/2006 | 14LE51    |
| Conjunction | 25/05/2001 | 04GE22    | Station (R) | 06/12/2006 | 25LE04    |
| Station (R) | 26/09/2001 | 14GE59    | Opposition  | 10/02/2007 | 21LE38    |
| Opposition  | 03/12/2001 | 11GE29    | Station (D) | 19/04/2007 | 18LE09    |
| Station (D) | 08/02/2002 | 08GE02    | Conjunction | 22/08/2007 | 28LE32    |
| Conjunction | 09/06/2002 | 18GE28    | Station (R) | 19/12/2007 | 08VI34    |
| Station (R) | 11/10/2002 | 29GE06    | Opposition  | 24/02/2008 | 05VI10    |
| Opposition  | 17/12/2002 | 25GE36    | Station (D) | 03/05/2008 | 01VI41    |
| Station (D) | 22/02/2003 | 22GE08    | Conjunction | 04/09/2008 | 11VI56    |
| Conjunction | 24/06/2003 | 02CN39    | Station (R) | 31/12/2008 | 21VI46    |
| Station (R) | 25/10/2003 | 13CN15    | Opposition  | 08/03/2009 | 18VI23    |
| Opposition  | 31/12/2003 | 09CN46    | Station (D) | 17/05/2009 | 14VI56    |
| Station (D) | 07/03/2004 | 06CN17    | Conjunction | 17/09/2009 | 25VI01    |
| Conjunction | 08/07/2004 | 16CN50    | Station (R) | 13/01/2010 | 04LI40    |
| Opposition  | 22/03/2010 | 01LI18    | Station (R) | 08/11/2004 | 27CN21    |
| Opposition  | 13/01/2005 | 23CN52    | Station (D) | 30/05/2010 | 27VI51    |
| Station (D) | 22/03/2005 | 20CN23    | Conjunction | 01/10/2010 | 07LI46    |
| Conjunction | 23/07/2005 | 00LE56    |             |            |           |

Table 8.14: The conjunctions, oppositions, and stations of Saturn during the years 2000–2010 AD. (R) indicates a retrograde station, and (D) a direct station.

## 9. The inferior planets

### 9.1 Determination of ecliptic longitude

Figure 9.1 compares and contrasts heliocentric and geocentric models of the motion of an inferior planet (that is, a planet that is closer to the Sun than the Earth),  $P$ , as seen from the Earth,  $G$ . The Sun is at  $S$ . As before, in the heliocentric model the Earth-planet displacement vector,  $\mathbf{P}$ , is the sum of the Earth-Sun displacement vector,  $\mathbf{S}$ , and the Sun-planet displacement vector,  $\mathbf{P}'$ . On the other hand, in the geocentric model  $\mathbf{S}$  gives the displacement of the guide-point,  $G'$ , from the Earth. Because  $\mathbf{S}$  is also the displacement of the Sun,  $S$ , from the Earth,  $G$ , it is clear that  $G'$  executes a Keplerian orbit about the Earth whose elements are the same as those of the apparent orbit of the Sun about the Earth. This implies that the Sun is coincident with  $G'$ . The ellipse traced out by  $G'$  is termed the deferent. The vector  $\mathbf{P}'$  gives the displacement of the planet,  $P$ , from the guide-point,  $G'$ . Because  $\mathbf{P}'$  is also the displacement of the planet,  $P$ , from the Sun,  $S$ , it is clear that  $P$  executes a Keplerian orbit about the guide-point whose elements are the same as those of the orbit of the planet about the Sun. The ellipse traced out by  $P$  about  $G'$  is termed the epicycle.

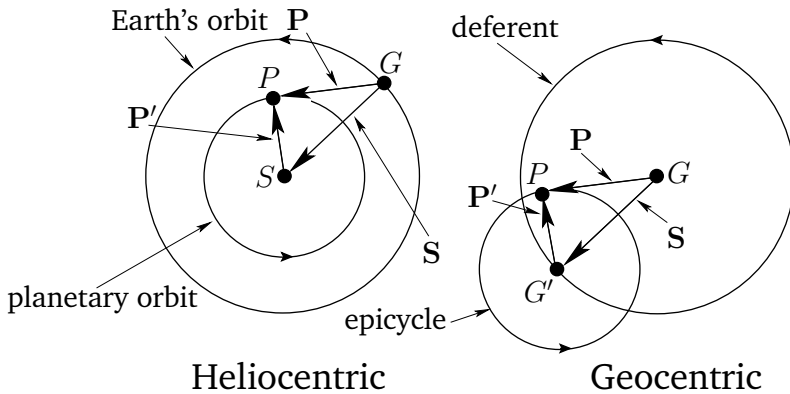


Figure 9.1: Heliocentric and geocentric models of the motion of an inferior planet. Here,  $S$  is the Sun,  $G$  the Earth, and  $P$  the planet. View is from the northern ecliptic pole.

As we have seen, the deferent of a superior planet has the same elements as the planet's orbit about the Sun, whereas the epicycle has the same elements as the Sun's apparent orbit about the Earth. On the other hand, the deferent of an inferior planet has the same elements as the Sun's apparent orbit about the Earth, whereas the epicycle has the same elements as the planet's orbit about the Sun. It follows that we can formulate a procedure for determining the ecliptic longitude of an inferior planet by simply taking the procedure used in the previous chapter for determining the ecliptic longitude of a superior planet and exchanging the roles of the Sun and the planet.

Our procedure is described in the following. As before, it is assumed that the ecliptic longi-

tude,  $\lambda_S$ , and the radial anomaly,  $\zeta_S$ , of the Sun have already been calculated. In the following,  $a, e, n, \tilde{n}, \bar{\lambda}_0$ , and  $M_0$  represent elements of the orbit of the planet in question about the Sun, whereas  $e_S$  is the eccentricity of the Sun's apparent orbit about the Earth. Again,  $a$  is the major radius of the planetary orbit in units in which the major radius of the Sun's apparent orbit about the Earth is unity. The requisite elements for all of the inferior planets at the J2000 epoch ( $t_0 = 2\,451\,545.0$  JD) are listed in Table 5.1. The ecliptic longitude of an inferior planet is specified by the following formulae:

$$\bar{\lambda} = \bar{\lambda}_0 + n(t - t_0), \quad (9.1)$$

$$M = M_0 + \tilde{n}(t - t_0), \quad (9.2)$$

$$q = 2e \sin M + (5/4)e^2 \sin 2M, \quad (9.3)$$

$$\zeta = e \cos M - e^2 \sin^2 M, \quad (9.4)$$

$$\mu = \bar{\lambda} + q - \lambda_S, \quad (9.5)$$

$$\bar{\theta} = \theta(\mu, \bar{z}) \equiv \tan^{-1} \left( \frac{\sin \mu}{a^{-1} \bar{z} + \cos \mu} \right), \quad (9.6)$$

$$\delta\theta_- = \theta(\mu, \bar{z}) - \theta(\mu, z_{\max}), \quad (9.7)$$

$$\delta\theta_+ = \theta(\mu, z_{\min}) - \theta(\mu, \bar{z}), \quad (9.8)$$

$$z = \frac{1 - \zeta_S}{1 - \zeta}, \quad (9.9)$$

$$\xi = \frac{\bar{z} - z}{\delta z}, \quad (9.10)$$

$$\theta = \Theta_-(\xi) \delta\theta_- + \bar{\theta} + \Theta_+(\xi) \delta\theta_+, \quad (9.11)$$

$$\lambda = \lambda_S + \theta. \quad (9.12)$$

Here,  $\bar{z} = (1 + e e_S)/(1 - e^2)$ ,  $\delta z = (e + e_S)/(1 - e^2)$ ,  $z_{\min} = \bar{z} - \delta z$ , and  $z_{\max} = \bar{z} + \delta z$ . The constants  $\bar{z}, \delta z, z_{\min}$ , and  $z_{\max}$  for each of the inferior planets are listed in Table 2.4. Finally, the functions  $\Theta_{\pm}$  are tabulated in Table 2.16.

For the case of Venus, the previous formulae are capable of matching NASA ephemeris data during the years 1995–2006 AD with a mean error of  $2'$  and a maximum error of  $10'$ . For the case of Mercury, given its relatively large eccentricity of 0.205636, it is necessary to modify the formulae slightly by expressing  $q$  and  $\zeta$  to third-order in the eccentricity:

$$q = [2e - (1/4)e^3] \sin M + (5/4)e^2 \sin 2M + (13/12)e^3 \sin 3M, \quad (9.13)$$

$$\begin{aligned} \zeta = & -(1/2)e^2 + [e - (3/8)e^3] \cos M + (1/2)e^2 \cos 2M \\ & + (3/8)e^3 \cos 3M. \end{aligned} \quad (9.14)$$

With this modification, the mean error is  $6'$  and the maximum error  $28'$ . Our modification of the planetary longitude model for the case of Mercury can be thought of as being equivalent to the additional epicycle that Ptolemy introduced into his model of Mercury's orbit. In



both cases, the amendment to the model is necessitated by Mercury's particularly large orbital eccentricity.

## 9.2 Determination of ecliptic longitude of Venus

The ecliptic longitude of Venus can be determined with the aid of Tables 9.1–9.3. Table 9.1 allows the mean longitude,  $\bar{\lambda}$ , and the mean anomaly,  $M$ , of Venus to be calculated as functions of time. Next, Table 9.2 permits the equation of center,  $q$ , and the radial anomaly,  $\zeta$ , to be determined as functions of the mean anomaly. Finally, Table 9.3 allows the quantities  $\delta\theta_-$ ,  $\bar{\theta}$ , and  $\delta\theta_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ .

The procedure for using the tables is as follows:

1. Determine the fractional Julian day number,  $t$ , corresponding to the date and time at which the ecliptic longitude is to be calculated with the aid of Tables 3.1–3.3. Form  $\Delta t = t - t_0$ , where  $t_0 = 2\,451\,545.0$  is the epoch.
2. Calculate the ecliptic longitude,  $\lambda_S$ , and radial anomaly,  $\zeta_S$ , of the Sun using the procedure set out in Section 5.2.
3. Enter Table 9.1 with the digit for each power of 10 in  $\Delta t$  and take out the corresponding values of  $\Delta\bar{\lambda}$  and  $\Delta M$ . If  $\Delta t$  is negative then the corresponding values are also negative. The value of the mean longitude,  $\bar{\lambda}$ , is the sum of all the  $\Delta\bar{\lambda}$  values plus value of  $\bar{\lambda}$  at the epoch. Likewise, the value of the mean anomaly,  $M$ , is the sum of all the  $\Delta M$  values plus the value of  $M$  at the epoch. Add as many multiples of  $360^\circ$  to  $\bar{\lambda}$  and  $M$  as is required to make them both fall in the range  $0^\circ$  to  $360^\circ$ . Round  $M$  to the nearest degree.
4. Enter Table 9.2 with the value of  $M$  and take out the corresponding value of the equation of center,  $q$ , and the radial anomaly,  $\zeta$ . It is necessary to interpolate if  $M$  is odd.
5. Form the epicyclic anomaly,  $\mu = \bar{\lambda} + q - \lambda_S$ . Add as many multiples of  $360^\circ$  to  $\mu$  as is required to make it fall in the range  $0^\circ$  to  $360^\circ$ . Round  $\mu$  to the nearest degree.
6. Enter Table 9.3 with the value of  $\mu$  and take out the corresponding values of  $\delta\theta_-$ ,  $\bar{\theta}$ , and  $\delta\theta_+$ . If  $\mu > 180^\circ$  then it is necessary to make use of the identities  $\delta\theta_\pm(360^\circ - \mu) = -\delta\theta_\pm(\mu)$  and  $\bar{\theta}(360^\circ - \mu) = -\bar{\theta}(\mu)$ .
7. Form  $z = (1 - \zeta_S)/(1 - \zeta)$ .
8. Obtain the values of  $\bar{z}$  and  $\delta z$  from Table 2.4. Form  $\xi = (\bar{z} - z)/\delta z$ .
9. Enter Table 2.16 with the value of  $\xi$  and take out the corresponding values of  $\Theta_-$  and  $\Theta_+$ . If  $\xi < 0$  then it is necessary to use the identities  $\Theta_+(\xi) = -\Theta_-(-\xi)$  and  $\Theta_-(\xi) = -\Theta_+(-\xi)$ .
10. Form the equation of the epicycle,  $\theta = \Theta_- \delta\theta_- + \bar{\theta} + \Theta_+ \delta\theta_+$ .

11. The ecliptic longitude,  $\lambda$ , is the sum of the ecliptic longitude of the Sun,  $\lambda_S$ , and the equation of the epicycle,  $\theta$ . If necessary convert  $\lambda$  into an angle in the range  $0^\circ$  to  $360^\circ$ . The decimal fraction can be converted into arc minutes using Table 5.2. Round to the nearest arc minute. The final result can be written in terms of the signs of the zodiac using the table in Section 2.6.

Two examples of this procedure are given in the following.

*Example 1:* May 5, 2005 AD, 00:00 UT

From Chapter 8,  $t - t_0 = 1\,950.5$  JD,  $\lambda_S = 44.602^\circ$ , and  $\zeta_S = -8.56 \times 10^{-3}$ . Making use of Table 9.1, we find:

| $t(\text{JD})$ | $\bar{\lambda}(^\circ)$ | $M(^\circ)$    |
|----------------|-------------------------|----------------|
| +1000          | 162.169                 | 162.130        |
| +900           | 1.952                   | 1.917          |
| +50            | 80.108                  | 80.107         |
| +5             | 0.801                   | 0.801          |
| Epoch          | 181.973                 | 49.237         |
|                | <u>427.003</u>          | <u>294.192</u> |
| Modulus        | 67.003                  | 294.192        |

Given that  $M \simeq 294^\circ$ , Table 9.2 yields

$$q(294^\circ) = -0.712^\circ, \quad \zeta(294^\circ) = 2.72 \times 10^{-3},$$

so

$$\mu = \bar{\lambda} + q - \bar{\lambda}_S = 67.003 - 0.712 - 44.602 = 21.689 \simeq 22^\circ.$$

It follows from Table 9.3 that

$$\delta\theta_-(22^\circ) = 0.126^\circ, \quad \bar{\theta}(22^\circ) = 9.212^\circ, \quad \delta\theta_+(22^\circ) = 0.129^\circ.$$

Now,

$$z = (1 - \zeta_S)/(1 - \zeta) = (1 + 8.56 \times 10^{-3})/(1 - 2.72 \times 10^{-3}) = 1.01131.$$

However, from Table 2.4,  $\bar{z} = 1.00016$  and  $\delta z = 0.02349$ , so

$$\xi = (\bar{z} - z)/\delta z = (1.00016 - 1.01131)/0.02349 \simeq -0.48.$$

According to Table 2.16,

$$\Theta_-(-0.48) = -0.355, \quad \Theta_+(-0.48) = -0.125,$$

so

$$\begin{aligned}\theta &= \Theta_- \delta\theta_- + \bar{\theta} + \Theta_+ \delta\theta_+ \\ &= -0.355 \times 0.126 + 9.212 - 0.125 \times 0.129 \\ &= 9.151^\circ.\end{aligned}$$

Finally,

$$\lambda = \bar{\lambda}_S + \theta = 44.602 + 9.151 = 53.753 \simeq 53^\circ 45'.$$

Thus, the ecliptic longitude of Venus at 00:00 UT on May 5, 2005 AD was 23TA45.

*Example 2:* December 25, 1800 AD, 00:00 UT

From Chapter 8,  $t - t_0 = -72\,690.5$  JD,  $\lambda_S = 273.055^\circ$ , and  $\zeta_S = 1.662 \times 10^{-2}$ . Making use of Table 9.1, we find:

| $t(\text{JD})$ | $\bar{\lambda}^\circ$ | $M^\circ$      |
|----------------|-----------------------|----------------|
| -70,000        | -191.810              | -189.128       |
| -2,000         | -324.337              | -324.261       |
| -600           | -241.301              | -241.278       |
| -90            | -144.195              | -144.192       |
| -.5            | -0.801                | -0.801         |
| Epoch          | 181.973               | 49.237         |
|                | <hr/> -720.471        | <hr/> -850.423 |
| Modulus        | 359.529               | 229.577        |

Given that  $M \simeq 230^\circ$ , Table 9.2 yields

$$q(230^\circ) = -0.592^\circ, \quad \zeta(230^\circ) = -4.38 \times 10^{-3},$$

so

$$\mu = \bar{\lambda} + q - \bar{\lambda}_S = 359.529 - 0.592 - 273.055 = 85.882 \simeq 86^\circ.$$

It follows from Table 9.3 that

$$\delta\theta_-(86^\circ) = 0.589^\circ, \quad \bar{\theta}(86^\circ) = 34.482^\circ, \quad \delta\theta_+(86^\circ) = 0.607^\circ.$$

Now,

$$z = (1 - \zeta_S)/(1 - \zeta) = (1 - 1.662 \times 10^{-2})/(1 + 4.38 \times 10^{-3}) = 0.97909,$$

so

$$\xi = (\bar{z} - z)/\delta z = (1.00016 - 0.97909)/0.02349 \simeq 0.90.$$

According to Table 2.16,

$$\Theta_-(0.90) = 0.045, \quad \Theta_+(0.90) = 0.855,$$

so

$$\begin{aligned} \theta &= \Theta_- \delta\theta_- + \bar{\theta} + \Theta_+ \delta\theta_+ \\ &= 0.045 \times 0.589 + 34.482 + 0.855 \times 0.607 \\ &= 35.027^\circ. \end{aligned}$$

Finally,

$$\lambda = \bar{\lambda}_S + \theta = 273.055 + 35.027 = 308.082 \simeq 308^\circ 5'.$$

Thus, the ecliptic longitude of Venus at 00:00 UT on December 25, 1800 AD was 8AQ5.

### 9.3 Conjunction and greatest elongation dates

The geocentric orbit of an inferior planet is similar to that of the superior planet shown in Figure 8.4, except for the fact that the Sun is coincident with guide-point  $G'$  in the former case. It follows that it is impossible for an inferior planet to have an opposition with the Sun (that is, for the Earth to lie directly between the planet and the Sun). However, inferior planets do have two different kinds of conjunctions with the Sun. A *superior conjunction* takes place when the Sun lies directly between the planet and the Earth. Conversely, an *inferior conjunction* takes place when the planet lies directly between the Sun and the Earth. It is clear from Figure 8.4 that a superior conjunction corresponds to  $\mu = 0^\circ$ , and an inferior conjunction to  $\mu = 180^\circ$ . Now, the equation of the epicycle,  $\theta$ , measures the angular separation between the planet and the Sun (because the Sun lies at the guide-point). It is evident from Figure 8.4 that  $\theta$  attains a maximum and a minimum value each time the planet revolves around its epicycle. In other words, there is a limit to how large the angular separation between an inferior planet and the Sun can become. The maximum value is termed the *greatest eastern elongation* of the planet, whereas the modulus of the minimum value is termed the *greatest western elongation*.

Tables 9.1–9.3 can be used to determine the dates of the conjunctions and greatest elongations of Venus. Consider the first superior conjunction after the epoch (January 1, 2000 AD). We can estimate the time at which this event occurs by approximating the epicyclic anomaly as the mean epicyclic anomaly:

$$\begin{aligned} \mu &\simeq \bar{\mu} = \bar{\lambda} - \bar{\lambda}_S = \bar{\lambda}_0 - \bar{\lambda}_{0S} + (n - n_S)(t - t_0) \\ &= 261.515 + 0.61652137(t - t_0). \end{aligned}$$

Thus,

$$t \simeq t_0 + (360 - 261.515)/0.61652137 \simeq t_0 + 160 \text{ JD}.$$

A calculation of the epicyclic anomaly at this time, using Tables 9.1–9.3, yields  $\mu = -1.267^\circ$ . Now, the actual conjunction takes place when  $\mu = 0^\circ$ . Hence, our final estimate is

$$t = t_0 + 160 + 1.267/0.61652137 = t_0 + 162.1 \text{ JD},$$

which corresponds to June 11, 2000 AD.

Consider the first inferior conjunction of Venus after the epoch. Our first estimate of the time at which this event takes place is

$$t \simeq t_0 + (540 - 261.515)/0.61652137 \simeq t_0 + 452 \text{ JD.}$$

A calculation of the epicyclic anomaly at this time yields  $\mu = 178.900^\circ$ . Now, the actual conjunction takes place when  $\mu = 180^\circ$ . Hence, our final estimate is

$$t = t_0 + 452 + 1.100/0.61652137 = t_0 + 453.8 \text{ JD,}$$

which corresponds to March 30, 2001 AD. Incidentally, it is clear from the previous analysis that the mean time period between successive superior, or inferior, conjunctions of Venus is  $360/0.61652137 = 583.9 \text{ JD}$ , which is equivalent to 1.60 years.

Consider the greatest elongations of Venus. We can approximate the equation of the epicycle as

$$\theta \simeq \bar{\theta} = \tan^{-1} \left( \frac{\sin \bar{\mu}}{\bar{a}^{-1} + \cos \bar{\mu}} \right), \quad (9.15)$$

where  $\bar{\mu}$  is the mean epicyclic anomaly, and  $\bar{a} = a/\bar{z}$ . It follows that

$$\frac{d\bar{\theta}}{d\bar{\mu}} = \frac{\bar{a}^{-1} \cos \bar{\mu} + 1}{1 + 2\bar{a}^{-1} \cos \bar{\mu} + \bar{a}^{-2}}. \quad (9.16)$$

Now,  $\bar{\theta}$  attains its maximum or minimum value when  $d\bar{\theta}/d\bar{\mu} = 0$ : *i.e.*, when

$$\bar{\mu} = \cos^{-1}(-\bar{a}). \quad (9.17)$$

For the case of Venus, we obtain  $\bar{\mu} = 136.3^\circ$  or  $223.7^\circ$ . The first solution corresponds to the greatest eastern elongation, and the second to the greatest western elongation. Substituting back into Equation (9.15), we find that  $\bar{\theta} = \pm 46.3^\circ$ . Hence, the mean value of the greatest eastern or western elongation of Venus is  $46.3^\circ$ . The mean time period between a greatest eastern elongation and the following inferior conjunction, or between an inferior conjunction and the following greatest western elongation, is  $(180 - 136.3)/0.61652137 \simeq 71 \text{ JD}$ . Unfortunately, the only option for accurately determining the dates at which the greatest elongations occur is to calculate the equation of the epicycle of Venus over a range of days centered 71 days before and after an inferior conjunction.

Table 9.4 shows the conjunctions, and greatest elongations of Venus for the years 2000–2015 AD, calculated using the previously described techniques.

#### 9.4 Determination of ecliptic longitude of Mercury

The ecliptic longitude of Mercury can be determined with the aid of Tables 9.5–9.7. Table 9.5 allows the mean longitude,  $\bar{\lambda}$ , and the mean anomaly,  $M$ , of Mercury to be calculated as functions of time. Next, Table 9.6 permits the equation of center,  $q$ , and the radial anomaly,  $\zeta$ , to be determined as functions of the mean anomaly. Finally, Table 9.7 allows the quantities  $\delta\theta_-$ ,

$\bar{\theta}$ , and  $\delta\theta_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ . The procedure for using the tables is analogous to the previously described procedure for using the Venus tables. One example of this procedure is given in the following.

*Example:* May 5, 2005 AD, 00:00 UT

From Chapter 8,  $t - t_0 = 1\,950.5$  JD,  $\lambda_S = 44.602^\circ$ , and  $\zeta_S = -8.56 \times 10^{-3}$ . Making use of Table 9.5, we find:

| $t(\text{JD})$ | $\bar{\lambda}(^\circ)$ | $M(^\circ)$    |
|----------------|-------------------------|----------------|
| +1000          | 132.377                 | 132.334        |
| +900           | 83.139                  | 83.101         |
| +50            | 204.619                 | 204.617        |
| +5             | 2.046                   | 2.046          |
| Epoch          | 252.087                 | 174.693        |
|                | <u>647.268</u>          | <u>596.791</u> |
| Modulus        | 314.268                 | 236.791        |

Given that  $M \simeq 237^\circ$ , Table 9.6 yields

$$q(237^\circ) = -16.974^\circ, \quad \zeta(237^\circ) = -1.367 \times 10^{-1},$$

so

$$\mu = \bar{\lambda} + q - \bar{\lambda}_S = 314.268 - 16.974 - 44.602 = 252.692 \simeq 253^\circ.$$

It follows from Table 9.7 that

$$\delta\theta_-(253^\circ) = -4.005^\circ, \quad \bar{\theta}(253^\circ) = -21.609^\circ, \quad \delta\theta_+(253^\circ) = -6.182^\circ.$$

Now,

$$z = (1 - \zeta_S)/(1 - \zeta) = (1 + 8.56 \times 10^{-3})/(1 + 1.367 \times 10^{-1}) = 0.8873.$$

However, from Table 2.4,  $\bar{z} = 1.04774$  and  $\delta z = 0.23216$ , so

$$\xi = (\bar{z} - z)/\delta z = (1.04774 - 0.8873)/0.23216 \simeq 0.69.$$

According to Table 2.16,

$$\Theta_-(0.69) = 0.107, \quad \Theta_+(0.69) = 0.583,$$

so

$$\begin{aligned} \theta &= \Theta_- \delta\theta_- + \bar{\theta} + \Theta_+ \delta\theta_+ \\ &= -0.107 \times 4.005 - 21.609 - 0.583 \times 6.182 \\ &= -25.642^\circ. \end{aligned}$$

Finally,

$$\lambda = \bar{\lambda}_S + \theta = 44.602 - 25.642 = 18.960 \simeq 18^\circ 58'.$$

Thus, the ecliptic longitude of Mercury at 00:00 UT on May 5, 2005 AD was 18AR58.

The conjunctions and elongations of Mercury can be investigated using analogous methods to those employed earlier to examine the conjunctions and elongations of Venus. We find that the mean time period between successive superior, or inferior, conjunctions of Mercury is 116 days. On average, the greatest eastern and western elongations of Mercury occur when the epicyclic anomaly takes the values  $\mu = 111.7^\circ$  and  $248.3^\circ$ , respectively. Furthermore, the mean value of the greatest eastern or western elongation is  $21.7^\circ$ . Finally, the mean time period between a greatest eastern elongation and the following inferior conjunction, or between the inferior conjunction and the following greatest western elongation, is 22 JD. The conjunctions and elongations of Mercury during the years 2000–2002 AD are shown in Table 9.8.

As before, the information contained in the mean motion tables, 9.1 and 9.5, and the differential anomaly tables, 9.2 and 9.6, is essentially equivalent to that contained in the “Tables of the mean longitudinal motion and anomalies of the five stars” (Κανόνες μέσων κινήσεων μήκους τε καὶ ἀνωμαλίας τῶν πέντε ἀστέρων) that appear in Section 4 of Book IX of the *Almagest*. Likewise, the information contained in the epicyclic anomaly tables, 9.3 and 9.7, is equivalent to that contained in the “Tables of the longitudinal corrections of the five planets” (Κανόνες τῆς κατὰ μήκος τῶν πέντε πλανωμένων διευκρινήσεως) that appear in Section 11 of Book XI of the *Almagest*. The computation of the greatest elongations of the inferior planets is discussed in Book XII of the *Almagest*.

## 9.5 Tables

| $\Delta t(\text{JD})$ | $\Delta\bar{\lambda}(^\circ)$ | $\Delta M(^\circ)$ | $\Delta\bar{F}(^\circ)$ | $\Delta t(\text{JD})$ | $\Delta\bar{\lambda}(^\circ)$ | $\Delta M(^\circ)$ | $\Delta\bar{F}(^\circ)$ |
|-----------------------|-------------------------------|--------------------|-------------------------|-----------------------|-------------------------------|--------------------|-------------------------|
| 10 000                | 181.687                       | 181.304            | 181.381                 | 1 000                 | 162.169                       | 162.130            | 162.138                 |
| 20 000                | 3.374                         | 2.608              | 2.761                   | 2 000                 | 324.337                       | 324.261            | 324.276                 |
| 30 000                | 185.062                       | 183.912            | 184.142                 | 3 000                 | 126.506                       | 126.391            | 126.414                 |
| 40 000                | 6.749                         | 5.216              | 5.523                   | 4 000                 | 288.675                       | 288.522            | 288.552                 |
| 50 000                | 188.436                       | 186.520            | 186.904                 | 5 000                 | 90.844                        | 90.652             | 90.690                  |
| 60 000                | 10.123                        | 7.824              | 8.284                   | 6 000                 | 253.012                       | 252.782            | 252.828                 |
| 70 000                | 191.810                       | 189.128            | 189.665                 | 7 000                 | 55.181                        | 54.913             | 54.966                  |
| 80 000                | 13.498                        | 10.432             | 11.046                  | 8 000                 | 217.350                       | 217.043            | 217.105                 |
| 90 000                | 195.185                       | 191.736            | 192.426                 | 9 000                 | 19.518                        | 19.174             | 19.243                  |
| 100                   | 160.217                       | 160.213            | 160.214                 | 10                    | 16.022                        | 16.021             | 16.021                  |
| 200                   | 320.434                       | 320.426            | 320.428                 | 20                    | 32.043                        | 32.043             | 32.043                  |
| 300                   | 120.651                       | 120.639            | 120.641                 | 30                    | 48.065                        | 48.064             | 48.064                  |
| 400                   | 280.867                       | 280.852            | 280.855                 | 40                    | 64.087                        | 64.085             | 64.086                  |
| 500                   | 81.084                        | 81.065             | 81.069                  | 50                    | 80.108                        | 80.107             | 80.107                  |
| 600                   | 241.301                       | 241.278            | 241.283                 | 60                    | 96.130                        | 96.128             | 96.128                  |
| 700                   | 41.518                        | 41.491             | 41.497                  | 70                    | 112.152                       | 112.149            | 112.150                 |
| 800                   | 201.735                       | 201.704            | 201.710                 | 80                    | 128.173                       | 128.170            | 128.171                 |
| 900                   | 1.952                         | 1.917              | 1.924                   | 90                    | 144.195                       | 144.192            | 144.192                 |
| 1                     | 1.602                         | 1.602              | 1.602                   | 0.1                   | 0.160                         | 0.160              | 0.160                   |
| 2                     | 3.204                         | 3.204              | 3.204                   | 0.2                   | 0.320                         | 0.320              | 0.320                   |
| 3                     | 4.807                         | 4.806              | 4.806                   | 0.3                   | 0.481                         | 0.481              | 0.481                   |
| 4                     | 6.409                         | 6.409              | 6.409                   | 0.4                   | 0.641                         | 0.641              | 0.641                   |
| 5                     | 8.011                         | 8.011              | 8.011                   | 0.5                   | 0.801                         | 0.801              | 0.801                   |
| 6                     | 9.613                         | 9.613              | 9.613                   | 0.6                   | 0.961                         | 0.961              | 0.961                   |
| 7                     | 11.215                        | 11.215             | 11.215                  | 0.7                   | 1.122                         | 1.121              | 1.121                   |
| 8                     | 12.817                        | 12.817             | 12.817                  | 0.8                   | 1.282                         | 1.282              | 1.282                   |
| 9                     | 14.420                        | 14.419             | 14.419                  | 0.9                   | 1.442                         | 1.442              | 1.442                   |

Table 9.1: Mean motion of Venus. Here,  $\Delta t = t - t_0$ ,  $\Delta\bar{\lambda} = \bar{\lambda} - \bar{\lambda}_0$ ,  $\Delta M = M - M_0$ , and  $\Delta\bar{F} = \bar{F} - \bar{F}_0$ . At epoch ( $t_0 = 2\,451\,545.0$  JD),  $\bar{\lambda}_0 = 181.973^\circ$ ,  $M_0 = 49.237^\circ$ , and  $\bar{F}_0 = 105.253^\circ$ .



| $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ |
|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|
| 0             | 0.000         | 0.678       | 90            | 0.777         | -0.005      | 180           | 0.000         | -0.678      | 270           | -0.777        | -0.005      |
| 2             | 0.027         | 0.677       | 92            | 0.776         | -0.028      | 182           | -0.027        | -0.677      | 272           | -0.776        | 0.019       |
| 4             | 0.055         | 0.676       | 94            | 0.774         | -0.052      | 184           | -0.054        | -0.676      | 274           | -0.775        | 0.043       |
| 6             | 0.082         | 0.674       | 96            | 0.772         | -0.075      | 186           | -0.080        | -0.674      | 276           | -0.773        | 0.066       |
| 8             | 0.109         | 0.671       | 98            | 0.768         | -0.099      | 188           | -0.107        | -0.671      | 278           | -0.770        | 0.090       |
| 10            | 0.136         | 0.667       | 100           | 0.764         | -0.122      | 190           | -0.134        | -0.668      | 280           | -0.766        | 0.113       |
| 12            | 0.163         | 0.663       | 102           | 0.758         | -0.145      | 192           | -0.160        | -0.663      | 282           | -0.761        | 0.137       |
| 14            | 0.189         | 0.657       | 104           | 0.752         | -0.168      | 194           | -0.186        | -0.658      | 284           | -0.755        | 0.160       |
| 16            | 0.216         | 0.651       | 106           | 0.745         | -0.191      | 196           | -0.212        | -0.652      | 286           | -0.748        | 0.183       |
| 18            | 0.242         | 0.644       | 108           | 0.737         | -0.214      | 198           | -0.238        | -0.645      | 288           | -0.741        | 0.205       |
| 20            | 0.268         | 0.636       | 110           | 0.728         | -0.236      | 200           | -0.263        | -0.637      | 290           | -0.732        | 0.228       |
| 22            | 0.293         | 0.628       | 112           | 0.718         | -0.258      | 202           | -0.289        | -0.629      | 292           | -0.722        | 0.250       |
| 24            | 0.318         | 0.618       | 114           | 0.707         | -0.279      | 204           | -0.313        | -0.620      | 294           | -0.712        | 0.272       |
| 26            | 0.343         | 0.608       | 116           | 0.695         | -0.301      | 206           | -0.338        | -0.610      | 296           | -0.701        | 0.293       |
| 28            | 0.367         | 0.597       | 118           | 0.683         | -0.322      | 208           | -0.362        | -0.599      | 298           | -0.688        | 0.315       |
| 30            | 0.391         | 0.586       | 120           | 0.670         | -0.342      | 210           | -0.385        | -0.588      | 300           | -0.675        | 0.335       |
| 32            | 0.414         | 0.573       | 122           | 0.656         | -0.362      | 212           | -0.409        | -0.576      | 302           | -0.662        | 0.356       |
| 34            | 0.437         | 0.560       | 124           | 0.641         | -0.382      | 214           | -0.431        | -0.563      | 304           | -0.647        | 0.376       |
| 36            | 0.460         | 0.547       | 126           | 0.625         | -0.401      | 216           | -0.453        | -0.550      | 306           | -0.631        | 0.395       |
| 38            | 0.481         | 0.532       | 128           | 0.609         | -0.420      | 218           | -0.475        | -0.536      | 308           | -0.615        | 0.414       |
| 40            | 0.502         | 0.517       | 130           | 0.592         | -0.438      | 220           | -0.496        | -0.521      | 310           | -0.598        | 0.433       |
| 42            | 0.523         | 0.502       | 132           | 0.574         | -0.456      | 222           | -0.516        | -0.506      | 312           | -0.580        | 0.451       |
| 44            | 0.543         | 0.485       | 134           | 0.555         | -0.473      | 224           | -0.536        | -0.490      | 314           | -0.562        | 0.468       |
| 46            | 0.562         | 0.468       | 136           | 0.536         | -0.490      | 226           | -0.555        | -0.473      | 316           | -0.543        | 0.485       |
| 48            | 0.580         | 0.451       | 138           | 0.516         | -0.506      | 228           | -0.574        | -0.456      | 318           | -0.523        | 0.502       |
| 50            | 0.598         | 0.433       | 140           | 0.496         | -0.521      | 230           | -0.592        | -0.438      | 320           | -0.502        | 0.517       |
| 52            | 0.615         | 0.414       | 142           | 0.475         | -0.536      | 232           | -0.609        | -0.420      | 322           | -0.481        | 0.532       |
| 54            | 0.631         | 0.395       | 144           | 0.453         | -0.550      | 234           | -0.625        | -0.401      | 324           | -0.460        | 0.547       |
| 56            | 0.647         | 0.376       | 146           | 0.431         | -0.563      | 236           | -0.641        | -0.382      | 326           | -0.437        | 0.560       |
| 58            | 0.662         | 0.356       | 148           | 0.409         | -0.576      | 238           | -0.656        | -0.362      | 328           | -0.414        | 0.573       |
| 60            | 0.675         | 0.335       | 150           | 0.385         | -0.588      | 240           | -0.670        | -0.342      | 330           | -0.391        | 0.586       |
| 62            | 0.688         | 0.315       | 152           | 0.362         | -0.599      | 242           | -0.683        | -0.322      | 332           | -0.367        | 0.597       |
| 64            | 0.701         | 0.293       | 154           | 0.338         | -0.610      | 244           | -0.695        | -0.301      | 334           | -0.343        | 0.608       |
| 66            | 0.712         | 0.272       | 156           | 0.313         | -0.620      | 246           | -0.707        | -0.279      | 336           | -0.318        | 0.618       |
| 68            | 0.722         | 0.250       | 158           | 0.289         | -0.629      | 248           | -0.718        | -0.258      | 338           | -0.293        | 0.628       |
| 70            | 0.732         | 0.228       | 160           | 0.263         | -0.637      | 250           | -0.728        | -0.236      | 340           | -0.268        | 0.636       |
| 72            | 0.741         | 0.205       | 162           | 0.238         | -0.645      | 252           | -0.737        | -0.214      | 342           | -0.242        | 0.644       |
| 74            | 0.748         | 0.183       | 164           | 0.212         | -0.652      | 254           | -0.745        | -0.191      | 344           | -0.216        | 0.651       |
| 76            | 0.755         | 0.160       | 166           | 0.186         | -0.658      | 256           | -0.752        | -0.168      | 346           | -0.189        | 0.657       |
| 78            | 0.761         | 0.137       | 168           | 0.160         | -0.663      | 258           | -0.758        | -0.145      | 348           | -0.163        | 0.663       |
| 80            | 0.766         | 0.113       | 170           | 0.134         | -0.668      | 260           | -0.764        | -0.122      | 350           | -0.136        | 0.667       |
| 82            | 0.770         | 0.090       | 172           | 0.107         | -0.671      | 262           | -0.768        | -0.099      | 352           | -0.109        | 0.671       |
| 84            | 0.773         | 0.066       | 174           | 0.080         | -0.674      | 264           | -0.772        | -0.075      | 354           | -0.082        | 0.674       |
| 86            | 0.775         | 0.043       | 176           | 0.054         | -0.676      | 266           | -0.774        | -0.052      | 356           | -0.055        | 0.676       |
| 88            | 0.776         | 0.019       | 178           | 0.027         | -0.677      | 268           | -0.776        | -0.028      | 358           | -0.027        | 0.677       |
| 90            | 0.777         | -0.005      | 180           | 0.000         | -0.678      | 270           | -0.777        | -0.005      | 360           | -0.000        | 0.678       |

Table 9.2: Differential anomalies of Venus.

| $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ |
|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|
| 0     | 0.000            | 0.000          | 0.000            | 45    | 0.267            | 18.694         | 0.274            | 90    | 0.629            | 35.875         | 0.649            | 135   | 1.344            | 46.305         | 1.408            |       |                  |                |                  |
| 1     | 0.006            | 0.420          | 0.006            | 46    | 0.273            | 19.100         | 0.281            | 91    | 0.640            | 36.217         | 0.660            | 136   | 1.369            | 46.320         | 1.434            |       |                  |                |                  |
| 2     | 0.011            | 0.839          | 0.012            | 47    | 0.280            | 19.505         | 0.288            | 92    | 0.650            | 36.557         | 0.671            | 137   | 1.393            | 46.317         | 1.461            |       |                  |                |                  |
| 3     | 0.017            | 1.259          | 0.017            | 48    | 0.286            | 19.910         | 0.294            | 93    | 0.661            | 36.893         | 0.682            | 138   | 1.418            | 46.294         | 1.489            |       |                  |                |                  |
| 4     | 0.023            | 1.679          | 0.023            | 49    | 0.293            | 20.314         | 0.301            | 94    | 0.672            | 37.227         | 0.693            | 139   | 1.444            | 46.252         | 1.517            |       |                  |                |                  |
| 5     | 0.028            | 2.098          | 0.029            | 50    | 0.300            | 20.717         | 0.308            | 95    | 0.683            | 37.558         | 0.705            | 140   | 1.470            | 46.188         | 1.546            |       |                  |                |                  |
| 6     | 0.034            | 2.518          | 0.035            | 51    | 0.307            | 21.119         | 0.315            | 96    | 0.694            | 37.886         | 0.717            | 141   | 1.496            | 46.102         | 1.575            |       |                  |                |                  |
| 7     | 0.040            | 2.937          | 0.041            | 52    | 0.313            | 21.521         | 0.322            | 97    | 0.706            | 38.210         | 0.729            | 142   | 1.523            | 45.992         | 1.605            |       |                  |                |                  |
| 8     | 0.045            | 3.357          | 0.046            | 53    | 0.320            | 21.921         | 0.329            | 98    | 0.718            | 38.531         | 0.741            | 143   | 1.550            | 45.857         | 1.636            |       |                  |                |                  |
| 9     | 0.051            | 3.776          | 0.052            | 54    | 0.327            | 22.321         | 0.336            | 99    | 0.729            | 38.849         | 0.753            | 144   | 1.577            | 45.695         | 1.667            |       |                  |                |                  |
| 10    | 0.057            | 4.195          | 0.058            | 55    | 0.334            | 22.720         | 0.344            | 100   | 0.742            | 39.164         | 0.766            | 145   | 1.605            | 45.505         | 1.698            |       |                  |                |                  |
| 11    | 0.062            | 4.614          | 0.064            | 56    | 0.341            | 23.119         | 0.351            | 101   | 0.754            | 39.474         | 0.779            | 146   | 1.632            | 45.284         | 1.730            |       |                  |                |                  |
| 12    | 0.068            | 5.033          | 0.070            | 57    | 0.348            | 23.516         | 0.358            | 102   | 0.766            | 39.781         | 0.792            | 147   | 1.660            | 45.032         | 1.762            |       |                  |                |                  |
| 13    | 0.074            | 5.452          | 0.076            | 58    | 0.355            | 23.912         | 0.366            | 103   | 0.779            | 40.084         | 0.805            | 148   | 1.688            | 44.745         | 1.794            |       |                  |                |                  |
| 14    | 0.079            | 5.870          | 0.082            | 59    | 0.363            | 24.308         | 0.373            | 104   | 0.792            | 40.383         | 0.818            | 149   | 1.715            | 44.422         | 1.827            |       |                  |                |                  |
| 15    | 0.085            | 6.289          | 0.087            | 60    | 0.370            | 24.702         | 0.381            | 105   | 0.805            | 40.677         | 0.832            | 150   | 1.742            | 44.060         | 1.859            |       |                  |                |                  |
| 16    | 0.091            | 6.707          | 0.093            | 61    | 0.377            | 25.095         | 0.388            | 106   | 0.818            | 40.968         | 0.846            | 151   | 1.769            | 43.657         | 1.892            |       |                  |                |                  |
| 17    | 0.097            | 7.125          | 0.099            | 62    | 0.385            | 25.487         | 0.396            | 107   | 0.832            | 41.253         | 0.860            | 152   | 1.795            | 43.210         | 1.924            |       |                  |                |                  |
| 18    | 0.102            | 7.543          | 0.105            | 63    | 0.392            | 25.879         | 0.404            | 108   | 0.846            | 41.534         | 0.875            | 153   | 1.820            | 42.716         | 1.955            |       |                  |                |                  |
| 19    | 0.108            | 7.960          | 0.111            | 64    | 0.400            | 26.269         | 0.411            | 109   | 0.860            | 41.810         | 0.890            | 154   | 1.844            | 42.173         | 1.986            |       |                  |                |                  |
| 20    | 0.114            | 8.378          | 0.117            | 65    | 0.407            | 26.658         | 0.419            | 110   | 0.874            | 42.081         | 0.905            | 155   | 1.867            | 41.577         | 2.015            |       |                  |                |                  |
| 21    | 0.120            | 8.795          | 0.123            | 66    | 0.415            | 27.045         | 0.427            | 111   | 0.889            | 42.346         | 0.920            | 156   | 1.888            | 40.923         | 2.043            |       |                  |                |                  |
| 22    | 0.126            | 9.212          | 0.129            | 67    | 0.423            | 27.432         | 0.435            | 112   | 0.904            | 42.606         | 0.936            | 157   | 1.906            | 40.210         | 2.069            |       |                  |                |                  |
| 23    | 0.131            | 9.628          | 0.135            | 68    | 0.431            | 27.817         | 0.443            | 113   | 0.919            | 42.860         | 0.952            | 158   | 1.921            | 39.433         | 2.092            |       |                  |                |                  |
| 24    | 0.137            | 10.045         | 0.141            | 69    | 0.439            | 28.201         | 0.452            | 114   | 0.934            | 43.108         | 0.968            | 159   | 1.933            | 38.587         | 2.112            |       |                  |                |                  |
| 25    | 0.143            | 10.461         | 0.147            | 70    | 0.447            | 28.583         | 0.460            | 115   | 0.950            | 43.349         | 0.985            | 160   | 1.942            | 37.669         | 2.128            |       |                  |                |                  |
| 26    | 0.149            | 10.876         | 0.153            | 71    | 0.455            | 28.964         | 0.468            | 116   | 0.966            | 43.585         | 1.002            | 161   | 1.945            | 36.675         | 2.140            |       |                  |                |                  |
| 27    | 0.155            | 11.292         | 0.159            | 72    | 0.463            | 29.344         | 0.477            | 117   | 0.983            | 43.813         | 1.019            | 162   | 1.943            | 35.599         | 2.146            |       |                  |                |                  |
| 28    | 0.161            | 11.707         | 0.165            | 73    | 0.471            | 29.722         | 0.485            | 118   | 1.000            | 44.034         | 1.037            | 163   | 1.934            | 34.437         | 2.145            |       |                  |                |                  |
| 29    | 0.167            | 12.121         | 0.172            | 74    | 0.480            | 30.099         | 0.494            | 119   | 1.017            | 44.248         | 1.055            | 164   | 1.918            | 33.186         | 2.137            |       |                  |                |                  |
| 30    | 0.173            | 12.536         | 0.178            | 75    | 0.488            | 30.474         | 0.503            | 120   | 1.034            | 44.453         | 1.074            | 165   | 1.893            | 31.840         | 2.119            |       |                  |                |                  |
| 31    | 0.179            | 12.950         | 0.184            | 76    | 0.497            | 30.847         | 0.512            | 121   | 1.052            | 44.651         | 1.093            | 166   | 1.859            | 30.396         | 2.091            |       |                  |                |                  |
| 32    | 0.185            | 13.363         | 0.190            | 77    | 0.506            | 31.219         | 0.521            | 122   | 1.070            | 44.840         | 1.112            | 167   | 1.814            | 28.850         | 2.050            |       |                  |                |                  |
| 33    | 0.191            | 13.776         | 0.196            | 78    | 0.514            | 31.589         | 0.530            | 123   | 1.089            | 45.021         | 1.132            | 168   | 1.758            | 27.200         | 1.996            |       |                  |                |                  |
| 34    | 0.197            | 14.189         | 0.203            | 79    | 0.523            | 31.958         | 0.539            | 124   | 1.108            | 45.191         | 1.152            | 169   | 1.688            | 25.442         | 1.926            |       |                  |                |                  |
| 35    | 0.203            | 14.601         | 0.209            | 80    | 0.532            | 32.324         | 0.548            | 125   | 1.127            | 45.353         | 1.173            | 170   | 1.604            | 23.577         | 1.840            |       |                  |                |                  |
| 36    | 0.209            | 15.013         | 0.215            | 81    | 0.541            | 32.689         | 0.558            | 126   | 1.147            | 45.503         | 1.194            | 171   | 1.505            | 21.604         | 1.735            |       |                  |                |                  |
| 37    | 0.216            | 15.424         | 0.222            | 82    | 0.551            | 33.052         | 0.567            | 127   | 1.167            | 45.644         | 1.216            | 172   | 1.391            | 19.526         | 1.612            |       |                  |                |                  |
| 38    | 0.222            | 15.834         | 0.228            | 83    | 0.560            | 33.412         | 0.577            | 128   | 1.188            | 45.772         | 1.238            | 173   | 1.261            | 17.347         | 1.468            |       |                  |                |                  |
| 39    | 0.228            | 16.245         | 0.235            | 84    | 0.569            | 33.771         | 0.587            | 129   | 1.209            | 45.889         | 1.260            | 174   | 1.116            | 15.071         | 1.305            |       |                  |                |                  |
| 40    | 0.234            | 16.654         | 0.241            | 85    | 0.579            | 34.127         | 0.597            | 130   | 1.230            | 45.994         | 1.284            | 175   | 0.956            | 12.707         | 1.123            |       |                  |                |                  |
| 41    | 0.241            | 17.063         | 0.248            | 86    | 0.589            | 34.482         | 0.607            | 131   | 1.252            | 46.085         | 1.307            | 176   | 0.783            | 10.266         | 0.922            |       |                  |                |                  |
| 42    | 0.247            | 17.472         | 0.254            | 87    | 0.599            | 34.834         | 0.617            | 132   | 1.275            | 46.163         | 1.331            | 177   | 0.598            | 7.760          | 0.707            |       |                  |                |                  |
| 43    | 0.254            | 17.880         | 0.261            | 88    | 0.609            | 35.183         | 0.628            | 133   | 1.297            | 46.226         | 1.356            | 178   | 0.404            | 5.202          | 0.478            |       |                  |                |                  |
| 44    | 0.260            | 18.287         | 0.267            | 89    | 0.619            | 35.530         | 0.638            | 134   | 1.321            | 46.274         | 1.382            | 179   | 0.204            | 2.610          | 0.242            |       |                  |                |                  |
| 45    | 0.267            | 18.694         | 0.274            | 90    | 0.629            | 35.875         | 0.649            | 135   | 1.344            | 46.305         | 1.408            | 180   | 0.000            | 0.000          | 0.000            |       |                  |                |                  |

Table 9.3: Epicyclic anomalies of Venus. All quantities are in degrees. Note that  $\bar{\theta}(360^\circ - \mu) = -\bar{\theta}(\mu)$ , and  $\delta\theta_\pm(360^\circ - \mu) = -\delta\theta_\pm(\mu)$ .

| Event                | Date       | $\lambda$ | Elongation |
|----------------------|------------|-----------|------------|
| Superior Conjunction | 11/06/2000 | 20GE46    |            |
| Greatest Elongation  | 17/01/2001 | 14PI23    | 47.1° E    |
| Inferior Conjunction | 30/03/2001 | 09AR36    |            |
| Greatest Elongation  | 08/06/2001 | 01TA39    | 45.8° W    |
| Superior Conjunction | 14/01/2002 | 23CP59    |            |
| Greatest Elongation  | 22/08/2002 | 15LI08    | 46.0° E    |
| Inferior Conjunction | 31/10/2002 | 07SC58    |            |
| Greatest Elongation  | 11/01/2003 | 03SG31    | 47.0° W    |
| Superior Conjunction | 18/08/2003 | 25LE20    |            |
| Greatest Elongation  | 29/03/2004 | 25TA04    | 46.0° E    |
| Inferior Conjunction | 08/06/2004 | 17GE52    |            |
| Greatest Elongation  | 17/08/2004 | 09CN29    | 45.8° W    |
| Superior Conjunction | 31/03/2005 | 10AR33    |            |
| Greatest Elongation  | 03/11/2005 | 28SG27    | 47.1° E    |
| Inferior Conjunction | 13/01/2006 | 23CP36    |            |
| Greatest Elongation  | 25/03/2006 | 18AQ07    | 46.5° W    |
| Superior Conjunction | 27/10/2006 | 04SC13    |            |
| Greatest Elongation  | 09/06/2007 | 03LE20    | 45.4° E    |
| Inferior Conjunction | 18/08/2007 | 24LE45    |            |
| Greatest Elongation  | 28/10/2007 | 18VI19    | 46.5° W    |
| Superior Conjunction | 09/06/2008 | 18GE41    |            |
| Greatest Elongation  | 14/01/2009 | 12PI03    | 47.1° E    |
| Inferior Conjunction | 27/03/2009 | 07AR19    |            |
| Greatest Elongation  | 05/06/2009 | 29AR25    | 45.8° W    |
| Superior Conjunction | 11/01/2010 | 21CP24    |            |
| Greatest Elongation  | 19/08/2010 | 12LI49    | 46.0° E    |
| Inferior Conjunction | 29/10/2010 | 05SC34    |            |
| Greatest Elongation  | 08/01/2011 | 01SG06    | 47.0° W    |
| Superior Conjunction | 16/08/2011 | 23LE13    |            |
| Greatest Elongation  | 27/03/2012 | 22TA50    | 46.0° E    |
| Inferior Conjunction | 06/06/2012 | 15GE43    |            |
| Greatest Elongation  | 15/08/2012 | 07CN18    | 45.8° W    |
| Superior Conjunction | 28/03/2013 | 08AR12    |            |
| Greatest Elongation  | 01/11/2013 | 26SG02    | 47.1° E    |
| Inferior Conjunction | 11/01/2014 | 21CP08    |            |
| Greatest Elongation  | 23/03/2014 | 15AQ44    | 46.5° W    |
| Superior Conjunction | 25/10/2014 | 01SC51    |            |
| Greatest Elongation  | 06/06/2015 | 01LE09    | 45.4° E    |
| Inferior Conjunction | 15/08/2015 | 22LE33    |            |
| Greatest Elongation  | 26/10/2015 | 16VI02    | 46.4° W    |

Table 9.4: The conjunctions and greatest elongations of Venus during the years 2000–2015 AD.

| $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(^{\circ})$ | $\Delta M(^{\circ})$ | $\Delta \bar{F}(^{\circ})$ | $\Delta t(\text{JD})$ | $\Delta \bar{\lambda}(^{\circ})$ | $\Delta M(^{\circ})$ | $\Delta \bar{F}(^{\circ})$ |
|-----------------------|----------------------------------|----------------------|----------------------------|-----------------------|----------------------------------|----------------------|----------------------------|
| 10 000                | 243.770                          | 243.344              | 243.422                    | 1 000                 | 132.377                          | 132.334              | 132.342                    |
| 20 000                | 127.541                          | 126.688              | 126.844                    | 2 000                 | 264.754                          | 264.669              | 264.684                    |
| 30 000                | 11.311                           | 10.032               | 10.266                     | 3 000                 | 37.131                           | 37.003               | 37.027                     |
| 40 000                | 255.081                          | 253.376              | 253.688                    | 4 000                 | 169.508                          | 169.338              | 169.369                    |
| 50 000                | 138.852                          | 136.720              | 137.110                    | 5 000                 | 301.885                          | 301.672              | 301.711                    |
| 60 000                | 22.622                           | 20.063               | 20.533                     | 6 000                 | 74.262                           | 74.006               | 74.053                     |
| 70 000                | 266.392                          | 263.407              | 263.955                    | 7 000                 | 206.639                          | 206.341              | 206.395                    |
| 80 000                | 150.162                          | 146.751              | 147.377                    | 8 000                 | 339.016                          | 338.675              | 338.738                    |
| 90 000                | 33.933                           | 30.095               | 30.799                     | 9 000                 | 111.393                          | 111.010              | 111.080                    |
|                       |                                  |                      |                            |                       |                                  |                      |                            |
| 100                   | 49.238                           | 49.233               | 49.234                     | 10                    | 40.924                           | 40.923               | 40.923                     |
| 200                   | 98.475                           | 98.467               | 98.468                     | 20                    | 81.848                           | 81.847               | 81.847                     |
| 300                   | 147.713                          | 147.700              | 147.703                    | 30                    | 122.771                          | 122.770              | 122.770                    |
| 400                   | 196.951                          | 196.934              | 196.937                    | 40                    | 163.695                          | 163.693              | 163.694                    |
| 500                   | 246.189                          | 246.167              | 246.171                    | 50                    | 204.619                          | 204.617              | 204.617                    |
| 600                   | 295.426                          | 295.401              | 295.405                    | 60                    | 245.543                          | 245.540              | 245.541                    |
| 700                   | 344.664                          | 344.634              | 344.640                    | 70                    | 286.466                          | 286.463              | 286.464                    |
| 800                   | 33.902                           | 33.868               | 33.874                     | 80                    | 327.390                          | 327.387              | 327.387                    |
| 900                   | 83.139                           | 83.101               | 83.108                     | 90                    | 8.314                            | 8.310                | 8.311                      |
|                       |                                  |                      |                            |                       |                                  |                      |                            |
| 1                     | 4.092                            | 4.092                | 4.092                      | 0.1                   | 0.409                            | 0.409                | 0.409                      |
| 2                     | 8.185                            | 8.185                | 8.185                      | 0.2                   | 0.818                            | 0.818                | 0.818                      |
| 3                     | 12.277                           | 12.277               | 12.277                     | 0.3                   | 1.228                            | 1.228                | 1.228                      |
| 4                     | 16.370                           | 16.369               | 16.369                     | 0.4                   | 1.637                            | 1.637                | 1.637                      |
| 5                     | 20.462                           | 20.462               | 20.462                     | 0.5                   | 2.046                            | 2.046                | 2.046                      |
| 6                     | 24.554                           | 24.554               | 24.554                     | 0.6                   | 2.455                            | 2.455                | 2.455                      |
| 7                     | 28.647                           | 28.646               | 28.646                     | 0.7                   | 2.865                            | 2.865                | 2.865                      |
| 8                     | 32.739                           | 32.739               | 32.739                     | 0.8                   | 3.274                            | 3.274                | 3.274                      |
| 9                     | 36.831                           | 36.831               | 36.831                     | 0.9                   | 3.683                            | 3.683                | 3.683                      |

Table 9.5: Mean motion of Mercury. Here,  $\Delta t = t - t_0$ ,  $\Delta \bar{\lambda} = \bar{\lambda} - \bar{\lambda}_0$ ,  $\Delta M = M - M_0$ , and  $\Delta \bar{F} = \bar{F} - \bar{F}_0$ . At epoch ( $t_0 = 2\,451\,545.0$  JD),  $\bar{\lambda}_0 = 252.087^{\circ}$ ,  $M_0 = 174.693^{\circ}$ , and  $\bar{F}_0 = 204.436^{\circ}$ .

| $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ | $M(^{\circ})$ | $q(^{\circ})$ | 100 $\zeta$ |
|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|---------------|---------------|-------------|
| 0             | 0.000         | 20.564      | 90            | 22.900        | -4.229      | 180           | 0.000         | -20.564     | 270           | -22.900       | -4.229      |
| 2             | 1.086         | 20.544      | 92            | 22.677        | -4.896      | 182           | -0.663        | -20.555     | 272           | -23.100       | -3.551      |
| 4             | 2.169         | 20.487      | 94            | 22.433        | -5.552      | 184           | -1.326        | -20.528     | 274           | -23.276       | -2.864      |
| 6             | 3.247         | 20.391      | 96            | 22.168        | -6.197      | 186           | -1.987        | -20.483     | 276           | -23.428       | -2.168      |
| 8             | 4.316         | 20.257      | 98            | 21.884        | -6.831      | 188           | -2.647        | -20.420     | 278           | -23.553       | -1.463      |
| 10            | 5.376         | 20.085      | 100           | 21.580        | -7.452      | 190           | -3.304        | -20.340     | 280           | -23.652       | -0.750      |
| 12            | 6.422         | 19.876      | 102           | 21.259        | -8.062      | 192           | -3.959        | -20.242     | 282           | -23.723       | -0.030      |
| 14            | 7.454         | 19.631      | 104           | 20.920        | -8.659      | 194           | -4.610        | -20.126     | 284           | -23.764       | 0.697       |
| 16            | 8.467         | 19.350      | 106           | 20.566        | -9.243      | 196           | -5.257        | -19.993     | 286           | -23.775       | 1.429       |
| 18            | 9.460         | 19.035      | 108           | 20.195        | -9.815      | 198           | -5.900        | -19.842     | 288           | -23.755       | 2.165       |
| 20            | 10.431        | 18.685      | 110           | 19.809        | -10.373     | 200           | -6.538        | -19.675     | 290           | -23.703       | 2.905       |
| 22            | 11.377        | 18.303      | 112           | 19.409        | -10.918     | 202           | -7.170        | -19.490     | 292           | -23.617       | 3.648       |
| 24            | 12.298        | 17.889      | 114           | 18.996        | -11.450     | 204           | -7.796        | -19.288     | 294           | -23.497       | 4.392       |
| 26            | 13.190        | 17.445      | 116           | 18.569        | -11.969     | 206           | -8.417        | -19.070     | 296           | -23.342       | 5.137       |
| 28            | 14.052        | 16.971      | 118           | 18.129        | -12.473     | 208           | -9.030        | -18.835     | 298           | -23.150       | 5.880       |
| 30            | 14.882        | 16.469      | 120           | 17.677        | -12.964     | 210           | -9.637        | -18.583     | 300           | -22.922       | 6.621       |
| 32            | 15.680        | 15.941      | 122           | 17.212        | -13.441     | 212           | -10.236       | -18.316     | 302           | -22.656       | 7.359       |
| 34            | 16.443        | 15.388      | 124           | 16.737        | -13.904     | 214           | -10.827       | -18.032     | 304           | -22.353       | 8.091       |
| 36            | 17.171        | 14.811      | 126           | 16.250        | -14.353     | 216           | -11.410       | -17.733     | 306           | -22.010       | 8.818       |
| 38            | 17.862        | 14.212      | 128           | 15.752        | -14.787     | 218           | -11.985       | -17.418     | 308           | -21.629       | 9.536       |
| 40            | 18.517        | 13.593      | 130           | 15.243        | -15.207     | 220           | -12.552       | -17.087     | 310           | -21.208       | 10.245      |
| 42            | 19.133        | 12.954      | 132           | 14.724        | -15.613     | 222           | -13.109       | -16.741     | 312           | -20.748       | 10.942      |
| 44            | 19.710        | 12.299      | 134           | 14.196        | -16.004     | 224           | -13.657       | -16.380     | 314           | -20.249       | 11.628      |
| 46            | 20.249        | 11.628      | 136           | 13.657        | -16.380     | 226           | -14.196       | -16.004     | 316           | -19.710       | 12.299      |
| 48            | 20.748        | 10.942      | 138           | 13.109        | -16.741     | 228           | -14.724       | -15.613     | 318           | -19.133       | 12.954      |
| 50            | 21.208        | 10.245      | 140           | 12.552        | -17.087     | 230           | -15.243       | -15.207     | 320           | -18.517       | 13.593      |
| 52            | 21.629        | 9.536       | 142           | 11.985        | -17.418     | 232           | -15.752       | -14.787     | 322           | -17.862       | 14.212      |
| 54            | 22.010        | 8.818       | 144           | 11.410        | -17.733     | 234           | -16.250       | -14.353     | 324           | -17.171       | 14.811      |
| 56            | 22.353        | 8.091       | 146           | 10.827        | -18.032     | 236           | -16.737       | -13.904     | 326           | -16.443       | 15.388      |
| 58            | 22.656        | 7.359       | 148           | 10.236        | -18.316     | 238           | -17.212       | -13.441     | 328           | -15.680       | 15.941      |
| 60            | 22.922        | 6.621       | 150           | 9.637         | -18.583     | 240           | -17.677       | -12.964     | 330           | -14.882       | 16.469      |
| 62            | 23.150        | 5.880       | 152           | 9.030         | -18.835     | 242           | -18.129       | -12.473     | 332           | -14.052       | 16.971      |
| 64            | 23.342        | 5.137       | 154           | 8.417         | -19.070     | 244           | -18.569       | -11.969     | 334           | -13.190       | 17.445      |
| 66            | 23.497        | 4.392       | 156           | 7.796         | -19.288     | 246           | -18.996       | -11.450     | 336           | -12.298       | 17.889      |
| 68            | 23.617        | 3.648       | 158           | 7.170         | -19.490     | 248           | -19.409       | -10.918     | 338           | -11.377       | 18.303      |
| 70            | 23.703        | 2.905       | 160           | 6.538         | -19.675     | 250           | -19.809       | -10.373     | 340           | -10.431       | 18.685      |
| 72            | 23.755        | 2.165       | 162           | 5.900         | -19.842     | 252           | -20.195       | -9.815      | 342           | -9.460        | 19.035      |
| 74            | 23.775        | 1.429       | 164           | 5.257         | -19.993     | 254           | -20.566       | -9.243      | 344           | -8.467        | 19.350      |
| 76            | 23.764        | 0.697       | 166           | 4.610         | -20.126     | 256           | -20.920       | -8.659      | 346           | -7.454        | 19.631      |
| 78            | 23.723        | -0.030      | 168           | 3.959         | -20.242     | 258           | -21.259       | -8.062      | 348           | -6.422        | 19.876      |
| 80            | 23.652        | -0.750      | 170           | 3.304         | -20.340     | 260           | -21.580       | -7.452      | 350           | -5.376        | 20.085      |
| 82            | 23.553        | -1.463      | 172           | 2.647         | -20.420     | 262           | -21.884       | -6.831      | 352           | -4.316        | 20.257      |
| 84            | 23.428        | -2.168      | 174           | 1.987         | -20.483     | 264           | -22.168       | -6.197      | 354           | -3.247        | 20.391      |
| 86            | 23.276        | -2.864      | 176           | 1.326         | -20.528     | 266           | -22.433       | -5.552      | 356           | -2.169        | 20.487      |
| 88            | 23.100        | -3.551      | 178           | 0.663         | -20.555     | 268           | -22.677       | -4.896      | 358           | -1.086        | 20.544      |
| 90            | 22.900        | -4.229      | 180           | 0.000         | -20.564     | 270           | -22.900       | -4.229      | 360           | -0.000        | 20.564      |

Table 9.6: Differential anomalies of Mercury.

| $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ | $\mu$ | $\delta\theta_-$ | $\bar{\theta}$ | $\delta\theta_+$ |
|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|-------|------------------|----------------|------------------|
| 0     | 0.000            | 0.000          | 0.000            | 45    | 1.711            | 11.702         | 2.403            | 90    | 3.450            | 20.277         | 5.113            | 135   | 4.257            | 19.475         | 7.325            |
| 1     | 0.038            | 0.270          | 0.052            | 46    | 1.749            | 11.941         | 2.459            | 91    | 3.486            | 20.395         | 5.177            | 136   | 4.237            | 19.267         | 7.327            |
| 2     | 0.075            | 0.540          | 0.104            | 47    | 1.788            | 12.179         | 2.515            | 92    | 3.522            | 20.509         | 5.241            | 137   | 4.214            | 19.048         | 7.324            |
| 3     | 0.113            | 0.809          | 0.156            | 48    | 1.827            | 12.415         | 2.572            | 93    | 3.557            | 20.618         | 5.305            | 138   | 4.188            | 18.818         | 7.317            |
| 4     | 0.150            | 1.079          | 0.208            | 49    | 1.866            | 12.650         | 2.628            | 94    | 3.592            | 20.722         | 5.368            | 139   | 4.159            | 18.578         | 7.304            |
| 5     | 0.188            | 1.348          | 0.260            | 50    | 1.905            | 12.882         | 2.685            | 95    | 3.627            | 20.822         | 5.432            | 140   | 4.127            | 18.326         | 7.286            |
| 6     | 0.225            | 1.618          | 0.313            | 51    | 1.944            | 13.114         | 2.742            | 96    | 3.662            | 20.917         | 5.496            | 141   | 4.092            | 18.064         | 7.262            |
| 7     | 0.263            | 1.887          | 0.365            | 52    | 1.983            | 13.343         | 2.799            | 97    | 3.696            | 21.007         | 5.559            | 142   | 4.053            | 17.791         | 7.233            |
| 8     | 0.301            | 2.156          | 0.417            | 53    | 2.022            | 13.571         | 2.857            | 98    | 3.729            | 21.091         | 5.623            | 143   | 4.011            | 17.506         | 7.197            |
| 9     | 0.338            | 2.425          | 0.469            | 54    | 2.061            | 13.797         | 2.914            | 99    | 3.762            | 21.171         | 5.686            | 144   | 3.966            | 17.210         | 7.155            |
| 10    | 0.376            | 2.693          | 0.521            | 55    | 2.100            | 14.021         | 2.972            | 100   | 3.795            | 21.246         | 5.749            | 145   | 3.917            | 16.903         | 7.106            |
| 11    | 0.414            | 2.961          | 0.574            | 56    | 2.139            | 14.244         | 3.030            | 101   | 3.827            | 21.315         | 5.812            | 146   | 3.865            | 16.585         | 7.050            |
| 12    | 0.451            | 3.229          | 0.626            | 57    | 2.178            | 14.464         | 3.088            | 102   | 3.858            | 21.378         | 5.874            | 147   | 3.809            | 16.255         | 6.986            |
| 13    | 0.489            | 3.497          | 0.678            | 58    | 2.217            | 14.683         | 3.146            | 103   | 3.889            | 21.436         | 5.937            | 148   | 3.749            | 15.914         | 6.915            |
| 14    | 0.527            | 3.764          | 0.731            | 59    | 2.257            | 14.899         | 3.205            | 104   | 3.919            | 21.488         | 5.999            | 149   | 3.686            | 15.561         | 6.836            |
| 15    | 0.564            | 4.031          | 0.783            | 60    | 2.296            | 15.113         | 3.263            | 105   | 3.949            | 21.534         | 6.060            | 150   | 3.619            | 15.197         | 6.749            |
| 16    | 0.602            | 4.298          | 0.836            | 61    | 2.335            | 15.326         | 3.322            | 106   | 3.977            | 21.575         | 6.121            | 151   | 3.548            | 14.822         | 6.654            |
| 17    | 0.640            | 4.564          | 0.889            | 62    | 2.374            | 15.536         | 3.381            | 107   | 4.005            | 21.609         | 6.182            | 152   | 3.473            | 14.436         | 6.549            |
| 18    | 0.678            | 4.829          | 0.941            | 63    | 2.413            | 15.743         | 3.441            | 108   | 4.032            | 21.636         | 6.242            | 153   | 3.394            | 14.039         | 6.436            |
| 19    | 0.716            | 5.094          | 0.994            | 64    | 2.453            | 15.949         | 3.500            | 109   | 4.059            | 21.658         | 6.301            | 154   | 3.311            | 13.630         | 6.314            |
| 20    | 0.753            | 5.359          | 1.047            | 65    | 2.492            | 16.152         | 3.560            | 110   | 4.084            | 21.673         | 6.360            | 155   | 3.224            | 13.211         | 6.182            |
| 21    | 0.791            | 5.622          | 1.100            | 66    | 2.531            | 16.353         | 3.620            | 111   | 4.109            | 21.681         | 6.418            | 156   | 3.133            | 12.780         | 6.041            |
| 22    | 0.829            | 5.886          | 1.153            | 67    | 2.570            | 16.551         | 3.680            | 112   | 4.132            | 21.682         | 6.475            | 157   | 3.039            | 12.339         | 5.890            |
| 23    | 0.867            | 6.148          | 1.206            | 68    | 2.609            | 16.747         | 3.740            | 113   | 4.155            | 21.676         | 6.532            | 158   | 2.940            | 11.888         | 5.729            |
| 24    | 0.905            | 6.410          | 1.259            | 69    | 2.649            | 16.940         | 3.801            | 114   | 4.176            | 21.663         | 6.587            | 159   | 2.838            | 11.426         | 5.558            |
| 25    | 0.943            | 6.672          | 1.312            | 70    | 2.688            | 17.131         | 3.862            | 115   | 4.197            | 21.643         | 6.641            | 160   | 2.732            | 10.955         | 5.377            |
| 26    | 0.981            | 6.932          | 1.366            | 71    | 2.727            | 17.319         | 3.923            | 116   | 4.216            | 21.616         | 6.695            | 161   | 2.622            | 10.474         | 5.186            |
| 27    | 1.019            | 7.192          | 1.419            | 72    | 2.766            | 17.504         | 3.984            | 117   | 4.233            | 21.580         | 6.747            | 162   | 2.508            | 9.983          | 4.985            |
| 28    | 1.057            | 7.451          | 1.473            | 73    | 2.805            | 17.686         | 4.045            | 118   | 4.250            | 21.538         | 6.797            | 163   | 2.391            | 9.483          | 4.774            |
| 29    | 1.095            | 7.709          | 1.526            | 74    | 2.844            | 17.865         | 4.107            | 119   | 4.265            | 21.487         | 6.846            | 164   | 2.271            | 8.974          | 4.554            |
| 30    | 1.133            | 7.967          | 1.580            | 75    | 2.882            | 18.042         | 4.169            | 120   | 4.278            | 21.428         | 6.894            | 165   | 2.147            | 8.457          | 4.324            |
| 31    | 1.172            | 8.223          | 1.634            | 76    | 2.921            | 18.215         | 4.231            | 121   | 4.291            | 21.361         | 6.940            | 166   | 2.019            | 7.932          | 4.084            |
| 32    | 1.210            | 8.479          | 1.688            | 77    | 2.960            | 18.385         | 4.293            | 122   | 4.301            | 21.286         | 6.984            | 167   | 1.889            | 7.399          | 3.835            |
| 33    | 1.248            | 8.734          | 1.742            | 78    | 2.999            | 18.552         | 4.355            | 123   | 4.310            | 21.202         | 7.026            | 168   | 1.756            | 6.859          | 3.578            |
| 34    | 1.286            | 8.987          | 1.796            | 79    | 3.037            | 18.716         | 4.417            | 124   | 4.317            | 21.109         | 7.067            | 169   | 1.620            | 6.312          | 3.312            |
| 35    | 1.325            | 9.240          | 1.851            | 80    | 3.075            | 18.876         | 4.480            | 125   | 4.322            | 21.008         | 7.105            | 170   | 1.481            | 5.759          | 3.038            |
| 36    | 1.363            | 9.491          | 1.905            | 81    | 3.114            | 19.033         | 4.543            | 126   | 4.326            | 20.898         | 7.140            | 171   | 1.340            | 5.200          | 2.757            |
| 37    | 1.402            | 9.742          | 1.960            | 82    | 3.152            | 19.186         | 4.606            | 127   | 4.327            | 20.778         | 7.173            | 172   | 1.197            | 4.636          | 2.469            |
| 38    | 1.440            | 9.991          | 2.015            | 83    | 3.190            | 19.336         | 4.669            | 128   | 4.326            | 20.649         | 7.204            | 173   | 1.052            | 4.067          | 2.174            |
| 39    | 1.479            | 10.240         | 2.070            | 84    | 3.227            | 19.482         | 4.732            | 129   | 4.323            | 20.511         | 7.231            | 174   | 0.905            | 3.494          | 1.875            |
| 40    | 1.517            | 10.487         | 2.125            | 85    | 3.265            | 19.625         | 4.795            | 130   | 4.318            | 20.363         | 7.256            | 175   | 0.756            | 2.917          | 1.570            |
| 41    | 1.556            | 10.732         | 2.180            | 86    | 3.302            | 19.763         | 4.859            | 131   | 4.311            | 20.206         | 7.277            | 176   | 0.607            | 2.337          | 1.261            |
| 42    | 1.594            | 10.977         | 2.236            | 87    | 3.340            | 19.898         | 4.922            | 132   | 4.301            | 20.038         | 7.295            | 177   | 0.456            | 1.755          | 0.949            |
| 43    | 1.633            | 11.220         | 2.291            | 88    | 3.377            | 20.029         | 4.986            | 133   | 4.289            | 19.861         | 7.309            | 178   | 0.304            | 1.171          | 0.634            |
| 44    | 1.672            | 11.462         | 2.347            | 89    | 3.413            | 20.155         | 5.049            | 134   | 4.274            | 19.673         | 7.319            | 179   | 0.152            | 0.586          | 0.317            |
| 45    | 1.711            | 11.702         | 2.403            | 90    | 3.450            | 20.277         | 5.113            | 135   | 4.257            | 19.475         | 7.325            | 180   | 0.000            | 0.000          | 0.000            |

Table 9.7: Epicyclic anomalies of Mercury. All quantities are in degrees. Note that  $\bar{\theta}(360^\circ - \mu) = -\bar{\theta}(\mu)$ , and  $\delta\theta_\pm(360^\circ - \mu) = -\delta\theta_\pm(\mu)$ .

| Event                | Date       | $\lambda$ | Elongation |
|----------------------|------------|-----------|------------|
| Superior Conjunction | 15/01/2000 | 25CP08    |            |
| Greatest Elongation  | 15/02/2000 | 13PI44    | 18.1° E    |
| Inferior Conjunction | 01/03/2000 | 11PI23    |            |
| Greatest Elongation  | 28/03/2000 | 10PI35    | 27.9° W    |
| Superior Conjunction | 09/05/2000 | 18TA59    |            |
| Greatest Elongation  | 09/06/2000 | 13CN27    | 24.2° E    |
| Inferior Conjunction | 06/07/2000 | 14CN39    |            |
| Greatest Elongation  | 27/07/2000 | 15CN03    | 19.7° W    |
| Superior Conjunction | 22/08/2000 | 29LE16    |            |
| Greatest Elongation  | 06/10/2000 | 09SC17    | 25.6° E    |
| Inferior Conjunction | 30/10/2000 | 06SC58    |            |
| Greatest Elongation  | 15/11/2000 | 04SC02    | 19.3° W    |
| Superior Conjunction | 25/12/2000 | 04CP18    |            |
| Greatest Elongation  | 28/01/2001 | 27AQ07    | 18.4° E    |
| Inferior Conjunction | 13/02/2001 | 24AQ23    |            |
| Greatest Elongation  | 11/03/2001 | 23AQ05    | 27.5° W    |
| Superior Conjunction | 23/04/2001 | 03TA22    |            |
| Greatest Elongation  | 22/05/2001 | 24GE13    | 22.6° E    |
| Inferior Conjunction | 16/06/2001 | 25GE26    |            |
| Greatest Elongation  | 09/07/2001 | 26GE18    | 21.1° W    |
| Superior Conjunction | 05/08/2001 | 13LE32    |            |
| Greatest Elongation  | 19/09/2001 | 22LI50    | 26.6° E    |
| Inferior Conjunction | 14/10/2001 | 20LI51    |            |
| Greatest Elongation  | 29/10/2001 | 17LI50    | 18.5° W    |
| Superior Conjunction | 04/12/2001 | 12SG45    |            |
| Greatest Elongation  | 12/01/2002 | 10AQ33    | 18.9° E    |
| Inferior Conjunction | 27/01/2002 | 07AQ42    |            |
| Greatest Elongation  | 21/02/2002 | 05AQ55    | 26.6° W    |
| Superior Conjunction | 07/04/2002 | 17AR27    |            |
| Greatest Elongation  | 04/05/2002 | 04GE54    | 21.0° E    |
| Inferior Conjunction | 27/05/2002 | 05GE48    |            |
| Greatest Elongation  | 21/06/2002 | 07GE04    | 22.8° W    |
| Superior Conjunction | 21/07/2002 | 28CN06    |            |
| Greatest Elongation  | 01/09/2002 | 06LI10    | 27.3° E    |
| Inferior Conjunction | 27/09/2002 | 04LI35    |            |
| Greatest Elongation  | 13/10/2002 | 01LI44    | 18.0° W    |
| Superior Conjunction | 14/11/2002 | 21SC39    |            |
| Greatest Elongation  | 26/12/2002 | 24CP01    | 19.8° E    |

Table 9.8: The conjunctions and greatest elongations of Mercury during the years 2000–2002 AD.





## 10. Planetary latitudes

### 10.1 Determination of ecliptic latitude of superior planet

Up to now, we have neglected the fact that the orbits of the five visible planets about the Sun are all slightly inclined to the plane of the ecliptic. Of course, these inclinations cause the ecliptic latitudes of the said planets to take small, but non-zero, values. In the following, we shall outline a model that is capable of predicting these values. Incidentally, the calculation of planetary latitudes is discussed in Book XIII of the *Almagest*, which is also the last book in the treatise. The planetary latitude model outlined in Book XIII is rather unwieldy, and not particularly accurate. Ptolemy greatly improved this model in his later works *Handy Tables* (Πρόχειροι κανόνες) and *Planetary Hypotheses* (ὑποθέσεις τῶν πλανωμένων).

Figure 10.1 shows the orbit of a superior planet. As we have already mentioned, the deferent and epicycle of such a planet have the same elements as the orbit of the planet in question around the Sun, and the apparent orbit of the Sun around the Earth, respectively. It follows that the deferent and epicycle of a superior planet are, respectively, inclined and parallel to the ecliptic plane. (Recall that the ecliptic plane corresponds to the plane of the Sun's apparent orbit about the Earth.) Let the plane of the deferent cut the ecliptic plane along the line  $NGN'$ . Here,  $N$  is the point at which the deferent passes through the plane of the ecliptic from south to north, in the direction of the mean planetary motion. This point is called the *ascending node*. Note that the line  $NGN'$  must pass through point  $G$ , because the Earth is common to the plane of the deferent and the ecliptic plane.

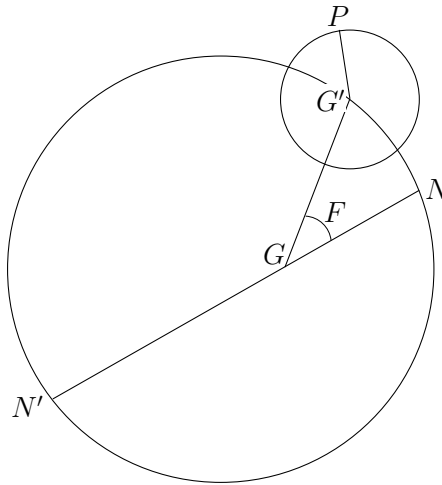


Figure 10.1: Orbit of a superior planet. Here,  $G$ ,  $G'$ ,  $P$ ,  $N$ ,  $N'$ , and  $F$  represent the Earth, the guide-point, the planet, the ascending node, the descending node, and the argument of latitude, respectively. The view is from northern ecliptic pole.

Now, it follows from simple geometry that the elevation of the guide-point  $G'$  above the ecliptic plane satisfies

$$v = r \sin i \sin F, \quad (10.1)$$

where  $r$  is the length  $GG'$ ,  $i$  the fixed inclination of the planetary orbit (and, hence, of the deferent) to the ecliptic plane, and  $F$  the angle  $NGG'$ . The angle  $F$  is termed the *argument of latitude*. We can write (see Chapter 8)

$$F = \bar{F} + q, \quad (10.2)$$

where  $\bar{F}$  is the *mean argument of latitude*, and  $q$  the equation of center of the deferent. Note that  $\bar{F}$  increases uniformly in time; that is,

$$\bar{F} = \bar{F}_0 + \tilde{n} (t - t_0). \quad (10.3)$$

Because the epicycle is parallel to the ecliptic plane, the elevation of the planet above the said plane is the same as that of the guide-point. Hence, from simple geometry, the ecliptic latitude of the planet satisfies

$$\beta = \frac{v}{r''}, \quad (10.4)$$

where  $r''$  is the length  $GP$ , and we have used the small angle approximation. However, it is apparent from Figure 8.3 that

$$r'' = (r^2 + 2 r r' \cos \mu + r'^2)^{1/2}, \quad (10.5)$$

where  $r'$  the length  $G'P$ , and  $\mu$  the equation of the epicycle. But, according to the analysis in Chapter 8,  $r/r' = a z$ , where  $a$  is the planetary major radius in units in which the major radius of the Sun's apparent orbit about the Earth is unity, and  $z$  is defined in Equation (8.4). Thus, we obtain

$$\beta = h \beta_0, \quad (10.6)$$

where

$$\beta_0(F) = \sin i \sin F \quad (10.7)$$

is termed the *deferential latitude*, and

$$h(\mu, z) = [1 + 2 (a z)^{-1} \cos \mu + (a z)^{-2}]^{-1/2} \quad (10.8)$$

the *epicyclic latitude correction factor*.

In the following,  $a, e, n, \tilde{n}, \bar{\lambda}_0, M_0, \bar{F}_0$ , and  $i$  are elements of the orbit of the planet in question about the Sun, and  $e_S, \zeta_S$ , and  $\lambda_S$  are elements of the Sun's apparent orbit about the Earth. The requisite elements for all of the superior planets at the J2000 epoch ( $t_0 = 2\,451\,545.0$  JD) are listed in Tables 5.1 and 10.1. Employing a quadratic interpolation scheme to represent  $F(\mu, z)$  (see Chapter 8), our procedure for determining the ecliptic latitude of a

superior planet is summed up by the following formulae:

$$\bar{\lambda} = \bar{\lambda}_0 + n(t - t_0), \quad (10.9)$$

$$M = M_0 + \tilde{n}(t - t_0), \quad (10.10)$$

$$\bar{F} = \bar{F}_0 + \tilde{n}(t - t_0), \quad (10.11)$$

$$q = 2e \sin M + (5/4)e^2 \sin 2M, \quad (10.12)$$

$$\zeta = e \cos M - e^2 \sin^2 M, \quad (10.13)$$

$$F = \bar{F} + q, \quad (10.14)$$

$$\beta_0 = \sin i \sin F, \quad (10.15)$$

$$\mu = \lambda_S - \bar{\lambda} - q, \quad (10.16)$$

$$\bar{h} = h(\mu, \bar{z}) \equiv [1 + 2(a\bar{z})^{-1} \cos \mu + (a\bar{z})^{-2}]^{-1/2}, \quad (10.17)$$

$$\delta h_- = h(\mu, \bar{z}) - h(\mu, z_{\max}), \quad (10.18)$$

$$\delta h_+ = h(\mu, z_{\min}) - h(\mu, \bar{z}), \quad (10.19)$$

$$z = \frac{1 - \zeta}{1 - \zeta_S}, \quad (10.20)$$

$$\xi = \frac{\bar{z} - z}{\delta z}, \quad (10.21)$$

$$h = \Theta_-(\xi) \delta h_- + \bar{h} + \Theta_+(\xi) \delta h_+, \quad (10.22)$$

$$\beta = h \beta_0. \quad (10.23)$$

Here,  $\bar{z} = (1 + e e_S)/(1 - e_S^2)$ ,  $\delta z = (e + e_S)/(1 - e_S^2)$ ,  $z_{\min} = \bar{z} - \delta z$ , and  $z_{\max} = \bar{z} + \delta z$ . The constants  $\bar{z}$ ,  $\delta z$ ,  $z_{\min}$ , and  $z_{\max}$  for each of the superior planets are listed in Table 8.1. Finally, the functions  $\Theta_{\pm}$  are tabulated in Table 8.2.

For the case of Mars, the previous formulae are capable of matching NASA ephemeris data during the years 1995–2006 AD with a mean error of  $0.3'$  and a maximum error of  $1.5'$ . For the case of Jupiter, the mean error is  $0.2'$  and the maximum error  $0.5'$ . Finally, for the case of Saturn, the mean error is  $0.05'$  and the maximum error  $0.08'$ .

## 10.2 Determination of ecliptic latitude of Mars

The ecliptic latitude of Mars can be determined with the aid of Tables 8.3, 10.2, and 10.3. Table 8.3 allows the mean argument of latitude,  $\bar{F}$ , of Mars to be calculated as a function of time. Next, Table 10.2 permits the deferential latitude,  $\beta_0$ , to be determined as a function of the true argument of latitude,  $F$ . Finally, Table 10.3 allows the quantities  $\delta h_-$ ,  $\bar{h}$ , and  $\delta h_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ .

The procedure for using the tables is as follows:

1. Determine the fractional Julian day number,  $t$ , corresponding to the date and time at

which the ecliptic latitude is to be calculated with the aid of Tables 3.1–3.3. Form  $\Delta t = t - t_0$ , where  $t_0 = 2\,451\,545.0$  is the epoch.

2. Calculate the planetary equation of center,  $q$ , ecliptic anomaly,  $\mu$ , and interpolation parameters  $\Theta_+$  and  $\Theta_-$  using the procedure set out in Chapter 8.
3. Enter Table 8.3 with the digit for each power of 10 in  $\Delta t$  and take out the corresponding values of  $\Delta \bar{F}$ . If  $\Delta t$  is negative then the corresponding values are also negative. The value of the mean argument of latitude,  $\bar{F}$ , is the sum of all the  $\Delta \bar{F}$  values plus the value of  $\bar{F}$  at the epoch.
4. Form the true argument of latitude,  $F = \bar{F} + q$ . Add as many multiples of  $360^\circ$  to  $F$  as is required to make it fall in the range  $0^\circ$  to  $360^\circ$ . Round  $F$  to the nearest degree.
5. Enter Table 10.2 with the value of  $F$  and take out the corresponding value of the deferential latitude,  $\beta_0$ . It is necessary to interpolate if  $F$  is odd.
6. Enter Table 10.3 with the value of  $\mu$  and take out the corresponding values of  $\delta h_-$ ,  $\bar{h}$ , and  $\delta h_+$ . If  $\mu > 180^\circ$  then it is necessary to make use of the identities  $\delta h_\pm(360^\circ - \mu) = \delta h_\pm(\mu)$  and  $\bar{h}(360^\circ - \mu) = \bar{h}(\mu)$ .
7. Form the epicyclic latitude correction factor,  $h = \Theta_- \delta h_- + \bar{h} + \Theta_+ \delta h_+$ .
8. The ecliptic latitude,  $\beta$ , is the product of the deferential latitude,  $\beta_0$ , and the epicyclic latitude correction factor,  $h$ . The decimal fraction can be converted into arc minutes using Table 5.2. Round to the nearest arc minute.

One example of this procedure is given in the following.

*Example:* May 5, 2005 AD, 00:00 UT

From Chapter 8,  $t - t_0 = 1\,950.5$  JD,  $q = -7.345^\circ$ ,  $\mu = 114.286^\circ$ ,  $\Theta_- = 0.101$ , and  $\Theta_+ = 0.619$ . Making use of Table 8.3, we find:

| $t(\text{JD})$ | $\bar{F}(^\circ)$ |
|----------------|-------------------|
| +1000          | 164.041           |
| +900           | 111.637           |
| +50            | 26.202            |
| +5             | 0.262             |
| Epoch          | 305.796           |
|                | <u>607.938</u>    |
| Modulus        | 247.938           |

Thus,

$$F = \bar{F} + q = 247.938 - 7.345 = 240.593 \simeq 241^\circ.$$

It follows from Table 10.2 that

$$\beta_0(241^\circ) = -1.615^\circ.$$

Because  $\mu \simeq 114^\circ$ , Table 10.3 yields

$$\delta h_-(114^\circ) = -0.017, \quad \bar{h}(114^\circ) = 1.056, \quad \delta h_+(114^\circ) = -0.027,$$

so

$$\begin{aligned} h &= \Theta_- \delta h_- + \bar{h} + \Theta_+ \delta h_+ \\ &= -0.101 \times 0.017 + 1.056 - 0.619 \times 0.027 \\ &= 1.038. \end{aligned}$$

Finally,

$$\beta = h \beta_0 = -1.038 \times 1.615 = -1.676 \simeq -1^\circ 41'.$$

Thus, the ecliptic latitude of Mars at 00:00 UT on May 5, 2005 AD was  $-1^\circ 41'$ .

### 10.3 Determination of ecliptic latitude of Jupiter

The ecliptic latitude of Jupiter can be determined with the aid of Tables 8.7, 10.4, and 10.5. Table 8.7 allows the mean argument of latitude,  $\bar{F}$ , of Jupiter to be calculated as a function of time. Next, Table 10.4 permits the deferential latitude,  $\beta_0$ , to be determined as a function of the true argument of latitude,  $F$ . Finally, Table 10.5 allows the quantities  $\delta h_-$ ,  $\bar{h}$ , and  $\delta h_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ . The procedure for using these tables is analogous to the previously described procedure for using the Mars tables. One example of this procedure is given in the following.

*Example:* May 5, 2005 AD, 00:00 UT:

From Chapter 8,  $t - t_0 = 1\,950.5$  JD,  $q = -0.091^\circ$ ,  $\mu = 208.192^\circ$ ,  $\Theta_- = -0.469$ , and  $\Theta_+ = -0.121$ . Making use of Table 8.7, we find:

| $t(\text{JD})$ | $\bar{F}(\circ)$ |
|----------------|------------------|
| +1000          | 83.081           |
| +900           | 74.773           |
| +50            | 4.154            |
| +5             | 0.042            |
| Epoch          | 293.660          |
|                | <u>455.710</u>   |
| Modulus        | 95.710           |

Thus,

$$F = \bar{F} + q = 95.710 - 0.091 = 95.619 \simeq 96^\circ.$$

It follows from Table 10.4 that

$$\beta_0(96^\circ) = 1.297^\circ.$$

Because  $\mu \simeq 208^\circ$ , Table 10.5 yields

$$\delta h_-(208^\circ) = 0.014, \quad \bar{h}(208^\circ) = 1.197, \quad \delta h_+(208^\circ) = 0.016,$$

so

$$\begin{aligned} h &= \Theta_- \delta h_- + \bar{h} + \Theta_+ \delta h_+ \\ &= -0.469 \times 0.014 + 1.197 - 0.121 \times 0.016 \\ &= 1.188. \end{aligned}$$

Finally,

$$\beta = h \beta_0 = 1.188 \times 1.297 = 1.541 \simeq 1^\circ 32'.$$

Thus, the ecliptic latitude of Jupiter at 00:00 UT on May 5, 2005 CE was  $1^\circ 32'$ .

#### 10.4 Determination of ecliptic latitude of Saturn

The ecliptic latitude of Saturn can be determined with the aid of Tables 8.11, 10.6, and 10.7. Table 8.11 allows the mean argument of latitude,  $\bar{F}$ , of Saturn to be calculated as a function of time. Next, Table 10.6 permits the deferential latitude,  $\beta_0$ , to be determined as a function of the true argument of latitude,  $F$ . Finally, Table 10.7 allows the quantities  $\delta h_-$ ,  $\bar{h}$ , and  $\delta h_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ . The procedure for using these tables is analogous to the previously described procedure for using the Mars tables. One example of this procedure is given in the following.

*Example:* May 5, 2005 AD, 00:00 UT:

From Chapter 8,  $t - t_0 = 1\,950.5$  JD,  $q = 2.561^\circ$ ,  $\mu = 286.625^\circ$ ,  $\Theta_- = 0.071$ , and  $\Theta_+ = 0.759$ . Making use of Table 8.11, we find:

| $t(\text{JD})$ | $\bar{F}(\circ)$ |
|----------------|------------------|
| +1000          | 33.478           |
| +900           | 30.130           |
| +50            | 1.674            |
| +5             | 0.017            |
| Epoch          | 296.482          |
|                | <u>361.781</u>   |
| Modulus        | 1.781            |

Thus,

$$F = \bar{F} + q = 1.781 + 2.561 = 4.342 \simeq 4^\circ.$$

It follows from Table 10.6 that

$$\beta_0(4^\circ) = 0.173^\circ.$$

Because  $\mu \simeq 287^\circ$ , Table 10.7 yields

$$\delta h_-(287^\circ) = -0.002, \quad \bar{h}(287^\circ) = 0.966, \quad \delta h_+(287^\circ) = -0.003,$$

so

$$\begin{aligned} h &= \Theta_- \delta h_- + \bar{h} + \Theta_+ \delta h_+ \\ &= -0.071 \times 0.002 + 0.966 - 0.759 \times 0.003 \\ &= 0.964. \end{aligned}$$

Finally,

$$\beta = h \beta_0 = 0.964 \times 0.173 = 0.167 \simeq 0^\circ 10'.$$

Thus, the ecliptic latitude of Saturn at 00:00 UT on May 5, 2005 CE was  $0^\circ 10'$ .

### 10.5 Determination of ecliptic latitude of inferior planet

Figure 10.2 shows the orbit of an inferior planet. As we have already mentioned, the epicycle and deferent of such a planet have the same elements as the orbit of the planet in question around the Sun, and the apparent orbit of the Sun around the Earth, respectively. It follows that the epicycle and deferent of an inferior planet are, respectively, inclined and parallel to the ecliptic plane. Let the plane of the epicycle cut the ecliptic plane along the line  $NG'N'$ . Here,  $N$  is the point at which the epicycle passes through the plane of the ecliptic from south to north, in the direction of the mean planetary motion. This point is called the *ascending node*. Note that the line  $NG'N'$  must pass through the guide-point,  $G'$ , because the Sun (which is coincident with the guide-point) is common to the plane of the planetary orbit and the ecliptic plane.

Now, it follows from simple geometry that the elevation of the planet  $P$  above the guide-point,  $G'$ , satisfies

$$v = r' \sin i \sin F, \quad (10.24)$$

where  $r'$  is the length  $G'P$ ,  $i$  the fixed inclination of the planetary orbit (and, hence, of the epicycle) to the ecliptic plane, and  $F$  the angle  $NG'P$ . The angle  $F$  is termed the argument of latitude. We can write (see Chapter 9)

$$F = \bar{F} + q, \quad (10.25)$$

where  $\bar{F}$  is the mean argument of latitude, and  $q$  the equation of center of the epicycle. Note that  $\bar{F}$  increases uniformly in time; that is,

$$\bar{F} = \bar{F}_0 + \bar{n}(t - t_0). \quad (10.26)$$

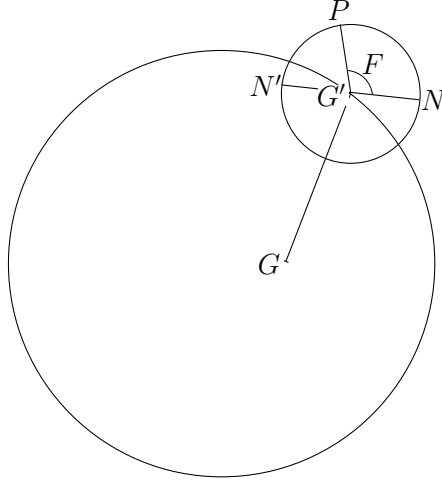


Figure 10.2: Orbit of an inferior planet. Here,  $G$ ,  $G'$ ,  $P$ ,  $N$ ,  $N'$ , and  $F$  represent the earth, the guide-point, the planet, the ascending node, the descending node, and the argument of latitude, respectively. The view is from northern ecliptic pole.

Because the deferent is parallel to the ecliptic plane, the elevation of the planet above the said plane is the same as that of the planet above the guide-point. Hence, from simple geometry, the ecliptic latitude of the planet satisfies

$$\beta = \frac{v}{r''}, \quad (10.27)$$

where  $r''$  is the length  $GP$ , and we have used the small angle approximation. However, it is apparent from Fig. 8.3 that

$$r'' = (r^2 + 2r r' \cos \mu + r'^2)^{1/2}, \quad (10.28)$$

where  $r$  the length  $GG'$ , and  $\mu$  the equation of the epicycle. But, according to the analysis in Chapter 9,  $r'/r = a/z$ , where  $a$  is the planetary major radius in units in which the major radius of the Sun's apparent orbit about the Earth is unity, and  $z$  is defined in Equation (9.9). Thus, we obtain

$$\beta = h \beta_0, \quad (10.29)$$

where

$$\beta_0(F) = a \sin i \sin F \quad (10.30)$$

is termed the *epicyclic latitude*, and

$$h(\mu, z) = [z^2 + 2a z \cos \mu + a^2]^{-1/2} \quad (10.31)$$

the *deferential latitude correction factor*.



In the following,  $a, e, n, \tilde{n}, \bar{\lambda}_0, M_0, \bar{F}_0$ , and  $i$  are elements of the orbit of the planet in question about the Sun, and  $e_S, \zeta_S$ , and  $\lambda_S$  are elements of the Sun's apparent orbit about the Sarth. The requisite elements for all of the superior planets at the J2000 epoch ( $t_0 = 2\,451\,545.0$  JD) are listed in Tables 5.1 and 10.1. Employing a quadratic interpolation scheme to represent  $F(\mu, z)$  (see Chapter 8), our procedure for determining the ecliptic latitude of a superior planet is summed up by the following formulae:

$$\bar{\lambda} = \bar{\lambda}_0 + n(t - t_0), \quad (10.32)$$

$$M = M_0 + \tilde{n}(t - t_0), \quad (10.33)$$

$$\bar{F} = \bar{F}_0 + \tilde{n}(t - t_0), \quad (10.34)$$

$$q = 2e \sin M + (5/4)e^2 \sin 2M, \quad (10.35)$$

$$\zeta = e \cos M - e^2 \sin^2 M, \quad (10.36)$$

$$F = \bar{F} + q, \quad (10.37)$$

$$\beta_0 = a \sin i \sin F, \quad (10.38)$$

$$\mu = \bar{\lambda} + q - \bar{\lambda}_S, \quad (10.39)$$

$$\bar{h} = h(\mu, \bar{z}) \equiv [\bar{z}^2 + 2a\bar{z} \cos \mu + a^2]^{-1/2}, \quad (10.40)$$

$$\delta h_- = h(\mu, \bar{z}) - h(\mu, z_{\max}), \quad (10.41)$$

$$\delta h_+ = h(\mu, z_{\min}) - h(\mu, \bar{z}), \quad (10.42)$$

$$z = \frac{1 - \zeta_S}{1 - \zeta}, \quad (10.43)$$

$$\xi = \frac{\bar{z} - z}{\delta z}, \quad (10.44)$$

$$h = \Theta_-(\xi) \delta h_- + \bar{h} + \Theta_+(\xi) \delta h_+, \quad (10.45)$$

$$\beta = h \beta_0. \quad (10.46)$$

Here,  $\bar{z} = (1 + e e_S)/(1 - e^2)$ ,  $\delta z = (e + e_S)/(1 - e^2)$ ,  $z_{\min} = \bar{z} - \delta z$ , and  $z_{\max} = \bar{z} + \delta z$ . The constants  $\bar{z}, \delta z, z_{\min}$ , and  $z_{\max}$  for each of the inferior planets are listed in Table 8.1. Finally, the functions  $\Theta_{\pm}$  are tabulated in Table 8.2.

For the case of Venus, the previous formulae are capable of matching NASA ephemeris data during the years 1995–2006 AD with a mean error of  $0.7'$  and a maximum error of  $1.8'$ . For the case of Mercury, with the augmentations to the theory described in Chapter 9, the mean error is  $1.6'$  and the maximum error  $5'$ .

### 10.6 Determination of ecliptic latitude of Venus

The ecliptic latitude of Venus can be determined with the aid of Tables 9.1, 10.8, and 10.9. Table 9.1 allows the mean argument of latitude,  $\bar{F}$ , of Venus to be calculated as a function of time. Next, Table 10.8 permits the epicyclic latitude,  $\beta_0$ , to be determined as a function of the

true argument of latitude,  $F$ . Finally, Table 10.9 allows the quantities  $\delta h_-$ ,  $\bar{h}$ , and  $\delta h_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ .

The procedure for using the tables is as follows:

1. Determine the fractional Julian day number,  $t$ , corresponding to the date and time at which the ecliptic latitude is to be calculated with the aid of Tables 3.1–3.3. Form  $\Delta t = t - t_0$ , where  $t_0 = 2\,451\,545.0$  is the epoch.
2. Calculate the planetary equation of center,  $q$ , ecliptic anomaly,  $\mu$ , and interpolation parameters  $\Theta_+$  and  $\Theta_-$  using the procedure set out in Chapter 9.
3. Enter Table 9.1 with the digit for each power of 10 in  $\Delta t$  and take out the corresponding values of  $\Delta \bar{F}$ . If  $\Delta t$  is negative then the corresponding values are also negative. The value of the mean argument of latitude,  $\bar{F}$ , is the sum of all the  $\Delta \bar{F}$  values plus the value of  $\bar{F}$  at the epoch.
4. Form the true argument of latitude,  $F = \bar{F} + q$ . Add as many multiples of  $360^\circ$  to  $F$  as is required to make it fall in the range  $0^\circ$  to  $360^\circ$ . Round  $F$  to the nearest degree.
5. Enter Table 10.8 with the value of  $F$  and take out the corresponding value of the epicyclic latitude,  $\beta_0$ . It is necessary to interpolate if  $F$  is odd.
6. Enter Table 10.9 with the value of  $\mu$  and take out the corresponding values of  $\delta h_-$ ,  $\bar{h}$ , and  $\delta h_+$ . If  $\mu > 180^\circ$  then it is necessary to make use of the identities  $\delta h_\pm(360^\circ - \mu) = \delta h_\pm(\mu)$  and  $\bar{h}(360^\circ - \mu) = \bar{h}(\mu)$ .
7. Form the deferential latitude correction factor,  $h = \Theta_- \delta h_- + \bar{h} + \Theta_+ \delta h_+$ .
8. The ecliptic latitude,  $\beta$ , is the product of the epicyclic latitude,  $\beta_0$ , and the deferential latitude correction factor,  $h$ . The decimal fraction can be converted into arc minutes using Table 5.2. Round to the nearest arc minute.

One example of this procedure is given in the following.

*Example:* May 5, 2005 AD, 00:00 UT

From Chapter 9,  $t - t_0 = 1\,950.5$  JD,  $q = -0.712^\circ$ ,  $\mu = 21.689^\circ$ ,  $\Theta_- = -0.355$ , and  $\Theta_+ = -0.125$ . Making use of Table 9.1, we find:

| $t(\text{JD})$ | $\bar{F}(\circ)$ |
|----------------|------------------|
| +1000          | 162.138          |
| +900           | 1.924            |
| +50            | 80.107           |
| +5             | 0.801            |
| Epoch          | 105.253          |
|                | <u>350.223</u>   |
| Modulus        | 350.223          |

Thus,

$$F = \bar{F} + q = 350.223 - 0.712 = 349.511 \simeq 350^\circ.$$

It follows from Table 10.8 that

$$\beta_0(350^\circ) = -0.423^\circ.$$

Because  $\mu \simeq 22^\circ$ , Table 10.9 yields

$$\delta h_-(22^\circ) = 0.008, \quad \bar{h}(22^\circ) = 0.591, \quad \delta h_+(22^\circ) = 0.008,$$

so

$$\begin{aligned} h &= \Theta_- \delta h_- + \bar{h} + \Theta_+ \delta h_+ \\ &= -0.355 \times 0.008 + 0.591 - 0.125 \times 0.008 \\ &= 0.587. \end{aligned}$$

Finally,

$$\beta = h \beta_0 = -0.587 \times 0.423 = -0.248 \simeq -0^\circ 15'.$$

Thus, the ecliptic latitude of Venus at 00:00 UT on May 5, 2005 AD was  $-0^\circ 15'$ .

### 10.7 Determination of ecliptic latitude of Mercury

The ecliptic latitude of Mercury can be determined with the aid of Tables 9.5, 10.10, and 10.11. Table 9.5 allows the mean argument of latitude,  $\bar{F}$ , of Mercury to be calculated as a function of time. Next, Table 10.10 permits the epicyclic latitude,  $\beta_0$ , to be determined as a function of the true argument of latitude,  $F$ . Finally, Table 10.11 allows the quantities  $\delta h_-$ ,  $\bar{h}$ , and  $\delta h_+$  to be calculated as functions of the epicyclic anomaly,  $\mu$ . The procedure for using the tables is analogous to the previously described procedure for using the Venus tables. One example of this procedure is given in the following.

*Example:* May 5, 2005 AD, 00:00 UT

From Chapter 9,  $t - t_0 = 1\,950.5$  JD,  $q = -16.974^\circ$ ,  $\mu = 252.692^\circ$ ,  $\Theta_- = 0.107$ , and  $\Theta_+ = 0.583$ . Making use of Table 9.5, we find:

| $t(\text{JD})$ | $\bar{F}(\circ)$ |
|----------------|------------------|
| +1000          | 132.342          |
| +900           | 83.108           |
| +50            | 204.617          |
| +5             | 2.046            |
| Epoch          | 204.436          |
|                | <u>626.549</u>   |
| Modulus        | 266.549          |

Thus,

$$F = \bar{F} + q = 266.549 - 16.974 = 249.575 \simeq 250^\circ.$$

It follows from Table 10.10 that

$$\beta_0(250^\circ) = -2.511^\circ.$$

Because  $\mu \simeq 253^\circ$ , Table 10.11 yields

$$\delta h_-(253^\circ) = 0.184, \quad \bar{h}(253^\circ) = 1.037, \quad \delta h_+(253^\circ) = 0.272,$$

so

$$\begin{aligned} h &= \Theta_- \delta h_- + \bar{h} + \Theta_+ \delta h_+ \\ &= 0.107 \times 0.184 + 1.037 + 0.583 \times 0.272 \\ &= 1.215. \end{aligned}$$

Finally,

$$\beta = h \beta_0 = -1.215 \times 2.511 = -3.051 \simeq -3^\circ 03'.$$

Thus, the ecliptic latitude of Mercury at 00:00 UT on May 5, 2005 CE was  $-3^\circ 03'$ .

### 10.8 Tables

| Object  | $i(^{\circ})$ | $\tilde{n}(^{\circ}/\text{day})$ | $\bar{F}_0(^{\circ})$ |
|---------|---------------|----------------------------------|-----------------------|
| Mercury | 6.9190        | 4.09234221                       | 204.436               |
| Venus   | 3.3692        | 1.60213807                       | 105.253               |
| Mars    | 1.8467        | 0.52404094                       | 305.796               |
| Jupiter | 1.3044        | 0.08308122                       | 293.660               |
| Saturn  | 2.4860        | 0.03347795                       | 296.482               |

Table 10.1: Additional Keplerian orbital elements for the five visible planets at the J2000 epoch (that is, 12:00 UT, January 1, 2000 CE, which corresponds to  $t_0 = 2\,451\,545.0$  JD). The elements are optimized for use in the time period 1800 CE to 2050 AD. Source: Jet Propulsion Laboratory (NASA), <http://ssd.jpl.nasa.gov/>.

| $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ | $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ |
|---------------|---------------------|---------------|---------------|---------------------|---------------|
| 000/180       | 0.000               | (180)/(360)   | 046/134       | 1.328               | (226)/(314)   |
| 002/178       | 0.064               | (182)/(358)   | 048/132       | 1.372               | (228)/(312)   |
| 004/176       | 0.129               | (184)/(356)   | 050/130       | 1.414               | (230)/(310)   |
| 006/174       | 0.193               | (186)/(354)   | 052/128       | 1.455               | (232)/(308)   |
| 008/172       | 0.257               | (188)/(352)   | 054/126       | 1.494               | (234)/(306)   |
| 010/170       | 0.321               | (190)/(350)   | 056/124       | 1.531               | (236)/(304)   |
| 012/168       | 0.384               | (192)/(348)   | 058/122       | 1.566               | (238)/(302)   |
| 014/166       | 0.447               | (194)/(346)   | 060/120       | 1.599               | (240)/(300)   |
| 016/164       | 0.509               | (196)/(344)   | 062/118       | 1.630               | (242)/(298)   |
| 018/162       | 0.571               | (198)/(342)   | 064/116       | 1.660               | (244)/(296)   |
| 020/160       | 0.631               | (200)/(340)   | 066/114       | 1.687               | (246)/(294)   |
| 022/158       | 0.692               | (202)/(338)   | 068/112       | 1.712               | (248)/(292)   |
| 024/156       | 0.751               | (204)/(336)   | 070/110       | 1.735               | (250)/(290)   |
| 026/154       | 0.809               | (206)/(334)   | 072/108       | 1.756               | (252)/(288)   |
| 028/152       | 0.867               | (208)/(332)   | 074/106       | 1.775               | (254)/(286)   |
| 030/150       | 0.923               | (210)/(330)   | 076/104       | 1.792               | (256)/(284)   |
| 032/148       | 0.978               | (212)/(328)   | 078/102       | 1.806               | (258)/(282)   |
| 034/146       | 1.032               | (214)/(326)   | 080/100       | 1.818               | (260)/(280)   |
| 036/144       | 1.085               | (216)/(324)   | 082/098       | 1.828               | (262)/(278)   |
| 038/142       | 1.137               | (218)/(322)   | 084/096       | 1.836               | (264)/(276)   |
| 040/140       | 1.187               | (220)/(320)   | 086/094       | 1.842               | (266)/(274)   |
| 042/138       | 1.235               | (222)/(318)   | 088/092       | 1.845               | (268)/(272)   |
| 044/136       | 1.283               | (224)/(316)   | 090/090       | 1.846               | (270)/(270)   |

Table 10.2: Deferential ecliptic latitude of Mars. The latitude is minus the value shown in the table if the argument is in parentheses.

| $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ |
|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|
| 0     | -0.025       | 0.604     | -0.028       | 45    | -0.025       | 0.652     | -0.029       | 90    | -0.025       | 0.836     | -0.031       | 135   | 0.015        | 1.410     | 0.003        |
| 1     | -0.025       | 0.604     | -0.028       | 46    | -0.025       | 0.654     | -0.029       | 91    | -0.025       | 0.843     | -0.031       | 136   | 0.018        | 1.433     | 0.006        |
| 2     | -0.025       | 0.604     | -0.028       | 47    | -0.025       | 0.656     | -0.029       | 92    | -0.024       | 0.850     | -0.031       | 137   | 0.021        | 1.457     | 0.009        |
| 3     | -0.025       | 0.604     | -0.028       | 48    | -0.025       | 0.659     | -0.029       | 93    | -0.024       | 0.857     | -0.031       | 138   | 0.025        | 1.482     | 0.013        |
| 4     | -0.025       | 0.605     | -0.028       | 49    | -0.025       | 0.661     | -0.029       | 94    | -0.024       | 0.865     | -0.031       | 139   | 0.028        | 1.507     | 0.017        |
| 5     | -0.025       | 0.605     | -0.028       | 50    | -0.025       | 0.664     | -0.030       | 95    | -0.024       | 0.872     | -0.031       | 140   | 0.032        | 1.533     | 0.021        |
| 6     | -0.025       | 0.605     | -0.028       | 51    | -0.025       | 0.666     | -0.030       | 96    | -0.024       | 0.880     | -0.031       | 141   | 0.037        | 1.560     | 0.026        |
| 7     | -0.025       | 0.605     | -0.028       | 52    | -0.025       | 0.669     | -0.030       | 97    | -0.024       | 0.888     | -0.031       | 142   | 0.041        | 1.588     | 0.031        |
| 8     | -0.025       | 0.606     | -0.028       | 53    | -0.025       | 0.672     | -0.030       | 98    | -0.023       | 0.896     | -0.031       | 143   | 0.046        | 1.616     | 0.036        |
| 9     | -0.025       | 0.606     | -0.028       | 54    | -0.025       | 0.674     | -0.030       | 99    | -0.023       | 0.904     | -0.031       | 144   | 0.051        | 1.646     | 0.043        |
| 10    | -0.025       | 0.606     | -0.028       | 55    | -0.025       | 0.677     | -0.030       | 100   | -0.023       | 0.912     | -0.030       | 145   | 0.057        | 1.676     | 0.049        |
| 11    | -0.025       | 0.607     | -0.028       | 56    | -0.025       | 0.680     | -0.030       | 101   | -0.023       | 0.921     | -0.030       | 146   | 0.063        | 1.708     | 0.056        |
| 12    | -0.025       | 0.607     | -0.028       | 57    | -0.026       | 0.683     | -0.030       | 102   | -0.022       | 0.930     | -0.030       | 147   | 0.069        | 1.740     | 0.064        |
| 13    | -0.025       | 0.608     | -0.028       | 58    | -0.026       | 0.686     | -0.030       | 103   | -0.022       | 0.939     | -0.030       | 148   | 0.076        | 1.773     | 0.073        |
| 14    | -0.025       | 0.609     | -0.028       | 59    | -0.026       | 0.689     | -0.030       | 104   | -0.022       | 0.948     | -0.030       | 149   | 0.084        | 1.807     | 0.083        |
| 15    | -0.025       | 0.609     | -0.028       | 60    | -0.026       | 0.693     | -0.030       | 105   | -0.021       | 0.958     | -0.030       | 150   | 0.092        | 1.843     | 0.093        |
| 16    | -0.025       | 0.610     | -0.028       | 61    | -0.026       | 0.696     | -0.030       | 106   | -0.021       | 0.968     | -0.029       | 151   | 0.100        | 1.879     | 0.104        |
| 17    | -0.025       | 0.611     | -0.028       | 62    | -0.026       | 0.699     | -0.030       | 107   | -0.021       | 0.978     | -0.029       | 152   | 0.109        | 1.916     | 0.117        |
| 18    | -0.025       | 0.611     | -0.028       | 63    | -0.026       | 0.703     | -0.030       | 108   | -0.020       | 0.988     | -0.029       | 153   | 0.118        | 1.955     | 0.130        |
| 19    | -0.025       | 0.612     | -0.028       | 64    | -0.026       | 0.706     | -0.030       | 109   | -0.020       | 0.999     | -0.029       | 154   | 0.128        | 1.994     | 0.145        |
| 20    | -0.025       | 0.613     | -0.028       | 65    | -0.026       | 0.710     | -0.030       | 110   | -0.019       | 1.010     | -0.028       | 155   | 0.139        | 2.034     | 0.161        |
| 21    | -0.025       | 0.614     | -0.028       | 66    | -0.026       | 0.714     | -0.030       | 111   | -0.019       | 1.021     | -0.028       | 156   | 0.151        | 2.075     | 0.178        |
| 22    | -0.025       | 0.615     | -0.028       | 67    | -0.026       | 0.718     | -0.030       | 112   | -0.018       | 1.032     | -0.027       | 157   | 0.163        | 2.117     | 0.197        |
| 23    | -0.025       | 0.616     | -0.028       | 68    | -0.026       | 0.722     | -0.030       | 113   | -0.018       | 1.044     | -0.027       | 158   | 0.175        | 2.160     | 0.218        |
| 24    | -0.025       | 0.617     | -0.028       | 69    | -0.026       | 0.726     | -0.031       | 114   | -0.017       | 1.056     | -0.027       | 159   | 0.189        | 2.203     | 0.240        |
| 25    | -0.025       | 0.618     | -0.029       | 70    | -0.026       | 0.730     | -0.031       | 115   | -0.016       | 1.069     | -0.026       | 160   | 0.202        | 2.247     | 0.265        |
| 26    | -0.025       | 0.619     | -0.029       | 71    | -0.026       | 0.734     | -0.031       | 116   | -0.015       | 1.082     | -0.025       | 161   | 0.217        | 2.292     | 0.291        |
| 27    | -0.025       | 0.621     | -0.029       | 72    | -0.026       | 0.738     | -0.031       | 117   | -0.015       | 1.095     | -0.025       | 162   | 0.232        | 2.337     | 0.319        |
| 28    | -0.025       | 0.622     | -0.029       | 73    | -0.026       | 0.743     | -0.031       | 118   | -0.014       | 1.108     | -0.024       | 163   | 0.248        | 2.382     | 0.349        |
| 29    | -0.025       | 0.623     | -0.029       | 74    | -0.026       | 0.747     | -0.031       | 119   | -0.013       | 1.122     | -0.023       | 164   | 0.264        | 2.427     | 0.382        |
| 30    | -0.025       | 0.625     | -0.029       | 75    | -0.026       | 0.752     | -0.031       | 120   | -0.012       | 1.137     | -0.023       | 165   | 0.280        | 2.472     | 0.416        |
| 31    | -0.025       | 0.626     | -0.029       | 76    | -0.026       | 0.757     | -0.031       | 121   | -0.011       | 1.151     | -0.022       | 166   | 0.297        | 2.517     | 0.453        |
| 32    | -0.025       | 0.627     | -0.029       | 77    | -0.026       | 0.762     | -0.031       | 122   | -0.010       | 1.167     | -0.021       | 167   | 0.314        | 2.560     | 0.491        |
| 33    | -0.025       | 0.629     | -0.029       | 78    | -0.025       | 0.767     | -0.031       | 123   | -0.008       | 1.182     | -0.020       | 168   | 0.331        | 2.603     | 0.530        |
| 34    | -0.025       | 0.631     | -0.029       | 79    | -0.025       | 0.772     | -0.031       | 124   | -0.007       | 1.198     | -0.019       | 169   | 0.348        | 2.644     | 0.571        |
| 35    | -0.025       | 0.632     | -0.029       | 80    | -0.025       | 0.777     | -0.031       | 125   | -0.006       | 1.215     | -0.017       | 170   | 0.364        | 2.683     | 0.612        |
| 36    | -0.025       | 0.634     | -0.029       | 81    | -0.025       | 0.782     | -0.031       | 126   | -0.004       | 1.232     | -0.016       | 171   | 0.380        | 2.721     | 0.654        |
| 37    | -0.025       | 0.636     | -0.029       | 82    | -0.025       | 0.788     | -0.031       | 127   | -0.003       | 1.249     | -0.015       | 172   | 0.395        | 2.755     | 0.694        |
| 38    | -0.025       | 0.637     | -0.029       | 83    | -0.025       | 0.793     | -0.031       | 128   | -0.001       | 1.267     | -0.013       | 173   | 0.408        | 2.787     | 0.734        |
| 39    | -0.025       | 0.639     | -0.029       | 84    | -0.025       | 0.799     | -0.031       | 129   | 0.001        | 1.286     | -0.011       | 174   | 0.421        | 2.816     | 0.770        |
| 40    | -0.025       | 0.641     | -0.029       | 85    | -0.025       | 0.805     | -0.031       | 130   | 0.003        | 1.305     | -0.009       | 175   | 0.432        | 2.840     | 0.804        |
| 41    | -0.025       | 0.643     | -0.029       | 86    | -0.025       | 0.811     | -0.031       | 131   | 0.005        | 1.325     | -0.007       | 176   | 0.442        | 2.861     | 0.833        |
| 42    | -0.025       | 0.645     | -0.029       | 87    | -0.025       | 0.817     | -0.031       | 132   | 0.007        | 1.345     | -0.005       | 177   | 0.449        | 2.878     | 0.857        |
| 43    | -0.025       | 0.647     | -0.029       | 88    | -0.025       | 0.823     | -0.031       | 133   | 0.010        | 1.366     | -0.003       | 178   | 0.455        | 2.890     | 0.874        |
| 44    | -0.025       | 0.649     | -0.029       | 89    | -0.025       | 0.830     | -0.031       | 134   | 0.012        | 1.388     | -0.000       | 179   | 0.458        | 2.897     | 0.885        |
| 45    | -0.025       | 0.652     | -0.029       | 90    | -0.025       | 0.836     | -0.031       | 135   | 0.015        | 1.410     | 0.003        | 180   | 0.459        | 2.899     | 0.889        |

Table 10.3: Epicyclic latitude correction factor for Mars.  $\mu$  is in degrees. Note that  $\bar{h}(360^\circ - \mu) = \bar{h}(\mu)$ , and  $\delta h_\pm(360^\circ - \mu) = \delta h_\pm(\mu)$ .

| $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ | $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ |
|---------------|---------------------|---------------|---------------|---------------------|---------------|
| 000/180       | 0.000               | (180)/(360)   | 046/134       | 0.938               | (226)/(314)   |
| 002/178       | 0.046               | (182)/(358)   | 048/132       | 0.969               | (228)/(312)   |
| 004/176       | 0.091               | (184)/(356)   | 050/130       | 0.999               | (230)/(310)   |
| 006/174       | 0.136               | (186)/(354)   | 052/128       | 1.028               | (232)/(308)   |
| 008/172       | 0.182               | (188)/(352)   | 054/126       | 1.055               | (234)/(306)   |
| 010/170       | 0.226               | (190)/(350)   | 056/124       | 1.081               | (236)/(304)   |
| 012/168       | 0.271               | (192)/(348)   | 058/122       | 1.106               | (238)/(302)   |
| 014/166       | 0.316               | (194)/(346)   | 060/120       | 1.130               | (240)/(300)   |
| 016/164       | 0.360               | (196)/(344)   | 062/118       | 1.152               | (242)/(298)   |
| 018/162       | 0.403               | (198)/(342)   | 064/116       | 1.172               | (244)/(296)   |
| 020/160       | 0.446               | (200)/(340)   | 066/114       | 1.192               | (246)/(294)   |
| 022/158       | 0.489               | (202)/(338)   | 068/112       | 1.209               | (248)/(292)   |
| 024/156       | 0.531               | (204)/(336)   | 070/110       | 1.226               | (250)/(290)   |
| 026/154       | 0.572               | (206)/(334)   | 072/108       | 1.240               | (252)/(288)   |
| 028/152       | 0.612               | (208)/(332)   | 074/106       | 1.254               | (254)/(286)   |
| 030/150       | 0.652               | (210)/(330)   | 076/104       | 1.266               | (256)/(284)   |
| 032/148       | 0.691               | (212)/(328)   | 078/102       | 1.276               | (258)/(282)   |
| 034/146       | 0.729               | (214)/(326)   | 080/100       | 1.284               | (260)/(280)   |
| 036/144       | 0.767               | (216)/(324)   | 082/098       | 1.292               | (262)/(278)   |
| 038/142       | 0.803               | (218)/(322)   | 084/096       | 1.297               | (264)/(276)   |
| 040/140       | 0.838               | (220)/(320)   | 086/094       | 1.301               | (266)/(274)   |
| 042/138       | 0.873               | (222)/(318)   | 088/092       | 1.303               | (268)/(272)   |
| 044/136       | 0.906               | (224)/(316)   | 090/090       | 1.304               | (270)/(270)   |

Table 10.4: Deferential ecliptic latitude of Jupiter. The latitude is minus the value shown in the table if the argument is in parentheses.

| $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ |
|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|
| 0     | -0.008       | 0.839     | -0.009       | 45    | -0.007       | 0.874     | -0.008       | 90    | -0.002       | 0.982     | -0.003       | 135   | 0.009        | 1.143     | 0.010        |
| 1     | -0.008       | 0.839     | -0.009       | 46    | -0.007       | 0.876     | -0.008       | 91    | -0.002       | 0.985     | -0.002       | 136   | 0.009        | 1.147     | 0.011        |
| 2     | -0.008       | 0.839     | -0.009       | 47    | -0.007       | 0.877     | -0.008       | 92    | -0.002       | 0.988     | -0.002       | 137   | 0.010        | 1.150     | 0.011        |
| 3     | -0.008       | 0.839     | -0.009       | 48    | -0.007       | 0.879     | -0.008       | 93    | -0.002       | 0.992     | -0.002       | 138   | 0.010        | 1.154     | 0.011        |
| 4     | -0.008       | 0.839     | -0.009       | 49    | -0.007       | 0.881     | -0.008       | 94    | -0.001       | 0.995     | -0.002       | 139   | 0.010        | 1.157     | 0.012        |
| 5     | -0.008       | 0.839     | -0.009       | 50    | -0.007       | 0.883     | -0.008       | 95    | -0.001       | 0.998     | -0.001       | 140   | 0.010        | 1.160     | 0.012        |
| 6     | -0.008       | 0.840     | -0.009       | 51    | -0.007       | 0.884     | -0.008       | 96    | -0.001       | 1.002     | -0.001       | 141   | 0.011        | 1.164     | 0.012        |
| 7     | -0.008       | 0.840     | -0.009       | 52    | -0.007       | 0.886     | -0.007       | 97    | -0.001       | 1.005     | -0.001       | 142   | 0.011        | 1.167     | 0.013        |
| 8     | -0.008       | 0.840     | -0.009       | 53    | -0.007       | 0.888     | -0.007       | 98    | -0.001       | 1.008     | -0.001       | 143   | 0.011        | 1.170     | 0.013        |
| 9     | -0.008       | 0.840     | -0.009       | 54    | -0.006       | 0.890     | -0.007       | 99    | -0.000       | 1.012     | -0.001       | 144   | 0.012        | 1.173     | 0.013        |
| 10    | -0.008       | 0.841     | -0.009       | 55    | -0.006       | 0.892     | -0.007       | 100   | -0.000       | 1.015     | -0.000       | 145   | 0.012        | 1.177     | 0.014        |
| 11    | -0.008       | 0.841     | -0.009       | 56    | -0.006       | 0.894     | -0.007       | 101   | 0.000        | 1.019     | -0.000       | 146   | 0.012        | 1.180     | 0.014        |
| 12    | -0.008       | 0.841     | -0.009       | 57    | -0.006       | 0.896     | -0.007       | 102   | 0.000        | 1.022     | 0.000        | 147   | 0.012        | 1.183     | 0.014        |
| 13    | -0.008       | 0.842     | -0.009       | 58    | -0.006       | 0.898     | -0.007       | 103   | 0.000        | 1.026     | 0.000        | 148   | 0.013        | 1.186     | 0.015        |
| 14    | -0.008       | 0.842     | -0.009       | 59    | -0.006       | 0.900     | -0.007       | 104   | 0.001        | 1.029     | 0.001        | 149   | 0.013        | 1.189     | 0.015        |
| 15    | -0.008       | 0.843     | -0.009       | 60    | -0.006       | 0.902     | -0.007       | 105   | 0.001        | 1.033     | 0.001        | 150   | 0.013        | 1.192     | 0.015        |
| 16    | -0.008       | 0.843     | -0.009       | 61    | -0.006       | 0.904     | -0.007       | 106   | 0.001        | 1.036     | 0.001        | 151   | 0.014        | 1.194     | 0.016        |
| 17    | -0.008       | 0.844     | -0.009       | 62    | -0.006       | 0.906     | -0.007       | 107   | 0.001        | 1.040     | 0.001        | 152   | 0.014        | 1.197     | 0.016        |
| 18    | -0.008       | 0.845     | -0.009       | 63    | -0.006       | 0.909     | -0.006       | 108   | 0.002        | 1.044     | 0.002        | 153   | 0.014        | 1.200     | 0.016        |
| 19    | -0.008       | 0.845     | -0.009       | 64    | -0.006       | 0.911     | -0.006       | 109   | 0.002        | 1.047     | 0.002        | 154   | 0.014        | 1.202     | 0.017        |
| 20    | -0.008       | 0.846     | -0.009       | 65    | -0.005       | 0.913     | -0.006       | 110   | 0.002        | 1.051     | 0.002        | 155   | 0.015        | 1.205     | 0.017        |
| 21    | -0.008       | 0.847     | -0.009       | 66    | -0.005       | 0.916     | -0.006       | 111   | 0.002        | 1.055     | 0.003        | 156   | 0.015        | 1.207     | 0.017        |
| 22    | -0.008       | 0.847     | -0.009       | 67    | -0.005       | 0.918     | -0.006       | 112   | 0.003        | 1.058     | 0.003        | 157   | 0.015        | 1.210     | 0.017        |
| 23    | -0.008       | 0.848     | -0.009       | 68    | -0.005       | 0.920     | -0.006       | 113   | 0.003        | 1.062     | 0.003        | 158   | 0.015        | 1.212     | 0.018        |
| 24    | -0.008       | 0.849     | -0.009       | 69    | -0.005       | 0.923     | -0.006       | 114   | 0.003        | 1.066     | 0.003        | 159   | 0.015        | 1.214     | 0.018        |
| 25    | -0.008       | 0.850     | -0.009       | 70    | -0.005       | 0.925     | -0.006       | 115   | 0.003        | 1.069     | 0.004        | 160   | 0.016        | 1.216     | 0.018        |
| 26    | -0.008       | 0.851     | -0.009       | 71    | -0.005       | 0.928     | -0.006       | 116   | 0.004        | 1.073     | 0.004        | 161   | 0.016        | 1.218     | 0.018        |
| 27    | -0.008       | 0.852     | -0.009       | 72    | -0.005       | 0.930     | -0.005       | 117   | 0.004        | 1.077     | 0.004        | 162   | 0.016        | 1.220     | 0.019        |
| 28    | -0.008       | 0.853     | -0.009       | 73    | -0.005       | 0.933     | -0.005       | 118   | 0.004        | 1.080     | 0.005        | 163   | 0.016        | 1.222     | 0.019        |
| 29    | -0.008       | 0.854     | -0.009       | 74    | -0.004       | 0.935     | -0.005       | 119   | 0.004        | 1.084     | 0.005        | 164   | 0.016        | 1.224     | 0.019        |
| 30    | -0.008       | 0.855     | -0.009       | 75    | -0.004       | 0.938     | -0.005       | 120   | 0.005        | 1.088     | 0.005        | 165   | 0.016        | 1.225     | 0.019        |
| 31    | -0.008       | 0.856     | -0.009       | 76    | -0.004       | 0.941     | -0.005       | 121   | 0.005        | 1.092     | 0.006        | 166   | 0.017        | 1.227     | 0.019        |
| 32    | -0.008       | 0.857     | -0.009       | 77    | -0.004       | 0.944     | -0.005       | 122   | 0.005        | 1.095     | 0.006        | 167   | 0.017        | 1.228     | 0.020        |
| 33    | -0.008       | 0.858     | -0.009       | 78    | -0.004       | 0.946     | -0.005       | 123   | 0.006        | 1.099     | 0.006        | 168   | 0.017        | 1.230     | 0.020        |
| 34    | -0.008       | 0.859     | -0.009       | 79    | -0.004       | 0.949     | -0.004       | 124   | 0.006        | 1.103     | 0.007        | 169   | 0.017        | 1.231     | 0.020        |
| 35    | -0.008       | 0.860     | -0.009       | 80    | -0.004       | 0.952     | -0.004       | 125   | 0.006        | 1.107     | 0.007        | 170   | 0.017        | 1.232     | 0.020        |
| 36    | -0.008       | 0.861     | -0.008       | 81    | -0.004       | 0.955     | -0.004       | 126   | 0.006        | 1.110     | 0.007        | 171   | 0.017        | 1.233     | 0.020        |
| 37    | -0.008       | 0.863     | -0.008       | 82    | -0.003       | 0.958     | -0.004       | 127   | 0.007        | 1.114     | 0.008        | 172   | 0.017        | 1.234     | 0.020        |
| 38    | -0.007       | 0.864     | -0.008       | 83    | -0.003       | 0.961     | -0.004       | 128   | 0.007        | 1.118     | 0.008        | 173   | 0.017        | 1.235     | 0.020        |
| 39    | -0.007       | 0.865     | -0.008       | 84    | -0.003       | 0.964     | -0.004       | 129   | 0.007        | 1.121     | 0.008        | 174   | 0.018        | 1.236     | 0.021        |
| 40    | -0.007       | 0.867     | -0.008       | 85    | -0.003       | 0.967     | -0.003       | 130   | 0.008        | 1.125     | 0.009        | 175   | 0.018        | 1.236     | 0.021        |
| 41    | -0.007       | 0.868     | -0.008       | 86    | -0.003       | 0.970     | -0.003       | 131   | 0.008        | 1.129     | 0.009        | 176   | 0.018        | 1.237     | 0.021        |
| 42    | -0.007       | 0.870     | -0.008       | 87    | -0.003       | 0.973     | -0.003       | 132   | 0.008        | 1.132     | 0.009        | 177   | 0.018        | 1.237     | 0.021        |
| 43    | -0.007       | 0.871     | -0.008       | 88    | -0.002       | 0.976     | -0.003       | 133   | 0.008        | 1.136     | 0.010        | 178   | 0.018        | 1.237     | 0.021        |
| 44    | -0.007       | 0.873     | -0.008       | 89    | -0.002       | 0.979     | -0.003       | 134   | 0.009        | 1.140     | 0.010        | 179   | 0.018        | 1.238     | 0.021        |
| 45    | -0.007       | 0.874     | -0.008       | 90    | -0.002       | 0.982     | -0.003       | 135   | 0.009        | 1.143     | 0.010        | 180   | 0.018        | 1.238     | 0.021        |

Table 10.5: Epicyclic latitude correction factor for Jupiter.  $\mu$  is in degrees. Note that  $\bar{h}(360^\circ - \mu) = \bar{h}(\mu)$ , and  $\delta h_\pm(360^\circ - \mu) = \delta h_\pm(\mu)$ .



| $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ | $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ |
|---------------|---------------------|---------------|---------------|---------------------|---------------|
| 000/180       | 0.000               | (180)/(360)   | 046/134       | 1.788               | (226)/(314)   |
| 002/178       | 0.087               | (182)/(358)   | 048/132       | 1.847               | (228)/(312)   |
| 004/176       | 0.173               | (184)/(356)   | 050/130       | 1.904               | (230)/(310)   |
| 006/174       | 0.260               | (186)/(354)   | 052/128       | 1.958               | (232)/(308)   |
| 008/172       | 0.346               | (188)/(352)   | 054/126       | 2.011               | (234)/(306)   |
| 010/170       | 0.432               | (190)/(350)   | 056/124       | 2.060               | (236)/(304)   |
| 012/168       | 0.517               | (192)/(348)   | 058/122       | 2.108               | (238)/(302)   |
| 014/166       | 0.601               | (194)/(346)   | 060/120       | 2.152               | (240)/(300)   |
| 016/164       | 0.685               | (196)/(344)   | 062/118       | 2.194               | (242)/(298)   |
| 018/162       | 0.768               | (198)/(342)   | 064/116       | 2.234               | (244)/(296)   |
| 020/160       | 0.850               | (200)/(340)   | 066/114       | 2.270               | (246)/(294)   |
| 022/158       | 0.931               | (202)/(338)   | 068/112       | 2.304               | (248)/(292)   |
| 024/156       | 1.011               | (204)/(336)   | 070/110       | 2.335               | (250)/(290)   |
| 026/154       | 1.089               | (206)/(334)   | 072/108       | 2.364               | (252)/(288)   |
| 028/152       | 1.167               | (208)/(332)   | 074/106       | 2.389               | (254)/(286)   |
| 030/150       | 1.243               | (210)/(330)   | 076/104       | 2.411               | (256)/(284)   |
| 032/148       | 1.317               | (212)/(328)   | 078/102       | 2.431               | (258)/(282)   |
| 034/146       | 1.390               | (214)/(326)   | 080/100       | 2.447               | (260)/(280)   |
| 036/144       | 1.461               | (216)/(324)   | 082/098       | 2.461               | (262)/(278)   |
| 038/142       | 1.530               | (218)/(322)   | 084/096       | 2.472               | (264)/(276)   |
| 040/140       | 1.597               | (220)/(320)   | 086/094       | 2.479               | (266)/(274)   |
| 042/138       | 1.663               | (222)/(318)   | 088/092       | 2.484               | (268)/(272)   |
| 044/136       | 1.726               | (224)/(316)   | 090/090       | 2.485               | (270)/(270)   |

Table 10.6: Deferential ecliptic latitude of Saturn. The latitude is minus the value shown in the table if the argument is in parentheses.

| $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ |
|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|
| 0     | -0.006       | 0.905     | -0.006       | 45    | -0.005       | 0.929     | -0.005       | 90    | -0.001       | 0.995     | -0.001       | 135   | 0.005        | 1.077     | 0.006        |
| 1     | -0.006       | 0.905     | -0.006       | 46    | -0.004       | 0.930     | -0.005       | 91    | -0.001       | 0.996     | -0.001       | 136   | 0.005        | 1.078     | 0.006        |
| 2     | -0.006       | 0.905     | -0.006       | 47    | -0.004       | 0.931     | -0.005       | 92    | -0.000       | 0.998     | -0.001       | 137   | 0.005        | 1.080     | 0.006        |
| 3     | -0.006       | 0.905     | -0.006       | 48    | -0.004       | 0.932     | -0.005       | 93    | -0.000       | 1.000     | -0.000       | 138   | 0.006        | 1.081     | 0.006        |
| 4     | -0.006       | 0.905     | -0.006       | 49    | -0.004       | 0.933     | -0.005       | 94    | -0.000       | 1.002     | -0.000       | 139   | 0.006        | 1.083     | 0.007        |
| 5     | -0.006       | 0.905     | -0.006       | 50    | -0.004       | 0.934     | -0.005       | 95    | -0.000       | 1.004     | -0.000       | 140   | 0.006        | 1.084     | 0.007        |
| 6     | -0.006       | 0.906     | -0.006       | 51    | -0.004       | 0.935     | -0.005       | 96    | 0.000        | 1.006     | -0.000       | 141   | 0.006        | 1.086     | 0.007        |
| 7     | -0.006       | 0.906     | -0.006       | 52    | -0.004       | 0.937     | -0.005       | 97    | 0.000        | 1.007     | 0.000        | 142   | 0.006        | 1.087     | 0.007        |
| 8     | -0.006       | 0.906     | -0.006       | 53    | -0.004       | 0.938     | -0.005       | 98    | 0.000        | 1.009     | 0.000        | 143   | 0.006        | 1.089     | 0.007        |
| 9     | -0.006       | 0.906     | -0.006       | 54    | -0.004       | 0.939     | -0.005       | 99    | 0.000        | 1.011     | 0.000        | 144   | 0.006        | 1.090     | 0.007        |
| 10    | -0.006       | 0.906     | -0.006       | 55    | -0.004       | 0.940     | -0.004       | 100   | 0.001        | 1.013     | 0.001        | 145   | 0.006        | 1.091     | 0.007        |
| 11    | -0.006       | 0.907     | -0.006       | 56    | -0.004       | 0.942     | -0.004       | 101   | 0.001        | 1.015     | 0.001        | 146   | 0.006        | 1.093     | 0.008        |
| 12    | -0.006       | 0.907     | -0.006       | 57    | -0.004       | 0.943     | -0.004       | 102   | 0.001        | 1.017     | 0.001        | 147   | 0.007        | 1.094     | 0.008        |
| 13    | -0.006       | 0.907     | -0.006       | 58    | -0.004       | 0.944     | -0.004       | 103   | 0.001        | 1.019     | 0.001        | 148   | 0.007        | 1.095     | 0.008        |
| 14    | -0.006       | 0.908     | -0.006       | 59    | -0.004       | 0.945     | -0.004       | 104   | 0.001        | 1.020     | 0.001        | 149   | 0.007        | 1.097     | 0.008        |
| 15    | -0.006       | 0.908     | -0.006       | 60    | -0.004       | 0.947     | -0.004       | 105   | 0.001        | 1.022     | 0.001        | 150   | 0.007        | 1.098     | 0.008        |
| 16    | -0.006       | 0.908     | -0.006       | 61    | -0.003       | 0.948     | -0.004       | 106   | 0.001        | 1.024     | 0.001        | 151   | 0.007        | 1.099     | 0.008        |
| 17    | -0.006       | 0.909     | -0.006       | 62    | -0.003       | 0.949     | -0.004       | 107   | 0.001        | 1.026     | 0.002        | 152   | 0.007        | 1.100     | 0.008        |
| 18    | -0.006       | 0.909     | -0.006       | 63    | -0.003       | 0.951     | -0.004       | 108   | 0.002        | 1.028     | 0.002        | 153   | 0.007        | 1.101     | 0.008        |
| 19    | -0.005       | 0.909     | -0.006       | 64    | -0.003       | 0.952     | -0.004       | 109   | 0.002        | 1.030     | 0.002        | 154   | 0.007        | 1.103     | 0.009        |
| 20    | -0.005       | 0.910     | -0.006       | 65    | -0.003       | 0.954     | -0.004       | 110   | 0.002        | 1.032     | 0.002        | 155   | 0.007        | 1.104     | 0.009        |
| 21    | -0.005       | 0.910     | -0.006       | 66    | -0.003       | 0.955     | -0.004       | 111   | 0.002        | 1.034     | 0.002        | 156   | 0.007        | 1.105     | 0.009        |
| 22    | -0.005       | 0.911     | -0.006       | 67    | -0.003       | 0.957     | -0.003       | 112   | 0.002        | 1.036     | 0.002        | 157   | 0.008        | 1.106     | 0.009        |
| 23    | -0.005       | 0.911     | -0.006       | 68    | -0.003       | 0.958     | -0.003       | 113   | 0.002        | 1.037     | 0.003        | 158   | 0.008        | 1.107     | 0.009        |
| 24    | -0.005       | 0.912     | -0.006       | 69    | -0.003       | 0.960     | -0.003       | 114   | 0.002        | 1.039     | 0.003        | 159   | 0.008        | 1.107     | 0.009        |
| 25    | -0.005       | 0.913     | -0.006       | 70    | -0.003       | 0.961     | -0.003       | 115   | 0.002        | 1.041     | 0.003        | 160   | 0.008        | 1.108     | 0.009        |
| 26    | -0.005       | 0.913     | -0.006       | 71    | -0.003       | 0.963     | -0.003       | 116   | 0.003        | 1.043     | 0.003        | 161   | 0.008        | 1.109     | 0.009        |
| 27    | -0.005       | 0.914     | -0.006       | 72    | -0.003       | 0.964     | -0.003       | 117   | 0.003        | 1.045     | 0.003        | 162   | 0.008        | 1.110     | 0.009        |
| 28    | -0.005       | 0.914     | -0.006       | 73    | -0.002       | 0.966     | -0.003       | 118   | 0.003        | 1.047     | 0.003        | 163   | 0.008        | 1.111     | 0.009        |
| 29    | -0.005       | 0.915     | -0.006       | 74    | -0.002       | 0.967     | -0.003       | 119   | 0.003        | 1.049     | 0.003        | 164   | 0.008        | 1.111     | 0.009        |
| 30    | -0.005       | 0.916     | -0.006       | 75    | -0.002       | 0.969     | -0.003       | 120   | 0.003        | 1.050     | 0.004        | 165   | 0.008        | 1.112     | 0.009        |
| 31    | -0.005       | 0.916     | -0.006       | 76    | -0.002       | 0.971     | -0.003       | 121   | 0.003        | 1.052     | 0.004        | 166   | 0.008        | 1.113     | 0.010        |
| 32    | -0.005       | 0.917     | -0.006       | 77    | -0.002       | 0.972     | -0.002       | 122   | 0.003        | 1.054     | 0.004        | 167   | 0.008        | 1.113     | 0.010        |
| 33    | -0.005       | 0.918     | -0.006       | 78    | -0.002       | 0.974     | -0.002       | 123   | 0.004        | 1.056     | 0.004        | 168   | 0.008        | 1.114     | 0.010        |
| 34    | -0.005       | 0.919     | -0.006       | 79    | -0.002       | 0.975     | -0.002       | 124   | 0.004        | 1.058     | 0.004        | 169   | 0.008        | 1.114     | 0.010        |
| 35    | -0.005       | 0.920     | -0.006       | 80    | -0.002       | 0.977     | -0.002       | 125   | 0.004        | 1.060     | 0.004        | 170   | 0.008        | 1.115     | 0.010        |
| 36    | -0.005       | 0.920     | -0.006       | 81    | -0.002       | 0.979     | -0.002       | 126   | 0.004        | 1.061     | 0.005        | 171   | 0.008        | 1.115     | 0.010        |
| 37    | -0.005       | 0.921     | -0.006       | 82    | -0.002       | 0.981     | -0.002       | 127   | 0.004        | 1.063     | 0.005        | 172   | 0.008        | 1.116     | 0.010        |
| 38    | -0.005       | 0.922     | -0.006       | 83    | -0.001       | 0.982     | -0.002       | 128   | 0.004        | 1.065     | 0.005        | 173   | 0.008        | 1.116     | 0.010        |
| 39    | -0.005       | 0.923     | -0.005       | 84    | -0.001       | 0.984     | -0.002       | 129   | 0.004        | 1.067     | 0.005        | 174   | 0.008        | 1.116     | 0.010        |
| 40    | -0.005       | 0.924     | -0.005       | 85    | -0.001       | 0.986     | -0.001       | 130   | 0.005        | 1.068     | 0.005        | 175   | 0.008        | 1.116     | 0.010        |
| 41    | -0.005       | 0.925     | -0.005       | 86    | -0.001       | 0.987     | -0.001       | 131   | 0.005        | 1.070     | 0.005        | 176   | 0.009        | 1.117     | 0.010        |
| 42    | -0.005       | 0.926     | -0.005       | 87    | -0.001       | 0.989     | -0.001       | 132   | 0.005        | 1.072     | 0.006        | 177   | 0.009        | 1.117     | 0.010        |
| 43    | -0.005       | 0.927     | -0.005       | 88    | -0.001       | 0.991     | -0.001       | 133   | 0.005        | 1.073     | 0.006        | 178   | 0.009        | 1.117     | 0.010        |
| 44    | -0.005       | 0.928     | -0.005       | 89    | -0.001       | 0.993     | -0.001       | 134   | 0.005        | 1.075     | 0.006        | 179   | 0.009        | 1.117     | 0.010        |
| 45    | -0.005       | 0.929     | -0.005       | 90    | -0.001       | 0.995     | -0.001       | 135   | 0.005        | 1.077     | 0.006        | 180   | 0.009        | 1.117     | 0.010        |

Table 10.7: Epicyclic latitude correction factor for Saturn.  $\mu$  is in degrees. Note that  $\bar{h}(360^\circ - \mu) = \bar{h}(\mu)$ , and  $\delta h_\pm(360^\circ - \mu) = \delta h_\pm(\mu)$ .

| $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ | $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ |
|---------------|---------------------|---------------|---------------|---------------------|---------------|
| 000/180       | 0.000               | (180)/(360)   | 046/134       | 1.752               | (226)/(314)   |
| 002/178       | 0.085               | (182)/(358)   | 048/132       | 1.810               | (228)/(312)   |
| 004/176       | 0.170               | (184)/(356)   | 050/130       | 1.866               | (230)/(310)   |
| 006/174       | 0.255               | (186)/(354)   | 052/128       | 1.919               | (232)/(308)   |
| 008/172       | 0.339               | (188)/(352)   | 054/126       | 1.970               | (234)/(306)   |
| 010/170       | 0.423               | (190)/(350)   | 056/124       | 2.019               | (236)/(304)   |
| 012/168       | 0.506               | (192)/(348)   | 058/122       | 2.066               | (238)/(302)   |
| 014/166       | 0.589               | (194)/(346)   | 060/120       | 2.109               | (240)/(300)   |
| 016/164       | 0.671               | (196)/(344)   | 062/118       | 2.151               | (242)/(298)   |
| 018/162       | 0.753               | (198)/(342)   | 064/116       | 2.189               | (244)/(296)   |
| 020/160       | 0.833               | (200)/(340)   | 066/114       | 2.225               | (246)/(294)   |
| 022/158       | 0.912               | (202)/(338)   | 068/112       | 2.258               | (248)/(292)   |
| 024/156       | 0.991               | (204)/(336)   | 070/110       | 2.289               | (250)/(290)   |
| 026/154       | 1.068               | (206)/(334)   | 072/108       | 2.316               | (252)/(288)   |
| 028/152       | 1.143               | (208)/(332)   | 074/106       | 2.341               | (254)/(286)   |
| 030/150       | 1.218               | (210)/(330)   | 076/104       | 2.363               | (256)/(284)   |
| 032/148       | 1.291               | (212)/(328)   | 078/102       | 2.382               | (258)/(282)   |
| 034/146       | 1.362               | (214)/(326)   | 080/100       | 2.399               | (260)/(280)   |
| 036/144       | 1.432               | (216)/(324)   | 082/098       | 2.412               | (262)/(278)   |
| 038/142       | 1.500               | (218)/(322)   | 084/096       | 2.422               | (264)/(276)   |
| 040/140       | 1.566               | (220)/(320)   | 086/094       | 2.430               | (266)/(274)   |
| 042/138       | 1.630               | (222)/(318)   | 088/092       | 2.434               | (268)/(272)   |
| 044/136       | 1.692               | (224)/(316)   | 090/090       | 2.436               | (270)/(270)   |

Table 10.8: Epicyclic ecliptic latitude of Venus. The latitude is minus the value shown in the table if the argument is in parentheses.

| $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ |
|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|
| 0     | 0.008        | 0.580     | 0.008        | 45    | 0.009        | 0.627     | 0.009        | 90    | 0.012        | 0.810     | 0.013        | 135   | 0.032        | 1.413     | 0.033        |
| 1     | 0.008        | 0.580     | 0.008        | 46    | 0.009        | 0.629     | 0.009        | 91    | 0.013        | 0.817     | 0.013        | 136   | 0.033        | 1.439     | 0.034        |
| 2     | 0.008        | 0.580     | 0.008        | 47    | 0.009        | 0.631     | 0.009        | 92    | 0.013        | 0.824     | 0.013        | 137   | 0.034        | 1.466     | 0.035        |
| 3     | 0.008        | 0.580     | 0.008        | 48    | 0.009        | 0.633     | 0.009        | 93    | 0.013        | 0.831     | 0.013        | 138   | 0.036        | 1.493     | 0.037        |
| 4     | 0.008        | 0.580     | 0.008        | 49    | 0.009        | 0.636     | 0.009        | 94    | 0.013        | 0.838     | 0.013        | 139   | 0.037        | 1.522     | 0.038        |
| 5     | 0.008        | 0.581     | 0.008        | 50    | 0.009        | 0.638     | 0.009        | 95    | 0.013        | 0.846     | 0.013        | 140   | 0.039        | 1.552     | 0.040        |
| 6     | 0.008        | 0.581     | 0.008        | 51    | 0.009        | 0.641     | 0.009        | 96    | 0.013        | 0.854     | 0.014        | 141   | 0.040        | 1.583     | 0.041        |
| 7     | 0.008        | 0.581     | 0.008        | 52    | 0.009        | 0.643     | 0.009        | 97    | 0.014        | 0.861     | 0.014        | 142   | 0.042        | 1.615     | 0.043        |
| 8     | 0.008        | 0.582     | 0.008        | 53    | 0.009        | 0.646     | 0.009        | 98    | 0.014        | 0.870     | 0.014        | 143   | 0.044        | 1.648     | 0.045        |
| 9     | 0.008        | 0.582     | 0.008        | 54    | 0.009        | 0.649     | 0.009        | 99    | 0.014        | 0.878     | 0.014        | 144   | 0.046        | 1.683     | 0.047        |
| 10    | 0.008        | 0.582     | 0.008        | 55    | 0.009        | 0.652     | 0.009        | 100   | 0.014        | 0.886     | 0.014        | 145   | 0.048        | 1.719     | 0.049        |
| 11    | 0.008        | 0.583     | 0.008        | 56    | 0.009        | 0.655     | 0.009        | 101   | 0.014        | 0.895     | 0.015        | 146   | 0.050        | 1.756     | 0.052        |
| 12    | 0.008        | 0.583     | 0.008        | 57    | 0.009        | 0.658     | 0.009        | 102   | 0.015        | 0.904     | 0.015        | 147   | 0.053        | 1.795     | 0.054        |
| 13    | 0.008        | 0.584     | 0.008        | 58    | 0.009        | 0.661     | 0.009        | 103   | 0.015        | 0.913     | 0.015        | 148   | 0.055        | 1.836     | 0.057        |
| 14    | 0.008        | 0.584     | 0.008        | 59    | 0.009        | 0.664     | 0.010        | 104   | 0.015        | 0.923     | 0.015        | 149   | 0.058        | 1.878     | 0.060        |
| 15    | 0.008        | 0.585     | 0.008        | 60    | 0.009        | 0.667     | 0.010        | 105   | 0.015        | 0.933     | 0.016        | 150   | 0.061        | 1.922     | 0.063        |
| 16    | 0.008        | 0.586     | 0.008        | 61    | 0.009        | 0.670     | 0.010        | 106   | 0.016        | 0.943     | 0.016        | 151   | 0.065        | 1.968     | 0.067        |
| 17    | 0.008        | 0.586     | 0.008        | 62    | 0.010        | 0.674     | 0.010        | 107   | 0.016        | 0.953     | 0.016        | 152   | 0.068        | 2.016     | 0.071        |
| 18    | 0.008        | 0.587     | 0.008        | 63    | 0.010        | 0.677     | 0.010        | 108   | 0.016        | 0.964     | 0.017        | 153   | 0.072        | 2.065     | 0.075        |
| 19    | 0.008        | 0.588     | 0.008        | 64    | 0.010        | 0.681     | 0.010        | 109   | 0.016        | 0.975     | 0.017        | 154   | 0.076        | 2.117     | 0.080        |
| 20    | 0.008        | 0.589     | 0.008        | 65    | 0.010        | 0.684     | 0.010        | 110   | 0.017        | 0.986     | 0.017        | 155   | 0.081        | 2.170     | 0.085        |
| 21    | 0.008        | 0.590     | 0.008        | 66    | 0.010        | 0.688     | 0.010        | 111   | 0.017        | 0.997     | 0.017        | 156   | 0.086        | 2.226     | 0.090        |
| 22    | 0.008        | 0.591     | 0.008        | 67    | 0.010        | 0.692     | 0.010        | 112   | 0.017        | 1.009     | 0.018        | 157   | 0.091        | 2.283     | 0.096        |
| 23    | 0.008        | 0.592     | 0.008        | 68    | 0.010        | 0.696     | 0.010        | 113   | 0.018        | 1.021     | 0.018        | 158   | 0.097        | 2.343     | 0.102        |
| 24    | 0.008        | 0.593     | 0.008        | 69    | 0.010        | 0.700     | 0.010        | 114   | 0.018        | 1.034     | 0.019        | 159   | 0.103        | 2.405     | 0.109        |
| 25    | 0.008        | 0.594     | 0.008        | 70    | 0.010        | 0.704     | 0.010        | 115   | 0.019        | 1.047     | 0.019        | 160   | 0.110        | 2.469     | 0.117        |
| 26    | 0.008        | 0.595     | 0.008        | 71    | 0.010        | 0.708     | 0.010        | 116   | 0.019        | 1.060     | 0.019        | 161   | 0.117        | 2.535     | 0.125        |
| 27    | 0.008        | 0.596     | 0.008        | 72    | 0.010        | 0.712     | 0.011        | 117   | 0.019        | 1.074     | 0.020        | 162   | 0.125        | 2.603     | 0.134        |
| 28    | 0.008        | 0.597     | 0.008        | 73    | 0.010        | 0.717     | 0.011        | 118   | 0.020        | 1.088     | 0.020        | 163   | 0.133        | 2.673     | 0.144        |
| 29    | 0.008        | 0.599     | 0.008        | 74    | 0.010        | 0.721     | 0.011        | 119   | 0.020        | 1.103     | 0.021        | 164   | 0.142        | 2.744     | 0.155        |
| 30    | 0.008        | 0.600     | 0.008        | 75    | 0.011        | 0.726     | 0.011        | 120   | 0.021        | 1.118     | 0.021        | 165   | 0.151        | 2.816     | 0.166        |
| 31    | 0.008        | 0.601     | 0.008        | 76    | 0.011        | 0.730     | 0.011        | 121   | 0.021        | 1.133     | 0.022        | 166   | 0.161        | 2.890     | 0.178        |
| 32    | 0.008        | 0.603     | 0.008        | 77    | 0.011        | 0.735     | 0.011        | 122   | 0.022        | 1.149     | 0.022        | 167   | 0.172        | 2.964     | 0.191        |
| 33    | 0.008        | 0.604     | 0.008        | 78    | 0.011        | 0.740     | 0.011        | 123   | 0.022        | 1.166     | 0.023        | 168   | 0.183        | 3.037     | 0.204        |
| 34    | 0.008        | 0.606     | 0.008        | 79    | 0.011        | 0.745     | 0.011        | 124   | 0.023        | 1.183     | 0.023        | 169   | 0.194        | 3.111     | 0.218        |
| 35    | 0.008        | 0.608     | 0.009        | 80    | 0.011        | 0.751     | 0.011        | 125   | 0.024        | 1.200     | 0.024        | 170   | 0.205        | 3.182     | 0.232        |
| 36    | 0.008        | 0.609     | 0.009        | 81    | 0.011        | 0.756     | 0.011        | 126   | 0.024        | 1.219     | 0.025        | 171   | 0.217        | 3.252     | 0.247        |
| 37    | 0.008        | 0.611     | 0.009        | 82    | 0.011        | 0.761     | 0.012        | 127   | 0.025        | 1.237     | 0.025        | 172   | 0.228        | 3.318     | 0.262        |
| 38    | 0.008        | 0.613     | 0.009        | 83    | 0.011        | 0.767     | 0.012        | 128   | 0.026        | 1.257     | 0.026        | 173   | 0.239        | 3.380     | 0.276        |
| 39    | 0.008        | 0.614     | 0.009        | 84    | 0.012        | 0.773     | 0.012        | 129   | 0.026        | 1.277     | 0.027        | 174   | 0.249        | 3.436     | 0.290        |
| 40    | 0.008        | 0.616     | 0.009        | 85    | 0.012        | 0.778     | 0.012        | 130   | 0.027        | 1.298     | 0.028        | 175   | 0.258        | 3.486     | 0.302        |
| 41    | 0.008        | 0.618     | 0.009        | 86    | 0.012        | 0.784     | 0.012        | 131   | 0.028        | 1.319     | 0.029        | 176   | 0.267        | 3.529     | 0.313        |
| 42    | 0.009        | 0.620     | 0.009        | 87    | 0.012        | 0.791     | 0.012        | 132   | 0.029        | 1.342     | 0.030        | 177   | 0.273        | 3.564     | 0.322        |
| 43    | 0.009        | 0.622     | 0.009        | 88    | 0.012        | 0.797     | 0.012        | 133   | 0.030        | 1.365     | 0.031        | 178   | 0.278        | 3.589     | 0.329        |
| 44    | 0.009        | 0.624     | 0.009        | 89    | 0.012        | 0.803     | 0.012        | 134   | 0.031        | 1.389     | 0.032        | 179   | 0.281        | 3.604     | 0.333        |
| 45    | 0.009        | 0.627     | 0.009        | 90    | 0.012        | 0.810     | 0.013        | 135   | 0.032        | 1.413     | 0.033        | 180   | 0.282        | 3.609     | 0.334        |

Table 10.9: Deferential latitude correction factor for Venus.  $\mu$  is in degrees. Note that  $\bar{h}(360^\circ - \mu) = \bar{h}(\mu)$ , and  $\delta h_\pm(360^\circ - \mu) = \delta h_\pm(\mu)$ .

| $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ | $F(^{\circ})$ | $\beta_0(^{\circ})$ | $F(^{\circ})$ |
|---------------|---------------------|---------------|---------------|---------------------|---------------|
| 000/180       | 0.000               | (180)/(360)   | 046/134       | 1.922               | (226)/(314)   |
| 002/178       | 0.093               | (182)/(358)   | 048/132       | 1.986               | (228)/(312)   |
| 004/176       | 0.186               | (184)/(356)   | 050/130       | 2.047               | (230)/(310)   |
| 006/174       | 0.279               | (186)/(354)   | 052/128       | 2.105               | (232)/(308)   |
| 008/172       | 0.372               | (188)/(352)   | 054/126       | 2.162               | (234)/(306)   |
| 010/170       | 0.464               | (190)/(350)   | 056/124       | 2.215               | (236)/(304)   |
| 012/168       | 0.556               | (192)/(348)   | 058/122       | 2.266               | (238)/(302)   |
| 014/166       | 0.646               | (194)/(346)   | 060/120       | 2.314               | (240)/(300)   |
| 016/164       | 0.736               | (196)/(344)   | 062/118       | 2.359               | (242)/(298)   |
| 018/162       | 0.826               | (198)/(342)   | 064/116       | 2.401               | (244)/(296)   |
| 020/160       | 0.914               | (200)/(340)   | 066/114       | 2.441               | (246)/(294)   |
| 022/158       | 1.001               | (202)/(338)   | 068/112       | 2.477               | (248)/(292)   |
| 024/156       | 1.087               | (204)/(336)   | 070/110       | 2.511               | (250)/(290)   |
| 026/154       | 1.171               | (206)/(334)   | 072/108       | 2.541               | (252)/(288)   |
| 028/152       | 1.254               | (208)/(332)   | 074/106       | 2.568               | (254)/(286)   |
| 030/150       | 1.336               | (210)/(330)   | 076/104       | 2.592               | (256)/(284)   |
| 032/148       | 1.416               | (212)/(328)   | 078/102       | 2.613               | (258)/(282)   |
| 034/146       | 1.494               | (214)/(326)   | 080/100       | 2.631               | (260)/(280)   |
| 036/144       | 1.570               | (216)/(324)   | 082/098       | 2.646               | (262)/(278)   |
| 038/142       | 1.645               | (218)/(322)   | 084/096       | 2.657               | (264)/(276)   |
| 040/140       | 1.717               | (220)/(320)   | 086/094       | 2.665               | (266)/(274)   |
| 042/138       | 1.788               | (222)/(318)   | 088/092       | 2.670               | (268)/(272)   |
| 044/136       | 1.856               | (224)/(316)   | 090/090       | 2.672               | (270)/(270)   |

Table 10.10: Epicyclic ecliptic latitude of Mercury. The latitude is minus the value shown in the table if the argument is in parentheses.

| $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ | $\mu$ | $\delta h_-$ | $\bar{h}$ | $\delta h_+$ |
|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|-------|--------------|-----------|--------------|
| 0     | 0.099        | 0.719     | 0.137        | 45    | 0.110        | 0.765     | 0.152        | 90    | 0.152        | 0.930     | 0.217        | 135   | 0.274        | 1.283     | 0.451        |
| 1     | 0.099        | 0.719     | 0.137        | 46    | 0.110        | 0.768     | 0.153        | 91    | 0.153        | 0.935     | 0.220        | 136   | 0.278        | 1.293     | 0.461        |
| 2     | 0.099        | 0.719     | 0.137        | 47    | 0.111        | 0.770     | 0.154        | 92    | 0.155        | 0.941     | 0.222        | 137   | 0.282        | 1.303     | 0.471        |
| 3     | 0.099        | 0.719     | 0.137        | 48    | 0.111        | 0.772     | 0.154        | 93    | 0.157        | 0.946     | 0.225        | 138   | 0.286        | 1.313     | 0.481        |
| 4     | 0.099        | 0.719     | 0.137        | 49    | 0.112        | 0.774     | 0.155        | 94    | 0.158        | 0.952     | 0.228        | 139   | 0.290        | 1.324     | 0.491        |
| 5     | 0.099        | 0.720     | 0.137        | 50    | 0.112        | 0.777     | 0.156        | 95    | 0.160        | 0.958     | 0.231        | 140   | 0.295        | 1.334     | 0.501        |
| 6     | 0.099        | 0.720     | 0.137        | 51    | 0.113        | 0.779     | 0.157        | 96    | 0.162        | 0.964     | 0.234        | 141   | 0.299        | 1.344     | 0.512        |
| 7     | 0.099        | 0.720     | 0.137        | 52    | 0.113        | 0.782     | 0.158        | 97    | 0.164        | 0.970     | 0.237        | 142   | 0.304        | 1.355     | 0.523        |
| 8     | 0.099        | 0.720     | 0.137        | 53    | 0.114        | 0.784     | 0.159        | 98    | 0.165        | 0.976     | 0.240        | 143   | 0.308        | 1.365     | 0.534        |
| 9     | 0.100        | 0.721     | 0.137        | 54    | 0.115        | 0.787     | 0.160        | 99    | 0.167        | 0.983     | 0.243        | 144   | 0.313        | 1.375     | 0.546        |
| 10    | 0.100        | 0.721     | 0.138        | 55    | 0.115        | 0.790     | 0.161        | 100   | 0.169        | 0.989     | 0.246        | 145   | 0.317        | 1.386     | 0.558        |
| 11    | 0.100        | 0.722     | 0.138        | 56    | 0.116        | 0.793     | 0.162        | 101   | 0.171        | 0.996     | 0.250        | 146   | 0.322        | 1.396     | 0.570        |
| 12    | 0.100        | 0.722     | 0.138        | 57    | 0.117        | 0.795     | 0.163        | 102   | 0.173        | 1.002     | 0.253        | 147   | 0.326        | 1.406     | 0.582        |
| 13    | 0.100        | 0.723     | 0.138        | 58    | 0.117        | 0.798     | 0.164        | 103   | 0.175        | 1.009     | 0.257        | 148   | 0.331        | 1.417     | 0.594        |
| 14    | 0.100        | 0.723     | 0.138        | 59    | 0.118        | 0.801     | 0.165        | 104   | 0.178        | 1.016     | 0.261        | 149   | 0.335        | 1.427     | 0.607        |
| 15    | 0.100        | 0.724     | 0.138        | 60    | 0.119        | 0.804     | 0.166        | 105   | 0.180        | 1.023     | 0.264        | 150   | 0.340        | 1.437     | 0.620        |
| 16    | 0.100        | 0.725     | 0.139        | 61    | 0.120        | 0.807     | 0.167        | 106   | 0.182        | 1.030     | 0.268        | 151   | 0.345        | 1.447     | 0.633        |
| 17    | 0.101        | 0.725     | 0.139        | 62    | 0.120        | 0.811     | 0.168        | 107   | 0.184        | 1.037     | 0.272        | 152   | 0.349        | 1.457     | 0.646        |
| 18    | 0.101        | 0.726     | 0.139        | 63    | 0.121        | 0.814     | 0.169        | 108   | 0.187        | 1.044     | 0.276        | 153   | 0.354        | 1.467     | 0.659        |
| 19    | 0.101        | 0.727     | 0.139        | 64    | 0.122        | 0.817     | 0.170        | 109   | 0.189        | 1.052     | 0.281        | 154   | 0.358        | 1.476     | 0.672        |
| 20    | 0.101        | 0.728     | 0.140        | 65    | 0.123        | 0.820     | 0.172        | 110   | 0.192        | 1.059     | 0.285        | 155   | 0.363        | 1.486     | 0.686        |
| 21    | 0.101        | 0.729     | 0.140        | 66    | 0.124        | 0.824     | 0.173        | 111   | 0.194        | 1.067     | 0.290        | 156   | 0.367        | 1.495     | 0.699        |
| 22    | 0.101        | 0.730     | 0.140        | 67    | 0.124        | 0.827     | 0.174        | 112   | 0.197        | 1.074     | 0.294        | 157   | 0.371        | 1.504     | 0.713        |
| 23    | 0.102        | 0.731     | 0.141        | 68    | 0.125        | 0.831     | 0.176        | 113   | 0.199        | 1.082     | 0.299        | 158   | 0.376        | 1.513     | 0.726        |
| 24    | 0.102        | 0.732     | 0.141        | 69    | 0.126        | 0.835     | 0.177        | 114   | 0.202        | 1.090     | 0.304        | 159   | 0.380        | 1.522     | 0.739        |
| 25    | 0.102        | 0.733     | 0.141        | 70    | 0.127        | 0.838     | 0.179        | 115   | 0.205        | 1.098     | 0.309        | 160   | 0.384        | 1.530     | 0.753        |
| 26    | 0.102        | 0.734     | 0.142        | 71    | 0.128        | 0.842     | 0.180        | 116   | 0.208        | 1.107     | 0.315        | 161   | 0.388        | 1.538     | 0.766        |
| 27    | 0.103        | 0.735     | 0.142        | 72    | 0.129        | 0.846     | 0.182        | 117   | 0.210        | 1.115     | 0.320        | 162   | 0.392        | 1.546     | 0.779        |
| 28    | 0.103        | 0.737     | 0.142        | 73    | 0.130        | 0.850     | 0.183        | 118   | 0.213        | 1.123     | 0.326        | 163   | 0.396        | 1.554     | 0.791        |
| 29    | 0.103        | 0.738     | 0.143        | 74    | 0.131        | 0.854     | 0.185        | 119   | 0.216        | 1.132     | 0.331        | 164   | 0.399        | 1.561     | 0.804        |
| 30    | 0.104        | 0.739     | 0.143        | 75    | 0.132        | 0.858     | 0.186        | 120   | 0.219        | 1.141     | 0.337        | 165   | 0.403        | 1.568     | 0.816        |
| 31    | 0.104        | 0.741     | 0.144        | 76    | 0.133        | 0.862     | 0.188        | 121   | 0.223        | 1.149     | 0.343        | 166   | 0.406        | 1.575     | 0.827        |
| 32    | 0.104        | 0.742     | 0.144        | 77    | 0.135        | 0.866     | 0.190        | 122   | 0.226        | 1.158     | 0.350        | 167   | 0.409        | 1.581     | 0.838        |
| 33    | 0.105        | 0.743     | 0.145        | 78    | 0.136        | 0.871     | 0.192        | 123   | 0.229        | 1.167     | 0.356        | 168   | 0.412        | 1.587     | 0.849        |
| 34    | 0.105        | 0.745     | 0.145        | 79    | 0.137        | 0.875     | 0.193        | 124   | 0.232        | 1.176     | 0.363        | 169   | 0.415        | 1.592     | 0.858        |
| 35    | 0.105        | 0.747     | 0.146        | 80    | 0.138        | 0.880     | 0.195        | 125   | 0.236        | 1.185     | 0.370        | 170   | 0.417        | 1.597     | 0.868        |
| 36    | 0.106        | 0.748     | 0.146        | 81    | 0.139        | 0.884     | 0.197        | 126   | 0.239        | 1.195     | 0.377        | 171   | 0.420        | 1.602     | 0.876        |
| 37    | 0.106        | 0.750     | 0.147        | 82    | 0.141        | 0.889     | 0.199        | 127   | 0.243        | 1.204     | 0.384        | 172   | 0.422        | 1.606     | 0.884        |
| 38    | 0.106        | 0.752     | 0.147        | 83    | 0.142        | 0.894     | 0.201        | 128   | 0.246        | 1.214     | 0.392        | 173   | 0.424        | 1.610     | 0.891        |
| 39    | 0.107        | 0.754     | 0.148        | 84    | 0.143        | 0.899     | 0.203        | 129   | 0.250        | 1.223     | 0.400        | 174   | 0.425        | 1.613     | 0.897        |
| 40    | 0.107        | 0.755     | 0.149        | 85    | 0.145        | 0.904     | 0.206        | 130   | 0.254        | 1.233     | 0.408        | 175   | 0.427        | 1.615     | 0.903        |
| 41    | 0.108        | 0.757     | 0.149        | 86    | 0.146        | 0.909     | 0.208        | 131   | 0.258        | 1.243     | 0.416        | 176   | 0.428        | 1.618     | 0.907        |
| 42    | 0.108        | 0.759     | 0.150        | 87    | 0.147        | 0.914     | 0.210        | 132   | 0.262        | 1.253     | 0.424        | 177   | 0.429        | 1.619     | 0.911        |
| 43    | 0.109        | 0.761     | 0.151        | 88    | 0.149        | 0.919     | 0.212        | 133   | 0.265        | 1.263     | 0.433        | 178   | 0.429        | 1.621     | 0.913        |
| 44    | 0.109        | 0.763     | 0.151        | 89    | 0.150        | 0.924     | 0.215        | 134   | 0.269        | 1.273     | 0.442        | 179   | 0.430        | 1.621     | 0.914        |
| 45    | 0.110        | 0.765     | 0.152        | 90    | 0.152        | 0.930     | 0.217        | 135   | 0.274        | 1.283     | 0.451        | 180   | 0.430        | 1.622     | 0.915        |

Table 10.11: Differential latitude correction factor for Mercury.  $\mu$  is in degrees. Note that  $\bar{h}(360^\circ - \mu) = \bar{h}(\mu)$ , and  $\delta h_\pm(360^\circ - \mu) = \delta h_\pm(\mu)$ .

## Technical terms

**Altitude:** The angle subtended at the observer by the radius vector connecting a celestial object to an observer on the Earth's surface, and the vector's projection onto the **horizontal plane**. Object's above/below the **horizon** have positive/negative altitudes.

**Altitude circle:** A **great circle** on the celestial sphere that passes through the local **zenith** at a given observation site on the Earth's surface.

**Anomaly:** Any deviation in an orbit from uniform circular motion that is concentric with the central body. Anomaly is also used as another word for angle.

**Apocenter:** Point on a **Keplerian orbit** that is farthest from the central body. If the central body is the Sun then the apocenter is generally termed the *aphelion*. Likewise, if the central body is the Earth then the apocenter is termed the *apogee*.

**Arctic circles:** Two latitude circles on the Earth's surface that are equidistant from the equator. Above the arctic circles, the Sun never sets for part of the year, and never rises for part of the year.

**Argument of latitude:** Angle subtended at the central body by the radius vectors connecting the central body to the orbiting body, and the central body to the **ascending node**, in a **Keplerian orbit**.

**Ascendent:** Point on **ecliptic circle** that is ascending at any given time on the eastern **horizon**.

**Ascending node:** Point on a **Keplerian orbit** at which the orbital plane crosses the **ecliptic plane** from south to north in the direction of motion of the orbiting body.

**Autumnal equinox:** The point at which the **ecliptic circle** crosses the **celestial equator** from north to south (in the direction of the Sun's apparent motion along the ecliptic).

**Azimuth:** Angle subtended at the observer by the projection of the vector connecting a celestial object to an observer on the Earth's surface onto the **horizontal plane**, and the vector connecting the north **compass point** to the observer. Azimuth increases clockwise (that is, from the north to the east) looking at the horizontal plane from above.

**Celestial axis:** An imaginary extension of the Earth's axis of rotation that pierces the **celestial sphere** at the two **celestial poles**. The sphere's **diurnal motion** is about this axis.

**Celestial coordinates:** Angular coordinate system whose fundamental plane is the **celestial plane**, and whose poles are the **celestial poles**. The polar and azimuthal angles in this system are called **declination** and **right ascension**, respectively.

**Celestial equator:** The intersection of the imagined extension of the Earth's **equatorial plane** with the **celestial sphere**.

**Celestial plane:** The plane containing the Earth's equator.

**Celestial poles:** The two points at which the **celestial axis** pierces the **celestial sphere**. The north celestial pole lies to the north of the **celestial plane**, whereas the south celestial pole lies to the south. The celestial poles are the only two points on the celestial sphere whose positions are unaffected by **diurnal motion**.

**Celestial sphere:** An imaginary sphere of infinite radius that is concentric with the Earth. All objects in the sky are thought of as attached to this sphere.

**Compass points:** At a given observation site on the Earth's surface, the north, east, south, and west compass points lie on the local horizon due north, east, south, and west, respectively, of the observer.

- Conjunction:** Two celestial objects are said to be in conjunction when they have the same **ecliptic longitude**. For an **inferior planet** in conjunction with the Sun, the conjunction is said to be *superior* if the planet is farther from the Earth than the Sun, and *inferior* if the Sun is farther from the Earth than the planet.
- Culmination:** A celestial object is said to culminate on a given day when it attains its maximum **altitude** in the sky.
- Declination:** Angle subtended at the Earth's center by the radius vector connecting a celestial object to the Earth's center, and the vector's projection onto the **celestial plane**. Object's to the north/south of the **celestial equator** have positive/negative declinations.
- Deferent:** Large circle centered on the Sun about which the **guide point** rotates in a **geocentric planetary orbit**.
- Deferential latitude:** **Ecliptic latitude** a **superior planet** has by virtue of the **inclination** of its **deferent**.
- Deferential latitude correction factor:** Correction to the **ecliptic latitude** of an **inferior planet** due to the finite size of its **deferent**.
- Diurnal motion:** Daily rotation of the **celestial sphere**, and the objects attached to it, from east to west (looking south in the Earth's northern hemisphere) about the **celestial axis**.
- Eccentricity:** Measure of the displacement along the **major axis** of the central body from the geometric center in a **Keplerian orbit**.
- Ecliptic axis:** Normal to the **ecliptic plane** that passes through the center of the Earth.
- Ecliptic circle:** Apparent path traced out by the Sun on the **celestial sphere** during the course of a year.
- Ecliptic coordinates:** Angular coordinate system whose fundamental plane is the **ecliptic plane**, and whose poles are the **ecliptic poles**.
- Ecliptic latitude:** Angle subtended at the Earth's center by the radius vector connecting a celestial object to the Earth's center, and the vector's projection onto the **ecliptic plane**. Objects to the north/south of the **ecliptic circle** have positive/ecliptic latitudes.
- Ecliptic longitude:** Angle subtended at the Earth's center by the projection of the vector connecting a celestial object to the Earth's center onto the **ecliptic plane**, and the vector connecting the **vernal equinox** to the Earth's center. Ecliptic longitude increases counterclockwise (that is, from the west to the east) looking at the ecliptic plane from the north.
- Ecliptic plane:** Plane containing the mean orbit of the Earth about the Sun.
- Ecliptic poles:** The two points at which the **ecliptic axis** pierces the **celestial sphere**. The north ecliptic pole lies to the north of the ecliptic plane, whereas the south ecliptic pole lies to the south.
- Elongation:** Difference in **ecliptic longitude** between two celestial objects.
- Epicycle:** Small circle, centered on the **guide-point**, about which a planet rotates in a **geocentric planetary orbit**.
- Epicyclic anomaly:** Angle subtended between the radius vectors connecting the Earth to the **guide-point**, and the guide-point to the planet, in a **geocentric planetary orbit**.
- Epicyclic latitude:** **Ecliptic latitude** an **inferior planet** has by virtue of the **inclination** of its **epicycle**.
- Epicyclic latitude correction factor:** Correction to the **ecliptic latitude** of a **superior planet** due to the finite size of its **epicycle**.
- Epoch:** Standard time at which the **orbital elements** of an orbiting body in the Solar System are specified.



**Equant:** Point about which the orbiting body appears to rotate uniformly in a **Keplerian orbit** of low **eccentricity**. The equant is diagrammatically opposite the central body with respect to the geometric center of the orbit.

**Equation of center:** Difference between the **true anomaly** and the **mean anomaly** in a **Keplerian orbit**.

**Equation of epicycle:** **Elongation** of a planet from its **guide-point** in a **geocentric planetary orbit**.

**Equation of time:** Time interval between **local noon** and **mean local noon**.

**Equinoxes:** The two opposite points on the **ecliptic circle** that the Sun reaches on the days of the year that day and night are equally long.

**Evection:** An **anomaly** of the Moon's orbit about the Earth that is associated with the perturbing influence of the Sun.

**Geocentric planetary orbit:** An orbit in which a planet rotates about a **guide-point** in a small circle called an **epicycle**, and the guide-point rotates about the Earth in a large circle called a **deferent**.

**Great circle:** A circle on the surface of a sphere produced by the intersection of a plane that bisects the sphere.

**Greatest elongation:** Greatest **elongation** of an *inferior planet* from the Sun. If the planet is to the east/west of the Sun then the elongation is called the greatest eastern/western elongation.

**Guide-point:** Center of an epicycle in a **geocentric planetary orbit**.

**Horizon:** Tangent plane to the Earth's surface, at a given observation site, that divides the **celestial sphere** into visible and invisible hemispheres.

**Horizontal coordinates:** Angular coordinate system whose fundamental plane is the **horizontal plane**, and whose poles are the **zenith** and **nadir**.

**Horizontal plane:** Plane containing the local horizon.

**Horoscope:** Point on the **ecliptic circle** that is ascending at a given time on the eastern **horizon**.

**Inclination:** Maximum angle subtended between the plane of a **Keplerian orbit** and the **ecliptic plane**.

**Inclination of ecliptic:** Inclination of the **ecliptic plane** to the **equatorial plane**.

**Inferior planet:** A planet that is closer to the Sun than the Earth.

**Julian day number:** Number ascribed to a particular day in a scheme in which days are numbered consecutively from January 1, 4713 BC (in Julian calendar), which is designated day zero. Julian days start at 12:00 UT.

**Keplerian orbit:** Ellipse that is confocal with the central object. The radius vector connecting the central and orbiting bodies sweeps out equal areas in equal time intervals.

**Local mean noon:** Instant in time at which the **mean Sun** attains its upper **transit**.

**Local noon:** Instant in time at which the Sun attains its upper **transit**.

**Longitude of ascending node:** Angle subtended at the central body by the radius vectors connecting the central body to the **ascending node**, and the central body to the **vernal equinox**, in a **Keplerian orbit**.

**Longitude of pericenter:** Angle subtended at the central body by the radius vectors connecting the central body to the **pericenter**, and the central body to the **vernal equinox**, in a **Keplerian orbit**.

**Major axis:** Longest diameter that passes through the geometric center of a **Keplerian orbit**.

- Major radius:** Half the length of the **major axis** of a **Keplerian orbit**.
- Mean anomaly:** Angle that would be subtended at the central body by the radius vectors connecting the central body to the orbiting body, and the central body to the **pericenter**, in a **Keplerian orbit**, if the orbiting body were to rotate about the central body with a uniform angular velocity.
- Mean argument of latitude:** Value the **argument of latitude** would have if the orbiting body in a **Keplerian orbit** were to rotate about the central body at a fixed angular velocity.
- Mean argument of latitude at epoch:** Value of the **mean argument of latitude** of a **Keplerian orbit** at the epoch.
- Mean (ecliptic) longitude:** Value the **ecliptic longitude** would have if the orbiting body in a **Keplerian orbit** were to rotate about the central body at a fixed angular velocity.
- Mean (ecliptic) longitude at epoch:** Value of the **mean longitude** of a **Keplerian orbit** at the epoch.
- Mean Solar day:** Time interval between successive **local mean noons**.
- Mean Solar time:** Time calculated using the **mean Sun**.
- Mean sun:** Fictitious body that travels around the **celestial equator** (from west to east looking south in the Earth's northern hemisphere) at a uniform rate, and completes one orbit every tropical year.
- Meridian plane:** Plane passing through the **zenith** and the north and south **compass points** at a given observation site on the Earth's surface.
- Minor axis:** The minor axis of a **Keplerian orbit** is the shortest diameter that passes through the geometric center.
- Minor radius:** The minor radius of a **Keplerian orbit** is half the length of the **minor axis**.
- Month:** A *sidereal month* (27.32166 days) is the mean time needed for the Moon to complete an orbital rotation around the Earth relative to the fixed stars. A *tropical month* (27.32158 days) is the mean time needed for the Moon's ecliptic latitude to increase by  $360^\circ$ , and is 6.8 seconds shorter than a sidereal month because of the precession of the equinoxes. A *synodic month* (29.53059 days) is the mean time interval between successive new Moons. An *anomalistic month* (27.55455 days) is the mean time interval between successive passages of the Moon through its perigee. A *draconic month* (27.21222 days) is the mean time interval between successive passages of the Moon through its ascending node.
- Nadir:** Point on the **celestial sphere** that is directly underfoot at a given observation site on the Earth's surface.
- Opposition:** Two celestial objects are said to be in opposition when their **ecliptic longitudes** differ by  $180^\circ$ .
- Orbital elements:** Eight quantities that completely specify a **Keplerian orbit**: that is, **major radius**, **eccentricity**, **rate of motion of mean longitude**, **rate of motion of mean anomaly**, **mean longitude at epoch**, **mean anomaly at epoch**, **inclination**, **rate of motion in mean argument of latitude**, **mean argument of latitude at epoch**.
- Parallactic angle:** Angle subtended between the **ecliptic circle** and an **altitude circle**.
- Parallax:** Apparent change in position of a nearby celestial object in the sky when it is viewed at different points on the Earth's surface.
- Pericenter:** Point on a **Keplerian orbit** that is closest to the central body. If the central body is the Sun then the pericenter is generally termed the *perihelion*. Likewise, if the central body is the Earth then the pericenter is termed the *perigee*.

**Precession of equinoxes:** A slow movement of the **vernal equinox** relative to the fixed stars that causes the **ecliptic longitude** of a fixed star to increase steadily at the rate of 50.3" per year.

**Prograde motion:** Motion of a **superior planet** in the sky in the same direction to that of its mean motion.

**Radial anomaly:** Difference between the length of the radius vector connecting the central body to the orbiting body, in a **Keplerian orbit**, and the **major radius**.

**Rate of motion in mean anomaly:** Time derivative of the **mean anomaly** of a **Keplerian orbit**.

**Rate of motion in mean argument of latitude:** Time derivative of the **mean argument of latitude** of a **Keplerian orbit**.

**Rate of motion in mean longitude:** Time derivative of the **mean longitude** of a **Keplerian orbit**.

**Retrograde motion:** Motion of a **superior planet** in the sky in the opposite direction to that of its mean motion.

**Right ascension:** Angle subtended at the Earth's center by the projection of the vector connecting a celestial body to the Earth's center onto the **celestial plane**, and the vector connecting the **vernal equinox** to the Earth's center. Right ascension increases counterclockwise (that is, from the west to the east) looking at the celestial plane from the north.

**Seasons:** Spring is the time interval between the **vernal equinox** and the **summer solstice**, summer the interval between the summer solstice and the **autumnal equinox**, autumn the interval between the autumnal equinox and the **winter solstice**, and winter the interval between the winter solstice and the next spring equinox.

**Sidereal day:** Time interval ( $23^h 56^m 4^s$ ) between successive upper **transits** of a fixed star.

**Sidereal hour:** One twenty-fourth of a **sidereal day**.

**Sidereal time:** Time calculated using the fixed stars.

**Solar day:** Time interval between successive **local noons**.

**Solar time:** Time calculated using the Sun.

**Solstices:** The two opposite points on the **ecliptic circle** that the Sun reaches on the longest and shortest days of the year.

**Station:** Point in the orbit of a **superior planet** at which it switches from **prograde** to **retrograde** motion, or vice versa. The former station is called a *retrograde station*, whereas the latter is called a *prograde station*.

**Summer solstice:** Most northerly point on the **ecliptic circle**.

**Superior planet:** A planet farther from the Sun than the Earth.

**Syzygy:** **Conjunction** or **opposition** of the Sun and the Moon.

**Transit:** On a given day, and at a given observation site on the Earth's surface, a celestial object is said to transit when it crosses the **meridian plane**. The object simultaneously attains either its highest or lowest altitude in the sky. The transit is called an upper/lower transit when the object attains its highest/lowest altitude.

**True anomaly:** Angle subtended at the central body by the radius vectors connecting the central body to the orbiting body, and the central body to the **pericenter**, in a **Keplerian orbit**.

**Tropics:** Two latitude circles on the Earth's surface that are equidistant from the equator. Between the tropics the Sun **culminates** both to the north and south of the **zenith** during the course of a year. Outside the tropics, the Sun culminates either only to the north or only to the south of the zenith.

**Universal time:** Time defined such that **mean local noon** coincides with 12:00 UT every day at an observation site of terrestrial longitude  $0^\circ$ .

**Vernal equinox:** Point at which the **ecliptic circle** crosses the **celestial equator** from south to north (in the direction of the Sun's apparent motion along the ecliptic).

**Winter solstice:** Most southerly point on the **ecliptic circle**.

**Year:** A *sidereal year* (365.256363004 days) is the time required for the Earth to complete an orbital rotation around the Sun relative to the fixed stars. A *tropical year* (365.2421875 days) is the time interval between successive passages of the Sun through the **vernal equinox**, and is 20.4 minutes shorter than a sidereal year.

**Zenith:** Point on the **celestial sphere** that is directly overhead at a given observation site on the Earth's surface.

**Zodiac:** The signs of the zodiac are conventional names given to  $30^\circ$  segments of the **ecliptic circle**.

## Index of symbols

|                                                     |                                                            |
|-----------------------------------------------------|------------------------------------------------------------|
| $A$ : azimuth.                                      | $M_S$ : mean anomaly of Sun.                               |
| $a$ : major radius, altitude.                       | $\mu$ : epicyclic anomaly, parallactic angle.              |
| $a_M$ : major radius of Moon's orbit.               | $n$ : rate of motion in mean longitude.                    |
| $a_S$ : major radius of Sun's orbit.                | $\tilde{n}$ : rate of motion in mean anomaly.              |
| $\alpha$ : right ascension.                         | $\tilde{n}$ : rate of motion in mean argument of latitude. |
| $\bar{\alpha}$ : right ascension of mean sun.       | $q$ : equation of center.                                  |
| $b$ : minor radius.                                 | $q_i$ : lunar longitudinal anomalies.                      |
| $\beta$ : ecliptic latitude.                        | $R_E$ : radius of Earth.                                   |
| $D$ : lunar-solar elongation.                       | $R_M$ : radius of Moon.                                    |
| $\bar{D}$ : mean lunar-solar elongation.            | $R_S$ : radius of Sun.                                     |
| $\tilde{D}$ : semi-mean lunar-solar elongation.     | $r$ : radial distance.                                     |
| $\delta$ : declination.                             | $\rho_S$ : angular radius of Sun.                          |
| $\delta_M$ : parallax of Moon.                      | $\rho_M$ : angular radius of Moon.                         |
| $e$ : eccentricity.                                 | $\rho_U$ : angular radius of Earth's umbra.                |
| $e_M$ : eccentricity of Moon's orbit.               | $t$ : time.                                                |
| $e_S$ : eccentricity of Sun's orbit.                | $t_0$ : epoch.                                             |
| $e$ : inclination of ecliptic to celestial equator. | $T$ : true anomaly.                                        |
| $E$ : elliptic anomaly.                             | $\tau$ : orbital period.                                   |
| $\zeta$ : radial anomaly.                           | $\varpi$ : longitude of perigee.                           |
| $\zeta_i$ : lunar radial anomalies.                 | $\Omega$ : longitude of ascending node.                    |
| $\theta$ : equation of epicycle.                    |                                                            |
| $F$ : argument of latitude.                         |                                                            |
| $\bar{F}$ : mean argument of latitude.              |                                                            |
| $\bar{F}_0$ : mean argument of latitude at epoch.   |                                                            |
| $\bar{F}_M$ : mean argument of latitude of Moon.    |                                                            |
| $\lambda$ : ecliptic longitude.                     |                                                            |
| $\lambda_S$ : ecliptic longitude of Sun.            |                                                            |
| $\bar{\lambda}$ : mean longitude.                   |                                                            |
| $\bar{\lambda}_0$ : mean longitude at epoch.        |                                                            |
| $\bar{\lambda}_M$ : mean longitude of Moon.         |                                                            |
| $\bar{\lambda}_S$ : mean longitude of Sun.          |                                                            |
| $L$ : terrestrial latitude.                         |                                                            |
| $M$ : mean anomaly.                                 |                                                            |
| $M_0$ : mean anomaly at epoch.                      |                                                            |
| $M_M$ : mean anomaly of Moon.                       |                                                            |



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